| SURFACE VEHICLE RECOMMENDED PRACTICE | J1939™-73 | DEC2024 |
|---|---|---|
| | Issued 1996-02<br>Revised 2024-12<br>Superseding J1939-73 AUG2022 | |
| Application Layer - Diagnostics | | |

# Application Layer - Diagnostics

## RATIONALE

This document has been revised to add CSERS PGs and their SPs in support of the November 2022 publication of 13 CCR 1971.1 in Table 1, to add recommendations for China 6 Non-Road in Table 1, to define SPN 3077 (Plug-in Hybrid Engine Ignition Cycle Counter) to DM57, to improve text for DM11, DM24, and DM58, and to define FMI 24 in Appendix A.

## FOREWORD

The SAE J1939 communications network is defined using a collection of individual SAE J1939 documents based upon the layers of the Open System Interconnect (OSI) model for computer communications architecture. The SAE J1939-73 document defines the SAE J1939 messages for diagnostic services for diagnostic information reporting and diagnostic repair, including services to satisfy regulated OBD requirements.

The SAE J1939 communications network is a high-speed ISO 11898-1 CAN-based communications network that supports real-time closed loop control functions, simple information exchanges, and diagnostic data exchanges between electronic control units (ECUs) physically distributed throughout the vehicle.

The SAE J1939 communications network is developed for use in heavy-duty environments and suitable for horizontally integrated vehicle industries. The SAE J1939 communications network is applicable for light-duty, medium-duty, and heavy-duty vehicles used on-road or off-road and for appropriate stationary applications that use vehicle-derived components (e.g., generator sets). Vehicles of interest include, but are not limited to, on-highway and off-highway trucks and their trailers, construction equipment, and agricultural equipment and implements. The physical layer aspects of SAE J1939 reflect its design goal for use in heavy-duty environments. Horizontally integrated vehicles involve the integration of different combinations of loose package components, such as engines and transmissions that are sourced from many different component suppliers. The SAE J1939 common communication architecture strives to offer an open interconnect system that allows the ECUs associated with different component manufacturers to communicate with each other.



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# 1. SCOPE

SAE J1939-73 defines the SAE J1939 messages to accomplish diagnostic services and identifies the diagnostic connector to be used for the vehicle service tool interface. Diagnostic messages (DMs) provide the utility needed when the vehicle is being repaired. Diagnostic messages are also used during vehicle operation by the networked electronic control modules to allow them to report diagnostic information and self-compensate as appropriate, based on information received. Diagnostic messages include services such as periodically broadcasting active diagnostic trouble codes, identifying operator diagnostic lamp status, reading or clearing diagnostic trouble codes, reading or writing control module memory, providing a security function, stopping/starting message broadcasts, reporting diagnostic readiness, monitoring engine parametric data, etc. California-, EPA-, or EU-regulated OBD requirements are satisfied with a subset of the specified connector and the defined messages.

# 2. REFERENCES

## 2.1 Applicable Documents

The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the latest issue of SAE publications shall apply.

### 2.1.1 SAE Publications

Available from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001, Tel: 877-606-7323 (inside USA and Canada) or +1 724-776-4970 (outside USA), www.sae.org.

SAE J1587 Electronic Data Interchange Between Microcomputer Systems in Heavy-Duty Vehicle Applications

SAE J1939 Serial Control and Communications Heavy-Duty Vehicle Network - Top Level Document

SAE J1939DA SAE J1939 Digital Annex

SAE J1939-03 On Board Diagnostics Implementation Guide

SAE J1939-5 Marine Stern Drive and Inboard Spark-Ignition Engine On-Board Diagnostics Implementation Guide

SAE J1939-13 Off-Board Diagnostic Connector

SAE J1939-21 Data Link Layer

SAE J1939-22 CAN FD Data Link Layer

SAE J1939-71 Vehicle Application Layer

SAE J1939-76 SAE J1939 Functional Safety Communications Protocol

SAE J1939-81 Network Management

SAE J1939-84 OBD Communications Compliance Test Cases for Heavy Duty Components and Vehicles

SAE J1979 E/E Diagnostic Test Modes

SAE J1979DA Digital Annex of E/E Diagnostic Test Modes

2.1.2 On-Board Diagnostics Regulations

California Code of Regulations, Title 13, Section 1956.8, California Standards and Test Procedures for New 2021 and Subsequent Model Heavy-Duty Zero-Emission Powertrains.

California Code of Regulations, Title 13, Section 1971, Engine Manufacturer Diagnostic System Requirements for 2007 and Subsequent Model-Year Heavy-Duty Engines.

California Code of Regulations, Title 13, Section 1971.1, On-Board Diagnostic System Requirements for 2010 and Subsequent Model-Year Heavy-Duty Engines (HD OBD). Year 2005/6 rule making.

California Code of Regulations, Title 13, Section 1971.1, On-Board Diagnostic System Requirements for 2010 and Subsequent Model-Year Heavy-Duty Engines (HD OBD). Year 2008/9 rule making.

California Code Regulations, Title 13, Section 1968.2, Malfunction and Diagnostic System Requirements for 2004 and Subsequent Model-Year Passenger Cars, Light-Duty Trucks, and Medium-Duty Vehicles and Engines (OBD II). California Air Resources Board (CARB) Publications are available from the Air Resources Board, Haagen-Smit Laboratory, 9528 Telstar Avenue, El Monte, CA 91731-2990.

California Code Regulations, Title 13, Section 1968.2, Malfunction and Diagnostic System Requirements for 2004 and Subsequent Model-Year Passenger Cars, Light-Duty Trucks, and Medium-Duty Vehicles and Engines (OBD II). California Air Resources Board (CARB) Publications are available from the Air Resources Board, Haagen-Smit Laboratory, 9528 Telstar Avenue, El Monte, CA 91731-2990. Year 2005/6 rule updates.

California Code Regulations, Title 13, Section 1968.2, Malfunction and Diagnostic System Requirements for 2004 and Subsequent Model-Year Passenger Cars, Light-Duty Trucks, and Medium-Duty Vehicles and Engines (OBD II). California Air Resources Board (CARB) Publications are available from the Air Resources Board, Haagen-Smit Laboratory, 9528 Telstar Avenue, El Monte, CA 91731-2990. Year 2008/9 rule updates.

European Directive 2005/55/EC of the European Parliament and of the Council of 28 September 2005 as implemented by Commission Directive 2005/78/EC and amended by Commission Directive 2006/51/EC.

European Directive 98/69/EC as amended by 99/102/EC, 2001/1/EC, 2001/100/EC and 2002/80/EC (Vehicles <7600 pounds).

European Union EU Reg 595/2009 and its implementing regulations, including Regulation 49, 06 Series of Amendments (these define EURO VI emissions and HD OBD, includes motor vehicles above 3500kg (i.e., M2, M3, N2, N3)).

European Union Heavy Duty OBD (Vehicles above 7600 pounds) 29 November 2005 (EC Directive 2005/78).

U.S. EPA, Title 40, CFR 86.005-17 On-Board Diagnostics, July 1, 2011 (OBD for Engines - for Vehicles 8500 to 14000 pounds).

U.S. EPA, Title 40, CFR 86.007-17 On-Board Diagnostics for Engines Used in Applications Less Than or Equal to 14000 pounds GVWR, July 1, 2011.

U.S. EPA, Title 40, CFR 86.010-18 On-Board Diagnostics for Engines Used in Applications Greater than 14000 pounds GVWR, July 1, 2011.

U.S. EPA, Title 40, CFR 86.1806-05 On-Board Diagnostics for Vehicles Less Than or Equal to 14000 pounds GVWR, July 1, 2011.

U.S. EPA, Title 40, CFR 86.1806-10 (OBD for Vehicles - 8500 to 14000 pounds), December 4, 2008.

World Wide Harmonized OBD proposed Global Technical Regulation Draft by the Economic Commission for Europe, reference ECE/TRANS/WP.29/GRPE/2006/8/Rev.1/27, March 2006.

China IV, HJ 1014-2020 Emissions Control Technical Requirements of Non-Road Diesel Mobile Machinery.

## 3. DEFINITIONS

Terms and definitions not found in this section are defined in SAE J1939 or in specific OBD regulations.

### 3.1 ACTIVE

The state used to indicate that a fault is currently occurring. A fault cannot be simultaneously transmitted as both “active” and “previously active.”

### 3.2 BROADCAST

Messages that are sent on a periodic basis without having to be solicited. In some cases, broadcasts can be normally off and solicited to come on, and then stay on until they are solicited to turn off (see DM13).

### 3.3 CALIBRATION

The software installed in a control module. This includes executable code and calibration data.

### 3.4 CONFIRMED

Confirmed is a term that is used to describe a category of DTCs where the diagnostic condition has been determined to be true. For example, when a system declares a malfunction is present, then it publishes it in DM1 as a confirmed and active DTC. When the system declares a malfunction is no longer present, then it publishes it in DM2 as a confirmed and previously active DTC. To satisfy OBD regulations, there can be additional requirements defined in a market regulation. In the U.S., the additional considerations led to the bifurcation of confirmed faults into the DM12 and DM23 displays.

### 3.5 CONFIRMED AND ILLUMINATING THE MIL (MIL-On DTC)

This terminology applies to a DTC that has matured to the point that it illuminates the MIL. Regulations do require that a DTC that is no longer active remain in this state for a certain number of drive cycles after the fault condition becomes inactive. These DTCs are reported in the DM12, as well as in DM1.

### 3.6 CONTINUOUSLY MONITORED SYSTEMS

Continuously monitored systems are those that are sampled at least two times per second. Note that some continuous monitors can require many conditions to be true before monitoring can be performed.

### 3.7 DIAGNOSTIC TROUBLE CODE

A 4-byte value that identifies the condition, the associated failure mode indicator (FMI), and its occurrence count.

### 3.8 DISCRIMINATORY

Discriminatory is a term used to differentiate the type of OBD operator illumination scheme that is used by the vehicle when controlling the malfunction indicator lamp (MIL or MI). Discriminatory and non-discriminatory operator illumination schemes for the MIL are defined in the WWH OBD global technical regulation. A non-discriminatory scheme will have the MIL-On continuously for all OBD malfunctions, while the discriminatory MIL scheme will only have the MIL-On continuous for malfunctions thought to cause the emissions to exceed the OBD emissions threshold.

### 3.9 FREEZE FRAME

A snapshot of a group of data stream signals based on the occurrence of a diagnostic trouble code.

### 3.10 KEY

The result of a set of mathematical operations performed upon a seed to provide a device with a means of authenticating a tool’s request.

### 3.11 MALFUNCTION INDICATOR LAMP

The MIL is used to report trouble codes that are emissions related. Trouble codes that are not emissions related shall not illuminate the MIL.

### 3.12 MEMORY ACCESS

This defines a set of messages (DM14 through DM18) and outlines the operational procedures for a tool (or device) wishing to read or write the memory, or storage space, of a device with or without data security.

### 3.13 NON-CONTINUOUSLY MONITORED SYSTEMS

System monitors that run once a trip or whenever conditions exist. Trip, in this context, is as defined by OBD regulations. It should be noted that there will be monitors that won't run every trip; e.g., cold-start aid monitors only run when the ambient temperature is below 10 °C (50 °F).

### 3.14 OBD DEVICE

This is a device that has declared in DM5 an OBD compliance value that indicates it satisfies OBD requirements. An OBD device can be a primary or secondary device. See Table H1.

### 3.15 OBJECT

Some entity within a memory and/or a space.

### 3.16 PASSWORD

The number sent when using a simple authentication technique wherein both the device and tool have a prior knowledge of the specific number and usually use equality as the verification.

### 3.17 PENDING

Pending defines a category of multi-trip DTCs. Pending DTC means a DTC that is stored by the OBD system, or diagnostic system, because a monitor has detected a malfunction may be present during the current or last completed driving cycle, but the malfunction has not matured to an active or confirmed category. For regulated systems, the definition of a pending DTC will vary according to the regulatory guidance.

### 3.18 PERMANENT

Permanent is a term used to label a confirmed and active (MIL-On) DTC that is recorded in memory and is not allowed to be erased by the OBD system until the monitoring algorithm has fully executed (i.e., has executed the minimum number of checks necessary for MIL illumination) and determined the malfunction is no longer present. Refer to California Code of Regulation 1971.1(d).

### 3.19 POINTER

A term used to label a device that identifies the memory location that should be read or written. Types of pointers include: direct memory address and directed spatial addressing (suspect parameter number [SPN] space, OEM proprietary space, and reserved to be assigned).

### 3.20 POTENTIAL

See PENDING.

### 3.21 PORT

Physical connection point(s) from a control module to a specific communications link (see DM13).

3.22 PREVIOUSLY ACTIVE

The state used to indicate that a fault has occurred but is not presently occurring. A fault cannot be simultaneously transmitted as both “active” and “previously active.”

3.23 RATIONALITY

Rationality fault diagnostic for an input component means verification of the accuracy of the input signal while in the range of what is physically possible (see Appendix A, region b versus regions f and g) when compared to all other available information. This is a term that the California Air Resources Board has defined in their OBD rules.

3.24 READINESS CODE

The readiness code status bits (i.e., those in DM5) are used in part or in whole to ensure a vehicle (engine) is ready for an emission system inspection. The definition of the complete or incomplete status is provided in regulation(s), including the impact of specific service actions on readiness status. If a status bit for a particular component or system is set to complete, then the OBD system has had the opportunity to run all of the diagnostics relevant to that particular component or system.

For example, on engine start-up, it is likely that an EGR monitor might require operation at speed and torque points beyond the idle condition. Therefore, if the system had its diagnostic trouble codes erased and then the engine was restarted, the readiness code for EGR would not be set until the condition for the EGR monitor tests had been executed and a pass/fail determination has been made. It is also expected that other tests, not just the EGR monitors, would need to be performed before the readiness code could be set. Once set to complete, the readiness code status bits shall not change to incomplete each time the vehicle (engine) is powered down.

See Appendix J for more detail.

3.25 SEED

A number sent by a device to a tool to obtain authentication of the tool’s right to access the device.

3.26 SPACE

A memory region containing a collection of objects.

3.27 USER_LEVEL

A number sent by a tool to a device along with an initial request to inform the device of some specific level of access that the tool wishes to gain.

4. ABBREVIATIONS

| | |
|---|---|
| CAL ID | Calibration Identification |
| CARB | California Air Resources Board |
| CM | SPN Conversion Method |
| CMAC | Cipher-Based Message Authentication Code |
| CVN | Calibration Verification Number |
| DM1 | Diagnostic Message 1, Active Diagnostic Trouble Codes (DTCs) |
| DM2 | Diagnostic Message 2, Previously Active DTCs |
| DM3 | Diagnostic Message 3, Diagnostic Data Clear/Reset for Previously Active DTCs |
| DM4 | Diagnostic Message 4, Freeze Frame Parameters |
| DM5 | Diagnostic Message 5, Diagnostic Readiness 1 |
| DM6 | Diagnostic Message 6, Emission-Related Pending DTCs |
| DM7 | Diagnostic Message 7, Command Non-Continuously Monitored Test |
| DM8 | Diagnostic Message 8, Test Results for Non-Continuously Monitored Systems |

| | |
|---|---|
| DM9 | Diagnostic Message 9, Oxygen Sensor Test Results |
| DM10 | Diagnostic Message 10, Non-Continuously Monitored Systems Test Identifiers Support |
| DM11 | Diagnostic Message 11, Diagnostic Data Clear/Reset for Active DTCs |
| DM12 | Diagnostic Message 12, Emission-Related MIL-On DTCs |
| DM13 | Diagnostic Message 13, Stop-Start Broadcast |
| DM14 | Diagnostic Message 14, Memory Access Request |
| DM15 | Diagnostic Message 15, Memory Access Response |
| DM16 | Diagnostic Message 16, Binary Data Transfer |
| DM17 | Diagnostic Message 17, Boot Load Data |
| DM18 | Diagnostic Message 18, Data Security |
| DM19 | Diagnostic Message 19, Calibration Information |
| DM20 | Diagnostic Message 20, Monitor Performance Ratio |
| DM21 | Diagnostic Message 21, Diagnostic Readiness 2 |
| DM22 | Diagnostic Message 22, Individual Clear/Reset of Active and Previously Active DTC |
| DM23 | Diagnostic Message 23, Emission Related Previously MIL-On DTCs |
| DM24 | Diagnostic Message 24, SPN Support |
| DM25 | Diagnostic Message 25, Expanded Freeze Frame |
| DM26 | Diagnostic Message 26, Diagnostic Readiness 3 |
| DM27 | Diagnostic Message 27, All Pending DTCs |
| DM28 | Diagnostic Message 28, Emission-Related Permanent DTCs |
| DM29 | Diagnostic Message 29, DTC Counts |
| DM30 | Diagnostic Message 30, Scaled Test Results |
| DM31 | Diagnostic Message 31, DTC to Lamp Association |
| DM32 | Diagnostic Message 32, Regulated Exhaust Emission Level Exceedance |
| DM33 | Diagnostic Message 33, Emission Increasing Auxiliary Emission Control Device Active Time |
| DM34 | Diagnostic Message 34, NTE Status |
| DM35 | Diagnostic Message 35, Immediate Fault Status |
| DM36 | Diagnostic Message 36, Harmonized Roadworthiness - Vehicle (HRWV) |
| DM37 | Diagnostic Message 37, Harmonized Roadworthiness - System (HRWS) |
| DM38 | Diagnostic Message 38, Harmonized Global Regulation Description (HGRD) |
| DM39 | Diagnostic Message 39, Harmonized Cumulative Continuous Malfunction Indicator - System (HCMI) |
| DM40 | Diagnostic Message 40, Harmonized B1 Failure Counts (HB1C) |
| DM41 | Diagnostic Message 41, DTCs- A, Pending |
| DM42 | Diagnostic Message 42, DTCs- A Confirmed and Active |
| DM43 | Diagnostic Message 43, DTCs- A, Previously Active |
| DM44 | Diagnostic Message 44, DTCs- B1, Pending |
| DM45 | Diagnostic Message 45, DTCs- B1, Confirmed and Active |
| DM46 | Diagnostic Message 46, DTCs- B1, Previously Active |
| DM47 | Diagnostic Message 47, DTCs- B2, Pending |
| DM48 | Diagnostic Message 48, DTCs- B2, Confirmed and Active |
| DM49 | Diagnostic Message 49, DTCs- B2, Previously Active |
| DM50 | Diagnostic Message 50, DTCs- C, Pending |
| DM51 | Diagnostic Message 51, DTCs- C, Confirmed and Active |
| DM52 | Diagnostic Message 52, DTCs- C, Previously Active |
| DM53 | Diagnostic Message 53, Active Service Only DTCs |
| DM54 | Diagnostic Message 54, Previously Active Service Only DTCs |
| DM55 | Diagnostic Message 55, Diagnostic Data Clear/Reset for All Service Only DTCs |

| | |
|---|---|
| DM56 | Diagnostic Message 56, Model Year and Certification Engine Family |
| DM57 | Diagnostic Message 57, OBD Information |
| DM58 | Diagnostic Message 58, Rationality Fault SP Data |
| DM59 | Diagnostic Message 59, Diagnostic Readiness 4 |
| DM60 | Diagnostic Message 60, Extended DTC Information |
| DTC | Diagnostic Trouble Code |
| EDC | Error Detection and/or Correction |
| EDCP | Error Detection and/or Correction Parameter |
| EI-AECD | Emissions Increasing Auxiliary Emissions Control Device |
| FMI | Failure Mode Indicator |
| FTP | Federal Test Procedure |
| GHG | Greenhouse Gas |
| GTR | Global Technical Regulation |
| MA | Memory Access |
| MI | Malfunction Indicator |
| MIL | Malfunction Indicator Lamp |
| NA | Not Applicable |
| OBD | On-Board Diagnostics |
| OBD II | On-Board Diagnostics II |
| OC | Occurrence Count |
| PG | Parameter Group |
| PGN | Parameter Group Number |
| PID | Parameter Identifier (SAE J1587 or SAE J1979) |
| SPN | Suspect Parameter Number |
| TID | Test Identifier |
| VVT | Variable Valve Timing and/or Control |
| WWH | World Wide Harmonized |
| ZEP | Zero Emissions Powertrain |
| ZEV | Zero Emissions Vehicle |

Refer to SAE J1939 for any terms and/or definitions not found in this document.

## 5. TECHNICAL REQUIREMENTS

### 5.1 General

The diagnostic definitions provided herein are intended to satisfy the needs of all potential users of the SAE J1939 network. These definitions are intended to be suitable for applications in any of the industry groups defined within SAE J1939. A broad range of capabilities are provided with provision made for future growth. Additional features, parameter groups, and parameter definitions will be defined over time; it is anticipated that this document will continuously evolve as long as the SAE J1939 network is an active recommended practice. Such growth will be implemented in such a way as to ensure backward compatibility with earlier versions. At the time of initial publication, many of these growth areas are identified but are yet to be defined. Such identification is provided so that the reader will be aware of those additions that are already planned for the document.

### 5.2 Overview of Diagnostic Requirements

The diagnostic requirements necessary to provide the type of capability our customers, our industry, and the regulatory bodies are demanding are outlined in 5.2.1. A description of the minimum requirements needed to satisfy regulatory requirements is contained in 5.2.2. A discussion of the general operating conditions for diagnostic procedures is defined in 5.2.3.

### 5.2.1 Diagnostic Capabilities Envisioned

The following capabilities will be defined in this and future publications of this document:

a. Security: Define a security scheme to be used on the serial data link that allows the industry standard service tools to be able to perform tasks that are necessary during service procedures. This will include accessing diagnostic information, accessing vehicle configuration information, and recalibrating control modules.

b. Connectors: Define the connector to be used for connection to the vehicle SAE J1939 network for service tools. The diagnostic connector is defined in SAE J1939-13.

c. Diagnostic Status Message Support: Provide a set of messages that allows the reading of fault information, clearing of fault information, monitoring of vehicle parameters, access to vehicle and component configuration, and other related information.

d. Diagnostic Test Support: Provide a capability that allows the service tool to put the various controllers into specific test modes in order to determine proper sub-system operation.

### 5.2.2 Suggested Diagnostic Support

#### 5.2.2.1 Emission Related Components

As a minimum capability, all controllers using SAE J1939 that must comply with regulated on-board diagnostics (OBD, OBD II, HD OBD, or EOBD, etc.) shall support the functions shown in Table 1. Additionally, these controllers must satisfy the requirements in SAE J1939-03 if required by the regional authority. See Table 2 for the legend defining the entries in the “Required by Regulation” column of Table 1.

***Table 1 - Emission-related component PGN and SPN support***

| Row # | Function | PGN | Acronym | SPN # | Required by Regulation (see Table 2) | Description |
|---|---|---|---|---|---|---|
| **Diagnostic Services** | | | | | | |
| 1 | Read DTCs and Lamps (MIL, RSL, AWL, Protect) | 65226 | DM1 | | A, B, C, E, F, G, H, J, K, L, N, O, P | All active DTCs and some lamps (MIL, RSL, AWL, protect) |
| 2 | Read DTCs | 65236 | DM12 | | A, B, H, J, K, L, N, P | Emission-related MIL-On DTCs |
| 3 | Read DTCs | 64949 | DM23 | | B, H, J, K, L, N, P[1] | Emission-related previously MIL-On DTCs |
| 4 | Read Pending DTCs | 65231 | DM6 | | A, B, E, F, G, H, J, K, L, N | Emission related pending DTCs |
| 5 | Read All Pending DTCs | 64898 | DM27 | | | All pending DTCs inclusive of emissions and non-emissions related |
| 6 | Read Permanent DTCs | 64896 | DM28 | | H, J | Emission related permanent DTCs |
| 7 | Read DTC Counts | 40448 | DM29 | | H, J, P | Number of regulated DTC counts (pending, permanent, MIL-On, PMIL-On) |
| 8 | Read DTCs- A, Pending | 64863 | DM41 | | L, N | DTCs where emissions exceed OBD threshold |
| 9 | Read DTCs- A, Confirmed and Active | 64862 | DM42 | | L, N | DTCs where emissions exceed OBD threshold |

| Row # | Function | PGN | Acronym | SPN # | Required by Regulation (see Table 2) | Description |
|---|---|---|---|---|---|---|
| 10 | Read DTCs- A, Previously Active | 64861 | DM43 | | L, N | DTCs where emissions exceed OBD threshold |
| 11 | Read DTCs- B1, Pending | 64860 | DM44 | | L, N | DTCs where emissions may exceed OBD threshold |
| 12 | Read DTCs- B1, Confirmed and Active | 64859 | DM45 | | L, N | DTCs where emissions may exceed OBD threshold |
| 13 | Read DTCs- B1, Previously Active | 64858 | DM46 | | L, N | DTCs where emissions may exceed OBD threshold |
| 14 | Read DTCs- B2, Pending | 64857 | DM47 | | L, N | DTCs where emissions do not exceed OBD threshold |
| 15 | Read DTCs- B2, Confirmed and Active | 64856 | DM48 | | L, N | DTCs where emissions do not exceed OBD threshold |
| 16 | Read DTCs- B2, Previously Active | 64855 | DM49 | | L, N | DTCs where emissions do not exceed OBD threshold |
| 17 | Read DTCs- C, Pending | 64854 | DM50 | | L, N | DTCs where emissions do not exceed emission standard |
| 18 | Read DTCs- C, Confirmed and Active | 64853 | DM51 | | L, N | DTCs where emissions do not exceed emission standard |
| 19 | Read DTCs- C, Previously Active | 64852 | DM52 | | L, N | DTCs where emissions do not exceed emission standard |
| 20 | Read DTCs- Immediate | 40704 | DM35 | | | Instantaneous status of diagnostic results |
| 21 | Read DTCs and Timers- | 41472 | DM32 | | K, N, P | DTCs and timers where OBD emissions threshold exceeded (e.g., NOx exceedance) |
| 22 | Command Test | 58112 | DM7 | | H, J, K, L, N, P[1] | Commanded Test; system, device, or component |
| 22a | Unscaled Test Results | 65232 | DM8 | | A, C, E, F, G | Test results |
| 22b | Scaled Test Results | 41984 | DM30 | | H, J, K, L, P[1] | Test results scaled |
| 23 | DTC to Lamp Association | 41728 | DM31 | | | Contains info supplementary to DM1 that can be requested in order to provide lamp info associated with each DTC |
| 24 | Clear DTCs | 65235 | DM11 | | A, B, C, E, F, G, H, J, K, L, N, O, P | Clear diagnostic information and active/previously active DTCs |
| 25 | Clear DTCs | 65228 | DM3 | | C | Clear diagnostic information and previously active DTCs |

[1] Recommended for China Non Road by SAE T&B Control and Communication Network J1939-73 Task Force

| Row # | Function | PGN | Acronym | SPN # | Required by Regulation (see Table 2) | Description |
|---|---|---|---|---|---|---|
| 26 | Freeze Frame Data | 65229 | DM4 | | A, C, E, F, G, K | Freeze frame definition and support (fixed format, DTC, and six specified parameters) |
| 27 | SPN Support | 64950 | DM24 | | B, H, J, L, N, O, P | SP support for data stream, expanded freeze frame, test results, and rationality fault SP data value |
| 28 | Expanded Freeze Frame | 64951 | DM25 | | B, H, J, L, N, P[1] | Expanded freeze frame (format allows DTC and manufacturer-specified number parameters) |
| 29 | Diagnostic Readiness | 65230 | DM5 | | A, B, C, E, F, G, H, J, K, L, N, O, P | OBD compliance, previously active, active DTC count, monitors supported, and their status (diagnostic readiness) |
| 30 | Diagnostic Readiness for this Trip | 64952 | DM26 | | B, E, H, J, N[2] | Monitors supported and their status for this trip |
| 31 | Monitor Performance Ratio | 49664 | DM20 | | B, H, J, N | Indicates how often monitors complete compared to vehicle operation |
| 32 | Emission Increasing - AECD Active Time | 41216 | DM33 | | H, J | Engine emissions increasing AECDs and associated timers (not required HD EPA) |
| 33 | NTE Status | 40960 | DM34 | | H, J | Engine emissions not-to-exceed status |
| 34 | Harmonized Roadworthiness - Vehicle | 64868 | DM36 | | L, N | Vehicle road worthiness status |
| 35 | Harmonized Roadworthiness - System | 64867 | DM37 | | L, N | Engine emissions road worthiness status |
| 36 | Harmonized Global Regulation Description | 64866 | DM38 | | L, N | Text description of WWH OBD version |
| 37 | Harmonized Continuous Cumulative Continuous Malfunction Indicator - System | 64865 | DM39 | | L, N | Cumulative MIL time and DTC-B1 time (largest) |
| 38 | Harmonized B1 Failure Counts | 64864 | DM40 | | L, N | DTCs-B1 and individual timers |
| 39 | Rationality Fault SP Data | 64475 | DM58 | | H, J | SP data value for an SP with a rationality diagnostics fault |
| 40 | Diagnostic Readiness 4 | 64300 | DM59 | | L, N | Continuous-MI time while engine operating, time elapsed since DTCs were erased |
| **Communication Services** | | | | | | |

[2] Only phases C and D of N.

| Row # | Function | PGN | Acronym | SPN # | Required by Regulation (see Table 2) | Description |
|---|---|---|---|---|---|---|
| 41 | Communication | 59904 | RQST | | A, B, C, E, F, G, H, J, K, L, N, O, P | Request PG |
| 42 | Communication | 59392 | ACKM | | A, B, C, E, F, G, H, J, K, L, N, O, P | Acknowledgment message |
| 43 | Communication | 60416 | TP.CMxx | | A, B, C, E, F, G, H, J, K, L, N, O, P | Transport protocol connection management |
| 44 | Communication | 60160 | TP.DT | | A, B, C, E, F, G, H, J, K, L, N, O, P | Transport protocol data transfer |
| **Data Stream** | | | | | | |
| 45 | Data Stream | 65260 | VI | 237 | A, B, C, E, F, G, H, J, K, L, N, O, P[1] | Vehicle identification number (VIN) |
| 46 | Data Stream | 54016 | DM19 | 1635 1634[1] | A, B, C, E, F, G, H, J, K, L, N, O, P | Calibration identification and calibration verification number (CAL ID and CVN) |
| 47 | Data Stream | 65262 | ETI | 110 | A, B, C, E, F, G, H, J, K, L, N, P | Engine coolant temperature |
| 48 | Data Stream | 65265 | CCVS | 84 | A, B, C, E, F, G, H, J, K, L, N, O, P | Wheel-based vehicle speed |
| 49 | Data Stream | 65270 | IC1 | 102 | A, B, C, E, F, G, H, J, K, L, N, P[1] | Engine intake manifold #1 pressure |
| 50 | Data Stream | 65270 | IC1 | 105 | A, B, C, E, F, G, H, J, K, L, N, P[1] | Engine intake manifold 1 temperature |
| 51 | Data Stream | 61443 | EEC2 | 91 | A, B, C, E, F, G, H, J, K, L, N, O, P[1] | Accelerator pedal position 1 |
| 52 | Data Stream | 61443 | EEC2 | 92 | A, B, C, E, F, G, H, J, K, L, N, P | Engine percent load at current speed |
| 53 | Data Stream | 61444 | EEC1 | 513 | A, B, C, E, F, G, H, J, K, L, N, P | Actual engine - percent torque |
| 54 | Data Stream | 61444 | EEC1 | 190 | A, B, C, E, F, G, H, J, K, L, N, P | Engine speed |
| 55 | Data Stream | 61444 | EEC1 | 899 | A, B, C, E, F, G, H, J, K, L, N, P[1] | Engine torque mode |
| 56 | Data Stream | 65159 | IT6 | 1436 | A, B, E, F, G, H, J, K, L, N, P[1] | Engine actual ignition timing |
| 57 | Data Stream | 49408 | DM21 | 3069 | A, B, E, F, G, H, J, K, L, P[1] | Distance traveled while MIL is activated |

| Row # | Function | PGN | Acronym | SPN # | Required by Regulation (see Table 2) | Description |
|---|---|---|---|---|---|---|
| 58 | Data Stream | 49408 | DM21 | 3294 | B, H, J, K, L, P[1] | Distance since diagnostic trouble codes cleared |
| 59 | Data Stream | 49408 | DM21 | 3295 | B, H, J, K, L, N, P[1] | Minutes run by engine while MIL is activated |
| 60 | Data Stream | 49408 | DM21 | 3296 | B, H, J, K, L, N, P[1] | Time since diagnostic trouble codes cleared |
| 61 | Data Stream | 64952 | DM26 | 3301 | B, H, J, K, L, N, P[1] | Time since engine start |
| 62 | Data Stream | 64952 | DM26 | 3302 | B, H, J, K, L, N, P[1] | Number of warm-ups since diagnostic trouble codes cleared |
| 63 | Data Stream | 64891 | AT1S1 | 5466 | H, J | Aftertreatment 1 diesel particulate filter soot load regeneration threshold |
| 64 | Data Stream | 60928 | AC | 2848 | H, J | Name of controller application |
| 65 | Data Stream | Various | | 5919 | O | Additional data stream parameters are captured in SAE J1939DA Appendix D under the SPN reference of SPN 5919 - HVESS Voltage Level |
| | Note: This and other SPNs are reported in DM24 to indicate the sets of SPNs to be supported by the HD ZEP. | | | | | |
| 66 | Data Stream | Various | | | B, H, J, K, L, N, P | Additional data stream parameters are required for OBD compliance, but they are dependent on the emission solution used for certification; these parameters are in SAE J1939 and/or SAE J1939-71 and/or SAE J1939DA |
| 67 | Data Stream | 65279 | OI | 5825 | P[3] | Driver Warning System Indicator Status |
| 68 | Data Stream | 64459 | EEPCSI | 8887 | P | Exhaust Particulate Control System Diagnostic Time |
| 69 | Data Stream | 64459 | EEPCSI | 8888 | P | Exhaust Particulate Control System Diagnostic Count |
| 70 | Data Stream | 65269 | AMB | 108 | H[3], J[3], P | Barometric Pressure |
| **Data Stream [2024MY]** | | | | | | |
| 71 | Data Stream | 65214 | EEC4 | 189 | H, J | Engine Rated Speed |
| 72 | Data Stream | 64711 | DM56 | 5845 | H, J | Certification Engine Family Name |
| 73 | Data Stream | 64841 | O2FT1 | 4236 | H, J | Short-term Fuel Trim - Bank 1 |
| 74 | Data Stream | 64841 | O2FT1 | 4240 | H[3], J[3] | Engine Exhaust Bank 1 O2 Sensor Closed Loop Operation |
| 75 | Data Stream | 64841 | O2FT1 | 4237 | H[3], J[3] | Engine Exhaust Bank 1 O2 Sensor Closed Loop. |
| 76 | Data Stream | 64706 | HSS1 | 7315 | H, J | Propulsion System Active |
| 77 | Data Stream | 61450 | EGF1 | 12758 | H, J | Target Fresh Air Mass Flow |
| 78 | Data Stream | 65263 | EFL/P1 | 101 | H, J | Engine Crankcase Pressure 1 |
| 79 | Data Stream | 64916 | EEC7 | 5444 | H, J | Engine Crankcase Breather Oil Separator Speed |
| 80 | Data Stream | 64251 | ESPC | 12744 | H, J | EVAP System Purge Pressure |
| 81 | Data Stream | 64247 | AT1HI3 | 12765 | H, J | Aftertreatment 1 Total Engine Runtime With No DEF Delivery |

[3] As an alternative to SP 1213.

| Row # | Function | PGN | Acronym | SPN # | Required by Regulation (see Table 2) | Description |
|---|---|---|---|---|---|---|
| 82 | Data Stream | 64920 | AT1HI1 | 3726 | H, J | Aftertreatment 1 Diesel Particulate Filter Total Number of Passive Regenerations |
| 83 | Data Stream | 64214 | EEC4 | 166 | H, J | Engine Rated Power |
| 84 | Data Stream | 64247 | AT1HI3 | 12766 | H, J | Aftertreatment 1 Total Time SCR Inlet Below 200 Celsius |
| 85 | Data Stream | 61691 | A1SCRDSR3 | 8442 | H, J | Aftertreatment 1 Diesel Exhaust Fluid Doser Valve 1 Command |
| 86 | Data Stream | 64488 | ASI2 | 12749 | H, J | Aftertreatment 1 DEF Dosing Mode |
| 87 | Data Stream | 64031 | ASI6 | 22719 | H, J | Aftertreatment 2 DEF Dosing Mode |
| 88 | Data Stream | 61475 | A1SCRDS1 | 4331 | H, J | Aftertreatment 1 Diesel Exhaust Fluid Actual Dosing Quantity |
| 89 | Data Stream | 64488 | ASI2 | 12748 | H, J | Aftertreatment Diesel Exhaust Fluid Use This Operating Cycle |
| 90 | Data Stream | 64250 | ATNXMS1 | 12753 | H, J | Aftertreatment 1 SCR 1 Target NH3 Storage Level |
| 91 | Data Stream | 64250 | ATNXMS1 | 12752 | H, J | Aftertreatment 1 SCR 1 Modeled/Actual Storage Level |
| 92 | Data Stream | 64830 | A1SCREGT1 | 4360 | H, J | Aftertreatment 1 SCR Intake Temperature |
| 93 | Data Stream | 64830 | A1SCREGT1 | 4363 | H, J | Aftertreatment 1 SCR Outlet Temperature |
| 94 | Data Stream | 61454 | AT1IG1 | 3220 | H, J | Engine Exhaust 1 NOx 1 Reading Stable |
| 95 | Data Stream | 61457 | AT2OG1 | 3269 | H, J | Aftertreatment 2 Outlet NOx 1 Reading Stable |
| 96 | Data Stream | 61450 | EGF1 | 2659 | H, J | Engine Exhaust Gas Recirculation 1 Mass Flow Rate |
| 97 | Data Stream | 37376 | HCDI1 | 5505 | H, J | Requested Fuel Mass Rate |
| 98 | Data Stream | 64488 | ASI2 | 12743 | H, J | Hydrocarbon Doser Duty Cycle |
| 99 | Data Stream | 64929 | AT1FC1 | 3480 | H, J | Aftertreatment 1 Fuel Pressure 1 |
| 100 | Data Stream | 61450 | EGF1 | 132 | H[4], J[3], P | Engine Intake Air Mass Flow Rate |
| 101 | Data Stream | 65110 | AT1T1I1 | 1761 | H[3], J[3], P | Aftertreatment 1 Diesel Exhaust Fluid Tank Volume |
| 102 | Data Stream | 65266 | LFE1 | 183 | H, J, P | Engine Fuel Rate |
| 103 | Data Stream | 64916 | EEC7 | 27 | H[3], J[3], P | Engine Exhaust Gas Recirculation 1 Valve Position |
| 104 | Data Stream | 64981 | EEC5 | 2791 | H[3], J[3], P | Engine Exhaust Gas Recirculation 1 Valve 1 Control 1 |
| 105 | Data Stream | 61454 | AT1IG1 | 3216 | H[3], J[3], P | Engine Exhaust 1 NOx 1 |
| 106 | Data Stream | 61454 | AT1IG1 | 3219 | H[3], J[3], P[1] | Aftertreatment 1 Intake Gas Sensor 1 at Temperature |
| 107 | Data Stream | 64657 | EEN | 6579 | H[3], J[3], P[1] | Engine Exhaust NOx |
| 108 | Data Stream | 61455 | AT1OG1 | 3226 | H[3], J[3], P | Aftertreatment 1 Outlet NOx 1 |
| 109 | Data Stream | 61674 | AT1OG3 | 7660 | H[3], J[3], P[1] | Aftertreatment 1 Outlet NOx 2 |
| 110 | Data Stream | 64946 | AT1IMG | 3251 | H, J, P | Aftertreatment 1 Diesel Particulate Filter Differential |

[4] If equipped.

| Row # | Function | PGN | Acronym | SPN # | Required by Regulation (see Table 2) | Description |
|---|---|---|---|---|---|---|
| 111 | Data Stream | 65247 | EEC3 | 514 | H, J, P | Nominal Friction - Percent Torque |
| 112 | Data Stream | 65251 | EC1 | 544 | H, J, P | Engine Reference Torque |
| 113 | Data Stream | 65279 | OI | 5825 | P[5] | Driver Warning System Indicator Status |
| 114 | Data Stream | 64459 | EEPCSI | 8887 | P | Exhaust Particulate Control System Diagnostic Time |
| 115 | Data Stream | 64459 | EEPCSI | 8888 | P | Exhaust Particulate Control System Diagnostic Count |
| 116 | Data Stream | 65276 | DD1 | 96 | P, H[5], J[6] | Fuel Level 1 |
| 117 | Data Stream | 64247 | AT1HI3 | 12759 | H, J | Aftertreatment 1 DPF Regen Complete Starting Engine Vehicle Distance 1 |
| 118 | Data Stream | 64247 | AT1HI3 | 12760 | H, J | Aftertreatment 1 DPF Regen Complete Ending Engine Vehicle Distance 1 |
| 119 | Data Stream | 64247 | AT1HI3 | 12761 | H, J | Aftertreatment 1 DPF Regen Complete Starting Engine Vehicle Distance 2 |
| 120 | Data Stream | 64247 | AT1HI3 | 12762 | H, J | Aftertreatment 1 DPF Regen Complete Ending Engine Vehicle Distance 2 |
| 121 | Data Stream | 64247 | AT1HI3 | 12763 | H, J | Aftertreatment 1 DPF Regen Complete Starting Engine Vehicle Distance 3 |
| 122 | Data Stream | 64247 | AT1HI3 | 12764 | H, J | Aftertreatment 1 DPF Regen Complete Ending Engine Vehicle Distance 3 |
| 123 | Data Stream | 64706 | HSS1 | 7315 | H | Propulsion System Active |
| 124 | Data Stream | 64706 | HSS1 | 7898 | H | External Energy Source Connection Status |
| 125 | Data Stream | 61584 | HVESSD1 | 5919 | H | HVESS Voltage Level |
| 126 | Data Stream | 61584 | HVESSD1 | 5920 | H | HVESS Current |
| **Data Stream - NOx Binning and GHG Tracking [2022 MY]** | | | | | | |
| 127 | Data Stream | 64258, 64259, 64260, 64261, 64262, 64263, 64264, 64265, 64266, 64267, 64268, 64269, 64270, 64271, 64272, 64273, 64274, 64275, 64276, 64277, 64278, 64279 | NTFCEA, NTEHEA, NTVMEA, NTEEEA, NTFCV, NTEHV, NTVMV, NTEEV, NTENV, NTSNV, NTFCS, NTEHS, NTVMS, NTEES, NTENS, NTSNS, NTFCA, NTEHA, NTVMA, NTEEA, NTENA, NTSNA | 12675 | H | NOx tracking |

[5] As an alternative to SP 1213.
[6] Only if used to enable/disable other diagnostics.

| Row # | Function | PGN | Acronym | SPN # | Required by Regulation (see Table 2) | Description |
|---|---|---|---|---|---|---|
| | Note: SPN 12675 (NOx Tracking Engine Activity Lifetime Fuel Consumption Bin 1 - Total) shall be reported in “Supported Parameter SPN” in DM24 to indicate NOx bin tracking messages are supported. | | | | | |
| 128 | Data Stream | 64255, 64256, 64257 | GHGTTS, GHGTTA, GHGTTL | 12691 | H | Greenhouse gas (GHG) technology tracking |
| | Note: Only SPN 12691 (GHG Tracking Lifetime Active Technology Index) shall be reported in “Supported Parameter SPN” in DM24 to indicate GHG technology tracking is supported. | | | | | |
| 129 | Data Stream | 64252, 64253, 64254 | GHGTL, GHGTS, GHGTA | 12730 | H | Greenhouse gas (GHG) tracking |
| | Note: Only SPN 12730 (GHG Tracking Lifetime Engine Run Time) shall be reported in “Supported Parameter SPN” in DM24 to indicate GHG tracking is supported. | | | | | |
| 130 | Data Stream | 64241, 64242, 64243 | PSATL PSATS PSATA | 12797 | H | Greenhouse gas (GHG) tracking specific to hybrids |
| | Note: Only SPN 12797 (Hybrid Lifetime Propulsion System Active Time) shall be reported in “Supported Parameter SPN” in DM24 to indicate GHG tracking for hybrids PSA PGs is supported. | | | | | |
| 131 | Data Stream | 64244, 64245, 64246 | HCDIOL HCDIOS HCDIOA | 12783 | H | Greenhouse gas (GHG) tracking specific to hybrids |
| | Note: Only SPN 12783 (Hybrid Lifetime Distance Traveled in Charge Depleting Operation with Engine off) shall be reported in “Supported Parameter SPN” in DM24 to indicate GHG tracking of hybrid plug-in PGs are supported. | | | | | |
| 132 | Data Stream | 64019, 64020 | CSERSC CSERSA | 22227 | H | Cold Start Emissions Reduction Strategy: Current Cycle and Average Data |
| | Note: Only SPN 22227 (Current Cycle Catalyst Heat Energy Until FTP Cold Start Tracking Time) shall be reported in “Supported Parameter SPN” in DM24 to indicate CSERS tracking for 2028MY era Diesel engines. | | | | | |

*Table 2 - Regulation definition legend*

| Identifier | Regulating Body | OBD Compliance (SPN 1220) (see Table 6) |
|---|---|---|
| A | CARB CCR 1968.1, (Vehicles <14000 pounds), April 21, 2003 | 01 |
| B | CARB CCR 1968.2, (Vehicles <14000 pounds), April 21, 2003 | 01 or 03 |
| C[7] | CARB CCR 1971, Engine Manufacturer Diagnostics (EMD), (Vehicles >14000 pounds), August 2004, support recommended by manufacturers | 17 |
| D | EU (Directive 98/69/EC as amended by 99/102/EC, 2001/1/EC, 2001/100/EC and 2002/80/EC) (Vehicles <7600 pounds) | 6 or 7 or 8 or 9 |
| E | EU (July 2003 Planned Audit to Directive 88/77/EEC) (Vehicles >7600 pounds) | 14 (Euro IV)<br>15 (Euro V) |
| F | EPA, Title 40, CFR 86.005-17, (Engines 8500 to 14000 pounds), December 28, 2000 | 02 or 03 |
| G | EPA, Title 40, CFR 86.1806-05, (Vehicles 8500 to 14000 pounds), December 28, 2000 | 02 or 03 |
| H | Title 13, California Code of Regulations, Section 1971.1, On-Board Diagnostic System Requirements for 2010 and Subsequent Model-Year Heavy-Duty Engines (HD OBD)<br>NOTE: The letter H covers the 2008, 2009, 2010, 2012, 2016, 2019, 2021, and 2022 publications by ARB, as well as EPA's 2008 and 2011 OBD updates to Title 40, CFR 86.1806-10, CFR 86.007-17, and CFR 86.010-18. | 18[8] or 19 or 20 or 34 or 35 |
| J | Title 13, California Code of Regulations, Section 1968.2, Malfunction and Diagnostic System Requirements for 2004 and Subsequent Model-Year Passenger Cars, Light-Duty Trucks, and Medium-Duty Vehicles and Engines (OBD II): Additions published in 2006, 2009, 2012, 2016, 2019, 2021, and 2022. | 22 or 34 |
| K | Directive 2005/55/EC of the European Parliament and of the Council of 28 September 2005 as implemented by Commission Directive 2005/78/EC and amended by Commission Directive 2006/51/EC | 23 |
| L | World Wide Harmonized On-Board Diagnostics, refer to: ECE/TRANS/WP.29/GRPE/2006/8/Rev.1/27 March 2006 | 21 |
| M | Refer to SAE J1939-5 for the specific regulation being complied to and the compliance details. | 25 |
| N | EU Reg 595/2009 and its implementing regulations, including Regulation 49, 06 Series of Amendments (these define EURO VI emissions and HD OBD). Identifier covers Euro VI phases A, B, C, and D, except when indicated otherwise. | 26 |
| O | Title 13, California Code of Regulations, Section 1956.8. California standards and test procedures for new 2021 and subsequent model heavy-duty zero-emission powertrains. | 27 |
| P | China IV, HJ 1014-2020 Emissions control technical requirements of non-road diesel mobile machinery | 50 |

[7] EMD does not mandate or require the use of any SAE J1939 services. However, due to the lack of data link standardization requirements, the SAE J1939 Committee recommends support of these services for EMD.

[8] For EMD-plus engines, the recommended services noted by letter H in Table 1 show a voluntary path for meeting service tool data link requirements.

#### 5.2.2.2 Non-Emission Related Components

To be determined in later revisions of this document.

### 5.2.3 General Conditions for Diagnostic Procedures

These guidelines are necessary to ensure proper operation of both the test equipment and the vehicle during diagnostic procedures. Test equipment, when using messages defined in this document, should not affect normal operation of the vehicle, except when that is the express purpose of the message.

The off-board test equipment may request data without knowledge of which module on the vehicle will respond. In this case, the SAE J1939-21 request PG would be directed to the global destination address for the desired information. Additionally, there are times where the desired information can be known to only be available from a specific device. When this is the case, the information flow is better managed with a request to a specific address other than "global." These guidelines should be followed in order to reduce network traffic. In some vehicles, multiple controllers can respond with the information requested. In addition, a single module can send multiple responses to a single request. Any test device requesting information shall, therefore, have provisions for receiving multiple responses.

The on-board systems should respond to a request as defined in SAE J1939-21. With multiple responses possible from a single request, this allows as much time as is necessary for all modules to access the data link and transmit their response(s). If there is no response within this time period (i.e., 0.25 second), the tool can either assume no response will be received or, if a response has already been received, that no more responses will be received.

A tool should always wait for a response from the previous request or "no response" time-out before sending another request. In no case should a request be sent in less than the time-out interval specified in SAE J1939-21 after the previous request. There are situations where the tool knows that it has received the desired information it needs; it may proceed to its next operation.

Destination-specific requests require a response. If a parameter group is not supported by the control module and a destination-specific request was used to request it from the control module, a NACK is required (refer to SAE J1939-21 PGN 59392). If the request for the parameter group was sent to a global destination address and a given device does not support it, then that device shall not NACK the request, unless granted an exception in this document or in SAE J1939-21. Response criteria for retry logic is identified in SAE J1939-21. Response to DM3 and DM11 requests for products subject to ARB 13 CCR 1971.1 deviate from SAE J1939-21 requirements as noted in the respective PG descriptions.

A tool shall expect that ECU resources may not permit an immediate response to a request for multipacket PGs. In particular, when a BAM session is active, a second BAM cannot be sent until the first is completed. For CTS/RTS responses, the ECU can have limited resources to handle multiple point to point transport protocol sessions. The control byte value of three or "busy" is used in the ACKNOWLEDGMENT PG when resource contention occurs.

Unless otherwise specified in SAE J1939-73, parameter values should be formatted in accordance with the parameter ranges as defined by SAE J1939-71, 5.1.3 and 5.1.4. The specific parameter definitions come from SAE J1939DA.

In this document, hexadecimal numbers are represented by a subscript lowercase letter "h" ($_h$), binary numbers are represented by a subscript lowercase letter "b" ($_b$), and decimal numbers have no subscript. In cases where the number base is obvious, the subscript is not included.

## 5.3 Security

One of the purposes of this recommended practice is to provide a standard protocol (a set of capabilities or diagnostic services) to allow users to access and modify memory areas inside a controller on the network. For these tools to be supported by the manufacturers of the ECUs that will be designed to connect to the SAE J1939 network, sufficient protection against "unauthorized" modifications shall be included. The messages described in 5.7.14 (DM14) though 5.7.18 (DM18) and their subsections are to be used for this purpose.

This security shall not be used to limit access to the capabilities defined in 5.7.1 (DM1) through 5.7.13 (DM13) and 5.7.19 (DM19) but is intended to allow manufacturers to limit the data that can be accessed by the user. The security systems outlined here represent a recommendation for ECU manufacturers and provide flexibility for them to tailor individual systems to their specific security needs. The vehicle modules addressed are those that are capable of having solid-state memory contents altered by an external command sent through this vehicle communication link. Improper memory content alteration could potentially damage the electronics, reduce the vehicle's compliance to legislated requirements, or breach the vehicle manufacturer's security interests.

Proper "unlocking" of the controller shall be a prerequisite to access certain critical on-board controller functions. Access to the on-board controller while in a "locked" mode is permitted only as determined by the controller's manufacturer. This can require that the user obtain specific codes or passwords directly from the manufacturer's representative and "unlocking" may only be possible when using product-specific software. This permits the controller to protect itself from unauthorized access.

The messages in 5.7.14 (DM14) through 5.7.18 (DM18) do not attempt to define capability as a requirement for any controller or to specify what information should be subject to any specific security measures; these decisions are left to the controller manufacturer. Implementation of the security system shall not prevent basic diagnostic communications between an external tool and the on-board controller.

Appendices C, D, and E contain additional information that will help implementers understand the intended use of these security processes for gaining access to controller memory and the several different modes available for limiting access areas of that memory.

## 5.4 Diagnostic Connector

The diagnostic connector is defined in SAE J1939-13.

## 5.5 Parameter Monitoring Requirements

The parameter definitions shall be those of the referenced SAE J1939DA. Any parameter that has been defined in an applications layer document and is included in a parameter group (PG) shall be used for diagnostics. Therefore, if a parameter has already been defined, it will not be redefined for diagnostic purposes. In some cases, it will be necessary to identify a closely related parameter, such as the value of the accelerator pedal sensor reading when the failure occurred rather than the current reading of the accelerator pedal sensor.

## 5.6 Diagnostic Trouble Code Definition

A diagnostic trouble code (DTC) is made up of four independent fields, as follows:

a. Suspect Parameter Number (SPN): 19 bits

b. Failure Mode Identifier (FMI): 5 bits

c. Occurrence Count (OC): 7 bits

d. SPN Conversion Method (CM): 1 bit

These independent parameters are not used together to form a number. They are merely a set of information that helps in understanding the failure that is being reported.

A diagnostic tool may also want to use the controller source address and the name to determine which controller is reporting the diagnostic information. This information is not needed to interpret the SPN, but it can be beneficial to have during the diagnostic process. Refer to SAE J1939 for the source address and name definitions.

Diagnostic trouble codes are transmitted as 4 bytes per trouble code. Those 4 bytes are interpreted as defined in 5.7.1. In an effort to provide continuity between the diagnostics defined in SAE J1587 to that of SAE J1939-73, the fault encoding format remains very similar. When possible, SAE J1587 PID numbers have been mapped one for one as SPNs.

Examples of diagnostic trouble codes (see Table 3):

EXAMPLE 1: This is an SAE J1587 parameter.

| | |
|---|---|
| SPN=91 | Suspect parameter is accelerator pedal position |
| FMI=3 | Failure mode is identified as voltage above normal |
| OC=5 | Occurrence count indicates trouble has occurred five times |
| CM= $0_b$ | (1 bit) |

EXAMPLE 2: This is not a parameter communicated as an SAE J1587 PID. Therefore, it is assigned a number above 511.

| | |
|---|---|
| SPN=656 | Suspect parameter is engine injector number six |
| FMI=3 | Failure mode is identified as voltage above normal |
| OC=2 | Occurrence count indicates trouble has occurred two times |
| CM= $0_b$ | (1 bit) |

EXAMPLE 3: Diagnostic trouble code as transmitted in diagnostic messages (e.g., DM1).

Given:
Parameter “pre-filter oil pressure,” suspect parameter number 1208
Failure mode identifier of three
Occurrence count of 10
SPN conversion method of zero
All fields of DTC sent in intel format (least significant byte first)

| | Decimal | Hexadecimal | Binary |
|---|---|---|---|
| SPN | 1208 | = $4B8_h$ | = 000 00000100 $10111000_b$ (19 bits) |
| FMI | 3 | = $3_h$ | = $00011_b$ (5 bits) |
| OC | 10 | = $A_h$ | = $0001010_b$ (7 bits) |
| CM | | | = $0_b$ (1 bit) |

***Table 3 - DTC representation in CAN data frame for DM1 (byte 3 closer to CAN identifier)***

<table>
<tr><td></td><td colspan="32">DTC</td></tr>
<tr><td></td><td colspan="8">Byte 3<br>8 least significant bits of SPN<br>(bit 8 most significant)</td><td colspan="8">Byte 4<br>second byte of SPN<br>(bit 8 most significant)</td><td colspan="8">Byte 5<br>3 most significant bits of SPN and the FMI (bit 8 SPN msb and bit 5 FMI msb)</td><td colspan="8">Byte 6</td></tr>
<tr><td></td><td colspan="19">SPN</td><td colspan="5">FMI</td><td>CM</td><td colspan="7">OC</td></tr>
<tr><td>J1939 Frame Format</td><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td></tr>
<tr><td></td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr>
</table>

## 5.7 Diagnostic Parameter Group Definitions

This section contains definitions of those parameter groups that will be used specifically for diagnostics. The format is a little different than SAE J1939-71 in that the parameter definitions will follow each parameter group definition as a subsection under that parameter group.

Parameter groups specified in SAE J1939-73 shall set the CAN data length code to 8 bytes. Unused bytes shall be set to 255 (FFh).

One of the goals of this diagnostic document is to satisfy the OBD requirements. One of the documents that contains many of the OBD requirements is SAE J1979. For that reason, Table 4 was created as a way of identifying how SAE J1939 satisfies the SAE J1979 requirements.

A summary listing of all diagnostic modes and PIDs from SAE J1979 and their corresponding SAE J1939 PGs is provided (see Table 4).

***Table 4 - Summary of diagnostic mode assignments***

| | SAE Motor Vehicle Council SAE J1979 Functions | | | SAE Truck and Bus Council SAE J1939 Support of Those Functions | |
|---|---|---|---|---|---|
| **Row #** | **SAE J1979 Description** | **SAE J1979 Mode** | **SAE J1979 PID** | **SAE J1939 DM (PGN)** | **PG Description** |
| 1 | Supported PIDs | $01_h$ request<br>$41_h$ response | 00 | DM24<br>(64950) | Systems supporting DM24 declare their emissions-related support for parametric data and the DM25 freeze frame.<br><br>SAE J1939-21 and SAE J1939-71 discuss methods to NACK data requests and to indicate non-supported data for systems that do not support DM24. |
| 2 | Number of DTCs, MIL status, and diagnostic monitors supported and their status | $01_h$ request<br>$41_h$ response | 01 | DM5<br>(65230) | OBD compliance, previously active and active DTC count, monitors supported, and their status (diagnostic readiness). |
| 3 | Parameters related to the engine operation | $01_h$ request<br>$41_h$ response | 3 to $1B_h$ | Various PGs | Normally provided PGs will be used to retrieve these parameters; for example, SAE J1939DA PGN 61444 contains engine speed. |
| 4 | Determine OBD type supported (OBD II-CARB, OBD-Federal, OBD and OBD II, OBD 1, other) | $01_h$ request<br>$41_h$ response | $1C_h$ | DM5<br>(65230) | Tells which OBD support is provided. |
| 5 | PIDs supported in freeze frame | $02_h$ request<br>$42_h$ response | 00 | DM4<br>(65229) | Freeze frame definition and support covered in DM4. |
| 6 | DTC that caused freeze frame | $02_h$ request<br>$42_h$ response | 02 | DM4<br>(65229) | Freeze frame PG tells what DTC caused it. |
| 7 | PID data value in freeze frame record | $02_h$ request<br>$42_h$ response | 03 to $FF_h$ | DM4<br>(65229) | Freeze frame PG contains all parameters (more than one freeze frame can be supported). |
| 8 | Emission-related DTCs (SAE J1979 service $03_h$ provides the union of DM12 and DM23 results) | $03_h$ request<br>$43_h$ response | 01 | DM12<br>(65236) | Emission-related MIL-On DTCs and lamp status information. |
| 9 | | | | DM1<br>(65226) or DM12<br>(65236) | Active DTCs and lamp status information with DM1 or with DM12 when required. |
| 10 | | | | DM2<br>(65227) or DM23<br>(64949) | Previously active DTCs and lamp status information or with DM23 when required. |

| | SAE Motor Vehicle Council SAE J1979 Functions | | | SAE Truck and Bus Council SAE J1939 Support of Those Functions | |
|---|---|---|---|---|---|
| Row # | SAE J1979 Description | SAE J1979 Mode | SAE J1979 PID | SAE J1939 DM (PGN) | PG Description |
| 11 | Clear/reset emission-related diagnostic information | $04_h$ request<br>$44_h$ response | NA | DM11<br>(65235) | Clear diagnostic information for active DTCs. |
| 12 | | | | DM3<br>(62228) | Clear diagnostic information for previously active DTCs. |
| 13 | Oxygen sensor monitoring test results | $05_h$ request<br>$45_h$ response | NA | DM9<br>(65233) | No planned message format definition. |
| 14 | On-board monitoring test results for specific monitored systems | $06_h$ request<br>$46_h$ response | NA | DM10<br>(65234) | Test IDs supported. |
| 15 | | | | DM7<br>(58112) | Invoke test. |
| 16 | | | | DM8<br>(65232) | Test results (see DM30 for scaled test results). |
| 17 | Emission-related DTCs detected during current or last completed driving cycle | $07_h$ request<br>$47_h$ response | NA | DM6<br>(65231) | Test results for pending DTCs. |
| 18 | Request control of on-board system, test, or component | $08_h$ request | NA | DM7<br>(58112) | Command on-board system, test, or component. |
| 19 | | $48_h$ response | | DM8<br>(65232) | Results commanded system, test, or component. |
| 20 | Calibration identification | $09_h$ request<br>$49_h$ response | Infotype $04_h$ | DM19<br>(54016) | Calibration information (bytes 5 to 20 are CAL ID). |
| 21 | Calibration verification number | $09_h$ request<br>$49_h$ response | Infotype $06_h$ | DM19<br>(54016) | Calibration information (bytes 1 to 4 are CVN). |
| 22 | Monitor performance ratio | $09_h$ request<br>$49_h$ response | Infotype $08_h$ | DM20<br>(49664) | Indicates how often monitors complete compared to vehicle operation. |
| 23 | Distance traveled while MIL is activated,<br>distance since DTCs cleared,<br>engine run time while MIL is activated,<br>engine run time since DTCs cleared | $01_h$ request<br>$41_h$ response | $21_h$<br>$31_h$<br>$4D_h$<br>$4E_h$ | DM21<br>(49408) | Diagnostic readiness 2, reports the diagnostic information relevant to a second PG conveying diagnostic readiness. |
| 24 | Previously active emission-related faults (SAE J1979 service $03_h$ provides the union of DM12 and DM23 results) | $03_h$ request<br>$43_h$ response | | DM23<br>(64949) | This DM contains DTCs that are confirmed but for which the MIL is off. |
| 25 | SPN support | $01_h$ request<br>$41_h$ response | $00_h$ | DM24<br>(64950) | This message is used to identify those SPNs supported by the product for test results, freeze frames, and data stream messages. |

| Row # | SAE Motor Vehicle Council SAE J1979 Functions<br>SAE J1979 Description | SAE J1979 Mode | SAE J1979 PID | SAE Truck and Bus Council SAE J1939 Support of Those Functions<br>SAE J1939 DM (PGN) | PG Description |
|---|---|---|---|---|---|
| 26 | Freeze frame data; DTC, and PID data values in freeze frame record | $02_h$ request<br>$42_h$ response | $02_h$ to $FF_h$ | DM25<br>(64951) | Freeze frame message providing more parameter support than the existing DM4. |
| 27 | Continuously monitored systems enabled/completed status:<br>time since engine start, number of warm-ups since DTCs cleared,<br>non-continuously monitored systems enabled status,<br>non-continuously monitored systems completed status | $01_h$ request<br>$41_h$ response | $41_h$<br>$1F_h$<br>$30_h$ | DM26<br>(64952) | Diagnostic readiness 3, conveys the pending status of OBD system monitors for the current drive cycle. |
| 28 | All pending DTCs | None | None | DM27<br>(64898) | This DM conveys all pending DTCs inclusive of emissions and non-emissions related. |
| 29 | Emission-related DTCs with permanent status | $0A_h$ request<br>$4A_h$ response | | DM28<br>(64896) | This DM conveys the number of permanent DTCs currently in this category. |
| 30 | Regulated DTC counts | Count included in individual DTC service responses | | DM29<br>(40448) | This DM conveys the number of regulated DTC counts (pending, permanent, MIL-On, PMIL-On). |
| 31 | Scaled test results | $05_h$ request<br>$4A_h$ response | | DM30<br>(41984) | This DM conveys test results scaling. |
| 32 | DTC to lamp association | None | None | DM31<br>(41728) | This message provides added capability of sharing information about the lamp color associated with each DTC being transmitted in a multiple DTC scenario. |

5.7.1 Active Diagnostic Trouble Codes (DM1)

DM1 provides diagnostic lamp status and diagnostic trouble codes (DTCs). Together, the lamp and DTC information convey the diagnostic condition of the transmitting electronic component to other components on the network. Occurrence counts may be provided as described in 5.7.1.15. Some additional requirements regarding the lamp status values reported in DM1 are provided in 5.7.1.3.

Requirements for DM1 reporting information vary based on whether the DM1 is from an OBD-regulated product. These requirements are further explained in 5.7.1.1 and 5.7.1.2.

Transmission Rate: A DM1 message shall be transmitted, regardless of the presence or absence of any DTC, once every second and on state change. To prevent a high message rate due to intermittent faults that have a very high frequency, it is recommended that no more than one state change per DTC per second be transmitted. For example, if a fault has been active for 1 second or longer, and then becomes inactive, a DM1 message shall be transmitted to reflect this state change. If a different DTC changes state within the 1-second update period, a new DM1 message is transmitted to reflect this new DTC.

Thus, a DTC that becomes active/inactive twice within a 1-second interval, such as shown in Example Case 1, would have one message identifying the DTC becoming active and one at the next periodic transmission identifying it being inactive. This message shall be sent every second or in response to a request. Note that this parameter group will require using the "multipacket transport" parameter group (refer to SAE J1939-21) when more than one active DTC exists.

DM1 shall be broadcast at 1.0 Hz rate, even when there are no active faults. This permits instrumentation to detect the loss of the lamp information and take appropriate action.

Data Length: Variable
Extended Data Page: 0
Data Page: 0
PDU Format: 254
PDU Specific: 202
Default Priority: 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used)
Parameter Group Number: 65226 (00FECA$_h$)

| | | | |
|---|---|---|---|
| Byte: 1 | bits 8-7 | Malfunction indicator lamp | see 5.7.1.4 |
| | bits 6-5 | Red stop lamp | see 5.7.1.5 |
| | bits 4-3 | Amber warning lamp | see 5.7.1.6 |
| | bits 2-1 | Protect lamp | see 5.7.1.7 |
| Byte: 2 | bits 8-7 | Flash malfunction indicator lamp | see 5.7.1.8 |
| | bits 6-5 | Flash red stop lamp | see 5.7.1.9 |
| | bits 4-3 | Flash amber warning lamp | see 5.7.1.10 |
| | bits 2-1 | Flash protect lamp | see 5.7.1.11 |
| Byte: 3 | bits 8-1 | SPN, 8 least significant bits of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: 4 | bits 8-1 | SPN, second byte of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: 5 | bits 8-6 | SPN, 3 most significant bits (most significant at bit 8) | see 5.7.1.12 |
| | bits 5-1 | FMI (most significant at bit 5) | see 5.7.1.13 |
| Byte: 6 | bit 8 | SPN conversion method | see 5.7.1.14 |
| | bits 7-1 | Occurrence count | see 5.7.1.15 |

NOTE: When the occurrence count is not available, it shall be set to all ones, which is a value of 127.

NOTE: When there is zero or one DTC to report, then unused bytes 7 and 8 of the CAN frame shall be set to 255 (as per SAE J1939-71).

NOTE: When there are no DTCs to report, each of bytes 3 through 6 shall be set to zero.

EXAMPLE 1: The following illustrates the message format for when there is more than one DTC.

Given:

a = lamp status
b = SPN
c = FMI
d = CM and OC

Message form will be as follows: a,b,c,d,b,c,d,b,c,d,b,c,d....etc. In this example, the transport protocol of SAE J1939-21 will have to be used to send the information because it requires more than 8 data bytes. Actually, any time there is more than one fault, the services of the transport protocol will have to be used.

EXAMPLE 2: The following illustrates the message format for when a request of the DM1 is made and there are zero active faults or when there are zero active faults and the message is being transmitted at its regular 1-second interval. The currently defined lamps (malfunction indicator lamp, red stop lamp, amber warning lamp, and protect lamp) should reflect the present state of the transmitting electronic component.

The recommended setting for bytes 3 through 6 is shown below. The required setting shall be used for engines and vehicles complying with government-regulated requirements (e.g., OBD, OBD II, EOBD, or HD OBD) and for any post-2006 product. The original publication of this recommended practice defined that bytes 3 through 6 should be set to all ones when there are zero faults. This particular implementation is no longer permitted. It provides context for existing implementations prior to the adoption of the recommended setting (March 2004). Use of all ones is shown as the legacy setting below but are no longer permitted.

Given:

| | | | | |
|---|---|---|---|---|
| Byte: | 1 | bits 8-7 | = 00 (example of reporting off) | |
| | | bits 6-5 | = 00 (example of reporting “off) | |
| | | bits 4-3 | = 00 (example of reporting off) | |
| | | bits 2-1 | = 00 (example of reporting off) | |
| Byte: | 2 | bits 8-7 | = 11 (example of reporting not available/don’t care) | |
| | | bits 6-5 | = 11 (example of reporting not available/don’t care) | |
| | | bits 4-3 | = 11 (example of reporting not available/don’t care) | |
| | | bits 2-1 | = 11 (example of reporting not available/don’t care) | |
| | | | **Legacy Setting**[9] | **Post 2004 Setting**[10] |
| Bytes: | 3-6 | | SPN = 524,287 - Indicates not available | = 0 |
| | | | FMI= 31 - Indicates not available | = 0 |
| | | | OC = 127- Indicates not available | = 0 |
| | | | CM = 1 - Indicates not available | = 0 |
| Byte: | 7 | | = 255 | = 255 |
| Byte: | 8 | | = 255 | = 255 |

Bytes 7 and 8 are shown set to 255 to demonstrate padding a 6-byte message with 2 bytes. Bytes 7 and 8, in this example, are not part of the repeating sequence for multiple diagnostic trouble codes.

EXAMPLE 3: Three cases are enumerated as follows to define the transmission rate requirements (see Figure 1). These diagrams are not intended to show how an OBD malfunction is to be handled (e.g., pending, confirmed, permanent, etc.) but does show the DM1 transmission rate requirements. See Appendix H for more signal charts related to DM6, DM12, DM23, and DM28 display sequencing.

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[9] Note that the recommendation that the legacy setting is no longer permitted has been documented in SAE J1939-73 since at least March 2004. As a result, no controller developed since 2006 is expected to use the legacy setting.
[10] Required for post-2006 products. Preferred for post-2004 products.

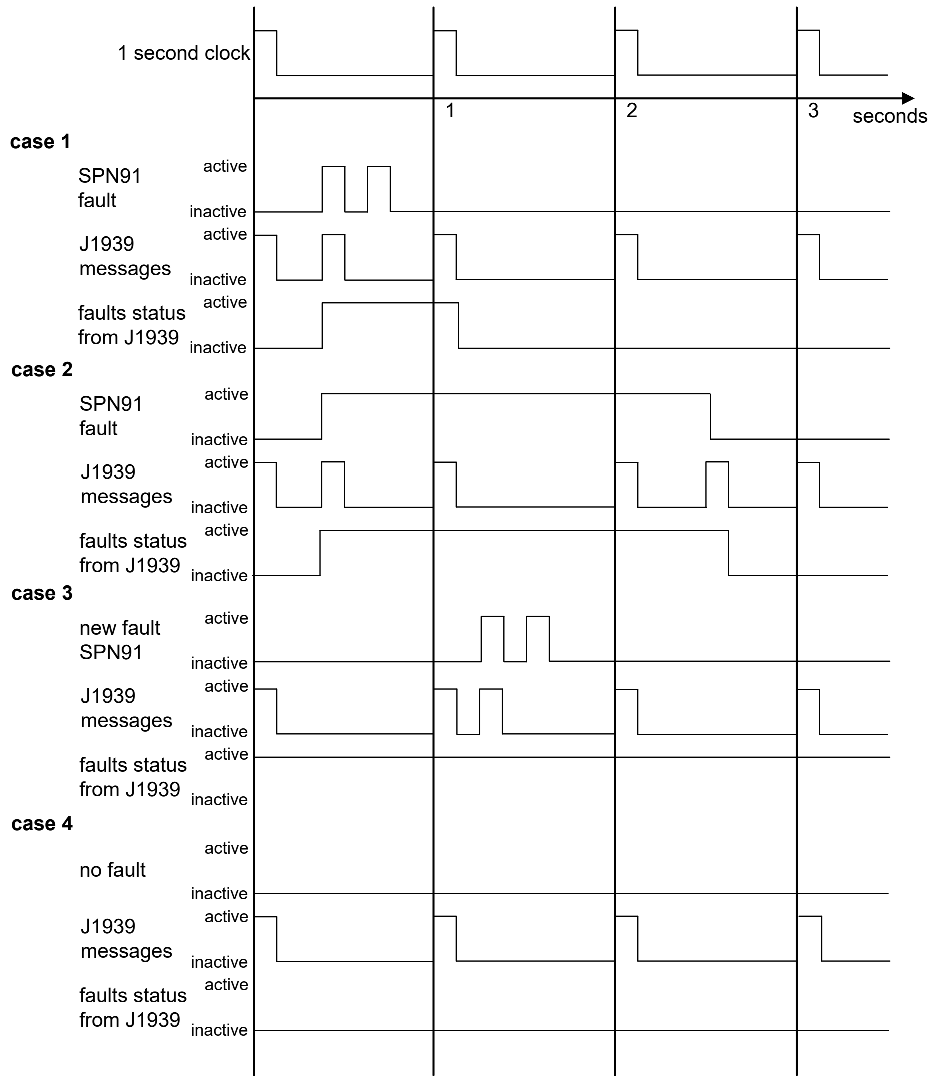


***Figure 1 - Defining the transmission rate requirements***

Case 1 illustrates that not every transition of a fault (active to inactive or inactive to active) results in a SAE J1939 message being sent. In this case, there are no other faults active when the example SPN 91 fault occurs. The SPN 91 fault is the accelerator pedal position parameter, which has an update faster than once a second. Therefore, the "SAE J1939 message" (DM1 message) will be sent every 1 second while this fault is active. Three observations should be made. First, note that the first SAE J1939 message is sent when the "SPN 91 fault" becomes active on the first occurrence and not when it goes inactive for the first occurrence or active/inactive for the second occurrence. The inactive state is sent once at the next normal 1-second update (T = 1 second). The second observation is that the "SAE J1939 message" (DM1) is required to be sent at the 1-second interval even though the fault is no longer active and the actual DM1 message will contain no active faults. This is done as the action to show the fault went away. The way this is done for this specific case (where there are no longer any active faults) is as shown in the preceding Example 2. If there were other active faults, they would have been sent in this message. The third observation is that if the second SPN 91 would have been a different SPN it would have been sent prior to the 1 second in a DM1 sent in between normal 1-second updates. The 1-second interval message would not contain this new SPN or SPN 91, assuming they both transitioned on and off before the 1-second message. Therefore, the 1-second DM1 message would still contain no faults.

Case 2 illustrates that the transition states can occur between the normal 1-second intervals. Therefore, a "SAE J1939 message" is sent in between time equals 0 and time equals 1 to indicate that the SPN 91 fault has gone active. It is sent per the normal 1-second update at the 1-second and 2-second points. It is sent at the time between 2 seconds and 3 seconds to convey the transition to the inactive state. To do this, the "SAE J1939 message" (DM1) is sent as shown in the preceding Example 2.

Case 3 shows the situation where there are already active faults in existence when SPN 91 becomes active. Note that the transition of SPN 91 to active state is sent between the 1-second and 2-second points. The message contains all active faults, not just the new one. The transition to the inactive state is sent during the normal 2-second update. This message would contain all active faults, and since SPN 91 went inactive, it would not be in this message.

Case 4 shows the situation where there are no faults in existence. The “SAE J1939 Message” is sent at the 1-second interval.

#### 5.7.1.1 DM1 DTC Information from Non-OBD-Regulated Devices

The information communicated by non-OBD-regulated systems is limited to the appropriate diagnostic lamp status and the currently active DTCs. SPN 3038 and SPN 1213 should be set to Not Available ($11_b$) for non-OBD-regulated systems. Non-OBD-regulated systems that are currently transmitting $00_b$ for SPN 3038 and SPN 1213 should transmit $11_b$ in future software updates.

#### 5.7.1.2 DM1 DTC Information from OBD-Regulated Devices

The information communicated is limited to diagnostic lamp status, the MIL-On emissions-related DTCs, and optionally the non-emissions-related currently active DTCs. The MIL-On emissions-related OBD DTCs for DM1 should match the content in the DM12 message. It is not required that MIL-On emissions-related DTCs precede non-emissions-related DTCs.

#### 5.7.1.3 DM1 Lamp Status

The defined lamps (malfunction indicator lamp, red stop lamp, amber warning lamp, and protect lamp) comprise a component’s lamp status, and the lamp status shall be conveyed at all times. Typically, when one of the lamps is indicated as on, DM1 will contain a DTC that explains the lamp. In some cases, the DTC can be reported by another DM instead of DM1. However, the component controlling the actual lamp illumination must consider the status from all components that provide these lamps before changing the display to the operator. The lamp information (malfunction indicator lamp, red stop lamp, amber warning lamp, and protect lamp) should reflect the present state of the transmitting electronic component. DM1 shall not convey temporary signals to provide for lamp test illumination, DTC numeric displays, or OBD readiness indication at key-on. When there are multiple DTCs with different lamp command requirements (e.g., SPN1213 is for the MIL) and lamp flash (e.g., SPN3038 is for the flash MIL), then the DTC with the MIL and fast flash takes priority over MIL with slow flash, which takes priority over the short MIL, which takes priority over the class C.

There are uses for additional lamp definitions to accomplish specific functions (e.g., a lamp that indicates when cruise control is actively controlling would require a separate lamp in another PG).

#### 5.7.1.4 Malfunction Indicator Lamp

A lamp used to relay only emissions-related trouble code information. This lamp is only illuminated when there is an emission-related trouble code active.

See Table 5 for the specified operation of the applicable lamp and flash SPNs.

In order to perform a lamp test or OBD readiness indication at key-on, the direct lamp control PG can be used. Refer to SAE J1939DA, PGN 64775, and SPN 5080.

| | |
|---|---|
| 00 | Lamp off |
| 01 | Lamp on (see Table 5 and Appendix I for additional explanation) |
| 10 | Short MIL for WWH OBD (for WWH OBD discriminatory display systems, not applicable for other OBD non-discriminatory display systems) |
| Type: | Status |
| Suspect Parameter Number: | 1213 |
| Reference: | 5.7.1, 5.7.2, 5.7.6, 5.7.12, 5.7.23, 5.7.27, 5.7.28, 5.7.35, 5.7.41, 5.7.42, 5.7.43, 5.7.44, 5.7.45, 5.7.46, 5.7.47, 5.7.48, 5.7.49, 5.7.50, 5.7.51, 5.7.52, 5.7.53, and 5.7.54 |

*Table 5 - Lamp command and lamp flash dependency definition*

| Possible Commanded Conditions | | Required Lamp Output Operation | | | |
|---|---|---|---|---|---|
| **Lamp SPNs: 1213, 623, 624, 987** | **Flash SPNs: 3038, 3039, 3040, 3041** | **MIL SPN 1213** | **RSL SPN 623** | **AWL SPN 624** | **Protect SPN 987** |
| 00 | 00 | Off, don't flash | Off | Off | Off |
| 00 | 01 | Off | Off | Off | Off |
| 00 | 10 | Off, class C previously active(1) | SAE Reserved | SAE Reserved | SAE Reserved |
| 00 | 11 | Off | Off | Off | Off |
| | | | | | |
| 01 | 00 | On, slow flash | On, slow flash | On, slow flash | On, slow flash |
| 01 | 01 | On, fast flash | On, fast flash | On, fast flash | On, fast flash |
| 01 | 10 | On, class C active(2) | SAE Reserved | SAE Reserved | SAE Reserved |
| 01 | 11 | On, don't flash | On, don't flash | On, don't flash | On, don't flash |
| | | | | | |
| 10 | 00 | Short MI previously active [1] | SAE Reserved | SAE Reserved | SAE Reserved |
| 10 | 01 | Short MI active | SAE Reserved | SAE Reserved | SAE Reserved |
| 10 | 10 | SAE Reserved | SAE Reserved | SAE Reserved | SAE Reserved |
| 10 | 11 | SAE Reserved | SAE Reserved | SAE Reserved | SAE Reserved |
| | | | | | |
| 11 | 00 | SAE Reserved | SAE Reserved | SAE Reserved | SAE Reserved |
| 11 | 01 | SAE Reserved | SAE Reserved | SAE Reserved | SAE Reserved |
| 11 | 10 | SAE Reserved | SAE Reserved | SAE Reserved | SAE Reserved |
| 11 | 11 | Don't Care | Don't Care | Don't Care | Don't Care |

(1) The values indicated for "class C not active" and "short MI not active" show optional values that are not required for compliance with Euro VI [UN/ECE R49 Annex 9B] MI activation modes. If observed, they imply MI Activation Mode 1. See Appendices I and K for additional discussion of SPN 1213 and Euro VI MI activation modes.
(2) MI Activation Mode 2.

5.7.1.5 Red Stop Lamp

This lamp is used to relay trouble code information that is of a severe-enough condition that it warrants stopping the vehicle. Also see Table 5 for the specified operation of the applicable lamp and flash SPNs.

| | |
|---|---|
| 00 | Lamp off |
| 01 | Lamp on |
| Type: | Status |
| Suspect Parameter Number: | 623 |
| Reference: | 5.7.1, 5.7.2, 5.7.6, 5.7.12, 5.7.23, 5.7.27, 5.7.28, 5.7.35, 5.7.41, 5.7.42, 5.7.43, 5.7.44, 5.7.45, 5.7.46, 5.7.47, 5.7.48, 5.7.49, 5.7.50, 5.7.51, 5.7.52, 5.7.53, and 5.7.54 |

5.7.1.6 Amber Warning Lamp

This lamp is used to relay trouble code information that is reporting a problem with the vehicle system but the vehicle need not be immediately stopped. Also see Table 5 for the specified operation of the applicable lamp and flash SPNs.

| | |
|---|---|
| 00 | Lamp off |
| 01 | Lamp on |
| Type: | Status |
| Suspect Parameter Number: | 624 |
| Reference: | 5.7.1, 5.7.2, 5.7.6, 5.7.12, 5.7.23, 5.7.27, 5.7.28, 5.7.35, 5.7.41, 5.7.42, 5.7.43, 5.7.44, 5.7.45, 5.7.46, 5.7.47, 5.7.48, 5.7.49, 5.7.50, 5.7.51, 5.7.52, 5.7.53, and 5.7.54 |

5.7.1.7 Protect Lamp

This lamp is used to relay trouble code information that is reporting a problem with a vehicle system that is most probably not electronic sub-system related. For instance, engine coolant temperature is exceeding its prescribed temperature range. See Table 5 for the specified operation of the applicable lamp and flash SPNs.

| | |
|---|---|
| 00 | Lamp off |
| 01 | Lamp on |
| Type: | Status |
| Suspect Parameter Number: | 987 |
| Reference: | 5.7.1, 5.7.2, 5.7.6, 5.7.12, 5.7.23, 5.7.27, 5.7.28, 5.7.35, 5.7.41, 5.7.42, 5.7.43, 5.7.44, 5.7.45, 5.7.46, 5.7.47, 5.7.48, 5.7.49, 5.7.50, 5.7.51, 5.7.52, 5.7.53, and 5.7.54 |

5.7.1.8 Flash Malfunction Indicator Lamp

This parameter provides the capability to flash the MIL. See Table 5 for the specified operation of the applicable lamp and flash SPNs. For OBD systems that are required to flash, the SPN for specific malfunctions shall use the “slow flash” rate.

| | |
|---|---|
| 00 | Slow flash (1 Hz, 50% duty cycle) |
| 01 | Fast flash (2 Hz or faster, 50% duty cycle) |
| 10 | Class C DTC (for WWH OBD discriminatory display systems, not applicable for other OBD non-discriminatory display systems) |
| 11 | Unavailable/do not flash |
| Type: | Status |
| Suspect Parameter Number: | 3038 |
| Reference: | 5.7.1, 5.7.2, 5.7.6, 5.7.12, 5.7.23, 5.7.27, 5.7.28, 5.7.35, 5.7.41, 5.7.42, 5.7.43, 5.7.44, 5.7.45, 5.7.46, 5.7.47, 5.7.48, 5.7.49, 5.7.50, 5.7.51, 5.7.52, 5.7.53, and 5.7.54 |

5.7.1.9 Flash Red Stop Lamp

This parameter provides the capability to flash the RSL. Also see Table 5 for the specified operation of the applicable lamp and flash SPNs.

| | |
|---|---|
| 00 | Slow flash (1 Hz, 50% duty cycle) |
| 01 | Fast flash (2 Hz or faster, 50% duty cycle) |
| 10 | SAE Reserved |
| 11 | Unavailable/do not flash |
| Type: | Status |
| Suspect Parameter Number: | 3039 |
| Reference: | 5.7.1, 5.7.2, 5.7.6, 5.7.12, 5.7.23, 5.7.27, 5.7.28, 5.7.35, 5.7.41, 5.7.42, 5.7.43, 5.7.44, 5.7.45, 5.7.46, 5.7.47, 5.7.48, 5.7.49, 5.7.50, 5.7.51, 5.7.52, 5.7.53, and 5.7.54 |

5.7.1.10 Flash Amber Warning Lamp

This parameter provides the capability to flash the AWL. See Table 5 for the specified operation of the applicable lamp and flash SPNs.

| | |
|---|---|
| 00 | Slow flash (1 Hz, 50% duty cycle) |
| 01 | Fast flash (2 Hz or faster, 50% duty cycle) |
| 10 | SAE Reserved |
| 11 | Unavailable/do not flash |
| Type: | Status |
| Suspect Parameter Number: | 3040 |
| Reference: | 5.7.1, 5.7.2, 5.7.6, 5.7.12, 5.7.23, 5.7.27, 5.7.28, 5.7.35, 5.7.41, 5.7.42, 5.7.43, 5.7.44, 5.7.45, 5.7.46, 5.7.47, 5.7.48, 5.7.49, 5.7.50, 5.7.51, 5.7.52, 5.7.53, and 5.7.54 |

#### 5.7.1.11 Flash Protect Lamp

This parameter provides the capability to flash the protect lamp. Also see Table 5 for the specified operation of the applicable lamp and flash SPNs.

| | |
|---|---|
| 00 | Slow flash (1 Hz, 50% duty cycle) |
| 01 | Fast flash (2 Hz or faster, 50% duty cycle) |
| 10 | SAE Reserved |
| 11 | Unavailable/do not flash |
| Type: | Status |
| Suspect Parameter Number: | 3041 |
| Reference: | 5.7.1, 5.7.2, 5.7.6, 5.7.12, 5.7.23, 5.7.27, 5.7.28, 5.7.35, 5.7.41, 5.7.42, 5.7.43, 5.7.44, 5.7.45, 5.7.46, 5.7.47, 5.7.48, 5.7.49, 5.7.50, 5.7.51, 5.7.52, 5.7.53, and 5.7.54 |

#### 5.7.1.12 Suspect Parameter Number

This 19-bit number is used to identify the item for which diagnostics are being reported. The SPN is used for multiple purposes, some of those that are specific to diagnostics are: (1) to identify a least-repairable sub-system that has failed; (2) to identify sub-systems and/or assemblies that may not have hard failures but are exhibiting abnormal operating performance; (3) to identify a particular event or condition to be reported; and (4) to report a component and nonstandard failure mode. SPNs are assigned to each individual parameter in a parameter group and to items that are relevant to diagnostics but are not a parameter in a parameter group. SPNs are independent of the source address for the message. However, the source address is necessary to determine which controller on the network performed the diagnosis.

The first 511 SPNs are reserved and will be assigned the exact same number as the parameter identifier (PID) used in SAE J1587. That is, the SPN for an accelerator problem will be reported as SPN 91, which is SAE J1587 PID 91. All other SPNs will be numbered sequentially starting at 512 and incrementing by one for each new assignment. Refer to the “SPs and PGs” worksheet in SAE J1939DA.

Proprietary suspect parameter numbers have been established to allow the reporting of manufacturer-specific diagnostics. The interpretation of the diagnostic trouble codes using proprietary SPNs varies by manufacturer and possibly by source address. As a result, it is possible, or even likely, that two (or more) manufacturers might use the same proprietary SPN to identify a component unique to their product. Thus, to completely interpret a DTC that uses a proprietary SPN, the source address and its associated manufacturer code from its SAE J1939 NAME are required. There are 8192 suspect parameter numbers defined for proprietary diagnostics. The SPNs for proprietary diagnostics cover the range 516096 to 524287. See Appendix F for the list of restrictions for the SPNs for proprietary diagnostics.

| | |
|---|---|
| Data Length: | 19 bits |
| Resolution: | 1 SPN/bit |
| Data Range: | 0 to 524287 |
| Type: | Status |
| Suspect Parameter Number: | 1214 |
| Reference: | 5.7.1, 5.7.2, 5.7.6, 5.7.12, 5.7.23, 5.7.27, 5.7.28, 5.7.35, 5.7.41, 5.7.42, 5.7.43, 5.7.44, 5.7.45, 5.7.46, 5.7.47, 5.7.48, 5.7.49, 5.7.50, 5.7.51, 5.7.52, 5.7.53, and 5.7.54 |

5.7.1.13 Failure Mode Identifier (FMI)

The FMI defines the type of failure detected in the sub-system identified by an SPN. Note that the failure may not be an electrical failure but may instead be a sub-system failure or condition needing to be reported to the service technician and maybe also to the operator. Conditions can include system events or status that need to be reported. The FMI, SPN, and SPN conversion method and occurrence count fields combine to form a given diagnostic trouble code. The “Reserved to be Assigned by SAE” FMIs will be assigned by the SAE J1939 Truck and Bus Control and Communications Network Committee if additional failure modes become necessary. The currently defined FMIs are listed in Appendix A.

| | |
|---|---|
| Data Length: | 5 bits |
| Resolution: | 1 FMI/bit |
| Data Range: | 0 to 31 |
| Type: | Status |
| Suspect Parameter Number: | 1215 |
| Reference: | 5.7.1, 5.7.2, 5.7.6, 5.7.12, 5.7.23, 5.7.27, 5.7.28, 5.7.35, 5.7.41, 5.7.42, 5.7.43, 5.7.44, 5.7.45, 5.7.46, 5.7.47, 5.7.48, 5.7.49, 5.7.50, 5.7.51, 5.7.52, 5.7.53, and 5.7.54 |

5.7.1.14 SPN Conversion Method

When this 1-bit field is equal to a zero, the SPN should be converted as it is defined in this document (see definition below for Version 4). The February 1996 version of SAE J1939-73 contained inadequate definitions to assure consistent implementations. Products implementing to the February 1996 version of the document will always have this bit set to a one. When this is the case, the SPN is in either Version 1, 2, or 3 format. The original publication of this recommended practice defined that this bit be set to one. This particular implementation is no longer permitted. It provides context for some implementations prior to the adoption of the recommended setting as zero (version 4 definition).

To clarify the ordering of bits and bytes within the SPN parameter (which is 19 bits long) and to keep that ordering consistent with other parameters in SAE J1939DA and SAE J1939-73, the bit order has been respecified. See Version 4 below for the recommended formatting. Version 4 is required for any device complying with 5.2.2.1 for emissions-related components.

To reduce problems in interpretation of the SPNs, the bit between the FMI field, and the occurrence count field, previously reserved, will be cleared to zero to identify use of the currently specified SPN bit pattern. This bit now comprises an SPN conversion method for the purpose of maintaining usability of those implementations that are already in use.

| | |
|---|---|
| Data Length: | 1 bit |
| Resolution: | Not applicable |
| Data Range: | 0 means convert SPNs per the Version 4 definition below<br>1 means convert SPNs per Version 1, 2, or 3 specified below<br>The four versions of interpretation are:<br>1. SPN assumed to be sent most significant bit first<br>2. SPN represented as Intel format for most significant 16 bits with 3 least significant bits of 19 bits in with FMI value<br>3. SPN represented as Intel format for all 19 bits (least significant sent first)<br>4. SPN represented as intel format for all 19 bits with the SPN conversion method set to 0 |
| Type: | Status |
| Suspect Parameter Number: | 1706 |
| Reference: | 5.7.1, 5.7.2, 5.7.6, 5.7.12, 5.7.23, 5.7.27, 5.7.28, 5.7.35, 5.7.41, 5.7.42, 5.7.43, 5.7.44, 5.7.45, 5.7.46, 5.7.47, 5.7.48, 5.7.49, 5.7.50, 5.7.51, 5.7.52, 5.7.53, and 5.7.54 |

Given:

| | | |
|---|---|---|
| SPN 1208 | $= 4B8_h$ | $= 000\ 00000100\ 10111000_b$ (19 bits) |
| FMI 3 | $= 3_h$ | $= 00011_b$ (5 bits) |
| OC 10 | $= A_h$ | $= 0001010_b$ (7 bits) |
| CM | | $= 0_b$ (1 bit) |

| Version 1. | DTC | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Byte 3<br>8 most significant bits of 16 most significant bits of SPN<br>(bit 8 most significant) | | | | | | | | Byte 4<br>8 least significant bits of 16 most significant bits of SPN<br>(bit 8 most significant) | | | | | | | | Byte 5<br>3 least significant bits of SPN and the FMI<br>(bit 8 SPN msb and bit 5 FMI msb) | | | | | | | | Byte 6 | | | | | | | |
| | SPN | | | | | | | | | | | | | | | | | | | FMI | | | | | CM | OC | | | | | | |
| J1939 Frame Format | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 0 |

| Version 2. | DTC | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Byte 3<br>8 least significant bits of 16 most significant bits of SPN<br>(bit 8 most significant) | | | | | | | | Byte 4<br>8 most significant bits of 16 most significant bits of SPN<br>(bit 8 most significant) | | | | | | | | Byte 5<br>3 least significant bits of SPN and the FMI<br>(bit 8 SPN msb and bit 5 FMI msb) | | | | | | | | Byte 6 | | | | | | | |
| | SPN | | | | | | | | | | | | | | | | | | | FMI | | | | | CM | OC | | | | | | |
| J1939 Frame Format | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| | 1 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 0 |

| Version 3. | DTC | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Byte 3<br>8 least significant bits of SPN<br>(bit 8 most significant) | | | | | | | | Byte 4<br>second byte of SPN<br>(bit 8 most significant) | | | | | | | | Byte 5<br>3 most significant bits of SPN and the FMI<br>(bit 8 SPN msb and bit 5 FMI msb) | | | | | | | | Byte 6 | | | | | | | |
| | SPN | | | | | | | | | | | | | | | | | | | FMI | | | | | CM | OC | | | | | | |
| J1939 Frame Format | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 0 |

| Version 4.* | DTC | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ***Recommended Version***<br>*Effective 1996, Version 4 shall be used for all future OBD applications. | Byte 3<br>8 least significant bits of SPN<br>(bit 8 most significant) | | | | | | | | Byte 4<br>second byte of SPN<br>(bit 8 most significant) | | | | | | | | Byte 5<br>3 most significant bits of SPN and the FMI<br>(bit 8 SPN msb and bit 5 FMI msb) | | | | | | | | Byte 6 | | | | | | | |
| | SPN | | | | | | | | | | | | | | | | | | | FMI | | | | | CM | OC | | | | | | |
| J1939 Frame Format | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 |

#### 5.7.1.15 Occurrence Count

The 7-bit occurrence count field contains the number of times a fault has been independently detected. The occurrence count is reported as one the first time the DTC is detected. The occurrence count will increment after the DTC has gone inactive and the same DTC subsequently returns to the active state. At this point, the occurrence count would be reported as two. This continues until the DTC has been independently detected 126 times. The occurrence count shall not be incremented from 126 to 127; it shall remain at 126 until cleared by DM3 or DM11. If an occurrence count is not available, then this field shall be set to all binary ones (127). The occurrence count is not incremented just due to an ignition key-off and ignition key-on. The diagnostic system shall have monitored the system or component (e.g., DTC) to see that it is no longer malfunctioning in order to declare it previously active.

| | |
|---|---|
| Data length: | 7 bits |
| Resolution: | 1 occurrence count/bit |
| Data range: | 0 to 126 (the value 127 is reserved for indicating not available) |
| Type: | status |
| Suspect parameter number: | 1216 |
| Reference: | 5.7.1, 5.7.2, 5.7.6, 5.7.12, 5.7.23, 5.7.27, 5.7.28, 5.7.35, 5.7.41, 5.7.42, 5.7.43, 5.7.44, 5.7.45, 5.7.46, 5.7.47, 5.7.48, 5.7.49, 5.7.50, 5.7.51, 5.7.52, 5.7.53, and 5.7.54 |

5.7.2 Previously Active Diagnostic Trouble Codes (DM2)

When supported, DM2 provides a list of all previously active (previously detected) DTCs. The list of DTCs must include all the emissions-related previously active DTCs provided in the device's DM23 for an HD OBD-compliant device, as well as all non-emissions-related previously active DTCs.

DM2 is provided using the "multipacket transport" facilities defined in SAE J1939-21 when there is more than one DTC to report. SPN conversion method 4 as described in 5.7.1.14 must be used by HD OBD devices and is recommended for all new designs.

The lamp information (malfunction indicator lamp, red stop lamp, amber warning lamp, and protect lamp) should reflect the present state of the transmitting electronic component. The lamp information shall not convey temporary signals to provide for lamp test illumination, DTC numeric displays, or OBD readiness indication at key-on.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | Variable |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 254 |
| PDU Specific: | 203 |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |
| Parameter Group Number: | 65227 (00FECB$_h$) |

| | | | | |
|---|---|---|---|---|
| Byte: | 1 | bits 8-7 | Malfunction indicator lamp | see 5.7.1.4 |
| | | bits 6-5 | Red stop lamp | see 5.7.1.5 |
| | | bits 4-3 | Amber warning lamp | see 5.7.1.6 |
| | | bits 2-1 | Protect lamp | see 5.7.1.7 |
| Byte: | 2 | bits 8-7 | Flash malfunction indicator lamp | see 5.7.1.8 |
| | | bits 6-5 | Flash red stop lamp | see 5.7.1.9 |
| | | bits 4-3 | Flash amber warning lamp | see 5.7.1.10 |
| | | bits 2-1 | Flash protect lamp | see 5.7.1.11 |
| Byte: | 3 | bits 8-1 | SPN, 8 least significant bits of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: | 4 | bits 8-1 | SPN, second byte of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: | 5 | bits 8-6 | SPN, 3 most significant bits (most significant at bit 8) | see 5.7.1.12 |
| | | bits 5-1 | FMI (most significant at bit 5) | see 5.7.1.13 |
| Byte: | 6 | bit 8 | SPN conversion method | see 5.7.1.14 |
| | | bits 7-1 | Occurrence count | see 5.7.1.15 |

NOTE: When the occurrence count is not available, it should be set to all ones, which is a value of 127.

NOTE: When there is zero or one DTC to report, then unused bytes 7 and 8 of the CAN frame shall be set to 255 (as per SAE J1939-71).

NOTE: When there are no DTCs to report, each of bytes 3 through 6 shall be set to zero.

EXAMPLE 1: The following illustrates the message format for when there is more than one DTC.

Given:
a = lamp status (LS)
b = SPN
c = FMI
d = CM and OC

Message form will be as follows: a,b,c,d,b,c,d,b,c,d,b,c,d....etc. In this example, the transport protocol of SAE J1939-21 will have to be used to send the information because it requires more than 8 data bytes. Actually, any time there is more than one fault, the services of the transport protocol will have to be used.

EXAMPLE 2: The following illustrates the message format for when a request of the DM2 is made and there are zero previously active faults. The currently defined lamps (malfunction indicator lamp, red stop lamp, amber warning lamp, and protect lamp) should reflect the present state of the transmitting electronic component. In this example, the amber lamp is identified as being on.

The original publication of this recommended practice defined that bytes 3 through 6 should be set to all ones when there are zero faults. This particular implementation is no longer permitted.

Given:

| | | |
|---|---|---|
| Byte: 1 | bits 8-7 | = 00 (example of reporting off) |
| | bits 6-5 | = 00 (example of reporting off) |
| | bits 4-3 | = 01 (example of reporting on) |
| | bits 2-1 | = 00 (example of reporting off) |
| Byte: 2 | bits 8-7 | = 11 (example of reporting not available/don't care) |
| | bits 6-5 | = 11 (example of reporting not available/don't care) |
| | bits 4-3 | = 11 (example of reporting not available/don't care) |
| | bits 2-1 | = 11 (example of reporting not available/don't care) |
| Bytes: 3-6 | SPN = 0 (required setting for reporting no diagnostic trouble code) | |
| | FMI = 0 (required setting for reporting no diagnostic trouble code) | |
| | OC = 0 (required setting for reporting no diagnostic trouble code) | |
| | CM = 0 (required setting for reporting no diagnostic trouble code) | |
| Byte: 7 | | = 255 |
| Byte: 8 | | = 255 |

5.7.3 Diagnostic Data Clear/Reset of Previously Active DTCs (DM3)

5.7.3.1 DM3 for Non-OBD Regulated Devices or Diagnostics

Products not subject to ARB 13 CCR 1971.1 are allowed to support DM3. When supported, all diagnostic information pertaining to the previously active trouble codes shall be erased when this PG is requested. The diagnostic data associated with active trouble codes will not be affected.

If the request is destination-specific (refer to SAE J1939-21 PGN 59392), upon completion of this operation or if there are no faults to clear, a positive acknowledgment shall be sent. If, for some reason, a device cannot perform the requested action when a destination-specific request is received, it is required to send a negative acknowledgment.

A positive or negative acknowledgment is not required in response to a global request.

Some applications need to provide a positive or negative acknowledgment to indicate the success or failure of a global request. (This is an exception, as discussed in Note 4 of Table 5 in SAE J1939-21.)

ECUs shall clear the DTCs with ignition ON and with the engine not running. A manufacturer may allow DTC clearing with the engine running.

5.7.3.2 DM3 for OBD Regulated Devices or Diagnostics

For products subject to ARB 13 CCR 1971.1, the DM3 service shall not be supported. As a result, the only acceptable response to a destination-specific DM3 request, for these products, shall be an acknowledgment PG (refer to SAE J1939-21 PGN 59392) with the control byte set to 1 (negative acknowledgment). Products should either provide no response to a globally addressed DM3 request or, to provide positive confirmation of the ARB requirement, an acknowledgment PG with the control byte set to 1 (negative acknowledgment).

Finally, for these products, no DTCs or related diagnostic information shall be cleared. DM3 shall not be used for other manufacturer-specific purposes.

5.7.3.3 Message Definition

Tool suppliers should be aware that it can take as long as 5 seconds to complete the clearing action.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | 0 |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 254 |
| PDU Specific: | 204 |
| Default Priority: | 6 |
| Parameter Group Number: | 65228 ($00FECC_h$) |
| Example: | |

Given:
1. A tool desires to clear the diagnostic data of the engine.
2. The engine is able to perform the requested action.

The tool shall send the request PGN 59904 directed specifically to the engine controller with the PGN 65228 as the requested PGN. The engine controller shall respond with the acknowledgment PGN 59392 indicating that the action was successfully completed for PGN 65228.

5.7.4 Freeze Frame Parameters (DM4)

A freeze frame is defined as the list of recorded parameters at the time a DTC was captured. The freeze frame recorded for each DTC will contain the required parameters first and then any manufacturer-specific information. It is possible that controllers will have more than one freeze frame available and can provide manufacturer-specific information. A freeze frame is specific to one DTC, and one DTC only has one freeze frame. This then limits the amount of freeze frame data per fault and for all faults that are included in this message to 1785 bytes (refer to SAE J1939-21 transport protocol).

This diagnostic message is best suited for systems that impact emissions and/or are powertrain related. However, the use of this message is not limited to just emission-related failures or just powertrain devices. It can be used to report non-emission-related or non-powertrain-related failures.

Implementers should refer to the applicable regulation for potential additional freeze frame requirements. For instance, some regulations might require the OBD freeze frame to have priority over non-OBD freeze frames.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | Variable |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 254 |
| PDU Specific: | 205 |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |
| Parameter Group Number: | 65229 (00FECD$_h$) |

| | | | | |
|---|---|---|---|---|
| Byte: | 1 | | Freeze frame length | See 5.7.4.1 |
| Byte: | 2 | bits 8-1 | SPN, 8 least significant bits of SPN (most significant at bit 8) | See 5.7.1.12 |
| Byte: | 3 | bits 8-1 | SPN, second byte of SPN (most significant at bit 8) | See 5.7.1.12 |
| Byte: | 4 | bits 8-6 | SPN, 3 most significant bits (most significant at bit 8) | See 5.7.1.12 |
| | | bits 5-1 | FMI (most significant at bit 5) | See 5.7.1.13 |
| Byte: | 5 | bit 8 | SPN conversion method | See 5.7.1.14 |
| | | bits 7-1 | Occurrence count | See 5.7.1.15 |
| Byte: | 6 | | Engine torque mode (SPN 899) | Refer to SAE J1939DA |
| Byte: | 7 | | Engine intake manifold #1 pressure (SPN 102) | Refer to SAE J1939DA |
| Byte: | 8 | | Engine speed (SPN 190) (LSB) | Refer to SAE J1939DA |
| Byte: | 9 | | Engine speed (SPN 190) (MSB) | |
| Byte: | 10 | | Engine percent load at current speed (SPN 92) | Refer to SAE J1939DA |
| Byte: | 11 | | Engine coolant temperature (SPN 110) | Refer to SAE J1939DA |
| Byte: | 12 | | Wheel-based vehicle speed (SPN 84) (LSB) | Refer to SAE J1939DA |
| Byte: | 13 | | Wheel-based vehicle speed (SPN 84) (MSB) | |
| Byte: | 14-n | | Manufacture-specific information | See 5.7.4.3 |

NOTE: When the occurrence count is not available, it should be set to all ones, which is a value of 127.

NOTE: If no DTCs (active or previously active) have been accumulated, then the response will be:

PGN = 65229

| | | |
|---|---|---|
| Byte: | 1 | = 0 |
| | 5-2 | = 0 |
| | 6 | = 255 |
| | 7 | = 255 |
| | 8 | = 255 |

When byte 1 is equal to zero, it identifies to the receiver that the other parameters in the message should not be interpreted. Also, notice that the values of the information put in bytes 1 through 5 are zero instead of all ones (binary), which typically indicate not available.

EXAMPLE: The following illustrates the message format for when there is more than one freeze frame.

Given:

a = freeze frame length
b = required parameters (bytes 2 through 13 for the first DTC and the corresponding bytes for each of the remaining DTCs)
c = manufacturer specific freeze frame information

Message form will be as follows: a,b,c,<u>a,b,c</u>,a,b,c,<u>a,b,c</u>,a,b,c....etc. The transport protocol of SAE J1939-21 will have to be used to send freeze frames because they are more than 8 data bytes.

5.7.4.1 Freeze Frame Length

The freeze frame length shall be equal to the number of bytes in the required parameters (that is bytes 2 through 13), plus the number of bytes in the manufacturer-specific parameters. That is: a = b + c.

| | |
|---|---|
| Data Length: | 8 bits |
| Resolution: | 1 byte/bit |
| Data Range: | 0 to 255 |
| Type: | Status |
| Suspect Parameter Number: | 1217 |
| Reference: | 5.7.4 |

EXAMPLE:

b = 12
c = 2..........oil pressure, intake manifold temperature
a = b + c
a = 12 + 2 = 14

5.7.4.2 Freeze Frame Parameters

The parameters collected in the freeze frame shall use the same scaling as is defined in the SAE J1939DA document. Freeze frame parameters consist of engine torque mode (SPN 899), engine intake manifold #1 pressure (SPN 102), engine speed (SPN 190), engine percent load at current speed (SPN 92), engine coolant temperature (SPN 110), and wheel-based vehicle speed (SPN 84) as recorded at the time of the failure.

5.7.4.3 Manufacturer-Specified Information

Manufacturer-specified information consists of additional parameters recorded at the time of the failure that may be useful in troubleshooting. The number of parameters is determined by each manufacturer.

| | |
|---|---|
| Data Length: | Varies per manufacturer-selected parameter |
| Resolution: | Varies per manufacturer-selected parameter |
| Data Range: | Varies per manufacturer-selected parameter |
| Type: | Varies per manufacturer-selected parameter |
| Suspect Parameter Number: | 4263 |
| Reference: | 5.7.4 |

5.7.5 Diagnostic Readiness 1 (DM5)

Reports the diagnostics information that relates to diagnostic readiness. When an HD OBD device (OBD compliance, SPN 1220, value is other than $05_h$ or $FF_h$) supports DM5 but does not provide any monitor support or monitoring status bits (SPN 1221, 1222, and 1223), the monitoring support and monitoring status bits are defined such that all bits are to be provided as binary zeros. Historically, non-OBD devices (OBD compliance, SPN 1220, is equal to $05_h$ or $FF_h$) have provided binary ones to indicate not available (per SAE J1939-71, Table 1) for SPN 1221, 1222, and 1223, when only SPNs 1218 and 1219 are supported.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | 8 |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 254 |
| PDU Specific: | 206 |
| Default Priority: | 6 |
| Parameter Group Number: | 65230 ($00FECE_h$) |

| | | | |
|---|---|---|---|
| Byte: | 1 | Active Diagnostic Trouble Code Count | see 5.7.5.1 |
| | 2 | Previously Active Diagnostic Trouble Code Count | see 5.7.5.2 |
| | 3 | OBD Compliance | see 5.7.5.3 |
| | 4 | Continuously Monitored Systems Support/Status | see 5.7.5.4 |
| | 5-6 | Non-Continuously Monitored Systems Support | see 5.7.5.5 |
| | 7-8 | Non-Continuously Monitored Systems Status | see 5.7.5.6 |

#### 5.7.5.1 Active Diagnostic Trouble Code Count

Identifies the number of active trouble codes that are present in a specific controller. If no DTCs are active, this field should be set to zero. The number of DTCs reported shall be equal to the count of DTCs reported in DM1. This count includes both emissions-related and non-emissions-related DTCs.

Data Length: 1 byte
Resolution: 1 trouble code/bit
Data Range: 0 to 250
Type: Measured
Suspect Parameter Number: 1218
Reference: 5.7.5

#### 5.7.5.2 Previously Active Diagnostic Trouble Code Count

Identifies the number of previously active trouble codes that are present in a specific controller. If no DTCs have been previously active, this field should be set to zero. The number of DTCs reported shall be equal to the count of DTCs reported in DM2. This count includes both emissions-related and non-emissions-related DTCs.

Data Length: 1 byte
Resolution: 1 trouble code/bit
Data Range: 0 to 250
Type: Measured
Suspect Parameter Number: 1219
Reference: 5.7.5

#### 5.7.5.3 OBD Compliance

Identifies the OBD compliance capability of the responding controller. Identifies the requirements level to which the controller was built. The OBD compliance assignments are shown in Table 6. The acronym column in Table 6 provides a recommended content for limited display devices. Non-regulated OBD products shall report the OBD compliance (SPN 1220) as a value of $05_h$. A value of $FF_h$ shall also be interpreted as a non-regulated OBD product. Devices reporting an OBD compliance code that indicates it satisfies OBD requirements are considered to be an OBD device (see 3.14). These OBD devices shall be evaluated in an OBD scan tool interface test.

Data Length: 1 byte
Resolution: See below
Data Range: 0 to 250
Type: Measured
Suspect Parameter Number: 1220
Reference: 5.7.5

*Table 6 - OBD compliance assignments*

| Value | Decimal | Acronym | Description |
|---|---|---|---|
| $00_h$ | 0 | | Reserved for assignment by SAE |
| $01_h$ | 1 | OBD II | OBD II (California Air Resources Board) |
| $02_h$ | 2 | OBD | OBD (Federal, EPA) |
| $03_h$ | 3 | OBD and OBD II | OBD and OBD II |
| $04_h$ | 4 | OBD I | OBD I |
| $05_h$ | 5 | | Not intended to meet OBD II requirements |
| $06_h$ | 6 | EOBD | EOBD |
| $07_h$ | 7 | EOBD and OBD II | EOBD and OBD II |
| $08_h$ | 8 | EOBD and OBD | EOBD and OBD |
| $09_h$ | 9 | EOBD, OBD, and OBD II | EOBD, OBD, and OBD II |
| $0A_h$ | 10 | JOBD | JOBD |
| $0B_h$ | 11 | JOBD and OBD II | JOBD and OBD II |
| $0C_h$ | 12 | JOBD and EOBD | JOBD and EOBD |
| $0D_h$ | 13 | JOBD, EOBD, and OBD II | JOBD, EOBD, and OBD II |
| $0E_h$ | 14 | | Heavy-duty vehicles (EURO IV) B1 |
| $0F_h$ | 15 | | Heavy-duty vehicles (EURO V) B2 |
| $10_h$ | 16 | | Heavy-duty vehicles (EURO EEC) C (gas engines) |
| $11_h$ | 17 | EMD | Engine manufacturer diagnostics (EMD) |
| $12_h$ | 18 | EMD+ | Engine manufacturer diagnostics enhanced (EMD+), ARB and EPA engine families that do not have to satisfy the more complicated OBD requirements |
| $13_h$ | 19 | HD OBD P | Heavy-duty/on-board diagnostics partial (CARB CCR 1971.1 and EPA 86.010-18) |
| $14_h$ | 20 | HD OBD | Heavy-duty/on-board diagnostics (CARB CCR 1971.1 and EPA 86.010-18) |
| $15_h$ | 21 | WWH OBD | Worldwide harmonized OBD |
| $16_h$ | 22 | OBD II | OBD II (California Air Resources Board, CCR 1968.2, 2007/2008/2009/2010 revisions) |
| $17_h$ | 23 | HD EOBD | Heavy-duty vehicles (EURO IV/V, revisions specified in 2005 and 2006) |
| $18_h$ | 24 | | Reserved for assignment by SAE |
| $19_h$ | 25 | OBD-M (SI-SD/I) | OBD-M compliance for spark-ignition sterndrive and inboard engines |
| $1A_h$ | 26 | EURO VI | Heavy-duty vehicles EURO VI (as described in UN/ECE R49 Annex 9) |
| $1B_h$ | 27 | HD-ZEP | California Air Resources Board, CCR 1956.8, 2019 |
| $1C_h$ | 28 | OBDBr-1 | Brazil OBD Phase 1 |
| $1D_h$ | 29 | OBDBr-2 | Brazil OBD Phase 2 and Phase 2+ |
| $1E_h$ | 30 | KOBD | Korean OBD |
| $1F_h$ | 31 | IOBD I | India OBD I |
| $20_h$ | 32 | IOBD II | India OBD II |
| $21_h$ | 33 | HD EOBD-VI | Heavy-duty Euro OBD Stage VI |
| $22_h$ | 34 | OBD, OBD II, HD OBD | OBD, OBD II, and heavy-duty/on-board diagnostics (CARB CCR 1971.1 and EPA 86.010-18) [Compliance codes 3 and 20] |
| $23_h$ | 35 | OBD, OBD II, HD OBD P | OBD, OBD II, and heavy-duty/on-board diagnostics Partial (CARB CCR 1971.1 and EPA 86.010-18) [Compliance codes 3 and 19] |
| $24_h$ to $28_h$ | 36 to 40 | Reserved for SAE | Reserved for SAE J1979 assignment |

| Value | Decimal | Acronym | Description |
|---|---|---|---|
| $29_h$ | 41 | CN-OBD-6 | China Nationwide Stage 6 |
| $2A_h$ | 42 | OBDBr-D | Brazil OBD diesel |
| $2B_h$ | 43 | CN-OBD-VI | China heavy duty VI |
| $2C_h$ | 44 | IOBD-I-BS6 | India OBD I Bharat Stage 6 |
| $2D_h$ | 45 | IOBD-II-BS6 | India OBD II Bharat Stage 6 |
| $2E_h$ | 46 | IHDOBD-BSVI | India HD OBD Bharat Stage VI |
| $2F_h$ | 47 | OBDBr-P8 | Brazil OBD Phase 8 |
| $30_h$ | 48 | HD-JOBD-II | Japan Heavy-Duty OBD II |
| $31_h$ | 49 | HD-KOBD-II | Korea Heavy-Duty OBD II |
| $32_h$ | 50 | CN-HDOBD-IV | China Tier IV Non Road OBD |
| $33_h$ | 51 | CARB ACC-II | Light Duty ZEV |
| $34_h$ to $37_h$ | 52 to 55 | Reserved for SAE | Assigned in SAE J1979DA for Motorcycle regulations |
| $38_h$ | 56 | HD ZEV | CARB Regulated [HD] ZEV |
| $39_h$ | 57 | BSV CEV | India Bharat Stage V – Construction Equipment Vehicle |
| $3A_h$ to $FA_h$ | 58 to 250 | Reserved for SAE | To be assigned by SAE/ISO |
| $FB_h$ to $FF_h$ | 251 to 255 | | Per SAE J1939-71, Table 1 definition |

#### 5.7.5.4 Continuously Monitored Systems Support/Status

Identifies the continuously monitored system support and status.

Data Length: 1 byte
Resolution: See below
Data Range: Bit mapped, see below
Type: Measured
Suspect Parameter Number: 1221
Reference: 5.7.5

Byte: Bit: Description:
4 8 Reserved for assignment by SAE (shall be reported as 0)
7 Comprehensive component monitoring status
6 Engine fuel system monitoring status
5 Misfire monitoring status
Where each status bit (bits 7, 6, 5) is interpreted:
0 = test complete, not supported
1 = test not complete
4 Reserved for assignment by SAE (shall be reported as 0)
3 Comprehensive component monitoring support
2 Engine fuel system monitoring support
1 Misfire monitoring support
Where each supported bit (bits 3, 2, 1) is interpreted:
0 = test not supported by this controller
1 = test supported by this controller

NOTE: A bit set to zero can mean test not supported. Controllers that do not support SPN 1221 should provide these bits as zeros. This is different than the typical SAE J1939 use of the value 1 to indicate not available. Any bits that are “Reserved for assignment by SAE” shall be reported as 0.

Data in SPN 1221 shall be ignored when OBD compliance, SPN 1220, is equal to $05_h$ or $FF_h$ (which indicates a non-OBD device). SPN 1221 may be set to $FF_h$ to indicate not available (per SAE J1939-71, Table 1) when OBD compliance, SPN 1220, is equal to $05_h$ or $FF_h$ (which indicates a non-OBD device).

5.7.5.5 Non-Continuously Monitored Systems Support

Identifies the non-continuously monitored systems support.

| | |
|---|---|
| Data Length: | 2 bytes (PG placement of data is specified below) |
| Resolution: | See below |
| Data Range: | Bit mapped, see below |
| Type: | Measured |
| Suspect Parameter Number: | 1222 |
| Reference: | 5.7.5 |

| Byte: | Bit: | Description: |
|---|---|---|
| 5 | 8 | EGR/VVT system monitoring support |
| | 7 | Exhaust Gas Sensor heater monitoring support [11] |
| | 6 | Exhaust Gas Sensor monitoring support[11] |
| | 5 | A/C system refrigerant monitoring support |
| | 4 | Secondary air system monitoring support |
| | 3 | Engine evaporative system monitoring support |
| | 2 | Heated catalyst monitoring support |
| | 1 | Catalyst monitoring support |
| 6 | 8-6 | Reserved for assignment by SAE (shall be reported as 0) |
| | 5 | NMHC converting catalyst monitoring support |
| | 4 | NOx converting catalyst and/or NOx adsorber monitoring support |
| | 3 | Diesel particulate filter (DPF) monitoring support |
| | 2 | Boost pressure control system monitoring support |
| | 1 | Cold start aid system monitoring support |

Where each bit is interpreted:

0 = test not supported by this controller
1 = test supported by this controller

NOTE: A bit set to zero means test not supported. Controllers that do not support SPN 1222 should provide these bits as zeros. This is different than the typical SAE J1939 use of the value 1 to indicate not available. Any bits that are “Reserved for assignment by SAE” shall be reported as 0.

NOTE: Data in SPN 1222 shall be ignored when OBD compliance, SPN 1220, is equal to 05h or FFh (these indicate a non-OBD device). SPN 1222 may be set to FFh to indicate not available (per SAE J1939-71, Table 1) when OBD compliance, SPN 1220, is equal to 05h or FFh (which indicates a non-OBD device).

5.7.5.6 Non-Continuously Monitored Systems Status

Identifies the non-continuously monitored systems status. Each bit identifies whether a particular test (or readiness group) is complete for a given controller.

| | |
|---|---|
| Data Length: | 2 bytes (PG placement of data is specified below) |
| Resolution: | See below |
| Data Range: | Bit mapped, see below |
| Type: | Measured |
| Suspect Parameter Number: | 1223 |
| Reference: | 5.7.5 |

| Byte: | Bit: | Description: |
|---|---|---|
| 7 | 8 | EGR/VVT system monitoring status |
| | 7 | Exhaust gas sensor heater monitoring status (see footnote [11]) |
| | 6 | Exhaust gas sensor monitoring status (see footnote [11]) |
| | 5 | A/C system refrigerant monitoring status |

[11] Exhaust gas sensor heater monitoring and exhaust gas sensor monitoring are names assigned to cover any and all of the following types of sensors: oxygen, NOx, particulate matter, individual instances of these sensors for upstream and downstream measurements, and uses of any and all of these sensors for bank 1 or bank 2.

| | | |
|---|---|---|
| | 4 | Secondary air system monitoring status |
| | 3 | Engine evaporative system monitoring status |
| | 2 | Heated catalyst monitoring status |
| | 1 | Catalyst monitoring status |
| 8 | 8-6 | Reserved for assignment by SAE (shall be reported as 0) |
| | 5 | NMHC converting catalyst monitoring status |
| | 4 | NOx converting catalyst and/or NOx adsorber monitoring status |
| | 3 | Diesel particulate filter (DPF) monitoring status |
| | 2 | Boost pressure control system monitoring status |
| | 1 | Cold start aid system monitoring status |

Where each bit is interpreted:

0 = test complete, or not supported

1 = test not complete

NOTE: A bit set to zero can mean test not supported. Controllers that do not support SPN 1223 should provide these bits as zeros. This is different than the typical SAE J1939 use of the value 1 to indicate not available. Any bits that are “Reserved for assignment by SAE” shall be reported as 0.

NOTE: Data in SPN 1223 shall be ignored when OBD compliance, SPN 1220, is equal to 05h or FFh (these indicate a non-OBD device). SPN 1223 may be set to FFh to indicate not available (per SAE J1939-71, Table 1) when OBD compliance, SPN 1220, is equal to 05h or FFh (which indicates a non-OBD device).

5.7.6 Emission-Related Pending Diagnostic Trouble Codes (DM6)

The purpose of this DM is to enable the external test equipment to obtain “pending” DTCs. The definition of a pending DTC varies according to the applicable regulatory guidance. Some regulations define pending to be a DTC detected during current or last completed driving cycle for emission-related components/systems. Other regulations define a pending DTC as one that is stored upon the initial detection of a malfunction (e.g., typically on a single driving cycle) prior to illumination of the MIL. Regulations may allow a DTC to be simultaneously pending and confirmed while others forbid this. DM6 is required for all emission OBD regulations. The intended use of this data is to assist the service technician after a vehicle repair, and after clearing diagnostic information, by reporting test results after a single driving cycle.

Reporting the pending DTCs is done using the same format as is used to report active DTCs.

The lamp information (malfunction indicator lamp, red stop lamp, amber warning lamp, and protect lamp) should reflect the present state of the transmitting electronic component. The lamp information shall not convey temporary signals to provide for lamp test illumination, DTC numeric displays, or OBD readiness indication at key-on.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | Variable |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 254 |
| PDU Specific: | 207 |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |
| Parameter Group Number: | 65231 (00FECF$_h$) |

| | | | | |
|---|---|---|---|---|
| Byte: 1 | bits 8-7 | Malfunction indicator lamp | see 5.7.1.4 |
| | bits 6-5 | Red stop lamp | see 5.7.1.5 |
| | bits 4-3 | Amber warning lamp | see 5.7.1.6 |
| | bits 2-1 | Protect lamp | see 5.7.1.7 |
| Byte: 2 | bits 8-7 | Flash malfunction indicator lamp | see 5.7.1.8 |
| | bits 6-5 | Flash red stop lamp | see 5.7.1.9 |
| | bits 4-3 | Flash amber warning lamp | see 5.7.1.10 |
| | bits 2-1 | Flash protect lamp | see 5.7.1.11 |

| | | | | |
|---|---|---|---|---|
| Byte: 3 | bits 8-1 | SPN, 8 least significant bits of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: 4 | bits 8-1 | SPN, second byte of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: 5 | bits 8-6 | SPN, 3 most significant bits (most significant at bit 8) | see 5.7.1.12 |
| | bits 5-1 | FMI (most significant at bit 5) | see 5.7.1.13 |
| Byte: 6 | bit 8 | SPN conversion method | see 5.7.1.14 |
| | bits 7-1 | Occurrence count | see 5.7.1.15 |

NOTE: When the occurrence count is not available, it should be set to all ones, which is a value of 127.

NOTE: When there is zero or one DTC to report, then unused bytes 7 and 8 of the CAN frame shall be set to 255 (as per SAE J1939-71).

NOTE: When there are no DTCs to report, each of bytes 3 through 6 shall be set to zero.

EXAMPLE 1: The following illustrates the message format for when there is more than one DTC.

Given:
a = lamp status (LS)
b = SPN
c = FMI
d = CM and OC

Message form is as follows: a,b,c,d,b,c,d,b,c,d,b,c,d....etc. In this example, the transport protocol of SAE J1939-21 has to be used to send the information because it requires more than 8 data bytes. Actually, any time there is more than one fault, the services of the transport protocol have to be used.

EXAMPLE 2: The following illustrates the message format for when a request of the DM6 is made and all test results indicate no trouble information. The currently defined lamps (malfunction indicator lamp, red stop lamp, amber warning lamp, and protect lamp) should reflect the present state of the transmitting electronic component. In this example, the amber lamp is identified as being on.

The original publication of this recommended practice defined that bytes 3 through 6 should be set to all ones when there are zero faults. This particular implementation is no longer permitted.

Given:

| | | |
|---|---|---|
| Byte: 1 | bits 8-7 = | 00 (example of reporting off) |
| | bits 6-5 = | 00 (example of reporting off) |
| | bits 4-3 = | 01 (example of reporting on) |
| | bits 2-1 = | 00 (example of reporting off) |
| Byte: 2 | bits 8-7 = | 11 (example of reporting not available/don’t care) |
| | bits 6-5 = | 11 (example of reporting not available/don’t care) |
| | bits 4-3 = | 11 (example of reporting not available/don’t care) |
| | bits 2-1 = | 11 (example of reporting not available/don’t care) |

Bytes: 3-6 SPN = 0 (required setting for reporting no diagnostic trouble code)

FMI = 0 (required setting for reporting no diagnostic trouble code)

OC = 0 (required setting for reporting no diagnostic trouble code)

CM = 0 (required setting for reporting no diagnostic trouble code)

Byte: 7 = 255

Byte: 8 = 255

### 5.7.7 Command Non-Continuously Monitored Test (DM7)

The purpose of this command in the diagnostic process is to: (1) Command manufacturer specific tests, (2) Command standard tests (if applicable), or (3) Command to return last measured scaled results. In all cases, either DM8 or DM30 response is required. For 2013 model year and subsequent engines, a DM30 response is recommended for all responses. The component manufacturer is responsible to assign test identifiers (TID) and component identifiers for tests of different systems and components. PGN 58112 (DM7) is used to invoke one of the manufacturer-defined test identifiers or invoke a standard test or request last measured results. Test results are reported, by test identifier, using PGN 65232 (DM8) for non-scaled test results. For test results that utilize standard scaling, see DM 30. DM7 also requests HD OBD rationality fault information, which is provided using PGN 64475 (DM58). If DM7 or the specific test identifier or the SPN and FMI is not supported, then a NACK is required (refer to SAE J1939-21 PGN 59392) to be returned.

| | |
|---|---|
| Transmission Rate: | Sent to a specific destination address whenever a test is desired |
| Data Length: | 8 |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 227 |
| PDU Specific: | Destination address (see Note below) |
| Default Priority: | 6 |
| Parameter Group Number: | 58112 (00E300$_h$) |

| | | | | |
|---|---|---|---|---|
| Byte: | 1 | | Test identifier | see 5.7.7.1 |
| Byte: | 2 | bits 8-1 | SPN identifying component/system-specific test<br>8 least significant bits of SPN (most significant at bit 8) | see 5.7.7.2 |
| Byte: | 3 | bits 8-1 | SPN identifying component/system-specific test<br>second byte of SPN (most significant at bit 8) | see 5.7.7.2 |
| Byte: | 4 | bits 8-6 | SPN identifying component/system-specific test<br>3 most significant bits (most significant at bit 8) | see 5.7.7.2 |
| | | bits 5-1 | FMI identifying component/system-specific test<br>(most significant at bit 5) | see 5.7.7.3 |
| Bytes: | 5-8 | | Reserved for assignment by SAE | |

NOTE: DM7 shall be sent to a specific address, not the global destination address. The DM8 or DM30 response shall be sent using RTS/CTS (refer to SAE J1939-21).

#### 5.7.7.1 Test Identifier

The TID has two methods to designate the test to be run. The first method uses the TID alone, and these test identifiers are manufacturer-defined test identifiers. For this first use, there are 64 valid test identifiers, 1 to 64. The second method uses the TID and the SPN/FMI to identify the test.

See Table 7 for values that are available to be used.

| | |
|---|---|
| Data Length: | 1 byte |
| Resolution: | See 5.7.8, 5.7.30, or 5.7.58 |
| Data Range: | 0 to 255 per table below |
| Type: | Status |
| Suspect Parameter Number: | 1224 |
| Reference: | 5.7.8, 5.7.30, or 5.7.58 |

*Table 7 - Test identifier assignments*

| Test Identifier Value | Name | Response PG | Test Results |
|---|---|---|---|
| 0 | Reserved for SAE assignment | NA | NA |
| 1 to 64 | Command manufacturer specific test | DM8 | Nonstandard scaling |
| 65 to 244 | Reserved for SAE assignment | TBA | NA |
| 245 | Return rationality fault SP data value | DM58 | Standard scaling |
| 246 | Return all scaled test results for all SPNs(3) | DM30 | Standard scaling |
| 247 | Return all scaled test results for one SPN(1) | DM30 | Standard scaling |
| 248 | Command manufacturer specific test | DM30 | Standard scaling |
| 249 | Command standard test(2) | DM30 | Standard scaling |
| 250 | Return last measured scaled results | DM30 | Standard scaling |
| 251 to 255 | Reserved for SAE assignment | NA | NA |

(1) Test Identifier 247 shall return all the test results for the SPN given in 5.7.7.2. An FMI value of 31 shall be used in 5.7.7.3 by scan tools to query for test results with TID 247. With the FMI value of 31 is used, the test results provided shall comprise all test results for the SPN given in 5.7.7.2 regardless of their assigned FMIs, including FMI 31. When there is no information regarding what specific FMIs are supported, the individual SPNs, indicated by DM24 as having test results, can therefore be queried using Test Identifier 247, the SPN displayed, and FMI 31. Test Identifier 250 is used when the results for only one test is desired where the test is defined by the SPN given in 5.7.7.2 and the FMI given in 5.7.7.3.

(2) SPN 4175 (Diesel Particulate Filter Active Regeneration Forced Status) with an FMI of 31 shall be used with TID 249 to provide a common means for scan-tool user requests to regenerate a DPF PM aftertreatment system. This capability is not required for OBD compliance.

(3) TID 246 - SPN 5846 (OBD test results) with an FMI of 31 shall be used with TID 246 to provide a common means for scan-tool user request for all test results. This method of request could require more than 200 bytes to report all results for all OBD test results. This capability is not required for OBD compliance.

5.7.7.2 SPN Identifying Component/System-Specific Test

This parameter identifies the component/system that will be tested.

| | |
|---|---|
| Data Length: | 19 bits |
| Resolution: | See below |
| Data Range: | Bit mapped, see below |
| Type: | Status |
| Suspect Parameter Number: | 4148 |
| Reference: | 5.7.7 |

5.7.7.3 FMI Identifying Component/System-Specific Test

This parameter identifies the failure mode identifier that represents the test(s) that shall be run on the component/system that will be tested.

TID 247 FMI 31 shall result in the return of all test results associated with the SPN given in 5.7.7.2, including the results for a DTC that has an FMI of 31. When the TID is 250, only the test results associated with the SPN given in 5.7.7.2 and the provided FMI is returned. When the TID is 245, the FMI is not used and should be set to 31.

| | |
|---|---|
| Data Length: | 5 bits |
| Resolution: | See below |
| Data Range: | Bit mapped, see below |
| Type: | Status |
| Suspect Parameter Number: | 4149 |
| Reference: | 5.7.7 |

### 5.7.8 Test Results for Non-Continuously Monitored Systems (DM8)

The purpose of this response PG is to report the test results for one of the non-continuously monitored tests invoked using DM7. The component manufacturer is responsible to assign test identifiers and component identifiers for tests of different systems and components. PGN 58112 (DM7) is used to invoke one of the manufacturer-defined test identifiers. Test results are reported by test identifier using PGN 65232 (DM8).

Transmission Rate: Sent in response to PGN 58112 when the results are available
A NACK is required if PG is not supported
(refer to SAE J1939-21 PGN 59392)
Data Length: Variable but sent in 8-byte sets (see example)
Extended Data Page: 0
Data Page: 0
PDU Format: 254
PDU Specific: 208
Default Priority: 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used)
Parameter Group Number: 65232 (00FED0$_h$)

| Byte | Description | Reference |
|---|---|---|
| 1 | Test identifier | see 5.7.7.1 |
| 2 | Test type/component identifier | see 5.7.8.1 |
| 4-3 | Test value | see 5.7.8.2 |
| 6-5 | Test limit maximum | see 5.7.8.3 |
| 8-7 | Test limit minimum | see 5.7.8.4 |

Further guidelines for the use of the test value, test limit maximum, and the test limit minimum to convey results for tests are enumerated in Table 8. For example, with a test where there is not a test limit maximum or test limit minimum, the results are determined from the test value alone (cases 1 to 4).

***Table 8 - Test results***

| Case # | Test Value | Test Maximum | Test Minimum | Interpretation |
|---|---|---|---|---|
| 1 | 0000$_h$ to FAFF$_h$ | FFFF$_h$ | FFFF$_h$ | Test Pass |
| 2 | FE00$_h$ (Error) | FFFF$_h$ | FFFF$_h$ | Test Fail |
| 3 | FB00$_h$ | FFFF$_h$ | FFFF$_h$ | Test Not Complete(1) |
| 4 | FB01$_h$ | FFFF$_h$ | FFFF$_h$ | Test Cannot Be Performed |
| 5 | 0003$_h$ | 0004$_h$ | 0001$_h$ | Test Pass |
| 6 | 0000$_h$ | 0004$_h$ | 0001$_h$ | Test Fail |
| 7 | 0005$_h$ | 0004$_h$ | 0001$_h$ | Test Fail |
| 8 | 0000$_h$ | FFFF$_h$ | 0001$_h$ | Test Fail |
| 9 | 0002$_h$ | FFFF$_h$ | 0001$_h$ | Test Pass |
| 10 | FAFF$_h$ | FAFE$_h$ | FFFF$_h$ | Test Fail |
| 11 | AF57$_h$ | AF59$_h$ | FFFF$_h$ | Test Pass |
| 12 | 0100$_h$ | FAFE$_h$ | 0100$_h$ | Test Pass |
| 13 | FAFE$_h$ | FAFE$_h$ | 0100$_h$ | Test Pass |

(1) Case 3 is required when an OBD test result is initialized and has not made a pass/fail decision.

Another observation is that if there is more than one test value to report on a given test, then the results will be sent using the transport protocol defined in SAE J1939-21. See the following example. If multiple test results are reported, then the component identifier parameter is used to distinguish the different result values. All test identifier values must be the same when multiple test results are reported in one DM8 response.

If the test performed does not have both a test limit minimum and maximum, then the appropriate limit value (maximum or minimum) should be set to all ones. SAE J1939-71 defines this to mean not available.

EXAMPLE:

| | |
|---|---|
| Given: | Assume three separate test values are desired to be communicated. |
| Where: | a = Test identifier, b = Test type/component identifier, c = Test value, d = Test limit maximum, e = Test limit minimum |

Message form will be as follows: a,b,c,d,e,a,b,c,d,e,a,b,c,d,e

The transport protocol of SAE J1939-21 will have to be used when there is more than one test value to send because 16 or more data bytes would be required. In this example, 24 bytes of data would need sent for three test values.

5.7.8.1 Test Type/Component Identifier

This parameter identifies the non-continuously monitored component identifier that was tested. These component identifiers are defined by the manufacturer. They are necessary when multiple components or systems are present on the vehicle and have the same definition of test identifier.

| | |
|---|---|
| Data Length: | 1 byte |
| Resolution: | See DM10 below |
| Data Range: | 1 to 64 (Note: 0 and 65 to 250 are reserved) |
| Type: | Measured |
| Suspect Parameter Number: | 1225 |
| Reference: | 5.7.8 |

5.7.8.2 Test Value

The test value collected during the test. If the test performed does not have both a test limit minimum and maximum, then the appropriate limit value (maximum or minimum) should be set to all ones. SAE J1939-71 defines this to mean not available.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | Not defined |
| Data Range: | 0 to 64255 (see Table 8 for use of $FB00_h$, $FB01_h$, and $FE00_h$) |
| Type: | Measured |
| Suspect Parameter Number: | 1226 |
| Reference: | 5.7.8 |

5.7.8.3 Test Limit Maximum

The test value must be less than or equal to test limit maximum in order for the test to pass. If the test performed does not have a test limit maximum, then the test limit maximum shall be set to all ones, which is defined by SAE J1939-71 to mean not available. The test limit maximum shall be scaled according to the SLOT definition in SAE J1939DA. The test limit maximum shall follow the conventions of Table 8.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | Not defined |
| Data Range: | 0 to 64255 (see Table 8 for use of $FFFF_h$) |
| Type: | Measured |
| Suspect Parameter Number: | 1227 |
| Reference: | 5.7.8 |

5.7.8.4 Test Limit Minimum

The test value must be greater than or equal to test limit minimum in order for the test to pass. If the test performed does not have a test limit minimum, then the test limit minimum shall be set to all ones, which is defined by SAE J1939-71 to mean not available. The test limit minimum shall be scaled according to the SLOT definition in SAE J1939DA. The test limit minimum shall follow the conventions of Table 8.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | Not defined |
| Data Range: | 0 to 64255 (see Table 8 for use of $FFFF_h$) |
| Type: | Measured |
| Suspect Parameter Number: | 1228 |
| Reference: | 5.7.8 |

5.7.9 Oxygen Sensor Test Results (DM9)

SAE J1939 will not specify an implementation for this DM. Oxygen sensor test and results should be communicated using DM7 and DM8.

5.7.10 Non-Continuously Monitored Systems Test Identifiers Support (DM10)

The purpose of this PG is to report the list of non-continuously monitored systems tests supported by the controller. The component manufacturer is responsible to assign test identifiers and component identifiers for tests of different systems and components. PGN 58112 (DM7) is used to invoke one of the manufacturer-defined test identifiers. Test results are reported by test identifier using PGN 65232 (DM8). Service tools can determine the supported tests by requesting PGN 65234 (DM10).

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | 8 |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 254 |
| PDU Specific: | 210 |
| Default Priority: | 6 |
| Parameter Group Number: | 65234 ($00FED2_h$) |

| | | |
|---|---|---|
| Bytes: 8-1 | Test Identifiers Supported | see 5.7.10.1 |

5.7.10.1 Test Identifiers Supported

Indicates the test identifiers that the controller supports. Each bit is assigned to one test. Up to 64 tests can be reported without having to use the transport protocol of SAE J1939-21. The assignment of a given test identifier to a given bit is manufacturer specific.

| | |
|---|---|
| Data Length: | 8 bytes (PG placement of data is specified below) |
| Resolution: | See below |
| Data Range: | 64 bits<br>Note: Bit mapped, each bit indicates an individual test identifier |
| Type: | Measured |
| Suspect Parameter Number: | 1229 |
| Reference: | 5.7.10 |

| Byte: | Bit: | Description: |
|---|---|---|
| 1 | 8 | Test 1 |
| | 7 | Test 2 |
| | 6 | Test 3 |
| | 5 | Test 4 |
| | 4 | Test 5 |
| | 3 | Test 6 |
| | 2 | Test 7 |

| | | |
|---|---|---|
| | 1 | Test 8 |
| 2 | 8 | Test 9 |
| | 7 | Test 10 |
| | 6 | Test 11 |
| | 5 | Test 12 |
| | 4 | Test 13 |
| | 3 | Test 14 |
| | 2 | Test 15 |
| | 1 | Test 16 |
| 3-8 | | Manufacturer assigned test 17 through 64 |

Where each bit is interpreted:

0 = test not supported
1 = test supported

See Table 9 for an example:

***Table 9 - Example - use of test identifiers supported***

| Test Identifier Representations | Byte 1 | Byte 2 | Byte 3 | Byte 4 | Byte 5 | Byte 6 | Byte 7 | Byte 8 |
|---|---|---|---|---|---|---|---|---|
| 1. Binary | $00000100_b$ | $00000001_b$ | $00000000_b$ | $00000101_b$ | $10100000_b$ | $00000000_b$ | $00000000_b$ | $00000001_b$ |
| 2. Hex | $04_h$ | $01_h$ | $00_h$ | $05_h$ | $A0_h$ | $00_h$ | $00_h$ | $01_h$ |
| 3. Test Identifiers | 6 | 16 | | 30, 32 | 33, 35 | | | 64 |

5.7.11 Diagnostic Data Clear/Reset for Active DTCs (DM11)

5.7.11.1 DM11 for Non-OBD-Regulated Devices or Diagnostics

Use of the DM11 is to clear all of the diagnostic information pertaining to the active diagnostic trouble codes. A request for DM11 is sent whenever the service tool wishes to clear/reset diagnostic data for active DTCs. This is expected to occur once the problem has been corrected.

Upon completion of this operation, or if there are no faults to clear, a positive acknowledgment shall be sent if the request is destination specific (refer to SAE J1939-21 PGN 59392). If, for some reason, a device cannot perform the requested action, then it is required to send a negative acknowledgment (refer to SAE J1939-21 PGN 59392).

A positive or negative acknowledgment is not required in response to a global request.

Some applications need to provide a positive or negative acknowledgment to indicate the success or failure of a global request. (This is an exception as discussed in Note 4 of Table 5 in SAE J1939-21.)

Tool suppliers and controller applications that are dependent on the diagnostic data from other controllers should be aware that it can take as long as 5 seconds to complete the clearing action.

5.7.11.2 DM11 for OBD-Regulated Devices or Diagnostics

For OBD-regulated products, DM11 is used to clear all diagnostic information as defined below.

All OBD devices shall clear all diagnostic information upon receipt of a globally addressed request (PGN 59904) for DM11 when the engine is not running.

OBD devices are allowed to clear all diagnostic information with the engine running if appropriate. However, if any OBD device erases any OBD data, all OBD devices need to erase all required data. Therefore, either nothing should be cleared in key on engine running or everything that would be cleared in key on engine off must be cleared.

No partial clearing within a device or clearing in only certain OBD devices is acceptable. OBD (refer to 13 CCR 1971.1) regulated products that receive a destination-specific request for DM11 shall transmit an acknowledgment PG (refer to SAE J1939-21 PGN 59392) with the control byte set to 1 (negative acknowledgment) and shall clear no diagnostic information.

Tool suppliers and controller applications that depend on diagnostic data from other controllers (including displays for SPN 6810) should be aware that it can take as long as 5 seconds to complete the clearing action.

All diagnostic information pertaining to the active DTCs for regulated OBD products includes:

- Number of diagnostic trouble codes (e.g., those read via DM5, DM29, etc.)
- Diagnostic trouble codes for pending, active, and previously active malfunctions
    - Active diagnostic trouble codes (can be read with DM1)
    - Previously active diagnostic trouble codes (can be read with DM2)
    - Emission-related pending diagnostic trouble codes (can be read with DM6)
    - Emission-related previously MIL-On diagnostic trouble codes (can be read with DM23)
    - All pending DTCs (can be read with DM27)
- Number of DTCs and readiness information
    - Number of active DTCs in ECU (can be read with DM5)
    - Number of previously active DTCs in ECU (can be read with DM5)
    - Readiness information (can be read with read with DM5)
    - Readiness information (can be read with DM26)
    - Number of MIL-On DTCs and previously MIL-On DTCs (can be read with DM29)
- Trouble code for freeze frame data (can be read with DM4 and/or DM25)
- Freeze frame data (can be read with DM4 and/or DM25)
- Status of system monitoring tests (can be read with DM26)
- All monitor test results (can be read with DM8 and DM30)
- Distance traveled while MIL is activated (can be read with DM21)
- Number of warm-ups since DTC cleared (can be read with DM26)
- Distance since diagnostic trouble codes cleared (can be read with DM21)
- Minutes run by the engine while MIL is activated (can be read with DM21)
- Time since diagnostic trouble codes cleared (can be read with DM21)
- Other manufacturer specific “clearing/resetting” actions may also occur in response to this request message

5.7.11.3 Message Definition

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21)<br>A NACK is required if PG is not supported<br>(refer to SAE J1939-21 PGN 59392) |
| Data length: | 0 |
| Extended data page: | 0 |
| Data Page: | 0 |
| PDU Format: | 254 |
| PDU Specific: | 211 |
| Default priority: | 6 |
| Parameter group number: | 65235 (00FED3$_h$) |

5.7.12 Emission-Related MIL-On Diagnostic Trouble Codes (DM12)

The information communicated is limited to the currently MIL-On emission-related DTCs preceded by the diagnostic lamp status. Both are used to notify other components on the network of the diagnostic condition of the transmitting electronic component. The data contains the lamp status and a list of diagnostic codes and occurrence counts for currently MIL-On emission-related DTCs.

The lamp information (malfunction indicator lamp, red stop lamp, amber warning lamp, and protect lamp) should reflect the present state of the transmitting electronic component. The lamp information shall not convey temporary signals to provide for lamp test illumination, DTC numeric displays, or OBD readiness indication at key-on.

This DM contains DTCs that are confirmed and active and, in general, for which the MIL is on. Specific regulations may permit the MIL to not be illuminated for some emissions-related confirmed and active DTCs.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request<br>(refer to SAE J1939-21 PGN 59392) |
| Data Length: | Variable |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 254 |
| PDU Specific: | 212 |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |
| Parameter Group Number: | 65236 (00FED4$_h$) |

| | | | | |
|---|---|---|---|---|
| Byte: | 1 | bits 8-7 | Malfunction indicator lamp | see 5.7.1.4 |
| | | bits 6-5 | Red stop lamp | see 5.7.1.5 |
| | | bits 4-3 | Amber warning lamp | see 5.7.1.6 |
| | | bits 2-1 | Protect lamp | see 5.7.1.7 |
| Byte: | 2 | bits 8-7 | Flash malfunction indicator lamp | see 5.7.1.8 |
| | | bits 6-5 | Flash red stop lamp | see 5.7.1.9 |
| | | bits 4-3 | Flash amber warning lamp | see 5.7.1.10 |
| | | bits 2-1 | Flash protect lamp | see 5.7.1.11 |
| Byte: | 3 | bits 8-1 | SPN, 8 least significant bits of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: | 4 | bits 8-1 | SPN, second byte of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: | 5 | bits 8-6 | SPN, 3 most significant bits (most significant at bit 8) | see 5.7.1.12 |
| | | bits 5-1 | FMI (most significant at bit 5) | see 5.7.1.13 |
| Byte: | 6 | bit 8 | SPN conversion method | see 5.7.1.14 |
| | | bits 7-1 | Occurrence count | see 5.7.1.15 |

NOTE: When the occurrence count is not available, it should be set to all ones, which is a value of 127.

NOTE: When there is zero or one DTC to report, then unused bytes 7 and 8 of the CAN frame shall be set to 255 (as per SAE J1939-71).

NOTE: When there are no DTCs to report, each of bytes 3 through 6 shall be set to zero.

EXAMPLE 1: The following illustrates the message format for when there is more than one DTC.

Given:
a = lamp status
b = SPN
c = FMI
d = CM and OC

Message form is as follows: a,b,c,d,b,c,d,b,c,d,b,c,d....etc. In this example, the transport protocol of SAE J1939-21 has to be used to send the information because it requires more than 8 data bytes. Actually, any time there is more than one fault, the services of the transport protocol have to be used.

EXAMPLE 2: The following illustrates the message format for when a request of the DM12 is made and there are zero active emissions faults. Note that the malfunction indicator lamp is off while any of the other three (red stop lamp, amber warning lamp, and protect lamp) could be on. In this example, all three are on.

The original publication of this recommended practice defined that bytes 3 through 6 should be set to all ones when there are zero faults. This particular implementation is no longer permitted.

Given:

Byte: 1
- bits 8-7 = 00 (example of reporting off)
- bits 6-5 = 01 (example of reporting on)
- bits 4-3 = 01 (example of reporting on)
- bits 2-1 = 01 (example of reporting on)

Byte: 2
- bits 8-7 = 11 (example of reporting not available/don’t care)
- bits 6-5 = 11 (example of reporting not available/don’t care)
- bits 4-3 = 11 (example of reporting not available/don’t care)
- bits 2-1 = 11 (example of reporting not available/don’t care)

Bytes: 3-6 SPN = 0 (required setting for reporting no diagnostic trouble code)

FMI = 0 (required setting for reporting no diagnostic trouble code)

OC = 0 (required setting for reporting no diagnostic trouble code)

CM = 0 (required setting for reporting no diagnostic trouble code)

Byte: 7 = 255

Byte: 8 = 255

### 5.7.13 Stop-Start Broadcast (DM13)

DM13 has two primary functions.

It can be used as a **command,** from either a tool or an ECU, directed to a single controller or to all controllers to request the receiving controller(s) to **stop** or **start** broadcast messages.

Additionally, it can be used by an ECU to **inform** other nodes that the sender is about to **suspend** its normal broadcast due to commands other than an SAE J1939 DM13 command received on that same network segment.

The broadcast messages stopped, started, or suspended may be on networks other than SAE J1939.

The following notes help to clarify use of this PG.

1. This command shall only be initiated when the vehicle is at 0 km/h and at 0 engine rpm.

2. All nodes shall “power-up” in their normal broadcasting mode. Therefore, if any node was “powered-down” while in a “stop broadcast” condition, it would revert to its normal operation on power-up.

3. This is not a message to ignore all communications. It is a message to minimize network traffic. It is recognized that some network messages may be required to continue even during the “stop broadcast” condition. If an unsafe or undesirable vehicle operating condition would result from the lack of normal messages, then this mode would cause all nonessential messages to be inhibited.

4. Requests that are generated during the “stop broadcast” state should be responded to. However, devices that are programmed to periodically issue requests should postpone these requests until the “stop broadcast” state is exited.

5. All devices that have been told to change state, plus those nodes affected by the absence of broadcast messages, could look for the “hold signal” as a plausible explanation for why the information is missing. In addition, all devices that have been told to change state shall monitor the “hold signal.” If the “hold signal” disappears for 6 seconds, then all applicable nodes shall revert back to the normal state.

6. Diagnostic trouble codes should not be recorded for failed communications due to broadcast PGs missing during the modified broadcast state. Network devices should look for the hold signal to be absent for more than 6 seconds before recording any applicable diagnostic trouble code.

7. When this command is used to disable broadcasts of information on other networks, it could result in diagnostic trouble codes being reported about this situation. Therefore, it is recommended that the use of this stop-start broadcast command be used with caution.

One of the uses for the “stop-start broadcast PG” is to reduce network traffic during certain diagnostic procedures. As an example, while calibrating a control module, the diagnostic tool will likely want to **STOP** the normal broadcasts of all network devices keeping in mind the comments made in the notes section above. Another use is that it allows the diagnostic tool to potentially emulate a remote device during a diagnostic procedure. In this case, the diagnostic tool could generate the messages that the remote device would normally generate.

| | |
|---|---|
| Transmission Rate: | Sent whenever a stop or start broadcast event is necessary. To maintain the modified state of the vehicle network(s) the commanding device shall send the hold signal once every 5 seconds.<br>A NACK is required if PG is not supported (refer to SAE J1939-21 PGN 59392).<br>Note that the NACK is only provided if PGN 57088 is directed to a specific destination address. |
| Data Length: | 8 |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 223 |
| PDU Specific: | DA |
| Default Priority: | 6 |
| Parameter Group Number: | 57088 ($00DF00_h$) |

Stop-Start Broadcast[1]

| | | | | |
|---|---|---|---|---|
| Byte: | 1 | bits 8-7 | Current data link | see 5.7.13.4 |
| | | bits 6-5 | SAE J1587 network | see 5.7.13.5 |
| | | bits 4-3 | SAE J1922 network | see 5.7.13.6 |
| | | bits 2-1 | SAE J1939 network #1, primary vehicle network | see 5.7.13.7 |
| Byte: | 2 | bits 8-7 | SAE J1939 network #2 | see 5.7.13.8 |
| | | bits 6-5 | ISO 9141 network | see 5.7.13.9 |
| | | bits 4-3 | SAE J1850 network | see 5.7.13.10 |
| | | bits 2-1 | Other, manufacturer-specified port | see 5.7.13.11 |
| Byte: | 3 | bits 8-7 | SAE J1939 network #3 | see 5.7.13.12 |
| | | bits 6-5 | Proprietary network #1 | see 5.7.13.16 |
| | | bits 4-3 | Proprietary network #2 | see 5.7.13.17 |
| | | bits 2-1 | SAE J1939 network #4 | see 5.7.13.18 |
| Byte: | 4 | bits 8-5 | Hold signal | see 5.7.13.13 |
| | | bits 4-1 | Suspend signal | see 5.7.13.14 |
| Bytes: | 5-6 | | Suspend duration | see 5.7.13.15 |
| Byte: | 7 | bits 8-7 | SAE J1939 network #5 | see 5.7.13.19 |
| | | bits 6-5 | SAE J1939 network #6 | see 5.7.13.20 |
| | | bits 4-3 | SAE J1939 network #7 | see 5.7.13.21 |
| | | bits 2-1 | SAE J1939 network #8 | see 5.7.13.22 |
| Byte: | 8 | bits 8-7 | SAE Reserved | |
| | | bits 6-5 | SAE J1939 network #9 | see 5.7.13.23 |
| | | bits 4-3 | SAE J1939 network #10 | see 5.7.13.24 |
| | | bits 2-1 | SAE J1939 network #11 | see 5.7.13.25 |

For each of the 2-bit fields in the stop-start broadcast command, they are interpreted as follows:

| Bits | Information |
|---|---|
| 00 | Stop broadcast |
| 01 | Start broadcast |
| 10 | SAE Reserved |
| 11 | Don’t care/take no action (leave as is) |

When DM13 is used as a command to **STOP** broadcast, the sequence of operation is to first direct DM13 to each (or all) device(s) for which the broadcast state is desired to be modified. The second step is to send DM13 to the global destination address with the appropriate bits set to indicate the “hold signal” is being communicated. Figure 2 illustrates the sequence of messages for a command to stop broadcast to two specific nodes to turn off all ports. Figure 3 illustrates the sequence of messages for a command to stop broadcast on all nodes and all ports. The hold signal allows the issuer of the DM13 message to not have to send DM13 to specific addresses but rather to the group of controllers that were modified or all devices. This reduces the number of messages that are required to keep the modified broadcast state of each individual controller active. This has benefit when the individual devices are commanded to turn off different communication ports.

This sequence does not apply when the DM13 is used to announce that the sender is about to **SUSPEND** or cease communication.

5.7.13.1 DM13 Usage Requirements to Modify the Broadcast State of an ECU

To modify the broadcast state of an ECU:

- Set DM13 destination address to the global address or to the specific address.

- Set the action for each communication port to the desired state (stop, start, or don’t care/take no action).

- Set the hold signal and suspend signal to not available.

The receiving device(s) shall modify its broadcast state per the command.

5.7.13.2 DM13 Usage Requirements to Maintain the Stop Broadcast State of an ECU

To maintain the stop broadcast state of an ECU:

- Set DM13 destination address to the global address.
- Set action for each communications port to don’t care/take no action (leave as is).
- Set the hold signal to “all devices” or “devices whose broadcast state has been modified.”
- Set the suspend signal to not available.

Device(s) with a stopped broadcast state shall maintain their stopped broadcast state. Other devices that rely on broadcast data from a stopped device should take appropriate action.

5.7.13.3 DM13 Usage Requirements to Alert Network Devices of Impending Suspended Broadcasts

To alert network devices of impending suspended broadcasts:

- Set DM13 destination address to the global address.
- Set all communication ports to “don’t care/take no action.”
- Set the hold signal to not available.

The receiving device(s) should not record DTCs due to the absence of broadcast message data.

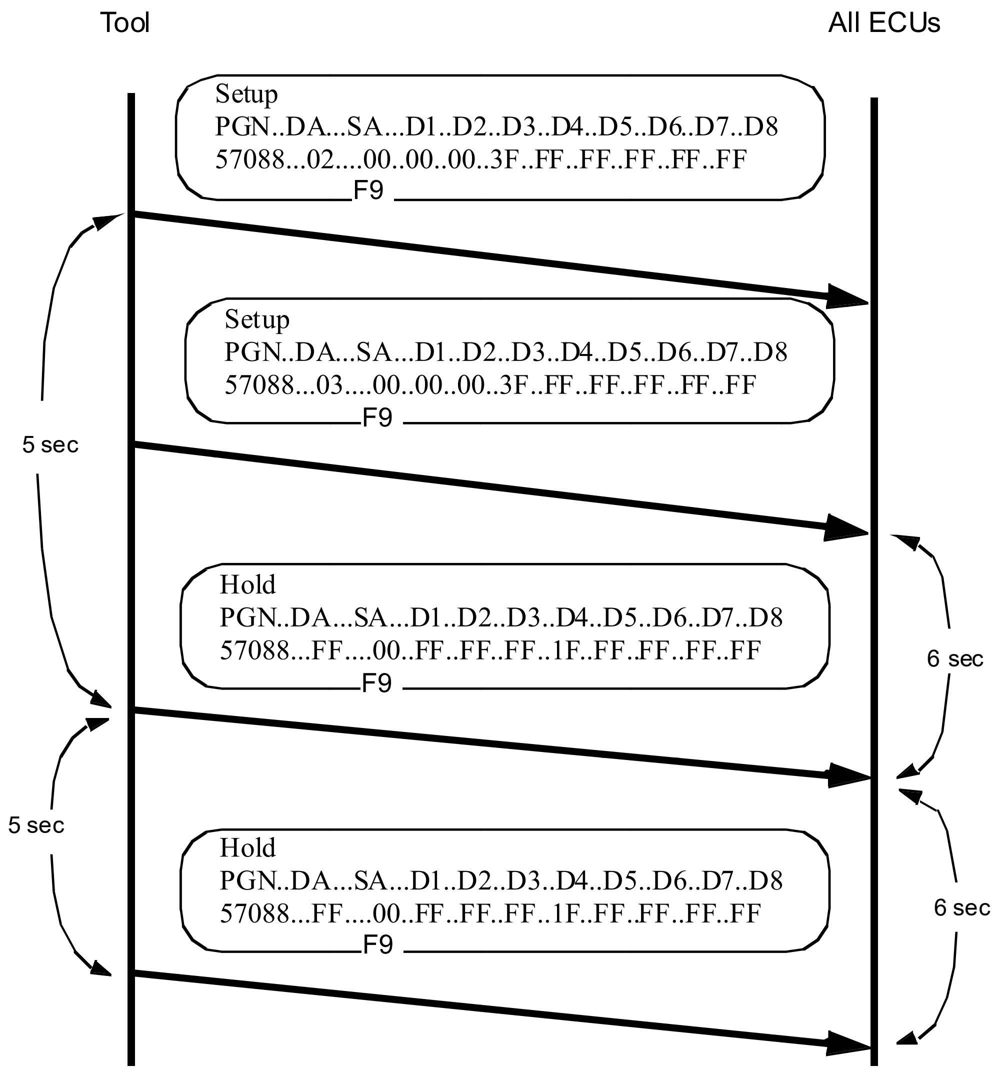


*Figure 2 - Stop-start broadcast to two specific nodes turning off all ports*

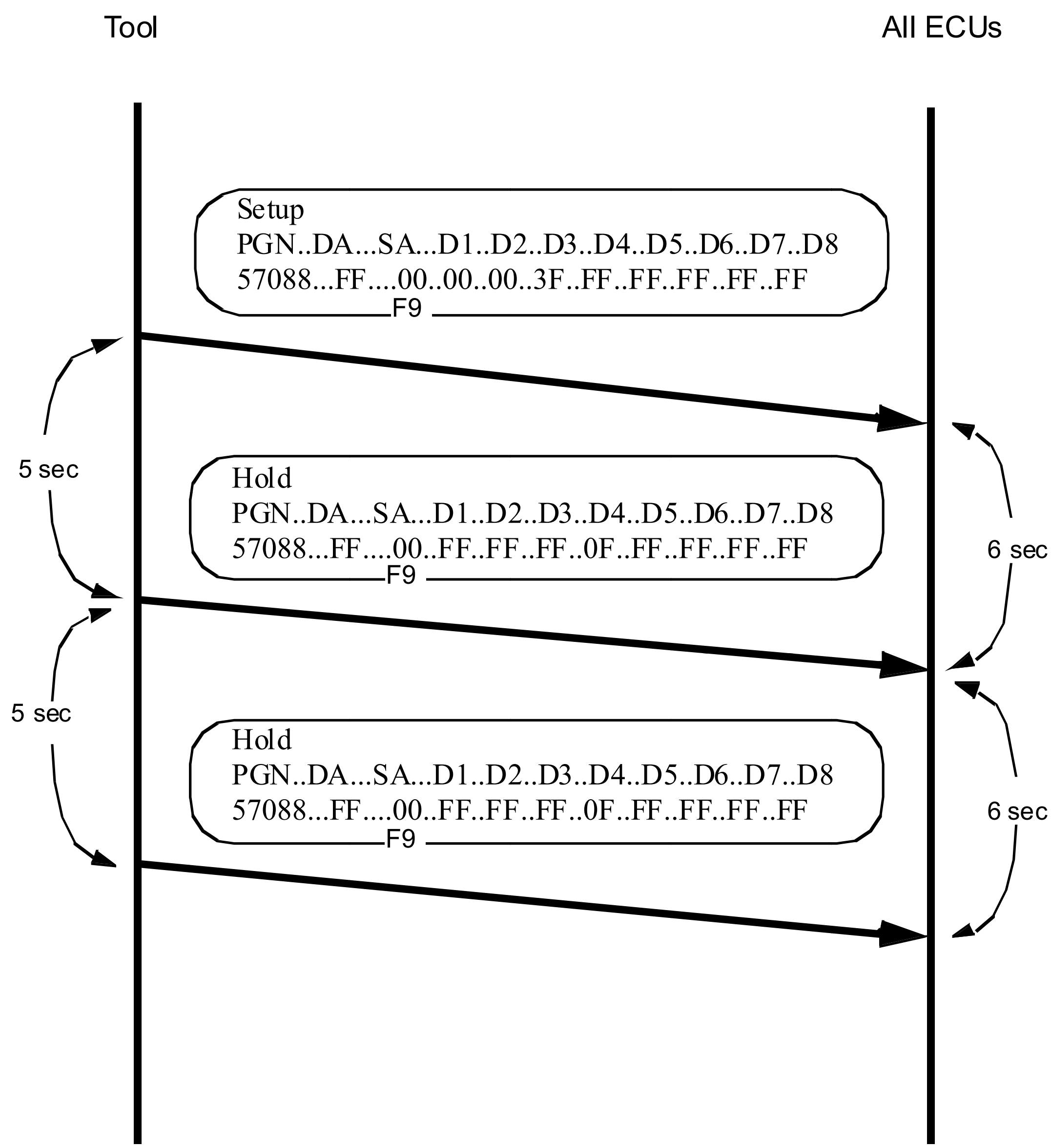


***Figure 3 - Stop-start broadcast to all nodes turning off all ports***

5.7.13.4 Current Data Link

Identifies the action to be performed on the communications port that this parameter was received on.

| | |
|---|---|
| 00 | Stop broadcast |
| 01 | Start broadcast |
| 10 | SAE Reserved |
| 11 | Don’t care/take no action (leave as is) |
| Type: | Status |
| Suspect Parameter Number: | 1230 |
| Reference: | 5.7.13 |

5.7.13.5 SAE J1587 Network

Identifies the action to be performed on the SAE J1587 communications port.

| | |
|---|---|
| 00 | Stop broadcast |
| 01 | Start broadcast |
| 10 | SAE Reserved |
| 11 | Don't care/take no action (leave as is) |
| Type: | Status |
| Suspect Parameter Number: | 608 |
| Reference: | 5.7.13 |

5.7.13.6 SAE J1922 Network

Identifies the action to be performed on the SAE J1922 communications port.

| | |
|---|---|
| 00 | Stop broadcast |
| 01 | Start broadcast |
| 10 | SAE Reserved |
| 11 | Don't care/take no action (leave as is) |
| Type: | Status |
| Suspect Parameter Number: | 622 |
| Reference: | 5.7.13 |

5.7.13.7 SAE J1939 Network #1, Primary Vehicle Network

Identifies the action to be performed on the SAE J1939 network #1 communications port.

| | |
|---|---|
| 00 | Stop broadcast |
| 01 | Start broadcast |
| 10 | SAE Reserved |
| 11 | Don't care/take no action (leave as is) |
| Type: | Status |
| Suspect Parameter Number: | 639 |
| Reference: | 5.7.13 |

5.7.13.8 SAE J1939 Network #2

Identifies the action to be performed on the SAE J1939 network #2 communications port.

| | |
|---|---|
| 00 | Stop broadcast |
| 01 | Start broadcast |
| 10 | SAE Reserved |
| 11 | Don't care/take no action (leave as is) |
| Type: | Status |
| Suspect Parameter Number: | 1231 |
| Reference: | 5.7.13 |

5.7.13.9 ISO 9141 Network

Identifies the action to be performed on the ISO 9141 communications port.

| | |
|---|---|
| 00 | Stop broadcast |
| 01 | Start broadcast |
| 10 | SAE Reserved |
| 11 | Don't care/take no action (leave as is) |
| Type: | Status |
| Suspect Parameter Number: | 1232 |
| Reference: | 5.7.13 |

5.7.13.10 SAE J1850 Network

Identifies the action to be performed on the SAE J1850 communications port.

| | |
|---|---|
| 00 | Stop broadcast |
| 01 | Start broadcast |
| 10 | SAE Reserved |
| 11 | Don’t care/take no action (leave as is) |
| Type: | Status |
| Suspect Parameter Number: | 1233 |
| Reference: | 5.7.13 |

5.7.13.11 Other, Manufacturer-Specified Port

Identifies the action to be performed on the “other, manufacturer-specified port” communications port.

| | |
|---|---|
| 00 | Stop broadcast |
| 01 | Start broadcast |
| 10 | SAE Reserved |
| 11 | Don’t care/take no action (leave as is) |
| Type: | Status |
| Suspect Parameter Number: | 1234 |
| Reference: | 5.7.13 |

5.7.13.12 SAE J1939 Network #3

Identifies the action to be performed on the SAE J1939 Network #3 communications port.

| | |
|---|---|
| 00 | Stop broadcast |
| 01 | Start broadcast |
| 10 | SAE Reserved |
| 11 | Don’t care/take no action (leave as is) |
| Type: | Status |
| Suspect Parameter Number: | 1235 |
| Reference: | 5.7.13 |

5.7.13.13 Hold Signal

Indicator to all nodes that the communication ports that have been acted upon by the “stop-start broadcast” PG are remaining in the modified state. Therefore all nodes should act accordingly. The hold signal is required to be broadcast every 5 seconds plus or minus one second. A device requesting stop broadcast shall send the hold signal every 5 seconds and if the message is not received for 6 seconds all applicable nodes revert back to their normal state.

**HOLD SIGNAL States**

| **Bit States for bits 8-5** | **Devices to take action** |
|---|---|
| 0000 | All devices |
| 0001 | Devices whose broadcast state has been modified |
| 0010 to 1110 | SAE Reserved |
| 1111 | Not available |
| Type: | Status |
| Suspect Parameter Number: | 1236 |
| Reference: | 5.7.13 |

5.7.13.14 Suspend Signal

Indicator to all nodes that broadcast messages are being suspended due to SAE J1939 DM13 commands or for other reasons such as commands on another network. Therefore, the receiving nodes should suspend time-out diagnostics for all messages from the transmitting devices. The suspend signal is to be broadcast once, but it may be repeated at the option of the transmitting device (if it is capable of doing so) to increase the chances of proper reception by repeating one or two times within the first second of the suspension. If it is able, the transmitter may also send a DM13 message with the suspend signal set to "1110" to indicate that it is returning to full broadcast status.

The suspend signal may not be needed if the suspension is due to DM13 commands on the current network. In this case, the hold signal is sufficient notification to other devices. If the DM13 stop broadcast signal was sent to a destination-specific device, that device may use the suspend signal as additional notification to other devices that it will be offline (and optionally for the indicated duration).

The suspend signal is not a command to suspend broadcast.

**SUSPEND SIGNAL States**

| **Bit States** | **Suspension Status** |
|---|---|
| 0000 | Indefinite suspension of all broadcasts |
| 0001 | Indefinite suspension of some messages |
| 0010 | Temporary suspension of all broadcasts |
| 0011 | Temporary suspension of some messages |
| 0100 TO 1101 | SAE reserved |
| 1110 | Resuming normal broadcast pattern |
| 1111 | Not available |
| Type: | Status |
| Suspect Parameter Number: | 2618 |
| Reference: | 5.7.13 |

5.7.13.15 Suspend Duration

Indicates the duration of a suspension of broadcast messages when that duration is known by the transmitting device. If the DM13 message is sent with the suspend signal value of 0010 or 0011, the value of this parameter will represent the duration of suspension in seconds. For the suspend signal values of 0000 to 0001, the suspend duration will be sent as not available and the duration will be indeterminate. Receivers will know when the suspension is over by the renewed presence of messages from the transmitter either for indefinite suspension or, in the case where the transmitter is permitted by an outside command, to return to full broadcast status. If it is able, the transmitter may also send a DM13 message with the suspend signal set to "1110" to indicate that it is returning to full broadcast status.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | Offset: 0 seconds, 1 s/bit |
| Data Range: | 0 to 64255 seconds (0 to 1070.9 minutes) |
| Type: | Status |
| Suspect Parameter Number: | 2619 |
| Reference: | 5.7.13 |

5.7.13.16 Proprietary Network #1

Identifies the action to be performed on the Proprietary Network #1 communications port.

| | |
|---|---|
| 00 | Stop broadcast |
| 01 | Start broadcast |
| 10 | SAE Reserved |
| 11 | Don't care/take no action (leave as is) |
| Type: | Status |
| Suspect Parameter Number: | 625 |
| Reference: | 5.7.13 |

5.7.13.17 Proprietary Network #2

Identifies the action to be performed on the Proprietary Network #2 communications port.

| | |
|---|---|
| 00 | Stop broadcast |
| 01 | Start broadcast |
| 10 | SAE Reserved |
| 11 | Don’t care/take no action (leave as is) |
| Type: | Status |
| Suspect Parameter Number: | 5588 |
| Reference: | 5.7.13 |

5.7.13.18 SAE J1939 Network #4

Identifies the action to be performed on the SAE J1939 Network #4 communications port.

| | |
|---|---|
| 00 | Stop broadcast |
| 01 | Start broadcast |
| 10 | SAE Reserved |
| 11 | Don’t care/take no action (leave as is) |
| Type: | Status |
| Suspect Parameter Number: | 1668 |
| Reference: | 5.7.13 |

5.7.13.19 SAE J1939 Network #5

Identifies the action to be performed on the SAE J1939 Network #5 communications port.

| | |
|---|---|
| 00 | Stop broadcast |
| 01 | Start broadcast |
| 10 | SAE Reserved |
| 11 | Don’t care/take no action (leave as is) |
| Type: | Status |
| Suspect Parameter Number: | 1669 |
| Reference: | 5.7.13 |

5.7.13.20 SAE J1939 Network #6

Identifies the action to be performed on the SAE J1939 Network #6 communications port.

| | |
|---|---|
| 00 | Stop broadcast |
| 01 | Start broadcast |
| 10 | SAE Reserved |
| 11 | Don’t care/take no action (leave as is) |
| Type: | Status |
| Suspect Parameter Number: | 1670 |
| Reference: | 5.7.13 |

5.7.13.21 SAE J1939 Network #7

Identifies the action to be performed on the SAE J1939 Network #7 communications port.

| | |
|---|---|
| 00 | Stop broadcast |
| 01 | Start broadcast |
| 10 | SAE Reserved |
| 11 | Don’t care/take no action (leave as is) |
| Type: | Status |
| Suspect Parameter Number: | 1671 |
| Reference: | 5.7.13 |

5.7.13.22 SAE J1939 Network #8

Identifies the action to be performed on the SAE J1939 Network #8 communications port.

| | |
|---|---|
| 00 | Stop broadcast |
| 01 | Start broadcast |
| 10 | SAE Reserved |
| 11 | Don’t care/take no action (leave as is) |
| Type: | Status |
| Suspect Parameter Number: | 1672 |
| Reference: | 5.7.13 |

5.7.13.23 SAE J1939 Network #9

Identifies the action to be performed on the SAE J1939 Network #9 communications port.

| | |
|---|---|
| 00 | Stop broadcast |
| 01 | Start broadcast |
| 10 | SAE Reserved |
| 11 | Don’t care/take no action (leave as is) |
| Type: | Status |
| Suspect Parameter Number: | 1673 |
| Reference: | 5.7.13 |

5.7.13.24 SAE J1939 Network #10

Identifies the action to be performed on the SAE J1939 Network #10 communications port.

| | |
|---|---|
| 00 | Stop broadcast |
| 01 | Start broadcast |
| 10 | SAE Reserved |
| 11 | Don’t care/take no action (leave as is) |
| Type: | Status |
| Suspect Parameter Number: | 1674 |
| Reference: | 5.7.13 |

5.7.13.25 SAE J1939 Network #11

Identifies the action to be performed on the SAE J1939 Network #11 communications port.

| | |
|---|---|
| 00 | Stop broadcast |
| 01 | Start broadcast |
| 10 | SAE Reserved |
| 11 | Don’t care/take no action (leave as is) |
| Type: | Status |
| Suspect Parameter Number: | 9511 |
| Reference: | 5.7.13 |

5.7.14 Memory Access Request (DM14)

The main use for the Memory_Access_Request message is by a tool wishing to alter the memory of a device. The tool uses this message to convey its request as well as any security information that must be passed to the device to prove the tool has authority to request said operation. The message may also be used to obtain the current status of a device in terms of the availability of said device’s memory access. The capabilities of the functions are outlined in Appendix B. The procedures for data interchange are outlined in Appendix C. A memory access state transition diagram (DM14 through DM18) is in Appendix C, Figure C1.

| | |
|---|---|
| Transmission Rate: | As needed |
| Data Length: | 8 |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 217 |
| PDU Specific: | DA |
| Default Priority: | 6 |
| Parameter Group Number: | 55552 ($00D900_h$) |

MEMORY_ACCESS_REQUEST

| | | | | |
|---|---|---|---|---|
| Byte: | 1 | bits 8-1 | Length/number requested<br>(Least significant 8 bits)<br>(Bit 1 is least significant bit) | see 5.7.14.5 |
| Byte: | 2 | bits 8-6 | Length/number requested<br>(Most significant 3 bits)<br>(Bit 8 is most significant bit) | see 5.7.14.5 |
| | | bit 5 | Pointer type | see 5.7.14.1 |
| | | bits 4-2 | Command | see 5.7.14.6 |
| | | bit 1 | SAE Reserved (sent as a one) | |
| Bytes: | 3-5 | | Pointer<br>(Byte 3 is least significant byte)<br>(Bit 1 is least significant bit) | see 5.7.14.3 |
| Byte: | 6 | | Pointer extension<br>(Most significant byte of pointer/pointer extension)<br>(Bit 8 is most significant bit) | see 5.7.14.2 |
| Byte: | 7-8 | | Key/User_Level | see 5.7.14.4 |

#### 5.7.14.1 Pointer Type

Indicates whether the pointer and pointer extension are direct memory addresses (pointer type identifier = 0) or if the pointer extension is identifying a particular SPACE with the pointer referencing a specific OBJECT within that particular SPACE (pointer type identifier = 1).

| | |
|---|---|
| Data Length: | 1 bit |
| Resolution: | Bitmap |
| Data Range: | 0 or 1 |
| 0 | Direct memory addressing |
| 1 | Directed spatial addressing |
| Type: | Status |
| Suspect Parameter Number: | 1641 |
| Reference: | 5.7.14 |

##### 5.7.14.1.1 Direct Memory Addressing

A pointer type value of 0 implies a memory access with a direct memory address (in bytes) with the pointer extension (8 bits) simply concatenated as the higher order bits and with the pointer (24 bits) as the lower order bits to form a single 32-bit address. (Parsing for device memory widths other than 1 byte is explained in 5.7.14.3.1.)

##### 5.7.14.1.2 Directed Spatial Addressing

A pointer type value of 1 implies a memory structure where the pointer extension provides identification of a particular space within memory and the pointer provides identification of a specific object within that particular space. This provides a form of directed spatial (object) addressing, where the user can control the meaning of the pointer used to interrogate a device. For this directed spatial addressing, half of the available spaces (128 of the 256 formed by the 8-bit pointer extension) will be reserved for assignment by SAE. The other half will be labeled proprietary and not constrained by the standard, allowing the manufacturer proprietary assignment. The first committee-assigned space is 0 (i.e., pointer type = 1 and pointer extension = 0), and it is assigned to be the space containing those parameters that can be identified by a particular SPN; this shall be referred to as the SPN space. Objects contained within the direct spatial address space may be variable length, and they may have different lengths.

5.7.14.2 Pointer Extension

This 8-bit parameter is either the high order 8 bits of a complete direct memory address, 5.7.14.1.1 (pointer type = 0), or the identifier of a particular SPACE, 5.7.14.1.2 (pointer type = 1) (see Table 10).

| | |
|---|---|
| Data Length: | 8 bits |
| Resolution: | Per definitions in this section |
| Data Range: | 0 to 255 |
| Type: | Status |
| Suspect Parameter Number: | 1643 |
| Reference: | 5.7.14 |

***Table 10 - Pointer extension states (if pointer type is "1")***

| Bit States | Pointer Extension States |
|---|---|
| $00000000_b$ | SPN SPACE |
| $00000001_b$ to $01111111_b$ | Reserved for assignment by SAE |
| $10000000_b$ to $11111111_b$ | OEM proprietary definition |

5.7.14.2.1 SPN Space

A pointer extension value of "0" in combination with a pointer type of "1" implies that the pointer is to the parameter identified by a specific SPN. Hence, the data will be the value of the parameter known by that SPN.

NOTE: Since some SPNs have no parameter associated with them and, in some cases, the device may not know the particular SPN's data, the memory access response may be "busy" with or without an error indicator.

The length of the data associated with each SPN is a function of that SPN, and the overall length of the response message data will be the sum of the byte lengths of the data for each of the SPNs, with the actual number of SPNs being determined from the value of length/number requested parameter. The five leading bits should be "$00000_b$" to fill the 24-bit pointer when using a 19-bit SPN. Later, at the committee's discretion, functions may be assigned to different values of these five bits. Therefore, they should be included both when interpreting a value and when sending a value.

5.7.14.2.2 Reserved for Assignment by SAE

Implies these values are not yet defined by the committee and are therefore not available for use. Future versions of this document will assign specific meanings.

5.7.14.2.3 OEM Proprietary Definition

Implies these values are available to OEMs for proprietary definition and use. (One such example use might be for memory block access.)

5.7.14.3 Pointer

If pointer type 0 is used, this 24-bit parameter, which has a value of 0 to 16777215 (0 to $FFFFFF_h$) with no reserved ranges, is concatenated with the 8-bit pointer extension to form a direct memory address. The address thus formed represents the first address to be accessed within the memory in units of bytes. If pointer type 1 is used, the pointer is to provide the identification of the specific OBJECT within whatever particular SPACE is being identified by the pointer extension. The direct memory address should be parsed as outlined in 5.7.14.3.1 if the device memory width is other than 1 byte.

| | |
|---|---|
| Data Length: | 24 bits |
| Resolution: | 1 byte/bit |
| Data Range: | 0 to 16777215 (0 to $FFFFFF_h$) |
| Type: | Status |
| Suspect Parameter Number: | 1644 |
| Reference: | 5.7.14 |

5.7.14.3.1 Memory Parsing

For all memory widths, the starting address is simply the pointer extension concatenated with the pointer (the pointer being the lower 24 bits and the extension the upper 8). For memory widths of 1 byte, there is a one-to-one mapping between data and the memory. Hence, the first data byte goes into the memory at the starting address, while the second data byte corresponds to the memory at the starting address plus one. For widths other than 1 byte, the data cannot map directly to the memory but must be used to assemble the necessary width. Hence, it will take as many data bytes per address as seven, plus the memory width in bits all divided by eight ([memory width + 7]/8). To maintain consistency with the rest of this standard, the first data byte should be used for the byte containing bits 1 to 8 at the starting address. The second data byte should be used for bits 9 to 16. This should continue for the number of bytes required; then the address should be incremented and those bytes filled. When the memory width is less than 1 byte (as might happen when addressing a 2-bit parameter through the SPN space), a whole byte is used to contain each object's data (in other words, no packing is to occur). The two bits will be placed in the least significant bits of the byte. For systems where the memory width is not an integer number of bytes, some bits in the highest byte are unused, reducing transfer efficiency but enabling all memory widths to be handled. Examples of address calculation and byte association (see 5.7.16.2) include:

- 8-bit-wide memory, pointer extension = $10_h$, pointer = $367800_h$ then the starting memory address is $10367800_h$ and the first byte of raw binary data would map directly into the memory at $10367800_h$, the second byte of raw binary data would then map into memory $10367801_h$, and so on until completed.

- 16-bit-wide memory, pointer extension = $10_h$, pointer = $367800_h$ then the starting memory address is $10367800_h$ and the first byte of raw binary data would map into bits 1 to 8 of the memory at $10367800_h$, while the second byte of raw binary data would map into bits 9 to 16 of the same memory. The third byte of raw binary data would then map into bits 1 to 8 of the memory at $10367801_h$, while the fourth byte of raw binary data would map into bits 9 to 16 of the memory at $10367801_h$.

- 32-bit-wide memory, pointer extension = $10_h$, pointer = $367800_h$ then the starting memory address is $10367800_h$ and the first byte of raw binary data would map into bits 1 to 8 of the memory at $10367800_h$, while the second byte of raw binary data would map into bits 9 to 16 of the memory at $10367800_h$, the third byte of raw binary data would then map into bits 17 to 24 of the same memory, and the fourth byte of raw binary data would map bits 25 to 32. The fifth byte of raw binary data would then map bits 1 to 8 of $10367801_h$, while the sixth byte of raw binary data would map into bits 9 to 16 of $10367801_h$, the seventh byte of raw binary data then mapping bits 17 to 24, and the eighth byte of raw binary data mapping to bits 25 to 32 of the memory.

- 12-bit-wide memory, pointer extension = $10_h$, pointer = $367800_h$ then the starting memory address is $10367800_h$ and the first byte of raw binary data would map into bits 1 to 8 there, while bits 9 to 12 of the second byte of raw binary data would map into bits 9 to 12 of $10367800_h$ (bits 13 to 16 are simply unused). The third byte of raw binary data would then map into bits 1 to 8 of $10367801_h$, while bits 9 to 12 of the fourth byte of raw binary data would map into bits 9 to 12 of $10367801_h$.

5.7.14.3.2 Handling of Pointer Offset

When the starting address created by concatenating the pointer extension and the pointer does not represent the beginning of an object, such as a memory block or memory word, the device shall be free to reject the requested memory access operation. If used by the manufacturer, then the appropriate error indicator/EDC parameter may be returned (see Table 13 and 5.7.15.3).

5.7.14.4 Key/User_Level

This is a 2-byte parameter that is used by the tool to primarily send a key to the device but can also be used by the tool to provide a password or a User_Level to the device if desired (see Appendix C). This Key/User_Level parameter can be used to send these independent variables since they will never be transmitted within the same message (a password or User_Level parameter would be sent at the beginning of an operation, while a key CANNOT be sent until after the receipt of a seed).

| | |
|---|---|
| Data Length: | 16 bits |
| Resolution: | Per definitions in this section |
| Data Range: | 0 to 65535 (0 to $FFFF_h$) |
| Type: | Status |
| Suspect Parameter Number: | 1645 |
| Reference: | 5.7.14 |

5.7.14.4.1 Key

The result of a set of mathematical operations performed upon a seed to provide a device with a means of authenticating a tool’s request (see Table 11).

***Table 11 - Key states (i.e., only for key and not user level)***

| Bit States | Key States |
|---|---|
| $0000_h$ | Use long seed or key from data security message |
| $0001_h$ to $FFFE_h$ | Key values |
| $FFFF_h$ | No key available |

5.7.14.4.1.1 Use Long Seed or Key from Data Security Message

The actual seed or key is in the data security message, and this is simply a flag.

5.7.14.4.1.2 Key Values

The actual values of the key.

5.7.14.4.1.3 No Key Available

There is no key at this time.

5.7.14.4.2 Seed

A number sent by a device to a tool to obtain authentication of the tool’s right to access the device. The tool shall return a key, which is a function of the seed, and the key matches the device’s expectations to obtain access.

5.7.14.4.3 Password

The number sent when using a simple authentication technique wherein both the device and tool have a prior knowledge of the specific number and usually use equality as the verification.

5.7.14.4.4 User_Level

A number sent by a tool to a device along with an initial request to inform the device of some specific level of access that the tool wishes to gain. In such a case, there is probably a following seed and key exchange. Usually, the seed and the mathematical operations to calculate the key from it would be a function of the User_Level requested.

5.7.14.4.5 Handling of Keys Larger Than 16 Bits

If a manufacturer feels a seed/key structure requires a key or seed longer than 16 bits for a particular device, this can be handled with the data security message (see 5.7.18) and the setting of the Key/User_Level and the seed parameters appropriately (see Tables 11 and 14). See Figures E9 and E10.

5.7.14.4.6 Acceptance Rules

The device will establish the set of rules governing acceptance of memory access requests (such as operational mode, User_Level versus memory (object) location versus requested operation, etc.). These rules may be manufacturer specific to prevent unauthorized modification of a device's memory. The command and the length/number requested with the memory access request message(s) should be constant throughout an entire sequence, or the device should reject the operation.

5.7.14.5 Length/Number Requested

This is an 11-bit parameter that identifies the amount of memory (i.e., the range within the memory) over which the tool desires an operation to be carried out when the command of the memory access request message is a read, write, boot load, or error detection and/or correction parameter (EDCP) generation. For these cases, the length is in bytes (cross-reference to 5.7.14.3.1) when the pointer type is "0" and in objects when the pointer type is "1" (for example, a length of two when referencing the SPN space with a pointer of 1648 would imply that you wanted the data of the two parameters with SPNs 1648 and 1649). When the command is erased, the length is the number of "blocks" of memory to be erased, with block size being specific to the device. When the command of the memory access request is operation failed, operation completed, or status request, the length is meaningless. The tool should therefore send it as "0," and the device should treat it as "DO NOT CARE."

| | |
|---|---|
| Data Length: | 11 bits |
| Resolution: | Pointer type 0 = 1 byte/bit<br>Pointer type 1 = 1 object/bit |
| Data Range: | 0 to 1784 |
| Type: | Status |
| Suspect Parameter Number: | 1640 |
| Reference: | 5.7.14 |

5.7.14.6 Command

This is a 3-bit parameter that allows the tool to send commands to the device. All memory access requests originate at a tool and are considered commands. Some of the values within the command have been overlaid with the same values in status, so perhaps a similar variable can be used.

| | |
|---|---|
| Data Length: | 3 bits |
| Resolution: | 1 bit/command |
| Data Range: | 0 to 7 |
| Type: | Status |
| Suspect Parameter Number: | 1642 |
| Reference: | 5.7.14 |

The following command values for memory access requests (i.e., tool to device) are defined:

| | |
|---|---|
| 0 - Erase | see 5.7.14.6.1 |
| 1 - Read | see 5.7.14.6.2 |
| 2 - Write | see 5.7.14.6.3 |
| 3 - Status Request | see 5.7.14.6.4 |
| 4 - Operation Completed | see 5.7.14.6.5 |
| 5 - Operation Failed | see 5.7.14.6.6 |
| 6 - Boot Load | see 5.7.14.6.7 |
| 7 - EDCP Generation | see 5.7.14.6.8 |

5.7.14.6.1 Erase

This command is a block erase where the length is defined as the number of blocks to erase. The length and width of a block are to be defined within the device and shall be known by the tool. The pointer is the address where erase should start. If the pointer is not on a block boundary, when corrected for memory width, then the erase is not to be allowed (i.e., the device is to respond with a "busy" with or without data in the error indicator/EDC parameter per the manufacturer's choice; see C.2.2.1).

5.7.14.6.2 Read

This command allows the transfer of the contents from a device memory to a tool. When this command is accepted, the device transfers the appropriate memory contents to the tool, including initiating a transport protocol session if necessary. See Figure E1 for an example message sequence used to accomplish a memory read operation with security (short form of security). See Figure E2 for an example message sequence used to accomplish a memory read operation without security. See Figure E3 for an example message sequence used to accomplish a multiple memory read operation with security (short form of security). Note that the operation complete message from the tool indicates the session is over from the tool's perspective. See Figure E7 for an example of a failed memory access read operation.

5.7.14.6.3 Write

This command allows the transfer of the new memory contents from a tool to a device. The device may use a smart write, which will force an erase before write if it is going to be necessary, due to the value being written and the memory type, and if any other memory that will be altered by the erase can be buffered and rewritten to its original value, affecting a transparent write operation. See Figure E5 for an example message sequence used to accomplish a memory write operation with security (short form of security).

5.7.14.6.4 Status Request

This command allows a tool to interrogate the device to determine the current status of operation. This enables a tool to determine what a device may currently be doing and/or why it has not heard a message indicating operation completed/failed from the device, when the tool itself believes sufficient time has elapsed for the operation. The device responds to this command with either operation completed, operation failed, proceed, or busy with a code indicating the current status or error condition within the feedback parameter.

5.7.14.6.5 Operation Completed

This command is sent during a close sequence. Operation completed is sent by the tool during the close sequence of an erase, read, write, boot load, or EDCP generation command to indicate to the device that the tool has heard the device's close and that the close sequence is completed. At the end of a read command, it further indicates that all of the expected memory contents were received. A device's receipt of an operation completed from a tool enables the device to consider the memory access finished. The device should have a time-out function such that on the failure to hear the expected operation completed (or operation failed) from the tool it assumes the memory access operation with the tool is complete. See Figure E4. The value for this time-out should be 100 ms with no worse than ±25 ms error. (See 5.7.15.1.3.)

5.7.14.6.6 Operation Failed

This command is sent by a tool only during the close sequence of a read command to indicate to the device that the expected memory contents were not received. (This initiates no further action from the device.)

If DM16 requires using TP when a non-recoverable transport session failure (e.g., session time-out) occurs, then the TP session shall be aborted first before sending the DM14 operation failed command.

5.7.14.6.7 Boot Load

This command allows a tool to transfer the execution of a device to some address and, if needed, write new values into this executable memory prior to transfer of execution. When no data is written, the device considers the operation simply a transfer of execution and continues operation with no change in network communications but with whatever other software changes the manufacturer has chosen to implement. When writing new data any values shall be executable upon a successful close, the device transfers execution to the address specified by the request initiating this operation. Once execution has transferred at the close of a boot load, the device is no longer required (it may do so if its designers choose) to operate upon any messages from nodes other than the specific tool that initiated the boot load. If the device no longer operates upon other messages, the tool is to be required to protect the address of the device from any address claim messages sent during the boot load process. If the close sequence of boot load, with data, indicates successful completion, then the tool sends data to the device using the boot load data PG until the tool determines that the boot load sequence has been completed. The tool then notifies the operator that the operation is complete so that the device (and probably the system and network) can be restarted at its power on self-test. There is no predefined close sequence for the end of the boot load data transfer provided by this standard. It is at the manufacturer's discretion to choose to have such a sequence.

5.7.14.6.8 EDCP Generation

This command allows a tool to request a device to generate a checksum or other form of memory error detection and correction parameter over some range of memory. It is expected that the tool shall have a prior knowledge of the length and generation procedure used by the device. Parameters greater than 24 bits in length are handled by a looping concatenation structure (see 5.7.15.2). The memory involved in this operation is the same as that defined within the read operation.

5.7.15 Memory Access Response (DM15)

The main use for the Memory_Access_Response is for a device to answer a tool that has attempted to access the memory within the device. With this message the device can request further security responses from the requestor (see Figure E6) as well as tell the requestor what is or is not allowed. The completion status of a memory operation may also be transferred with this message. A memory access state transition diagram (DM14 through DM18) is in Appendix C.

| | |
|---|---|
| Transmission Rate: | As needed |
| Data Length: | 8 |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 216 |
| PDU Specific: | DA |
| Default Priority: | 6 |
| Parameter Group Number: | 55296 (00D800$_h$) |

MEMORY_ACCESS_RESPONSE

| | | | | |
|---|---|---|---|---|
| Byte: | 1 | bits 8-1 | Length/number allowed<br>(Least significant 8 bits)<br>(Bit 1 is least significant bit) | see 5.7.15.5 |
| Byte: | 2 | bits 8-6 | Length/number allowed<br>(Most significant 3 bits)<br>(Bit 8 is most significant bit) | see 5.7.15.5 |
| | | bit 5 | SAE Reserved | |
| | | bits 4-2 | Status | see 5.7.15.1 |
| | | bit 1 | SAE Reserved | |
| Bytes: | 3-5 | | Error indicator/EDC parameter<br>(Byte 3 is least significant byte)<br>(Bit 1 is least significant bit) | see 5.7.15.3 |
| Byte: | 6 | | EDCP extension<br>(When used as an EDCP extension,<br>this is the most significant byte)<br>(Bit 8 is most significant bit) | see 5.7.15.2 |
| Bytes: | 7-8 | | Seed | see 5.7.15.4 |

5.7.15.1 Status

This is a 3-bit parameter that allows the device to return its status. All memory access requests originate at a tool and are considered commands. All memory access responses originate at a device and are considered status. The device may choose to send further information on its status within the error indicator/EDC parameter (see 5.7.15.2 and 5.7.15.3).

| | |
|---|---|
| Data Length: | 3 bits |
| Resolution: | 1 status value/bit |
| Data Range: | 0 to 7 |
| Type: | Status |
| Suspect Parameter Number: | 1646 |
| Reference: | 5.7.15 |

The following status values for memory access responses (i.e., device to tool) are defined:

| | |
|---|---|
| 0 - Proceed | see 5.7.15.1.1 |
| 1 - Busy | see 5.7.15.1.2 |
| 2 - SAE Reserved | |
| 3 - SAE Reserved | |
| 4 - Operation Completed | see 5.7.15.1.3 |
| 5 - Operation Failed | see 5.7.15.1.4 |
| 6 - SAE Reserved | |
| 7 - SAE Reserved | |

5.7.15.1.1 Proceed

This status is sent from a device to indicate that a specific tool may continue with the sequence of a memory access operation the tool had requested. When sent as a response to a status request command, this means the device is not presently engaged in any memory access operation (i.e., is not busy). Optionally, at the manufacturer’s preference, the error indicator/EDC parameter may contain the error indicator for the previous operation the device had performed (should be $FFFFFF_h$ otherwise).

5.7.15.1.2 Busy

This status is sent from the device to indicate to a tool that there is a condition that prevents the sequence from continuing. The length/number allowed parameter will be zero, the error indicator/EDC parameter will contain a value indicating the condition that is preventing the memory access from continuing, with the manufacturer having a choice of how detailed the error indicator is (see 5.7.15.3), and the value of the other parameters will be treated as Do Not Care. When issued as a response to a memory access status request command, this means the device may still be busy and in the process of completing a requested operation (this includes, but is not limited to: transmitting/receiving data required for an operation, erasing memory, or programming memory). See Figure E8 for an example use of the busy indication.

5.7.15.1.3 Operation Completed

This status is sent during a close sequence or in response to a status request command. Operation completed is sent as status from the device during the close sequence of an erase, read, write, boot load, or EDCP generation command to indicate that the request was successfully completed. There may be an EDC value contained within the error indicator/EDC parameter. This status is the start of the close sequence for all successful commands that operate upon a device’s memory. A device’s receipt of an operation completed from a tool enables the device to consider the memory access finished. (See 5.7.14.6.5.) The device should have a time-out such that on failure to hear the expected operation completed (or operation failed) from the tool it closes the session. The value for this time-out should be 100 ms with no worse than ±25 ms error. See Figure E4 for an example where the tool does not send the required operation complete message. When the operation completed message is sent by a device in response to a status request, it indicates that the last operation was successfully completed only if the close sequence has not been completed. Once the close sequence is completed for an operation, a device no longer needs to maintain any data about that operation and may send a status of proceed, with or without the error indicator value from the previous operation, in response to a status request.

5.7.15.1.4 Operation Failed

This status is sent during a close sequence or in response to a status request command. Operation failed is sent as status from the device during the close sequence of an erase, write, boot load, or EDCP generation command to indicate that the request was unsuccessful. The error indicator/EDC parameter should contain an error indicator. When sent in response to a status request, it indicates that the last operation failed only if the close sequence has not been completed. Once the close sequence is completed for an unsuccessful operation, the device sends the proceed status in response to a status request.

5.7.15.2 EDCP Extension

This is an 8-bit parameter used to identify how to handle the data in the error indicator/EDC parameter. This EDCP extension parameter is used within the memory access response message (device to tool). Meaning shall be determined from a table of predefined values (see Table 12). If there is no error indicator/EDC parameter being sent, then this (EDCP extension) parameter shall be properly set ($11111111_b$). The use of the error indicator/EDC parameter is at the manufacturer's discretion, but it shall be properly set relative to this parameter. For example, suppose the unit is not willing to reveal the current cause of an error for security reasons, then if this EDCP extension is set to $00000110_b$, then the error indicator/EDC parameter shall be set to $000001_h$ to indicate the error is not identified (see Table 12).

| | |
|---|---|
| Data Length: | 8 bits |
| Resolution: | 1 state/bit |
| Data Range: | 0 to 255 (0 to $FF_h$) |
| Type: | Status |
| Suspect Parameter Number: | 1647 |
| Reference: | 5.7.15 |

***Table 12 - EDCP extension states***

| Bit States | EDCP Extension States |
|---|---|
| $00000000_b$ | Completed: All of the EDC parameter has been sent |
| $00000001_b$ | Reserved: To be assigned by SAE |
| $00000010_b$ | More: Concatenate the following data as higher order EDC parameter |
| $00000011_b$ | More: Concatenate the following data as lower order EDC parameter |
| $00000100_b$ to $00000101_b$ | Reserved: To be assigned by SAE |
| $00000110_b$ | Data in error indicator/EDC parameter is an error indicator |
| $00000111_b$ | Data in error indicator/EDC parameter is an error indicator and data in seed is an expected time to completion |
| $00001000_b$ to $11111110_b$ | Reserved: To be assigned by SAE |
| $11111111_b$ | No error indicator/EDC parameter available |

5.7.15.2.1 Completed

An EDCP extension value of "0" implies all of the EDC parameter has been sent within the error indicator/EDC parameter.

5.7.15.2.2 More - Concatenate as Higher

Implies the following EDCP components should be concatenated as the next HIGHER order with those previously received.

5.7.15.2.3 More - Concatenate as Lower

Implies the following EDCP components should be concatenated as the next LOWER order with those previously received.

5.7.15.2.4 Error Indicator/EDC Parameter Data Is an Error Indicator

Implies the following data is an error indicator value and not a component of an EDCP.

5.7.15.2.5 Error Indicator/EDC Parameter is an Error Indicator and Data in Seed is an Expected Time to Completion

Implies the following data is an error indicator value and not a component of an EDCP, as well as the seed parameter, contains an expected time to completion. Time value shall have a resolution of 0.1 s/bit.

5.7.15.2.6 No EDCP Available

Implies there are no EDCP components available in this system, and could easily imply that an EDCP is not even used.

5.7.15.3 Error Indicator/EDC Parameter

This is a 24-bit parameter that has two uses. One is to transfer a checksum, CRC, or other type of EDC parameter (or any segment thereof) from a device to a tool within the memory access response message. The second use is to send an error indicator any time the device is not able to complete or act upon a tool's request. Some error indicator states are predefined (see Table 13), although it is up to the manufacturer to decide if a particular error will be identified. (If identified, the predefined value is to be used.) The tool is responsible for knowing the EDC parameter generation techniques used by the device. The tool is also responsible for the verification that the EDCP is correct. The EDCP is sent within the close sequence (see Appendix C) at the completion of each operation. Since some users may wish an EDCP greater than 24 bits, there is provision to form a larger value by concatenation. In such cases, the EDCP extension parameter is used to determine the direction of concatenation and the completion of the concatenation sequence. An EDCP extension value of "all 1s" implies that the EDCP, as well as the extension, is not available and is not really being used by the device. In such cases, the value in the EDCP has no meaning (see Table 13).

| | |
|---|---|
| Data Length: | 24 bits |
| Resolution: | Per definitions in this section |
| Data Range: | 0 to 16,777,215 (0 to FFFFFF$_h$) |
| Type: | Status |
| Suspect Parameter Number: | 1648 |
| Reference: | 5.7.15 |

***Table 13 - Error indicator states (only when EDCP extension = 6)***

| Bit States | Error Indicator States |
|---|---|
| 000000$_h$ | No error |
| 000001$_h$ | Error NOT identified |
| 000002$_h$ | Currently processing for someone else |
| 000003$_h$ to 00000F$_h$ | Reserved: To be assigned by SAE |
| 000010$_h$ | Currently processing erase request |
| 000011$_h$ | Currently processing read request |
| 000012$_h$ | Currently processing write request |
| 000013$_h$ | Currently processing status request |
| 000014$_h$ | Reserved: To be assigned by SAE |
| 000015$_h$ | Reserved: To be assigned by SAE |
| 000016$_h$ | Currently processing boot load request |
| 000017$_h$ | Currently processing EDCP generation request |
| 000018$_h$ to 00001E$_h$ | Reserved: To be assigned by SAE |
| 00001F$_h$ | Currently processing unspecified request from this address |
| 000020$_h$ | EDC parameter not correct for data stream |
| 000021$_h$ | RAM did not verify on write |
| 000022$_h$ | FLASH did not verify on write |
| 000023$_h$ | PROM did not verify on write |
| 000024$_h$ | Internal failure preventing request (i.e., within the ECU) |
| 000025$_h$ to 0000FF$_h$ | Reserved: To be assigned by SAE |
| 000100$_h$ | Addressing or DATA general error |
| 000101$_h$ | Addressing Error: Address not on a valid boundary (block, word, object, etc.) |

| Bit States | Error Indicator States |
|---|---|
| $000102_h$ | Addressing Error: Length not valid for memory structure and operation |
| $000103_h$ | Addressing Error: Required memory exceeded available memory |
| $000104_h$ | Addressing Error: Requested operation requires prior erase of DATA memory |
| $000105_h$ | Addressing Error: Requested operation requires prior erase of PROGRAM memory |
| $000106_h$ | Addressing Error: Requested operation requires prior execution transfer and erase of PROGRAM memory |
| $000107_h$ | Addressing Error: Requested address for boot loader execution transfer is NOT within executable memory |
| $000108_h$ | Addressing Error: Requested address for boot loader execution transfer is NOT on valid boundary |
| $000109_h$ | DATA Error: Data does NOT conform to expected or allowed value ranges |
| $00010A_h$ | DATA Error: NAME does NOT conform to expected value |
| $00010B_h$ to $000FFF_h$ | Reserved: To be assigned by SAE |
| $001000_h$ | Security error general |
| $001001_h$ | Security Error: Invalid password |
| $001002_h$ | Security Error: Invalid user level |
| $001003_h$ | Security Error: Invalid key {seed} |
| $001004_h$ | Security Error: NOT in diagnostic mode |
| $001005_h$ | Security Error: NOT in engineering or development mode |
| $001006_h$ | Security Error: Engine running |
| $001007_h$ | Security Error: Vehicle NOT in “park” or otherwise NOT stationary |
| $001008_h$ to $00FFFF_h$ | Reserved: To be assigned by SAE |
| $010000_h$ | Abort from external to normal software process |
| $010001_h$ | Too Many Retries: Module exceeding a set number of retries |
| $010002_h$ | NO response in the time allowed |
| $010003_h$ | Transport of data NOT initiated within the time allowed |
| $010004_h$ | Transport of data NOT completed within the time allowed |
| $010005_h$ to $FFFFFE_h$ | Reserved: To be assigned by SAE |
| $FFFFFF_h$ | No error indicator available |

5.7.15.3.1 No Error

An error indicator value of “0” implies no error was detected by the device. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.2 Error Not Identified

Implies the device could (or would) NOT identify the specific error preventing continued operation. This value is to be used by the manufacturer when the device is unable (or unwilling), whether by design or failure, to generate a more detailed summary of the fault or condition preventing continued operation on the given memory access request. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.3 Currently Processing for Someone Else

Implies that the device is processing a memory access for some other address than the one which just requested. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.4 Currently Processing Some Request for This Device

There are several errors that could be the result of the device already being busy processing a memory access request from this address. Since it was thought by some that it would be nice to know the specific operation underway, several errors have been assigned. It is planned that these errors will be grouped between the values $10_h$ and $1F_h$. The specific request can then be identified by the lower nibble of lower byte of error indicator/EDC parameter as:

5.7.15.3.4.1 Currently Processing Erase Request

Implies that the device is processing a memory access erase from this address already. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.4.2 Currently Processing Read Request

Implies that the device is processing a memory access read from this address already. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.4.3 Currently Processing Write Request

Implies that the device is processing a memory access write from this address already. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.4.4 Currently Processing Status Request

Implies that the device is processing a memory access status request from this address already. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.4.5 Currently Processing Boot Load Request

Implies that the device is processing a memory access boot load from this address already. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.4.6 Currently Processing EDCP Generation Request

Implies that the device is processing a memory access EDCP generation request from this address already. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.4.7 Currently Processing Unspecified Request

Implies that the device is not identifying the specific request it is presently processing but is identifying that it is from this address. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.5 Some Failure

There are several errors that can be sent to indicate that an operation has failed. Some errors do not indicate the exact problem but more the event. Members of this form of error will be grouped between values $20_h$ and $FF_h$. The presently assigned errors are:

5.7.15.3.5.1 EDC Parameter Not Correct for Data Stream

Implies that EDC was not correct for the data. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.5.2 RAM Did Not Verify on Write

Identifies that some failure has caused RAM not to verify following a write. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.5.3 FLASH Did Not Verify on Write

Identifies that some failure has caused a FLASH memory not to verify following a write. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.5.4 PROM Did Not Verify on Write

Identifies that some failure has caused a PROM memory not to verify following a write. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.6 Addressing or Data Errors

There are several errors that imply that something was wrong with the addressing of the request or the data sent for the request. These errors have been grouped with a lower value of $100_h$, and the specific error can be parsed on the value of the lower byte as:

5.7.15.3.6.1 Addressing or Data General Error

Identifies that the failure has been within the addressing or data but that it cannot be identified further. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.6.2 Addressing Error - Length Not Valid for Memory Structure and Operation

Identifies that the failure has been a length that is not compatible with the memory and/or the particular operation attempted upon said memory. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.6.3 Addressing Error - Required Memory Exceeded Available Memory

Identifies that the failure has been a request for which there is not sufficient memory available. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.6.4 Addressing Error - Requested Operation Requires Prior Erase of DATA Memory

Identifies that the failure has been a request for which there needed to be an erase of some DATA memory prior to the requested operation. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.6.5 Addressing Error - Requested Operation Requires Prior Erase of PROGRAM Memory

Identifies that the failure has been a request for which there needed to be an erase of some PROGRAM memory prior to the requested operation. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.6.6 Addressing Error - Requested Operation Requires Prior Execution Transfer and Erase of PROGRAM Memory

Identifies that the failure has been a request for which there needed to be a transfer of execution to some other program segment and an erase of some PROGRAM memory prior to the requested operation. (Note: The error indicator is only valid when EDCP extension = 6; see section 5.7.15.3.)

5.7.15.3.6.7 Addressing Error - Requested Address for Boot Loader Execution Transfer is NOT within Executable Memory

Identifies that the failure has been a request to transfer execution to some address not in an executable memory. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.6.8 Addressing Error - Requested Address for Boot Loader Execution Transfer is NOT on A Valid Boundary

Identifies that the failure has been a request to transfer execution to some address not on a valid boundary within executable memory. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.7 Security Error

There are several errors that imply that something was wrong with the security used within the request. These errors have been grouped with a lower value of $1000_h$, and the specific error can be parsed on the value of the lower byte as:

5.7.15.3.7.1 Security Error General

Identifies that the failure has been within the security, but that it is not (or cannot be) identified any further. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.7.2 Security Error - Invalid Password

Identifies that the failure has been an invalid password for the requested operation. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.7.3 Security Error - Invalid User Level

Identifies that the failure has been an invalid user level for the requested operation. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.7.4 Security Error - Invalid Key {Seed

Identifies that the failure has been an invalid key returned for the seed that was provided for the requested operation. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.7.5 Security Error - NOT in Diagnostic Mode

Identifies that the failure has been that the unit is not in some diagnostic mode prior to the requested operation. This is an allowable manufacturer additional requirement. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.7.6 Security Error - NOT in Engineering or Development Mode

Identifies that the requested operation requires that the unit be in an engineering or development mode prior to the requested operation. This is an allowable manufacturer additional requirement. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.7.7 Security Error - Engine Running

Identifies that the requested operation requires the engine to be stopped prior to the requested operation. This is an allowable manufacturer additional requirement. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.7.8 Security Error - Vehicle NOT in PARK or Otherwise NOT Stationary

Identifies that the requested operation requires the vehicle to be in park or otherwise not able to move prior to the requested operation. This is an allowable manufacturer additional requirement. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.8 Time-Out Errors

There are several errors that imply that something has taken too long or too many tries and the unit has given up. These errors have been grouped with a lower value of $10000_h$, and the specific error can be parsed on the value of the lower byte as:

5.7.15.3.8.1 Abort from External to Normal Software Process

Identifies that some event within the unit has caused an abort of this software process. Hence, this memory access operation has also been terminated. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.8.2 Too Many Retries

Identifies that the failure has been an excessive number of attempts were made without the desired event occurring. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.8.3 No Response in Time Allowed

Identifies that there has been a time-out within the process, although no further identification of the time-out is possible. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.8.4 Transport of Data NOT Initiated Within the Time Allowed

Identifies that there has been a time-out within the process, and that it was in waiting for the establishment of the transport session to send the data. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.8.5 Transport of Data NOT Completed Within the Time Allowed

Identifies that there has been a time-out within the process, and that it has taken too long for the transport session to complete sending the data. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.3.9 No Error Indicator Available

Implies there is no error indicator AVAILABLE at this time. (Note: The error indicator is only valid when EDCP extension = 6; see 5.7.15.3.)

5.7.15.4 Seed

This is a 16-bit parameter that is used by the device primarily to send a seed to a tool when using a seed/key type security system. It is also used by the device to signal the tool that the device is satisfied that a complete key has been received or that the data security message is expected to contain the seed data. This parameter can also contain an expected time to completion when the EDCP extension is 7 (see 5.7.15.3). The seed is to be the mathematical basis upon which any key is calculated. The device verifies the validity of the key {seed} from the tool and enable memory access operations appropriately. See also 5.7.14.4 and Table 11.

| | |
|---|---|
| Data Length: | 16 bits |
| Resolution: | Per definitions in this section |
| Data Range: | 0 to 65535 (0 to $FFFF_h$) |
| Type: | Status |
| Suspect Parameter Number: | 1599 |
| Reference: | 5.7.15 |

***Table 14 - Seed states***

| Bit States | Seed States |
|---|---|
| 0 | Seed completed: Begin sending key |
| 1 | Use long seed or key from data security message |
| 2 to $FFFE_h$ | Seed values |
| $FFFF_h$ | No further key required of tool |

5.7.15.4.1 Seed Completed

A specific value to indicate that no further seed or seed segments is going to be provided by the device; see 5.7.14.4.5 and 5.7.15.4.5. The tool should understand it is to begin the operation supposing that the operation is allowed within the Status parameter and that a non-zero length has been allowed.

5.7.15.4.2 Use Long Seed or Key from Data Security Message

The actual seed or key is in the data security message.

5.7.15.4.3 Seed Values

The actual numeric values that can be used for seeds.

5.7.15.4.4 No Further Key Required of Tool

A specific value to indicate that no key or further key segments is required of the tool to begin this operation; see 5.7.14.4.5 and 5.7.15.4.1.

5.7.15.4.5 Handling of Keys Larger Than 16 Bits

If a manufacturer feels a seed/key structure requires a key or seed longer than 16 bits for a particular device, this can be handled with the data security message (see 5.7.18) and the setting of the Key/User_Level and the seed parameters appropriately (see Tables 9 and 12).

5.7.15.4.6 Acceptance Rules

The device establishes the set of rules governing acceptance of memory access requests (such as operational mode, User_Level versus memory (OBJECT) location versus requested operation, etc.). These rules may be manufacturer specific to prevent unauthorized modification of a device’s memory. The manufacturer also has the option to allow an initial operation by a tool to establish a security level and then let the device honor multiple requests from the same tool (i.e., the network node whose source address matches the source address originally used by the tool opening the original memory access operation, the manufacturer has the option to further check the NAME to address association) without further security operations. This optional re-entrant security is to be considered completely ended when the tool sends a “close” or the device time-outs waiting for the tool’s “close.”

5.7.15.4.7 Expected Time to Completion Values

The expected time to completion of an operation when the device is already processing a request. Numeric values are in milliseconds.

5.7.15.5 Length/Number Allowed

When the status of the memory access response message is a proceed, an 11-bit parameter identifies the amount of memory (i.e., the range of memory) over which the device is willing to allow a particular operation to be carried out. For “proceed,” the length value is either in bytes or objects (see 5.7.14.5). When the status of the memory access request is busy, operation failed, or operation completed, the length is meaningless. The device should therefore send it as “0” and the tool should treat it as “DO NOT CARE.”

| | |
|---|---|
| Data Length: | 11 bits |
| Resolution: | 1 byte/bit or object/bit or other |
| Data Range: | 0 to 1784 |
| Type: | Status |
| Suspect Parameter Number: | 1649 |
| Reference: | 5.7.15 |

5.7.16 Binary Data Transfer (DM16)

Used primarily to transfer data for the memory access commands. A memory access state transition diagram (DM14 through DM18) is in Appendix C.

| | |
|---|---|
| Transmission Rate: | As needed |
| Data Length: | Variable (8 to n) |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 215 |
| PDU Specific: | DA |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |
| Parameter Group Number: | 55040 (00D700h) |

BINARY_DATA_TRANSFER

| | | | |
|---|---|---|---|
| Byte: | 1 | Number of occurrences of raw binary data | See 5.7.16.1 |
| Bytes: | 2-8 | Raw binary data | See 5.7.16.2 |
| Bytes: | 9-n | Raw binary data - when multipacketed | See 5.7.16.2 |

5.7.16.1 Number of Occurrences of Raw Binary Data

This is an 8-bit (1-byte) parameter to be sent within the binary data transfer PG to provide information on the number of raw binary data parameters that will follow when the message is single packet. Its value is between 1 and 7 when the binary data transfer PG is not multipacketed. If the message is multipacketed, then the value of this parameter is to be 255 (FFh). When the message is multipacketed, the number of occurrences of the raw binary data parameter shall be determined from the total message size parameter (refer to SAE J1939-21) sent in the connection request message.

| | |
|---|---|
| Data Length: | 8 bits |
| Resolution: | 1 byte/bit |
| Data Range: | 1 to 7 or 255<br>Values 0 or 8 to 254 are not used |
| Type: | Status |
| Suspect Parameter Number: | 1650 |
| Reference: | 5.7.16 |

5.7.16.2 Raw Binary Data

This is a 1-byte parameter representing the value for 1 byte of memory. It can have any value between 0 and 255 (0 and FFh) with no reserved values. The number of occurrences of this parameter within a message can range from 1 to 1784 (one less than the 1785 transport limit to account for the parameter - number of occurrences of raw binary data). When more than seven occurrences are to be sent, a transport protocol session shall be used (remember, 1 of the 8 message data bytes was used for the number of occurrences of raw binary data parameter). The number of occurrences of raw binary data parameter shall be used to determine the message length when single packeted. In this case, the number of occurrences of raw binary data parameter provides the number of raw binary data parameters being sent. This value plus one is the number of data bytes within the single packet message. When there are greater than seven occurrences of the raw binary data parameter to be sent, transport protocol will be needed and it will be necessary to send the sequence number (refer to SAE J1939-21, 3.10.12) from the transport session. Hence, the first transport packet will have the sequence number, along with the "number of occurrences of raw binary data" parameter and six occurrences of this (raw binary data) parameter. In each subsequent transport packet there will be the sequence number and seven occurrences of this (raw binary data) parameter. The sequence number shall be used to calculate the occurrence number of each of the raw binary data parameters. Also, as outlined in SAE J1939-21, the last packet, although 8 bytes in length, may contain fewer than seven occurrences of this parameter, and the total message size parameter (sent in the session connection message) shall be used to identify when the end of data is reached. An example of the positioning of the raw binary data within the messages is shown in Tables 15 through Table 17. For an example of parsing memory with widths other than 8 bits, see 5.7.14.3.1.

| | |
|---|---|
| Data Length: | 8 bits |
| Resolution: | Not applicable |
| Data Range: | 0 to 255 (0 to FFh) |
| Type: | Status |
| Suspect Parameter Number: | 1651 |
| Reference: | 5.7.16 |

***Table 15 - Message appearance when multipacketed***

| Message | CAN ID | CAN DB1 | CAN DB2 | CAN DB3 | CAN DB4 | CAN DB5 | CAN DB6 | CAN DB7 | CAN DB8 |
|---|---|---|---|---|---|---|---|---|---|
| First packet of a transport session | Transport Protocol - Data Transfer Message | Sequence Number J1939-21 3.10.12 | Number of Occurrences of Raw Binary Data (value = $FF_h$) | Raw Binary Data - #1 | Raw Binary Data - #2 | Raw Binary Data - #3 | Raw Binary Data - #4 | Raw Binary Data - #5 | Raw Binary Data - #6 |
| Second packet of a transport session | Transport Protocol - Data Transfer Message | Sequence Number J1939-21 3.10.12 | Raw Binary Data - #7 | Raw Binary Data - #8 | Raw Binary Data - #9 | Raw Binary Data - #10 | Raw Binary Data - #11 | Raw Binary Data - #12 | Raw Binary Data - #13 |
| Last packet of a transport session | Transport Protocol - Data Transfer Message | Sequence Number J1939-21 3.10.12 | Raw Binary Data - #(Total Message Length - 2) | Raw Binary Data - #(Total Message Length - 1) | $FF_h$ | $FF_h$ | $FF_h$ | $FF_h$ | $FF_h$ |

***Table 16 - Message appearance - seven occurrences of raw binary data (i.e., without transport)***

| Message | CAN ID | CAN DB1 | CAN DB2 | CAN DB3 | CAN DB4 | CAN DB5 | CAN DB6 | CAN DB7 | CAN DB8 |
|---|---|---|---|---|---|---|---|---|---|
| When not a transport session | Binary Data Transfer Message | Number of Occurrences of Raw Binary Data (value = $07_h$) | Raw Binary Data - #1 | Raw Binary Data - #2 | Raw Binary Data - #3 | Raw Binary Data - #4 | Raw Binary Data - #5 | Raw Binary Data - #6 | Raw Binary Data - #7 |

***Table 17 - Message appearance - four occurrences of raw binary data (i.e., without transport)***

| Message | CAN ID | CAN DB1 | CAN DB2 | CAN DB3 | CAN DB4 | CAN DB5 | CAN DB6 | CAN DB7 | CAN DB8 |
|---|---|---|---|---|---|---|---|---|---|
| When not a transport session with less than seven occurrences of raw binary data | Binary Data Transfer Message | Number of Occurrences of Raw Binary Data (value = $04_h$) | Raw Binary Data - #1 | Raw Binary Data - #2 | Raw Binary Data - #3 | Raw Binary Data - #4 | $FF_h$ | $FF_h$ | $FF_h$ |

#### 5.7.17 Boot Load Data (DM17)

Used primarily to load boot data/program into a device when a memory access boot load command has been issued. A memory access state transition diagram (DM14 through DM18) is in Appendix C. The CAN data length code of the message is set to 8 bytes to deliberately avoid the use of transport protocol and thereby reduce the program overhead that would need to be functional within a device while its program is being reloaded.

| | |
|---|---|
| Transmission Rate: | As needed |
| Data Length: | 8 |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 214 |
| PDU Specific: | DA |
| Default Priority: | 6 |
| Parameter Group Number: | 54784 (00D600h) |

BOOT_LOAD_DATA

| | | | |
|---|---|---|---|
| Byte: | 1-8 | Boot load data | see 5.7.17.1 |

##### 5.7.17.1 Boot Load Data

This is a 1-byte parameter using the same SLOT as the raw binary data. There shall be eight occurrences of this parameter in the message. The meaning of this parameter is proprietary. The structure used to reference the program and verify the data is also proprietary.

| | |
|---|---|
| Data length: | 8 bits |
| Resolution: | Not applicable |
| Data range: | 0 to 255 (0 to FFh) |
| Type: | Status |
| Suspect parameter number: | 1652 |
| Reference: | 5.7.17 |

#### 5.7.18 Data Security (DM18)

The data security parameter group is used to send security entities of a given type and length. These entities are data produced by or used for applications of cryptography and supporting procedures to ensure data security. Also included is the provision to provide a long seed and long key to be used with memory access functions. The capabilities of the long seed and key are outlined in Appendix D. A memory access state transition diagram (DM14 through DM18) is in Appendix C.

| | |
|---|---|
| Transmission Rate: | As needed |
| Data Length: | Variable (8 to n) |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 212 |
| PDU Specific: | DA |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |
| Parameter Group Number: | 54272 (00D400h) |

DATA_SECURITY

| | | | | |
|---|---|---|---|---|
| Byte: | 1 | bits 8-1 | Security entity length<br>(Least significant 8 bits)<br>(Bit 1 is least significant bit) | see 5.7.18.2 |
| Byte: | 2 | bits 8-5 | Security entity length<br>(Most significant 4 bits)<br>(Bit 8 is most significant bit) | see 5.7.18.2 |
| | | bits 4-1 | Security entity type | see 5.7.18.1 |
| Bytes: | 3-n | | Data security parameter<br>(Least significant byte is Byte 3)<br>(Bit 1 is least significant bit)<br>(Most significant byte is Byte n)<br>(Bit 8 is most significant bit) | see 5.7.18.3 |

5.7.18.1 Security Entity Type

This 4-bit parameter that indicates whether the data in the following security entity parameter is to be used as a long seed, long key, session key, or certificate (see Table 18).

| | |
|---|---|
| Data Length: | 4 bits |
| Resolution: | 1 type/bit |
| Data Range: | 0 to 15 (see Table 18) |
| Type: | Status |
| Suspect Parameter Number: | 1479 |
| Reference: | 5.7.18 |

***Table 18 - Security entity types***

| Bit States | Security Entity Type |
|---|---|
| $0000_b$ | Data is long seed |
| $0001_b$ | Data is long key |
| $0010_b$ | Data is session key |
| $0011_b$ | Data is certificate |
| $0100_b$ to $1111_b$ | Reserved: To be assigned by SAE |

5.7.18.1.1 Data is Long Seed

A security entity type value of $0000_b$ implies that the data in the following data security parameter is to be used as a long seed. It is most likely then going from a device to a tool.

5.7.18.1.2 Data is Long Key

A security entity type value of $0001_b$ implies that the data in the following data security parameter is to be used as a long key. In general this would imply previous receipt of a long seed upon which to base the long key. Also, the direction would typically be from a tool to a device.

5.7.18.1.3 Data is Session Key

A security entity type value of $0010_b$ implies that the data in the following data security parameter is to be used as a session key. The session key is sent encrypted by using a secret key (symmetric encryption) or the public key of the addressed ECU (asymmetric encryption). The addressed ECU has to decrypt the session key before it can be used. The length of the decrypted session key is 8 bytes. In the case of using asymmetric encryption, the session key is put into the first 8 bytes of the data string to be encrypted, followed by 8 bytes, each filled with $FF_h$, and arbitrary numbers for the remaining bytes. This provides a mechanism for the receiving ECU to check if its decryption was successful.

5.7.18.1.4 Data is Certificate

A security entity type value of $0011_b$ implies that the data in the following data security parameter is to be used as a certificate.

5.7.18.2 Security Entity Length

This 12-bit parameter contains the length, in bytes, of the data security parameter.

| | |
|---|---|
| Data Length: | 12 bits |
| Resolution: | 1 byte/bit |
| Data Range: | 0 to 1785 |
| Type: | Status |
| Suspect Parameter Number: | 1596 |
| Reference: | 5.7.18 |

5.7.18.3 Data Security Parameter

This parameter is used to send the data for the data security message. There are presently four different items defined. The data security parameter shall be sent least significant byte first.

| | |
|---|---|
| Data Length: | Variable (length given in the security entity length parameter) |
| Resolution: | 1 byte/bit |
| Data Range: | 0 to 1785 |
| Type: | Status |
| Suspect Parameter Number: | 1597 |
| Reference: | 5.7.18 |

5.7.18.3.1 Long Seed

When the security entity type value is $0000_b$, the data is a long seed. The long seed is a number. The number is sent (sometimes randomly) when requesting message or application authentication to rule out replay attacks. (See Appendix D.)

5.7.18.3.2 Long Key

When the security entity type value is $0001_b$, the data is a long key. The long key is a number. This number represents a mathematical function of a previously received long seed sent when attempting to justify one's request for a message or application. (See Appendix D.)

5.7.18.3.3 Session Key

When the security entity type value is $0010_b$, the data is a session key. In this application, the long seed/key data parameter shall be interpreted only if it contains a session key.

5.7.18.3.4 Certificate

When the security entity type value is $0011_b$, the data is a certificate. Parameter group to be sent on request from an ECU authorized by a certification authority to send authentication messages. Acceptance of the certificate is a prerequisite for the receiving unit to send a session key. The certificate is only needed when the establishment of a session key is based on an asymmetric encryption procedure. For symmetric encryption, the installation of the secret key and the algorithm used is not specified here. The content of the certificate is given by ISO/IEC 9594-8 with the subject being the sender of the message. The certificate contains the public key of the sender.

5.7.19 Calibration Information (DM19)

Provides information about the calibration to an interrogating tool (see Figure 4).

If DM19 is requested using the destination-specific method, or using the global method before computation of the calibration verification number is complete, then the responder shall send the acknowledgment PG with the mode set to three to indicate that the tool should request DM19 at a later time. The tool should wait 30 seconds and retry until successful.

Some regulations require that the last computed value be stored and reported while a current cycle calculation is underway.

OBD implementers should see Appendix H for more information on when DM19 support is required.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | Variable |
| Extended Data Page: | 0 |
| Function: | Provide information about the calibration to scan tool |
| Data Page: | 0 |
| PDU Format: | 211 |
| PDU Specific: | DA |
| Default Priority: | 7 |
| Parameter Group Number: | 54016 ($00D300_h$) |

| | | | |
|---|---|---|---|
| Bytes: | 1-4 | Calibration verification number<br>(Byte 1 is least significant byte) | see 5.7.19.1 |
| Bytes: | 5-20 | Calibration identification | see 5.7.19.2 |

Definitions:
a = Calibration verification number (CVN)
b = Calibration identification (CAL ID)

Message format shall be as follows: a1, b1, a2, b2, … ai, bi for i [> 0] CVN and CAL ID pairs.

For each pair of ai, bi, the value of bi, shall be corresponding CAL ID for the CVN given by ai. Figure 5 shows an example with three a, b pairs of a CVN followed by its CAL IDs. When a controller is reporting more than one CAL ID and CVN pair, it must maintain the CAL ID and CVN position in the set reported so that the receiving device can associate the correct CAL ID and CVN with the proper device.

If any value of any pair, ai, bi, cannot be obtained due to incomplete calibration processes, communications failures, or other electrical failures, or where the calculation of the CVN cannot be completed in the time as allowed in regulations, then the value of all zeros ($00_h$) shall be used for each such CVN, and high values (i.e., $FF_h$) shall be used for each such CAL ID. The count of pairs, i, conveyed in the message shall reflect the known count of CVN and CAL ID pairs for the OBD system or component. See the note following Figure 5 for an example.

Use of more than one CAL ID and CVN pair may require regulatory approval prior to their use in some jurisdictions.

The transport protocol of SAE J1939-21 is used since DM19 requires more than 8 data bytes to convey response.

5.7.19.1 Calibration Verification Number

Four-byte checksum of the entire calibration. Includes code and data. Excludes parameters that exist only in RAM, nonvolatile parameters that change values during the life cycle of the module (hours of operation, miles, number of on/off cycles, freeze frame data, etc.), or non-emissions-related parameters that can be changed by the operator (offsets for real-time clocks, user selectable preferences, etc.). If the checksum is less than 4 bytes, it must be padded with $00_h$ (the $00_h$ pad is placed in the most significant byte(s) when needed). The checksum algorithm shall be more robust than a two's complement checksum. Implementers should refer to the applicable regulation for potential additional checksum algorithm requirements. For instance, some regulations might require the calibration verification number to use more sophisticated algorithms that use polynomials or roll functions such that it is very difficult to "tweak" other calibration values to get back to the original CVN value.

| | |
|---|---|
| Data Length: | 4 bytes |
| Resolution: | Not applicable |
| Data Range: | 0 to 4294967295 (00 00 00 $00_h$ to FF FF FF $FF_h$) |
| Type: | Hexadecimal |
| Suspect Parameter Number: | 1634 |
| Reference: | 5.7.19 |

5.7.19.2 Calibration Identification

Sixteen-byte CAL ID number. Uniquely identifies the software installed in the control module. The CAL ID must be unique but does not need to be 16 bytes long. If the CAL ID is less than 16 bytes, those unused bytes are reported at the end of the CAL ID as $00_h$ (the $00_h$ pad is placed in the least significant bytes of the CAL ID when needed). The $00_h$, if needed, is added to the end of the ASCII character string for CAL ID.

In the instance where a CAL ID and CVN has not ever been reported by a subnetwork device, then the controller responding with the information to the scan tool shall report the CAL ID as 16 bytes of $FF_h$ and the CVN as all zeros in the position in the string that would have been sent to the scan tool.

| | |
|---|---|
| Data Length: | 16 bytes |
| Resolution: | Not applicable |
| Data Range: | 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 $00_h$ to FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF $FF_h$ |
| Type: | ASCII and limited to printable characters only, except that $00_h$ used as padding at the end for a CAL ID less than 16 bytes |
| Suspect Parameter Number: | 1635 |
| Reference: | 5.7.19 |

EXAMPLE: Figure 4 shows an example of how a 3-byte CVN $ABCDEF_h$ and a 10-byte CAL ID “CONTENDER1” would be sent. The hexadecimal representation for the ASCII “CONTENDER1” is:

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| ASCII: | C | O | N | T | E | N | D | E | R | 1 |
| Hex: | 43 | 4F | 4E | 54 | 45 | 4E | 44 | 45 | 52 | 31 |

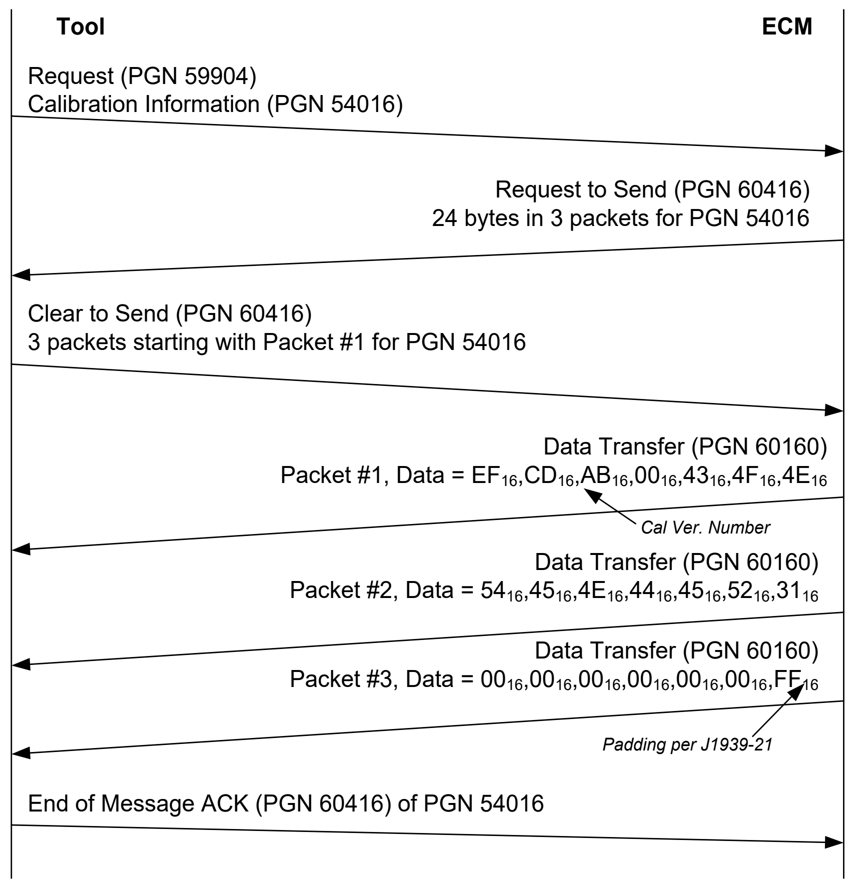


***Figure 4 - Example of tool requesting the CAL ID and CVN from an ECM***

NOTE: The CVN is sent LSB-MSB per SAE J1939-71 practice for numeric values so the last byte is the padding per SAE J1979. Note also that CAL ID is sent per SAE J1939-71 standard practice for ASCII values. Finally, note on this example that the entire calibration information PG is 20 bytes long so the last byte in the data transfer is $FF_h$ per SAE J1939-21.

EXAMPLE: Figure 5 shows an example how an ECM sends three pairs of 3-byte CVN ABCDEF$_h$ and a 10-byte CAL ID "CONTENDER1." The hexadecimal representation for the ASCII "CONTENDER1," "CONTENDER2," and "CONTENDER3" is shown below:

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| ASCII: | C | O | N | T | E | N | D | E | R | 1 |
| Hex: | 43 | 4F | 4E | 54 | 45 | 4E | 44 | 45 | 52 | 31 |
| ASCII: | C | O | N | T | E | N | D | E | R | 2 |
| Hex: | 43 | 4F | 4E | 54 | 45 | 4E | 44 | 45 | 52 | 32 |
| ASCII: | C | O | N | T | E | N | D | E | R | 3 |
| Hex: | 43 | 4F | 4E | 54 | 45 | 4E | 44 | 45 | 52 | 33 |

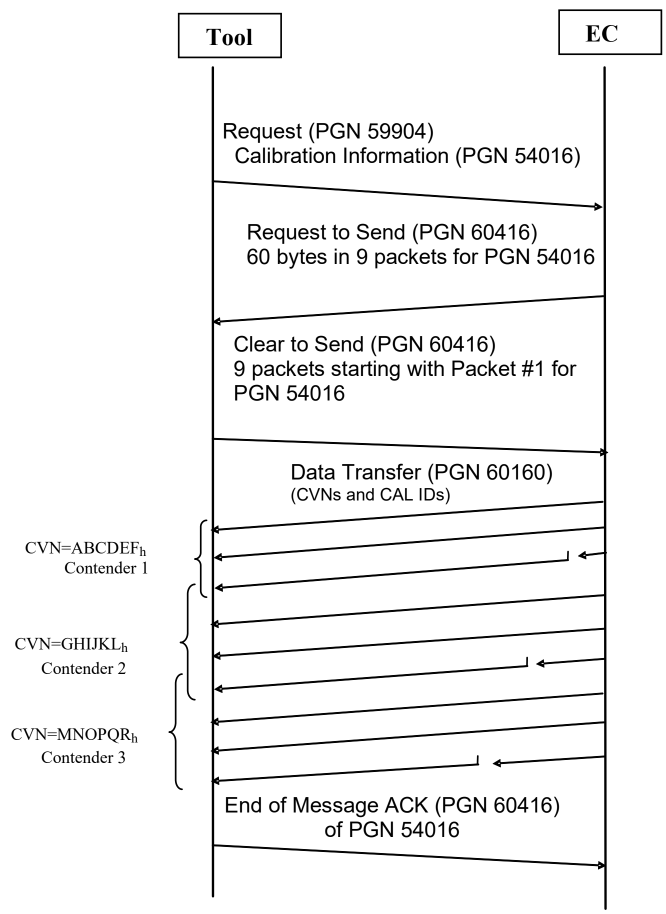


***Figure 5 - Example of tool requesting the CAL IDs and CVNs from an ECM with more than one to report***

The overlapping brackets in Figure 5 illustrate that the CAL ID/CVN pairs are sent back to back and therefore the end of the "CONTENDER1" is in the same packet that starts CAL ID for "CONTENDER2" and the packet for the last part of "CONTENDER2" shares a packet with the start of the CAL ID for "CONTENDER3."

5.7.20 Monitor Performance Ratio (DM20)

Legislated on-board diagnostics requirements specify that manufacturers must monitor all emission and OBD system-related components throughout the expected life of the vehicle. Manufacturers are to monitor all components that impact engine emissions. The monitor performance ratio indicates how often the OBD system monitors particular components compared to the amount of vehicle operation.

The ratio for each parameter is defined as the numerator divided by the denominator. The requirements for incrementing the numerator and denominator are defined on an individual monitor basis.

Only the applicable monitor performance data are required to be reported according to the regulations. However, the engine ignition cycle counter and the OBD monitoring conditions encountered counts (e.g., CARB's general denominator) are required in all transmissions of DM20.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request.<br>(refer to SAE J1939-21 PGN 59392) |
| Data Length: | Variable |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 194 |
| PDU Specific: | Destination address |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |
| Parameter Group Number: | 49664 ($00C200_h$) |

| | | |
|---|---|---|
| Bytes: 1-2 | Engine ignition cycle counter | see 5.7.20.1 |
| Bytes: 3-4 | OBD monitoring conditions encountered counts | see 5.7.20.2 |
| Bytes: 5-7 | SPN of applicable system monitor | see 5.7.20.3 |
| Bytes: 8-9 | Applicable system monitor numerator | see 5.7.20.4 |
| Bytes: 10-11 | Applicable system monitor denominator | see 5.7.20.5 |

Definitions:

a = Engine ignition cycle counter
b = OBD monitoring conditions encountered counts
c = SPN which defines the monitor ratio being reported
d = Monitor ratio numerator
e = Monitor ratio denominator

Message format shall be as follows: a,b,c,d,e,<u>c,d,e</u>,c,d,e,<u>c,d,e</u>,c,d,e,....etc. The transport protocol of SAE J1939-21 is used when DM20 requires more than 8 data bytes to convey response. DM20 can be sent as a single CAN frame if no ratios are being reported. When no ratios are being reported, the message format shall be a, b, bytes 5 through 8 are sent to fill out the CAN frame (receivers should ignore bytes 5 through 8 in this case).

Table 19 identifies some of the possible performance monitor ratios that a system may report. It also identifies those required to be reported to satisfy emissions OBD regulatory requirements (see Table 2). Table 19 has Columns A through G.

Column A is simply the row number.

Column B is the mapping of the in-use ratio to regulatory requirement.

EXAMPLE: If Column E = H or J or H, J, a system or sub-system monitor shall be tracked separately and be part of the reported in-use ratio for that system; e.g., SPN 4792 shall report the lowest in-use ratio of catalyst efficiency and shall not consider improper reductant monitor in-use ratio to be reported in SPN 4792.

Column C is the regulatory name for the system/sub-system monitor.

Column C has different formats for the text used to represent system or sub-system. The capital text is the system monitor and the italic text under each capital text is the sub-system. Sub-system is indented to the right of the system.

EXAMPLE: Row #6 Column C has capital text, NOx CONVERTING CATALYST. That means this is a system monitor. And rows #7 and #8 have italic text, *catalyst efficiency and improper reductant.* That means these are sub-system monitors of NOx CONVERTING CATALYST system monitor.

Column D is a mapping of separately tracked in-use ratios for a system/sub-system.

EXAMPLE: If Column D has the text "required," that means the system/sub-system must be tracked and reported separately. If Column D has the text "independent and optional," that means two things: (1) it shall not be part of the regulated and reported in-use ratio for that system, and (2) the system/sub-system monitor may be tracked separately and reported in a different SPN if a manufacturer chooses to do so.

Column E specifies whether system or sub-system is applicable to diesel (D), gasoline (G), or both (DG).

Column F is the SPN to report in-use ratio for a particular/specific system monitor.

Some of the cells in Column F are merged to represent that only the lowest in-use ratio of that group of sub-systems will be reported in the SPN. If there are multiple banks or multiple systems in a bank, the worst in-use ratio will be reported in the SPN specified in Column F.

EXAMPLE: The EGR system's in-use ratio is the lowest numerical ratio of the monitor ratios that are tracked separately for each required monitor (e.g., slow response, cooler performance, catalyst) that detects malfunctions of the EGR system, and this lowest numerical ratio shall be reported in SPN 3058. If there are multiple banks with separate EGR, the worst in-use ratio will be reported in SPN 3058.

NOTE: If the cell is labeled with SPN # and the column, SPN Name, is blank, then the manufacturer can select the SPN or SPNs to use from the SAE J1939 standard <u>but must not use any of the SPNs defined in Table 19 as "required" in Column D, or otherwise specified for another purpose.</u> Where there is an independent and optional notation and an SPN # is defined, that is one possible SPN to use for that in-use ratio. There may be others that could be applicable as well. For example, misfire for engine injector cylinder #03, SPN 653.

Column G is SPN name.

***Table 19 - SAE J1939-73 monitor performance***

| A | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|
| Row # | Regulation Reference (See Table 2) | Regulatory Required System & Sub-system Monitor | In-Use Ratio Reporting Requirement | Engine Type [D]iesel, [G]as, [DG] Diesel & Gas | Reported SPN # | SPN Name | Note |
| 1 | H,J | NMHC CONVERTING CATALYST MONITOR | | D | 5322 | Aftertreatment NMHC Converting Catalyst System Monitor | |
| 2 | *H,J* | *Conversion Efficiency* | Required | D | | | |
| 3 | *H,J* | *Other Aftertreatment Assistance Functions* | Required | D | | | |
| 4 | H,J | CATALYST MONITOR | | G | | | |
| 5 | *H,J* | *Conversion Efficiency* | Required | G | 3050 | Catalyst Bank 1 System Monitor | |
| 6 | *H,J* | *Conversion Efficiency* | Required | G | 3051 | Catalyst Bank 2 System Monitor | Required if equipped |
| 7 | H,J | NOX CONVERTING CATALYST | | D | 4792 | Aftertreatment 1 Selective Catalytic Reduction System | |
| 8 | *H,J* | *Conversion Efficiency* | Required | D | | | |
| 9 | *H,J* | *Reductant Delivery Performance* | Independent & Optional | D | SPN # | | |
| 10 | *H,J* | *Insufficient Reductant* | Independent & Optional | D | SPN # | | |
| 11 | *H,J* | *Improper Reductant* | Independent & Optional | D | SPN # | | |
| 12 | *H,J* | *Feedback Control (Time Interval, Default or Open Loop Operation, Control Limit )* | Independent & Optional | D | SPN # | | |
| 13 | H,J | NOX ADSORBER | | D | 5308 | Aftertreatment 1 NOx Adsorber Catalyst System Monitor | |
| 14 | *H,J* | *NOx Adsorber Capability* | Required | D | | | |
| 15 | *H,J* | *Active/Intrusive Injection* | Independent & Optional | D | SPN # | | |

| A | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|
| Row # | Regulation Reference (See Table 2) | Regulatory Required System & Sub-system Monitor | In-Use Ratio Reporting Requirement | Engine Type [D]iesel, [G]as, [DG] Diesel & Gas | Reported SPN # | SPN Name | Note |
| 16 | H,J | Feedback Control (Time Interval, Default or Open Loop Operation, Control Limit ) | Independent & Optional | D | SPN # | | |
| 17 | H,J | EXHAUST GAS SENSOR [See footnote 3] - for CI engines | | D | 5318 | Aftertreatment Exhaust Gas Sensor System Monitor | Discontinue use for Primary and Secondary Oxygen Sensor Monitors with 2024+ MY SI engines |
| 18 | H,J | Sensor Performance Faults - Threshold | Required | D | | | |
| 19 | H,J | Sensor Performance - OBD Monitoring Device | Required | D | | | |
| 20 | H,J | Sensor Performance Faults - Upstream Air-Fuel Ratio Sensors | Required | D | | | |
| 21 | H,J | Sensor performance faults - Downstream Air-Fuel Ratio Sensors | Required | D | | | |
| 22 | H,J | Sensor Performance faults - NOx and PM Sensors | Required | D | | | |
| 23 | H,J | Monitoring capability - Upstream Air-Fuel Ratio Sensors (EGS) | Independent & Optional | D | SPN # | | |
| 24 | H,J | Monitoring capability - Downstream Air-Fuel Ratio Sensors (EGS) | Independent & Optional | D | SPN # | | |
| 25 | H,J | Monitoring capability - NOx and PM Sensors (EGS) | Independent & Optional | D | SPN # | | |
| 26 | H,J | Circuit Faults - Upstream Air-Fuel Ratio Sensors (EGS) | Independent & Optional | D | SPN # | | |
| 27 | H,J | Circuit Faults - Downstream Air-Fuel Ratio Sensors (EGS) | Independent & Optional | D | SPN # | | |
| 28 | H,J | Circuit Faults - NOx and PM Sensors (EGS) | Independent & Optional | D | SPN # | | |
| 29 | H,J | Feedback Faults - Upstream Air-Fuel Ratio Sensors (EGS) | Independent & Optional | D | SPN # | | Other than SI O2 Sensors |
| 30 | H,J | Feedback Faults - Downstream Air-Fuel Ratio Sensors (EGS) | Independent & Optional | D | SPN # | | Other than SI O2 Sensors |
| 31 | H,J | Feedback Faults - NOx and PM Sensors (EGS) | Independent & Optional | D | SPN # | | |
| 32 | H,J | Sensor - Heater Performance (EGS) | Independent & Optional | D | SPN # | | |
| 33 | H,J | Sensor - Heater Circuit Faults (EGS) | Independent & Optional | D | SPN # | | |
| 34 | H,J | EXHAUST GAS SENSOR [See footnote 3] - for SI engines | | G | 5318 | Aftertreatment Exhaust Gas Sensor System Monitor | Discontinue use for Primary and Secondary Oxygen Sensor Monitors with 2024+ MY SI engines<br><br>Continue use for SI NOx Sensors<br><br>Use SPs 3056, 3057, 21227, and 21228 for O2 sensor monitor ratio reporting with 2024+ MY SI engines |
| 35 | H,J | Sensor Performance Faults - Threshold | Required | G | | | |
| 36 | H,J | Sensor Performance - OBD Monitoring Device | Required | G | | | |
| 37 | H,J | Sensor Performance faults - NOx and PM Sensors | Required | G | | | |
| 38 | H,J | Air-Fuel Ratio Sensors (EGS)-Bank 1 | Required | G | 3056 | Engine Exhaust Bank 1 O2 Sensor Monitor | |
| 39 | H,J | Air-Fuel Ratio Sensors (EGS) - Bank 2 | Required | G | 3057 | Engine Exhaust Bank 2 O2 Sensor Monitor | Required if equipped |
| 40 | H,J | Secondary Air-Fuel Ratio Sensors (EGS) - Bank 1 | Required | G | 21227 | Engine Exhaust Bank 1 Secondary O2 Sensor Monitor | Required if equipped |
| 41 | H,J | Secondary Air-Fuel Ratio Sensors (EGS) - Bank 2 | Required | G | 21228 | Engine Exhaust Bank 2 Secondary O2 Sensor Monitor | Required if equipped |
| 42 | H,J | Circuit Faults - Air-Fuel Ratio Sensors (EGS) | Independent & Optional | G | SPN # | | |
| 43 | H,J | Circuit Faults - NOx and PM Sensors (EGS) | Independent & Optional | G | SPN # | | |
| 44 | H,J | Feedback Faults - Air-Fuel Ratio Sensors (EGS) | Independent & Optional | G | SPN # | | |
| 45 | H,J | Feedback Faults - NOx and PM Sensors (EGS) | Independent & Optional | G | SPN # | | |
| 46 | H,J | Sensor - Heater Performance (EGS) | Independent & Optional | G | SPN # | | |
| 47 | H,J | Sensor - Heater Circuit Faults (EGS) | Independent & Optional | G | SPN # | | |
| 48 | H,J | EVAPORATIVE SYSTEM | | G | 3053 | Engine Evaporative System Monitor | |
| 49 | H | 0.150-Inch Leak Detection | Required | G | | | |
| 50 | J | 0.020-Inch Leak Detection | Required | G | | | |
| 51 | H,J | Purge Flow | Independent & Optional | G | SPN # | | |
| 52 | H,J | 0.040-Inch Leak Detection | Independent & Optional | G | SPN # | | |
| 53 | H,J | 0.090-Inch Leak Detection | Independent & Optional | G | SPN # | | |
| 54 | H,J | EGR SYSTEM | | DG | 3058 | | |

| A | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|
| **Row #** | **Regulation Reference (See Table 2)** | **Regulatory Required System & Sub-system Monitor** | **In-Use Ratio Reporting Requirement** | **Engine Type [D]iesel, [G]as, [DG] Diesel & Gas** | **Reported SPN #** | **SPN Name** | **Note** |
| 55 | *H,J* | *Slow Response* | Required | D | | Engine Exhaust Gas Recirculation System Monitor | |
| 56 | *H,J* | *Cooler Performance* | Required | D | | | |
| 57 | *H,J* | *Flow Rate* | Required | G | | | |
| 58 | *H,J* | *Low Flow* | Required | D | | | Required for 2024 + MY engines |
| 59 | *H,J* | *High Flow* | Required | D | | | Required for 2024 + MY engines |
| 60 | *H,J* | *Feedback Control (Time Interval, Default or Open Loop Operation, Control Limit )* | Required | D | | | Required for 2024 + MY engines |
| 61 | *H,J* | *EGR Catalyst Performance* | Independent & Optional | D | SPN # | | |
| 62 | H,J | VARIABLE VALVE TIMING AND/OR CONTROL SYSTEM | | DG | 3306 | Variable Valve Timing and/or Control | |
| 63 | *H,J* | *Flow Rate - VVT* | Required | G | | | |
| 64 | *H,J* | *Target Error - VVT* | Required | DG | | | |
| 65 | *H,J* | *Slow Response - VVT* | Required | DG | | | |
| 66 | *H,J* | *Proper Functional Response - VVT* | Required | DG | | | |
| 67 | H,J | SECONDARY AIR SYSTEM | | G | 3054 | Secondary Air System Monitor | |
| 68 | *H,J* | *Performance* | Required | G | | | |
| 69 | H,J | PM FILTER | | D | 3064 | Aftertreatment Diesel Particulate Filter System Monitor | |
| 70 | *H,J* | *Filtering Performance* | Required | D | | | |
| 71 | *H,J* | *Frequent Regeneration* | Required | D | | | Required for 2024 + MY engines |
| 72 | *H,J* | *Incomplete Regeneration* | Independent & Optional | D | SPN # | | |
| 73 | *H,J* | *NMHC Conversion* | Independent & Optional | D | SPN # | | |
| 74 | *H,J* | *Missing Substrate* | Required | D | 3064 | Aftertreatment Diesel Particulate Filter System Monitor | Required for 2024 + MY engines |
| 75 | *H,J* | *Active/Intrusive Injection* | Required | D | | | Required for 2024 + MY engines |
| 76 | *H,J* | *Feedback Control (Time Interval, Default or Open Loop Operation, Control Limit )* | Independent & Optional | D | SPN # | | |
| 77 | H,J | BOOST PRESSURE CONTROL SYSTEM | | D | 5321 | Engine Intake Manifold Pressure System Monitor | |
| 78 | *H,J* | *Slow Response* | Required | D | | | |
| 79 | *H,J* | *Charge Air Cooler Performance/Charge Air Undercooling* | Required | D | | | |
| 80 | *H,J* | *Under Boost* | Required | D | | | Required for 2024 + MY engines |
| 81 | *H,J* | *Over Boost* | Required | D | | | Required for 2024 + MY engines |
| 82 | *H,J* | *Feedback Control (Time Interval, Default or Open Loop Operation, Control Limit )* | Required | D | | | Required for 2024 + MY engines |
| 83 | H,J | ENGINE IGNITION CYCLE COUNTER | Required | DG | 3048 | Engine Ignition Cycle Counter | |
| 84 | H,J | OBD MONITORING CONDITIONS ENCOUNTERED COUNTS | Required | DG | 3049 | OBD Monitoring Conditions Encountered Counts | |
| 85 | H,J | ENGINE MISFIRE MONITORING | Independent & Optional | DG | 3052 | Engine Misfire System Monitor | |
| 86 | H,J | ENGINE FUEL SYSTEM MONITORING | | D | 3055 | Engine Fuel System Monitor | Discontinue use of SP 3055 for Air Fuel Ratio Cylinder Imbalance Monitor with MY2924_ SI Engines. Use SPs 21229 and 21230 instead |
| 87 | *H,J* | *Injection Quantity* | Required | D | | | |
| 88 | *H,J* | *Injection Timing* | Required | D | | | |
| 89 | *H,J* | *Fuel System Pressure Control* | Independent & Optional | D | SPN # | | |
| 90 | *H,J* | *Feedback Control (Time Interval, Default or Open Loop Operation, Control Limit )* | Independent & Optional | D | SPN # | | |
| 91 | *H,J* | *Fuel Delivery System* | Independent & Optional | G | SPN # | | |
| 92 | *H,J* | *Secondary Feedback Control System* | Independent & Optional | G | SPN # | | |
| 93 | *H,J* | *Air-Fuel Ratio Cylinder Imbalance - Bank 1* | Required | G | 21229 | Air Fuel Ratio Cylinder Imbalance Bank 1 Monitor | Use for 2024+ MY SI engines instead of SP 3055, if equipped |

| A | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|
| Row # | Regulation Reference (See Table 2) | Regulatory Required System & Sub-system Monitor | In-Use Ratio Reporting Requirement | Engine Type [D]iesel, [G]as, [DG] Diesel & Gas | Reported SPN # | SPN Name | Note |
| 94 | *H,J* | *Air-Fuel Ratio Cylinder Imbalance - Bank 2* | Required | G | 21230 | Air Fuel Ratio Cylinder Imbalance Bank 2 Monitor | Use for 2024+ MY SI engines instead of SP 3055, if equipped |
| 95 | *H,J* | *Adaptive Feedback Control* | Independent & Optional | G | SPN # | | |
| 96 | *H,J* | *Feedback Control (Failed to Close Within Time Interval)* | Independent & Optional | G | SPN # | | |
| 97 | H,J | ENGINE POSITIVE CRANKCASE VENTILATION SYSTEM MONITOR | Independent & Optional | DG | 3059 | Engine Positive Crankcase Ventilation System Monitor | |
| 98 | *H,J* | *Disconnection* | Independent & Optional | DG | SPN # | | |
| 99 | H,J | ENGINE COOLING SYSTEM MONITORING | Independent & Optional | DG | 3060 | Engine Cooling System Monitor | |
| 100 | *H,J* | *Thermostat* | Independent & Optional | DG | SPN # | | |
| 101 | *H,J* | *ECT Sensor (Circuit Continuity, Time to Reach Close Loop, or Feedback Enable Condition)* | Independent & Optional | DG | SPN # | | |
| 102 | *H,J* | *ECT Sensor (Stuck in Range)* | Independent & Optional | DG | SPN # | | |
| 103 | H,J | ENGINE COLD START EMISSION REDUCTION STRATEGY SYSTEM MONITOR | Independent & Optional | DG | 3061 | Engine Cold Start Emission Reduction Strategy System Monitor | |
| 104 | *H,J* | *Proper Response of System or Component* | Independent & Optional | DG | SPN # | | |
| 105 | *H,J* | *Performance to OBD Emission Threshold* | Independent & Optional | DG | SPN # | | |
| 106 | *H,J* | *Functional or Individual Components* | Independent & Optional | DG | SPN # | | |
| 107 | H,J | AIR CONDITIONING SYSTEM COMPONENT MONITOR | Independent & Optional | G | 3062 | Air Conditioning System Component Monitor | |
| 108 | *H,J* | *Performance to OBD Emission Threshold* | Independent & Optional | G | SPN # | | |
| 109 | *H,J* | *Functional or Individual Components* | Independent & Optional | G | SPN # | | |
| 110 | J | DIRECT OZONE REDUCTION SYSTEM MONITOR | Independent & Optional | G | 3063 | Direct Ozone Reduction System Monitor | |

#### 5.7.20.1 Engine Ignition Cycle Counter

The engine ignition cycle counter is defined as a single counter that defines the number of operator vehicle ignition switch cycles. Requirements for incrementing the ignition cycle counter are specified in the appropriate legislative documents.

NOTE: This counter shall increment for Hybrid (and Battery Electric) Vehicles where vehicle operation (defined by propulsion system active, SPN 7315) is observed, even if an installed internal combustion engine did not start. See 5.7.57.2 for the Plug-in Hybrid Engine Ignition Cycle Counter (SPN 3077), which counts engine operation for plug-in hybrid vehicles.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | 1 cycle/bit, 0 offset |
| Data Range: | 0 to 65535 |
| Type: | Measured |
| Suspect Parameter Number: | 3048 |
| Reference: | 5.7.20 |

5.7.20.2 OBD Monitoring Conditions Encountered Counts

OBD monitoring conditions encountered counts displays the number of times that the vehicle has been operated in the specified OBD monitoring conditions (e.g., CARB defines this as the general denominator). Requirements for incrementing the OBD monitoring conditions encountered counts are specified in the appropriate legislative documents.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | 1 count/bit, 0 offset |
| Data Range: | 0 to 65535 |
| Type: | Status |
| Suspect Parameter Number: | 3049 |
| Reference: | 5.7.20 |

5.7.20.3 SPN of Applicable System Monitor

This 3-byte field will contain the SPN of the system monitor for which monitor ratio is being reported. The SPN will be positioned in the least significant 19 bits of the 3-byte field. See Column F of Table 19 for allowable contents.

| | |
|---|---|
| Data Length: | 3 bytes (the most significant 5 bits will be set to all ones) |
| Resolution: | Not applicable |
| Data Range: | 0 to 524287 |
| Type: | Status |
| Suspect Parameter Number: | 3066 |
| Reference: | 5.7.20 |

5.7.20.4 Applicable System Monitor Numerator

The number of times a vehicle has been operated such that all conditions necessary for the applicable system monitor to detect a malfunction have been encountered (e.g., CARB numerator requirements). When SPN 3067 and SPN 3068 are both reported as $FFFF_h$, the monitor given in SPN 3066 is not supported by this ECU.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | 1 count/bit, 0 offset |
| Data Range: | 0 to 65535 |
| Type: | Status |
| Suspect Parameter Number: | 3067 |
| Reference: | 5.7.20 |

5.7.20.5 Applicable System Monitor Denominator

The number of times a vehicle has been operated that constitutes a driving cycle where this applicable system monitor could be operated per regulatory requirements (e.g., CARB denominator requirements). When SPN 3067 and SPN 3068 are both reported as FFFFh, the monitor given in SPN 3066 is not supported by this ECU.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | 1 count/bit, 0 offset |
| Data Range: | 0 to 65535 |
| Type: | Status |
| Suspect Parameter Number: | 3068 |
| Reference: | 5.7.20 |

#### 5.7.21 Diagnostic Readiness 2 (DM21)

Reports the diagnostic information relevant to a second PG conveying diagnostic readiness. See DM5, 5.7.5.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | 8 bytes |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 193 |
| PDU Specific: | Destination address |
| Default Priority: | 6 |
| Parameter Group Number: | 49408 ($00C100_h$) |

| | | |
|---|---|---|
| Bytes: 1-2 | Distance traveled while MIL is activated | see 5.7.21.1 |
| Bytes: 3-4 | Distance since diagnostic trouble codes cleared | see 5.7.21.2 |
| Bytes: 5-6 | Minutes run by engine while MIL is activated | see 5.7.21.3 |
| Bytes: 7-8 | Time since diagnostic trouble codes cleared | see 5.7.21.4 |

##### 5.7.21.1 Distance Traveled While MIL is Activated

The distance accumulated while the MIL is activated. See the rollover clearing requirements defined in legislative documentation referenced in 2.1.2 of this document.

Requirements relative to SPN 3069:

- Reset to $0000_h$ when MIL state changes from deactivated to activated by this ECU.
- Accumulate distance in kilometers if MIL is lamp on.
- Do not change value while MIL is lamp off.
- Reset to $0000_h$ if diagnostic information is cleared either by DM11 or at least 40 warm-up cycles without MIL lamp on.
- Do not wrap to $0000_h$ if value is 64255, but clamp the value at 64255.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | 1 km/bit; 0 km offset |
| Data Range: | 0 to 64255 km units |
| Type: | Measured |
| Suspect Parameter Number: | 3069 |
| Reference: | 5.7.21 |

##### 5.7.21.2 Distance Since Diagnostic Trouble Codes Cleared

Distance accumulated since emission-related DTCs were cleared (via an external test equipment or, possibly, a battery disconnect). This parameter (SP) is not associated with any particular emission-related DTC. It is simply an indication for I/M (inspection/maintenance) of the last time an external test equipment was used to clear emission-related DTCs. If greater than 64255 km have occurred, the distance shall remain at 64255 km and not wrap to zero.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | 1 km/bit; 0 km offset |
| Data Range: | 0 to 64255 km units |
| Type: | Measured |
| Suspect Parameter Number: | 3294 |
| Reference: | 5.7.21 |

5.7.21.3 Minutes Run by Engine While MIL is Activated

Accumulated count (in minutes) while the MIL is activated (on). Conditions include: reset to $0000_h$ when MIL state changes from deactivated to activated by this ECU; accumulate counts in minutes if MIL is activated (ON); do not change value while MIL is not activated (OFF); reset to $0000_h$ if diagnostic information is cleared either by DM11 or 40 warm-up cycles without MIL activated; do not wrap to $0000_h$ if value is 64255. For WWH and EURO VI, this time shall be accumulated any time there is a class A “confirmed and active” DTC or Class B1 “confirmed and active” with counter greater than 200 hours. Requirements relative to SPN 3295:

- Reset to $0000_h$ when MIL state changes from deactivated to activated by this ECU.
- Accumulate counts in minutes if MIL is lamp on and engine is running.
- Do not change value while MIL is lamp off.
- Reset to $0000_h$ if diagnostic information is cleared either by DM11 or at least 40 warm-up cycles without MIL lamp on.
- Do not wrap to $0000_h$ if value is 64255, but clamp the value at 64255.

For hybrid vehicles or for vehicles that employ engine shutoff strategies (e.g., engine shutoff at idle), SPN 3295 shall increment while at least one of the following are met:

- The ignition switch is turned to the on position and the engine is running.
- The vehicle can be started in electric-only mode, after the ignition switch is turned to the on position and the propulsion system is active (SPN 7315 = 01).
- The engine is turned off by the vehicle control system during normal operation.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | 1 minute, 0 minute offset |
| Data Range: | 0 to 64255 minutes |
| Type: | Measured |
| Suspect Parameter Number: | 3295 |
| Reference: | 5.7.21 |

5.7.21.4 Time Since Diagnostic Trouble Codes Cleared

Time accumulated, with the ignition on and engine running, since emission-related DTCs were cleared (via an external test equipment or, possibly, a battery disconnect). This SP is not associated with any particular emission-related DTC. It is simply an indication for I/M (inspection/maintenance) of the last time external test equipment was used to clear emission-related DTCs. If greater than 64255 minutes have occurred, SPN 3296 shall remain at 64255 minutes and not wrap to zero.

For hybrid vehicles or for vehicles that employ engine shutoff strategies (e.g., engine shutoff at idle), SPN 3296 shall increment while at least one of the following conditions are met:

- The ignition switch is turned to the on position and the engine is running.
- The vehicle can be started in electric-only mode, after the ignition switch is turned to the on position and propulsion system is active (SPN 7315 = 01).

- The engine is turned off by the vehicle control system during normal operation.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | 1 minute, 0 minute offset |
| Data Range: | 0 to 64255 minutes |
| Type: | Measured |
| Suspect Parameter Number: | 3296 |
| Reference: | 5.7.21 |

5.7.22 Individual Clear/Reset of Active and Previously Active DTC (DM22)

All of the diagnostic information pertaining to the specified diagnostic trouble code should be erased when the CLR_PA_REQ or CLR_ACT_REQ action of this PG is requested. This PG is used to provide the DTC clear/reset services offered with DM3 and DM11 but for individual DTCs. When the individual clear of a previously active DTC is performed, the diagnostic data associated with active trouble codes will not be affected. Upon the completion of a requested clear/reset operation, a positive acknowledgment using CLR_PA_ACK or CLR_ACT_ACK, respectively, is required.

DM22 is a command-type message. It is not queried using SAE J1939-21 request (PGN EA00), and use of the SAE J1939-21 acknowledgment PG (PGN E800) is not expected for replies to DM22. Instead, devices that support DM22 should send a DM22 response. Devices that don't support DM22 (at all) are not required to provide a response using DM22. As a result, responses to DM22 queries can time out, according to SAE J1939-21 rules for response time (Tr). Similarly, retries for DM22 shall be limited to the three-attempt convention described in SAE J1939-21.

NOTE: A DM22 CLR_PA_REQ or CLR_ACT_REQ may or may not generate a response from the target device. No response would be expected if the target node does not support the PG. The target node might filter out the message at the hardware level to reduce CPU utilization or the target node may have no receive handler for this message, effectively discarding the DM22 with no action. A DM22 response (DM22 CLR_PA_ACK, CLR_PA_NACK, CLR_ACT_ACK, or CLR_ACT_NACK) is only expected if the target device supports this PG.

NOTE: A response to a DM 22 CLR_PA_REQ or CLR_ACT_REQ will be one of a DM22 CLR_PA_ACK, CLR_PA_NACK, CLR_ACT_ACK, or CLR_ACT_NACK. The response should not be an acknowledge PGN (E800).

Devices that comply with HD OBD or OBD II shall not clear any individual HD OBD defined DTC.

The SPN format shall follow version 4 as specified in 5.7.1.14.

| | |
|---|---|
| Transmission Rate: | As needed |
| Data Length: | 8 bytes |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 195 |
| PDU Specific: | Destination address |
| Default Priority: | 6 |
| Parameter Group Number: | 49920 (00C300$_h$) |

| | | | | |
|---|---|---|---|---|
| Byte: | 1 | | Individual DTC clear/reset control byte | see 5.7.22.1 |
| Byte: | 2 | | Control byte specific indicator for individual DTC clear | see 5.7.22.2 |
| Bytes: | 3-5 | | Reserved for assignment by SAE | |
| Byte: | 6 | bits 8-1 | SPN, 8 least significant bits of SPN (most significant at bit 8) | see 5.7.1.12 |
| | 7 | bits 8-1 | SPN, second byte of SPN (most significant at bit 8) | see 5.7.1.12 |
| | 8 | bits 8-6 | SPN, 3 most significant bits (most significant at bit 8) | see 5.7.1.12 |
| | | bits 5-1 | FMI (most significant at bit 5) | see 5.7.1.13 |

Data ranges for parameters used by this group function:

Control bytes: 1 through 3 and 17 through 19 are defined below, while 0, 4 through 16, and 20 through 250 are reserved for SAE assignment.

Request to Clear/Reset Previously Active DTC (DM22.CLR_PA_REQ)

| | | | | |
|---|---|---|---|---|
| Bytes: | 1 | Control byte = 1, Request to clear/reset previously active DTC (CLR_PA_REQ) | | see 5.7.22.1 |
| | 2-5 | | Reserved for assignment by SAE (fill with $FF_h$) | |
| | 6 | bits 8-1 | SPN, 8 least significant bits of SPN (most significant at bit 8) | see 5.7.1.12 |
| | 7 | bits 8-1 | SPN, second byte of SPN (most significant at bit 8) | see 5.7.1.12 |
| | 8 | bits 8-6 | SPN, 3 most significant bits (most significant at bit 8) | see 5.7.1.12 |
| | | bits 5-1 | FMI (most significant at bit 5) | see 5.7.1.13 |

Positive Acknowledge of Clear/Reset Previously Active DTC (DM22.CLR_PA_ACK)

| | | | | |
|---|---|---|---|---|
| Bytes: | 1 | Control byte = 2, Positive acknowledge of previously active DTC Clear/Reset (CLR_PA_ACK) | | see 5.7.22.1 |
| | 2-5 | | Reserved for assignment by SAE (Fill with $FF_h$) | |
| | 6 | bits 8-1 | SPN, 8 least significant bits of SPN (most significant at bit 8) | see 5.7.1.12 |
| | 7 | bits 8-1 | SPN, second byte of SPN (most significant at bit 8) | see 5.7.1.12 |
| | 8 | bits 8-6 | SPN, 3 most significant bits (most significant at bit 8) | see 5.7.1.12 |
| | | bits 5-1 | FMI (most significant at bit 5) | see 5.7.1.13 |

Negative Acknowledge of Clear/Reset Previously Active DTC (DM22.CLR_PA_NACK)

| | | | | |
|---|---|---|---|---|
| Bytes: | 1 | Control byte = 3, Negative acknowledge of previously active DTC clear/reset (CLR_PA_NACK) | | see 5.7.22.1 |
| | 2 | | Control byte specific indicator (see Table 21) | see 5.7.22.2 |
| | 3-5 | | Reserved for assignment by SAE (fill with $FF_h$) | |
| | 6 | bits 8-1 | SPN, 8 least significant bits of SPN (most significant at bit 8) | see 5.7.1.12 |
| | 7 | bits 8-1 | SPN, second byte of SPN (most significant at bit 8) | see 5.7.1.12 |
| | 8 | bits 8-6 | SPN, 3 most significant bits (most significant at bit 8) | see 5.7.1.12 |
| | | bits 5-1 | FMI (most significant at bit 5) | see 5.7.1.13 |

Request to Clear/Reset Active DTC (DM22. CLR_ACT_REQ)

| | | | | |
|---|---|---|---|---|
| Bytes: | 1 | Control byte = 17, Request to clear/reset active DTC (CLR_ACT_REQ) | | see 5.7.22.1 |
| | 2-5 | | Reserved for assignment by SAE (fill with $FF_h$) | |
| | 6 | bits 8-1 | SPN, 8 least significant bits of SPN (most significant at bit 8) | see 5.7.1.12 |
| | 7 | bits 8-1 | SPN, second byte of SPN (most significant at bit 8) | see 5.7.1.12 |
| | 8 | bits 8-6 | SPN, 3 most significant bits (most significant at bit 8) | see 5.7.1.12 |
| | | bits 5-1 | FMI (most significant at bit 5) | see 5.7.1.13 |

Positive Acknowledge of Clear/Reset Active DTC (DM22. CLR_ACT_ACK)

| | | | | |
|---|---|---|---|---|
| Bytes: | 1 | Control byte = 18, Positive acknowledge of active DTC clear/reset (CLR_ACT_ACK) | | see 5.7.22.1 |
| | 2-5 | | Reserved for assignment by SAE (fill with $FF_h$) | |
| | 6 | bits 8-1 | SPN, 8 least significant bits of SPN (most significant at bit 8) | see 5.7.1.12 |
| | 7 | bits 8-1 | SPN, second byte of SPN (most significant at bit 8) | see 5.7.1.12 |
| | 8 | bits 8-6 | SPN, 3 most significant bits (most significant at bit 8) | see 5.7.1.12 |
| | | bits 5-1 | FMI (most significant at bit 5) | see 5.7.1.13 |

Negative Acknowledge of Clear/Reset Previously Active DTC (DM22.CLR_ACT_NACK)

| | | | | |
|---|---|---|---|---|
| Bytes: | 1 | Control byte = 19, Negative acknowledge of active DTC Clear/reset (CLR_ACT_NACK) | | see 5.7.22.1 |
| | 2 | | Control byte specific indicator (see Table 21) | see 5.7.22.2 |
| | 3-5 | | Reserved for assignment by SAE (fill with $FF_h$) | |
| | 6 | bits 8-1 | SPN, 8 least significant bits of SPN (most significant at bit 8) | see 5.7.1.12 |
| | 7 | bits 8-1 | SPN, second byte of SPN (most significant at bit 8) | see 5.7.1.12 |
| | 8 | bits 8-6 | SPN, 3 most significant bits (most significant at bit 8) | see 5.7.1.12 |
| | | bits 5-1 | FMI (most significant at bit 5) | see 5.7.1.13 |

#### 5.7.22.1 Individual DTC Clear/Reset Control Byte

A numeric indication of the message function and content within the individual DTC clear message. See Table 20.

| | |
|---|---|
| Data Length: | 1 byte |
| Resolution: | See Table 20 |
| Data Range: | 0 to 250 |
| Type: | Status |
| Suspect Parameter Number: | 3034 |
| Reference: | 5.7.22 |

***Table 20 - Individual DTC clear/reset control byte***

| Control Byte Values | Individual DTC Clear/Reset Control Byte |
|---|---|
| 0 | Reserved for assignment by SAE |
| 1 | Request to clear/reset a specific previously active DTC |
| 2 | Positive acknowledge of clear/reset of a specific previously active DTC |
| 3 | Negative acknowledge of clear/reset of a specific previously active DTC |
| 4 to 16 | Reserved for assignment by SAE |
| 17 | Request to clear/reset a specific active DTC |
| 18 | Positive acknowledge of clear/reset of a specific active DTC |
| 19 | Negative acknowledge of clear/reset of a specific active DTC |
| 20 to 250 | Reserved for assignment by SAE |
| 251 to 255 | Per SAE J1939-71, Table 1 definition |

#### 5.7.22.2 Control Byte Specific Indicator for Individual DTC Clear

A numeric value with interpretation that is specific to the control byte value within the individual DTC clear message.

Data Length: 1 byte
Resolution: See Table 21
Data Range: 0 to 250
Type: Status
Suspect Parameter Number: 3035
Reference: 5.7.22

NOTE: For control byte values 3 and 19 of an individual DTC clear/reset request, see the negative acknowledge indicators for individual DTC clear table for interpretation (see Table 21).

***Table 21 - Negative acknowledge indicators for individual DTC clear***

| Bit States | Negative Acknowledge Indicators for Individual DTC Clear |
|---|---|
| 0 | General negative acknowledge |
| 1 | Access denied (security denied access) |
| 2 | Diagnostic trouble code unknown/does not exist |
| 3 | Diagnostic trouble code no longer previously active |
| 4 | Diagnostic trouble code no longer active |
| 5 to 250 | Reserved for assignment by SAE |
| 251 to 255 | Per SAE J1939-71, Table 1 definition |

EXAMPLE 1: Tool request to transmission to clear a previously active DTC that is currently previously active.

18C303F9 01 FF FF FF FF 00 97 03 (Tool requests clearing of previously active DTC with SPN 1208 and FMI 3)

18C3F903 02 FF FF FF FF 00 97 03 (Transmission acknowledges previously active DTC with SPN 1208 and FMI 3 is cleared)

EXAMPLE 2: Tool request to transmission to clear a previously active DTC that become active.

18C303F9 01 FF FF FF FF 00 97 03 (Tool requests clearing of previously active DTC with SPN 1208 and FMI 3)

18C3F903 03 03 FF FF FF 00 97 03 (Transmission acknowledges previously active DTC with SPN 1208 and FMI 3 is cleared)

EXAMPLE 3: Tool request to transmission to clear an active DTC that the engine does not support.

18C303F9 17 FF FF FF FF 00 97 02 (Tool requests clearing of previously active DTC with SPN 1208 and FMI 2)

18C3F903 19 02 FF FF FF 00 97 02 (Transmission sends negative acknowledge of DTC with SPN 1208 and FMI 2)

EXAMPLE 4: Tool request to transmission that does not support DM22 (possibly an HD OBD engine) to clear an active DTC.

18C303F9 17 FF FF FF FF 00 97 03 (Tool requests clearing of previously active DTC with SPN 1208 and FMI 3)

18C3F903 19 00 FF FF FF 00 97 03 (Transmission transmits general negative acknowledgment), or

18C3F903 19 01 FF FF FF 00 97 03 (Transmission transmits access denied), or

No response (ECU does not support this message)

5.7.23 Emission-Related Previously MIL-On Diagnostic Trouble Codes (DM23)

The information communicated is limited to the current previously MIL-On emission-related diagnostic trouble codes preceded by the diagnostic lamp status.

The lamp information (malfunction indicator lamp, red stop lamp, amber warning lamp, and protect lamp) should reflect the present state of the transmitting electronic component. The lamp information shall not convey temporary signals to provide for lamp test illumination, DTC numeric displays, or OBD readiness indication at key-on.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | Variable |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 253 |
| PDU Specific: | 181 |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |
| Parameter Group Number: | 64949 (00FDB5$_h$) |

| | | | | |
|---|---|---|---|---|
| Byte: | 1 | bits 8-7 | Malfunction indicator lamp | see 5.7.1.4 |
| | | bits 6-5 | Red stop lamp | see 5.7.1.5 |
| | | bits 4-3 | Amber warning lamp | see 5.7.1.6 |
| | | bits 2-1 | Protect lamp | see 5.7.1.7 |
| Byte: | 2 | bits 8-7 | Flash malfunction indicator lamp | see 5.7.1.8 |
| | | bits 6-5 | Flash red stop lamp | see 5.7.1.9 |
| | | bits 4-3 | Flash amber warning lamp | see 5.7.1.10 |
| | | bits 2-1 | Flash protect lamp | see 5.7.1.11 |
| Byte: | 3 | bits 8-1 | SPN, 8 least significant bits of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: | 4 | bits 8-1 | SPN, second byte of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: | 5 | bits 8-6 | SPN, 3 most significant bits (most significant at bit 8) | see 5.7.1.12 |
| | | bits 5-1 | FMI (most significant at bit 5) | see 5.7.1.13 |
| Byte: | 6 | bit 8 | SPN conversion method (shall be sent as a 0) | see 5.7.1.14 |
| | | bits 7-1 | Occurrence count | see 5.7.1.15 |

NOTE: When the occurrence count is not available, it should be set to all ones, which is a value of 127.

NOTE: When there is zero or one DTC to report, then unused bytes 7 and 8 of the CAN frame shall be set to 255 (as per SAE J1939-71).

NOTE: When there are no DTCs to report, each of bytes 3 through 6 shall be set to zero.

EXAMPLE 1: The following illustrates the message format for when there is more than one diagnostic trouble code.

Given:

a = lamp status
b = SPN
c = FMI
d = CM and OC

Message form will be as follows: a,b,c,d,b,c,d,b,c,d,b,c,d....etc. In this example, the transport protocol of SAE J1939-21 will have to be used to send the information because it requires more than 8 data bytes. Actually, any time there is more than one fault, the services of the transport protocol will have to be used.

EXAMPLE 2: The following illustrates the required message format for reporting DM23 when there are zero active faults. The currently defined lamps (malfunction indicator lamp, red stop lamp, amber warning lamp, and protect lamp) shall reflect the present state of the transmitting electronic component.

The required settings for bytes 3 through 6 for reporting no DTC information is shown below. Implementations are required to set bytes 3 through 6 to all zeros and bytes 7 and 8 to all ones when there are no trouble codes to report

Given:

| | | |
|---|---|---|
| Byte: 1 | bits 8-7 | = 00 (example of reporting off) |
| | bits 6-5 | = 00 (example of reporting off) |
| | bits 4-3 | = 00 (example of reporting off) |
| | bits 2-1 | = 00 (example of reporting off) |
| Byte: 2 | bits 8-7 | = 11 (example of reporting not available/don’t care) |
| | bits 6-5 | = 11 (example of reporting not available/don’t care) |
| | bits 4-3 | = 11 (example of reporting not available/don’t care) |
| | bits 2-1 | = 11 (example of reporting not available/don’t care) |

Required Setting:

| | | |
|---|---|---|
| Bytes: 3-6 | SPN | = 0 (required setting for reporting no diagnostic trouble code) |
| | FMI | = 0 (required setting for reporting no diagnostic trouble code) |
| | OC | = 0 (required setting for reporting no diagnostic trouble code) |
| | CM | = 0 (required setting for reporting no diagnostic trouble code) |
| Byte: 7 | | = 255 |
| Byte: 8 | | = 255 |

### 5.7.24 SPN Support (DM24)

This message is used to identify those parameters (by SPN) supported by the product for test results (using DM7/DM30), expanded freeze frame (using DM25), rationality data (using DM7/DM58), and data stream messages (SPNs supported by guidance). DM24 content shall remain constant under all normal engine and vehicle operating modes. The data stream messages are those PGs that contain data for the SPs reported in DM24. Specifically, DM24 content shall be the same whether the engine is running or not running. The request for DM24 for the regulated OBD system shall be directed to OBD devices (see 3.14) that indicate OBD compliance (SPN 1220) in DM5. At least one device that provides a positive value for OBD compliance (SPN 1220) shall support DM24 for MY2013 and later engines. All devices that provide a positive value for OBD compliance on a vehicle may support DM24.

Unregulated devices may also support DM24 for non-regulated freeze frame content, test results, rationality data, and data stream support.

Transmission Rate: On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)
A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392)
The request message for DM24 shall only be sent using a destination-specific address.
Data Length: 8 bytes (variable, typical engine could be 50 parameters x 4 = 200 bytes)
Extended Data Page: 0
Data Page: 0
PDU Format: 253
PDU Specific: 182
Default Priority: 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used)
Parameter Group Number: 64950 ($00FDB6_h$)

| | | | |
|---|---|---|---|
| Byte: 1 | bits 8-1 | Supported parameter SPN, 8 least significant bits of SPN (most significant at bit 8) | see 5.7.24.1 |
| Byte: 2 | bits 8-1 | Supported parameter SPN, second byte of SPN (most significant at bit 8) | see 5.7.24.1 |
| Byte: 3 | bits 8-6 | Supported parameter SPN, 3 most significant bits (most significant at bit 8) | see 5.7.24.1 |
| | bit 5 | Reserved for assignment by SAE (set to 1) | |
| | bit 4 | Supported in rationality fault SP data value | see 5.7.24.2.4 |
| | bit 3 | Supported in scaled test results | see 5.7.24.2.3 |
| | bit 2 | Supported in data stream | see 5.7.24.2.2 |
| | bit 1 | Supported in expanded freeze frame | see 5.7.24.2.1 |
| Byte: 4 | bits 8-1 | SP data length | see 5.7.24.3 |

Given:
A = Supported parameter SPN
B = Supported parameter support type
C = SP data length

Message form will be as follows: a,b,c,a,b,c,a,b,c....etc.

***Table 22 - SPN representation in CAN data frame for DM24***

<table>
<tr><td colspan="8">Byte 1: Eight Least Significant Bits of SPN (Bit 8 Most Significant)</td><td colspan="8">Byte 2: Second Byte of SPN (Bit 8 Most Significant)</td><td colspan="3">Byte 3: Three Most Significant Bits of SPN (Bit 8 MSB)</td><td colspan="5">Byte 3: Bits 5-1 Supported Parameter Support Type</td></tr>
<tr><td colspan="19">SPN</td><td colspan="5">Support Types</td></tr>
<tr><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td></tr>
</table>

NOTE: DM24 provides space to identify up to 446 SPNs. It may not be possible to include all the desired SPNs provided by an individual device or controller application in a given DM24 response. The following priority rules are recommended to manage the number of SPNs provided in a DM24 response.

1. Include all SPNs provided in the DM25 freeze frame response in the order that the SPN appears in the freeze frame. Mark these SPNs with bit 1 set to 0.

2. Include all SPNs queried with DM7 for DM30 test results that are not included in 1. Mark these SPNs and the SPNs reused from 1 with bit 3 set to 0.

3. Include all SPNs queried with DM7 for DM58 data that are not included in 1 or 2. Mark these SPNs and the SPNs reused from 1 and 2 with bit 4 set to 0.

4. Include lead SPNs identified in Table 1 for NOx Binning and GHG Tracking, if the arrays are supported. Mark these SPNs with bit 2 set to 0

5. Include one supported SPN for each PG provided to meet the device's data stream requirements that is not included in 1, 2, 3, or 4. Where there is insufficient space for all desired SPNs, prioritize one SPN for any desired On Request PGs over one SPN for a broadcast PG. Mark these SPNs with bit 2 set to 0.

Providing only one SPN to represent all the supported SPNs in a PG, the principle described in 5, is recommended to manage the number of DM24 packets as this reduces transport protocol resources allocations in devices and tools, in addition to J1939-73 CAN bus utilization.

An SPN shall appear only once in a device's DM24 response. All support roles shall be noted for the SPN's support bit settings. Data that is provided in a freeze frame by one device may be provided by the data stream of a second device of a distributed [OBD] system. Test data support (bit 3) and rationality data support may also be distributed in DM24 responses.

5.7.24.1 Supported Parameter SPN

This parameter identifies a parameter (by its SPN) that is supported by the ECU in data stream, expanded freeze frame, scaled test results, or rationality fault data. Any SP not supported for at least one of these purposes shall not be reported in the response message. The reported SPs may be limited to those required to satisfy regulated OBD requirements. Be aware that some of the reported supported data stream SPs can come from different source addresses than the device reporting the DM24 response. Additionally, the SPs reported in the parameter shall be limited to those identified in SAE J1939DA, excluding those specified in SAE J1939-73. An exception to this rule is if the SPN is in SAE J1939-73 and must be provided as part of an OBD freeze frame, then it shall be contained in DM24 (e.g., time since engine start) and can indicate data stream, freeze frame, and test results, if appropriate.

Manufacturers may populate the Supported Parameter SPN with a superset list of SPNs for multiple product lines. An individual product can have some SPNs in the SPN list reported with the SPN Support Type as all ones (meaning not supported).

| | |
|---|---|
| Data Length: | 19 bits |
| Resolution: | 1 SPN per bit |
| Data Range: | 0 to 524287 (00 00 $00_h$ to 7F FF $FF_h$) |
| Type: | Status |
| Suspect Parameter Number: | 3297 |
| Reference: | 5.7.24 |

NOTE: See Table 1 for the conventions to be used for NOx Binning and GHG Tracking SPs. These PG for NOx Binning and GHG Tracking are divided into four groups and only one SP per group shall be displayed in the DM24 Response. GHG technology tracking (row 68) and GHG Tracking Hybrid Messages (row 69) groups shall be omitted when they are not supported by the responding device.

5.7.24.2 Supported Parameter Support Type

This parameter identifies the type of support for the parameter identified by its SPN in Supported Parameter SPN (SPN 3297). The type of support for a parameter shall indicate support for at least one support type: data stream, expanded freeze frame (DM25), scaled test results (DM30), or rationality fault SP data (DM58).

*Table 23 - Supported parameter support type*

| Support Type Position | Definition |
|---|---|
| Bit 1 | Supported in expanded freeze frame (i.e., DM25) |
| Bit 2 | Supported in data stream |
| Bit 3 | Supported in scaled test results (i.e., DM30) |
| Bit 4 | Supported in rationality fault SP data (i.e., DM58) |
| Bit 5 | SAE Reserved (set to one) |

Data Length: 5 bits
Resolution: Not applicable
Data Range: 0 to 31 ($00_h$ to $1F_h$)
Type: Status
Suspect Parameter Number: 3298
Reference: 5.7.24

5.7.24.2.1 Supported in Expanded Freeze Frame

This parameter indicates if the parameter, identified by its SPN in 'SPN Supported', is supported in the expanded freeze frame message, i.e. DM25.

0 = Supported in expanded freeze frame message
1 = Not supported in expanded freeze frame message

Data Length: 1 bit
Data Range: 0 to 1
Type: Status
Suspect Parameter Number: 4100
Reference: 5.7.24

5.7.24.2.2 Supported in Data Stream

This parameter indicates if the parameter, identified by its SPN in “SPN Supported,” is supported in the data stream messages.

0 = Supported in data stream messages
1 = Not supported in data stream messages

Data Length: 1 bit
Data Range: 0 to 1
Type: Status
Suspect Parameter Number: 4101
Reference: 5.7.24

5.7.24.2.3 Supported in Scaled Test Results

This parameter indicates if the applicable parameter, identified by its SPN in “SPN Supported,” is supported in the scaled test results message, DM30.

0 = Supported in scaled test results message
1 = Not supported in scaled test results message

Data Length: 1 bit
Data Range: 0 to 1
Type: Status
Suspect Parameter Number: 4102
Reference: 5.7.24

5.7.24.2.4 Supported in Rationality Fault SP Data Value

This parameter indicates if the parameter, identified by its SPN in "SPN Supported," is supported in the rationality fault SP data, DM58 message (see 5.7.58).

0 = Supported in rationality fault SP data message
1 = Not supported in rationality fault SP data message

| | |
|---|---|
| Data Length: | 1 bit |
| Resolution: | Not applicable |
| Data Range: | 0 to 1 |
| Type: | Status |
| Suspect Parameter Number: | 8557 |
| Reference: | 5.7.24 |

5.7.24.3 SP Data Length

This parameter indicates the number of data bytes required for reporting the data for the Supported Parameter, such as in the freeze frame.

The SP Data Length enables compatibility of old and new OBD tools with vehicle OBD systems. For instance, if the vehicle supports a newer parameter, the SP Data Length allows an older tool to still parse the data, such as expanded freeze frame data, and bypass the unresolved parameter data. SP data value scaling is per the applicable SAE J1939 specification (SAE J1939-71, SAE J1939DA, SAE J1939-75, etc.). SP Data Length shall be reported as "1" for parameters defined with less than 8 bits.

| | |
|---|---|
| Data Length: | 1 byte |
| Resolution: | 1 data byte |
| Data Range: | 0 to 250 |
| Type: | Status |
| Suspect Parameter Number: | 4103 |
| Reference: | 5.7.24 |

NOTE: The value in SPN 4103 has no standardized meaning and should be ignored when the corresponding value of SPN 4100 is a binary one, which means not supported.

5.7.25 Expanded Freeze Frame (DM25)

Freeze frame message providing more flexible parameter support than the existing DM4.

A freeze frame is defined as the list of recorded parameters at the time a diagnostic trouble code was captured. The freeze frame recorded for each diagnostic trouble code will contain the required parameters first and then any manufacturer-specific information. It is possible that controllers will have more than one freeze frame available, and each may have some manufacturer-specific information. A freeze frame is specific to one diagnostic trouble code, and one diagnostic trouble code only has one freeze frame. This then limits the amount of freeze frame data per fault and for all faults that are included in this message to 1785 bytes (refer to SAE J1939-21 transport protocol).

This diagnostic message was created for systems that impact emissions and/or are powertrain related. However, the use of this message is not limited to just emission-related failures or just powertrain devices. It can be used to report non-emission-related or non-powertrain-related freeze frame failures.

The order of the freeze frame data parameters will be per the order defined in DM24. The parameter length for each individual parameter in the freeze frame is also determined from the information provided in DM24.

Implementers should refer to the applicable regulation for potential additional freeze frame requirements. For instance, some regulations might require the OBD freeze frame to have priority over non-OBD freeze frames.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392).<br>The request message for DM25 shall only be sent using a destination-specific address. |
| Data Length: | Variable |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 253 |
| PDU Specific: | 183 |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |
| Parameter Group Number: | 64951 (00FDB7$_h$) |

| | | | | |
|---|---|---|---|---|
| Byte: | 1 | | Expanded freeze frame length | see 5.7.25.1 |
| Byte: | 2 | bits 8-1 | SPN, 8 least significant bits of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: | 3 | bits 8-1 | SPN, second byte of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: | 4 | bits 8-6 | SPN, 3 most significant bits (most significant at bit 8) | see 5.7.1.12 |
| | | bits 5-1 | FMI (most significant at bit 5) | see 5.7.1.13 |
| Byte: | 5 | bit 8 | SPN Conversion method | see 5.7.1.14 |
| | | bits 7-1 | Occurrence count | see 5.7.1.15 |
| Byte: | 6 - n | | Expanded freeze frame data | see 5.7.25.2 |

EXAMPLE 1: The following illustrates the message format for when there is more than one freeze frame.

Given:

a = expanded freeze frame length
b = DTC associated with freeze frame data (i.e., bytes 2 to 5 above)
c = data for freeze frame supported parameters

Message form will be as follows: a,b,c,<u>a,b,c</u>,a,b,c,<u>a,b,c</u>,a,b,c....etc. The transport protocol of SAE J1939-21 will have to be used to send freeze frames because they are more than 8 data bytes.

EXAMPLE 2: Illustration of a DM25 expanded freeze frame message (see Figure 6) when supplied with the data identified in Table 24.

Given:

1 = SPN 91 (Accelerator Pedal Position 1)
2 = Failure Mode of 3, Voltage Above Normal, is occurring
3 = DM24 (Response per Table 22)

***Table 24 - Example DM24 response***

| | Supported Parameter SPN (SPN 3297) | SP Data Length (SPN 4103) | Supported in Extended Freeze Frame (SPN 4100) |
|---|---|---|---|
| Engine Speed | 190 | 2 | Yes |
| Engine Coolant Temperature | 110 | 0 | No |
| Wheel-Based Vehicle Speed | 84 | 1 | Yes |
| Accelerator Pedal Position 1 | 91 | 1 | Yes |

NOTE: The value given in SPN 4103, number of data bytes in freeze frame, has no standardized meaning and should be ignored when the corresponding value of SPN 4100, supported in extended freeze frame, is a binary one (No).

Results: DM25 as shown in Figure 6.

<table>
<tr><td></td><td colspan="6">DTC</td><td colspan="4">Expanded Freeze Frame Data</td></tr>
<tr><td>Byte 1</td><td>Byte 2</td><td>Byte 3</td><td colspan="2">Byte 4</td><td colspan="2">Byte 5</td><td>Byte 6</td><td>Byte 7</td><td>Byte 8</td><td>Byte 9</td></tr>
<tr><td>Expanded Freeze Frame Length</td><td colspan="3">SPN</td><td>FMI</td><td>CM</td><td>Occurrence Count</td><td colspan="2">Data for SPN 190</td><td>Data for SPN 84</td><td>Data for SPN 91</td></tr>
<tr><td>$8_{10}$</td><td colspan="3">$91_{10}$</td><td>3</td><td>0</td><td>1</td><td colspan="2">$8000_{10}$</td><td>$100_{10}$</td><td>$254_{10}$</td></tr>
</table>

***Figure 6 - Example DM25***

EXAMPLE 3: The following illustrates the DM25 message content when there are zero freeze frames to report.

The DM25 message content when there are zero freeze frames will report the value zero ($00_h$) for the expanded freeze frame length, all zeros ($00_h$) for each of the bytes 2 through 5, and all ones ($FF_h$) for each of the bytes 6 through 8.

Byte 1 = 0
Bytes 2-5
SPN = 0
FMI = 0
CM = 0
OC = 0
Byte 6 = 255
Byte 7 = 255
Byte 8 = 255

#### 5.7.25.1 Expanded Freeze Frame Length

The freeze frame length is the number of bytes to convey the DTC and all supported parameter data in the freeze frame.

| | |
|---|---|
| Data Length: | 8 bits |
| Resolution: | 1 byte/bit |
| Data Range: | 0 to 255 |
| Type: | Status |
| Suspect Parameter Number: | 3300 |
| Reference: | 5.7.25 |

EXAMPLE: This example is showing a very short, expanded freeze frame just to illustrate the use of the “expanded freeze frame length.”

Given:
a = expanded freeze frame length
b = DTC associated with freeze frame data (i.e., bytes 2 through 5 above)
c = extended freeze frame data (for supported parameters)

Length is calculated as follows:
b = 4
c = 4..........engine speed, vehicle speed, accelerator
a = b + c
a = 4 + 4 = 8

#### 5.7.25.2 Expanded Freeze Frame Data

The Expanded Freeze Frame data contains the data pertaining to each of the Supported Parameters reported in DM24 with Support Type indicating expanded freeze frame support. The order and number of bytes per SPN is determined from the DM24 response.

| | |
|---|---|
| Data Length: | determined from the DM24 response |
| Resolution: | determined from the DM24 response |
| Data Range: | determined from the DM24 response |
| Type: | determined from the DM24 response |
| Suspect Parameter Number: | 1545 |
| Reference: | 5.7.25 |

In DM25, an entire byte is used for each parameter if the parameter itself is less than 8 bits long. The SPN data value shall be placed into the byte right justified to the least significant bit, regardless of its defined position in the normal data PG. Any remaining bits of the data byte shall be reported as 0. For example, if Accelerator Pedal Kick-Down Switch (SPN 559) is reported as a DM25 freeze frame parameter, then its data shall be positioned in bits 1 and 2 of the byte in DM25, even though the data for this parameter is reported in bit 3 and bit 4 of PGN 61443. The remaining unused bits of this byte in the DM25 message will be filled with “0.” Each of the remaining six most significant bits of this freeze frame byte (bits 2 to 8) will be filled with “0.”

### 5.7.26 Diagnostic Readiness 3 (DM26)

This message conveys information useful in determining whether the OBD system has a defect or not. This specific message conveys the pending status of OBD system monitors for the current drive cycle. It, along with the data from DM5 and DM21, is used to formulate OBD system readiness. Note that this parameter group will be sent using the “multipacket transport” parameter group as specified in SAE J1939-21 when applicable.

The bits in Continuously Monitored Systems Enabled/Completed Status (SPN 3303), Non-Continuously Monitored Systems Enabled Status (SPN 3304), and Non-Continuously Monitored Systems Completed Status (SPN 3305) shall report two pieces of information for each monitor (or readiness group):

1. Monitor enable status for the current driving cycle. This bit shall indicate when a monitor is disabled in a manner such that there is no likely or reasonable way for the driver to operate the vehicle to allow the monitor run. Typical examples are:

    - Engine-off soak not long enough (e.g., cold start temperature conditions not satisfied)
    - Monitor maximum time limit or number of attempts/aborts exceeded
    - Ambient air temperature too low or too high
    - BARO too low (high altitude)

    The monitor shall not indicate “disabled” for operator-controlled conditions such as rpm, load, accelerator position, minimum time limit not exceeded, etc.

2. Monitor completion status for the current driving/monitoring cycle. Status shall be reset to “not complete” upon starting a new monitoring cycle. Note that some monitoring cycles can include various engine-operating conditions; other monitoring cycles begin after the ignition key is turned off. Some status bits on a given vehicle can utilize engine-running monitoring cycles while others can utilize engine-off monitoring cycles. Resetting the bits to “not complete” upon starting the engine will accommodate most engine-running and engine-off monitoring cycles; however, manufacturers are free to define their own monitoring cycles.

NOTE: DM26 bits shall be utilized for all non-continuous monitors that are supported and change completion status in DM5. If a non-continuous monitor is not supported or always shows “complete,” the corresponding DM26 bits shall indicate disabled and complete. DM26 bits may be utilized at the vehicle manufacturer’s discretion for all continuous monitors that are supported with the exception of the bit that shall always show CCM (comprehensive component monitoring) as enabled for spark ignition and compression ignition engines.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | Variable (presently 8 bytes) |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 253 |
| PDU Specific: | 184 |
| Default Priority: | 6 |
| Parameter Group Number: | 64952 (00FDB8$_h$) |

| Bytes | Description | Reference |
|---|---|---|
| 1-2 | Time since engine start | see 5.7.26.1 |
| 3 | Number of warm-ups since diagnostic trouble codes cleared | see 5.7.26.2 |
| 4 | Continuously monitored systems enable/completed status | see 5.7.26.3 |
| 5-6 | Non-continuously monitored systems enable status | see 5.7.26.4 |
| 7-8 | Non-continuously monitored systems complete status | see 5.7.26.5 |

5.7.26.1 Time Since Engine Start

For non-hybrid vehicles, the time since key-on that the engine has been running. SPN 3301 shall increment while the engine is running. It shall freeze if the engine stalls. SPN 3301 shall be reset to zero during every control module power-up and when entering the key-on, engine off position. SPN 3301 is limited to 64255 seconds and shall not wrap around to zero.

For hybrid vehicles or for vehicles that employ engine shutoff strategies (e.g., engine shutoff at idle), SPN 3301 shall accumulate time after the ignition switch is turned to the on position and the engine is running, or, if the vehicle can be started in electric-only mode, SPN 3301 shall increment after the propulsion system is active (SPN 7315 = 01). It shall continue to increment even if the engine is turned off by the vehicle control system. SPN 3301 shall be reset to zero during every control module power-up and when entering the key-on, engine off position. SPN 3301 is limited to 64255 seconds and shall not wrap around to zero.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | 1 s/bit |
| Data Range: | 0 to 64255 seconds |
| Type: | Measured |
| Suspect Parameter Number: | 3301 |
| Reference: | 5.7.26 |

5.7.26.2 Number of Warm-Ups Since Diagnostic Trouble Codes Cleared

Number of OBD warm-up cycles since all DTCs were cleared (via an external test equipment or, possibly, a battery disconnect). A warm-up is defined in the OBD regulations to be sufficient vehicle operation such that coolant temperature rises by at least 22.2 °C (40 °F) from engine starting and reaches a minimum temperature of 71.1 °C (160 °F) (or 60 °C [140 °F] for diesels). This parameter is not associated with any particular DTC. It is simply an indication for I/M of the last time an external test equipment was used to clear DTCs. If greater than 250 warm-ups have occurred, WARM_UPS (i.e., number of warm-ups since diagnostic trouble codes cleared) shall remain at 250 and not wrap to zero.

| | |
|---|---|
| Data Length: | 1 byte |
| Resolution: | 1 trouble code/bit |
| Data Range: | 0 to 250 |
| Type: | Measured |
| Suspect Parameter Number: | 3302 |
| Reference: | 5.7.26 |

5.7.26.3 Continuously Monitored Systems Enabled/Completed Status

This parameter identifies the continuously monitored systems enabled/completed support and status.

Data Length: 1 byte
Resolution: See below
Data Range: Bit mapped, see below
Type: Measured
Suspect Parameter Number: 3303
Reference: 5.7.26

| Byte: | Bit: | Description: | |
|---|---|---|---|
| 4 | 8 | Reserved for assignment by SAE (shall be reported as 0) | |
| | 7 | Comprehensive component completed | see 5.7.26 |
| | 6 | Engine Fuel System monitoring completed | see 5.7.26 |
| | 5 | Misfire monitoring completed | see 5.7.26 |

Where each completed bit (bits 7, 6, 5) is interpreted:
0 = monitor complete this monitoring cycle, or not supported (YES)
1 = monitor not complete this monitoring cycle (NO)

See DM5 to determine which monitors are supported.

| Bit: | Description: |
|---|---|
| 4 | Reserved for assignment by SAE (shall be reported as 0) |
| 3 | Comprehensive component monitoring enabled |
| 2 | Engine fuel system monitoring enabled |
| 1 | Misfire monitoring enabled |

Where each enabled bit (bits 3, 2, 1) is interpreted:
0 = monitor disabled for rest of this monitoring cycle or not supported (NO)
1 = monitor enabled for this monitoring cycle (YES)

NOTE: A bit set to zero can mean test not supported. This is different than the typical SAE J1939 use of the value 1 to indicate not available. Any bits that are “reserved for assignment by SAE” shall be reported as 0.

5.7.26.4 Non-Continuously Monitored Systems Enabled Status

Enable status of non-continuous monitors this monitoring cycle.

Data Length: 2 bytes
(PG placement of data is specified below)
Resolution: See below
Data Range: Bit mapped, see below
Type: Measured
Suspect Parameter Number: 3304
Reference: 5.7.26

| Byte: | Bit: | Description: |
|---|---|---|
| 5 | 8 | EGR/VVT system monitoring enabled |
| | 7 | Exhaust gas sensor heater monitoring enabled (see footnote 11) |
| | 6 | Exhaust gas sensor monitoring enabled (see footnote 11) |
| | 5 | A/C system refrigerant monitoring enabled |
| | 4 | Secondary air system monitoring enabled |
| | 3 | Engine evaporative system monitoring enabled |
| | 2 | Heated catalyst monitoring enabled |
| | 1 | Catalyst monitoring enabled |
| 6 | 8-6 | Reserved for assignment by SAE (shall be reported as 0) |
| | 5 | NMHC converting catalyst monitoring enabled |
| | 4 | NOx converting catalyst and/or NOx adsorber monitoring enabled |
| | 3 | Diesel particulate filter (DPF) monitoring enabled |
| | 2 | Boost pressure control system monitoring enabled |
| | 1 | Cold start aid system monitoring enabled |

Where each enable bit is interpreted:

0 = test monitor disabled for rest of this monitoring cycle (NO)
1 = monitor enabled for this monitoring cycle (YES)

NOTE: A bit set to zero can mean test not supported. This is different than the typical SAE J1939 use of the value 1 to indicate not available. Any bits that are “reserved for assignment by SAE” shall be reported as 0.

5.7.26.5 Non-Continuously Monitored Systems Completed Status

Completion status of non-continuous monitors this monitoring cycle. Each bit identifies whether a particular test is complete for a given controller.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | See below |
| Data Range: | Bit mapped, see below |
| Type: | Measured |
| Suspect Parameter Number: | 3305 |
| Reference: | 5.7.26 |

| Byte: | Bit: | Description: |
|---|---|---|
| 7 | 8 | EGR/VVT system monitoring status |
| | 7 | Exhaust gas sensor heater monitoring status (see footnote 11) |
| | 6 | Exhaust gas sensor monitoring status (see footnote 11) |
| | 5 | A/C system refrigerant monitoring status |
| | 4 | Secondary air system monitoring status |
| | 3 | Engine evaporative system monitoring status |
| | 2 | Heated catalyst monitoring status |
| | 1 | Catalyst monitoring status |
| 8 | 8-6 | Reserved for assignment by SAE (shall be reported as 0) |
| | 5 | NMHC converting catalyst monitoring status |
| | 4 | NOx converting catalyst and/or NOx adsorber monitoring status |
| | 3 | Diesel particulate filter (DPF) monitoring status |
| | 2 | Boost pressure control system monitoring status |
| | 1 | Cold start aid system monitoring status |

Where each bit is interpreted:

0 = monitor complete this monitoring cycle, or not supported (yes)
1 = monitor not complete this monitoring cycle (no)

NOTE: A bit set to zero can mean test not supported. This is different than the typical SAE J1939 use of the value 1 to indicate not available. Any bits that are “reserved for assignment by SAE” shall be reported as 0.

5.7.27 All Pending DTCs (DM27)

The purpose of this DM is to enable the external test equipment to obtain all “pending” diagnostic trouble codes detected during current or last completed driving cycle for all components/systems, including emission-related components/systems. DM27 can be used for all DTCs and is independent of DM6. The intended use of this data is to assist the service technician after a vehicle repair, and after clearing diagnostic information, by reporting test results after a single driving cycle. Test results reported by this service do not necessarily indicate a faulty component/system. All Pending DTCs include both (any) emissions-related (reported in DM6) and (any) non-emissions-related DTCs.

Reporting the pending DTCs is done using the same format as is used to report active DTCs.

The lamp information (malfunction indicator lamp, red stop lamp, amber warning lamp, and protect lamp) should reflect the present state of the transmitting electronic component. The lamp information shall not convey temporary signals to provide for lamp test illumination, DTC numeric displays, or OBD readiness indication at key-on.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | Variable |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 253 |
| PDU Specific: | 130 |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |
| Parameter Group Number: | 64898 (00FD82h) |

| | | | | |
|---|---|---|---|---|
| Byte: | 1 | bits 8-7 | Malfunction indicator lamp | see 5.7.1.4 |
| | | bits 6-5 | Red stop lamp | see 5.7.1.5 |
| | | bits 4-3 | Amber warning lamp | see 5.7.1.6 |
| | | bits 2-1 | Protect lamp | see 5.7.1.7 |
| Byte: | 2 | bits 8-7 | Flash malfunction indicator lamp | see 5.7.1.8 |
| | | bits 6-5 | Flash red stop lamp | see 5.7.1.9 |
| | | bits 4-3 | Flash amber warning lamp | see 5.7.1.10 |
| | | bits 2-1 | Flash protect lamp | see 5.7.1.11 |
| Byte: | 3 | bits 8-1 | SPN, 8 least significant bits of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: | 4 | bits 8-1 | SPN, second byte of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: | 5 | bits 8-6 | SPN, 3 most significant bits (most significant at bit 8) | see 5.7.1.12 |
| | | bits 5-1 | FMI (most significant at bit 5) | see 5.7.1.13 |
| Byte: | 6 | bit 8 | SPN conversion method | see 5.7.1.14 |
| | | bits 7-1 | Occurrence count | see 5.7.1.15 |

NOTE: When the occurrence count is not available, it should be set to all ones, which is a value of 127.

NOTE: When there is zero or one DTC to report, then unused byte 7 and byte 8 of the CAN frame shall be set to 255 (as per SAE J1939-71).

NOTE: When there are no DTCs to report, each of bytes 3 through 6 shall be set to zero.

EXAMPLE 1: The following illustrates the message format for when there is more than one diagnostic trouble code.

Given:

a = lamp status (LS)
b = SPN
c = FMI
d = CM and OC

Message form is as follows: a,b,c,d,<u>b,c,d</u>,b,c,d,<u>b,c,d</u>....etc. In this example, the transport protocol of SAE J1939-21 has to be used to send the information because it requires more than 8 data bytes. Actually, any time there is more than one fault, the services of the transport protocol have to be used.

EXAMPLE 2: The following illustrates the required message format for reporting DM27 when there are zero pending faults. The currently defined lamps (malfunction indicator lamp, red stop lamp, amber warning lamp, and protect lamp) shall reflect the present state of the transmitting electronic component.

The required settings for bytes 3 through 6 for reporting no DTC information is shown below. Implementations are required to set bytes 3 through 6 to all zeros and byte 7 and byte 8 to all ones when there are no trouble codes to report.

Given:

| | | |
|---|---|---|
| Byte: 1 | bits 8-7 | = 00 (example of reporting off) |
| | bits 6-5 | = 00 (example of reporting off) |
| | bits 4-3 | = 00 (example of reporting off) |
| | bits 2-1 | = 00 (example of reporting off) |
| Byte: 2 | bits 8-7 | = 11 (example of reporting not available/don't care) |
| | bits 6-5 | = 11 (example of reporting not available/don't care) |
| | bits 4-3 | = 11 (example of reporting not available/don't care) |
| | bits 2-1 | = 11 (example of reporting not available/don't care) |

Required Setting

| | | |
|---|---|---|
| Bytes: 3-6 | SPN | = 0 (required setting for reporting no diagnostic trouble code) |
| | FMI | = 0 (required setting for reporting no diagnostic trouble code) |
| | OC | = 0 (required setting for reporting no diagnostic trouble code) |
| | CM | = 0 (required setting for reporting no diagnostic trouble code) |
| Byte: 7 | | = 255 |
| Byte: 8 | | = 255 |

5.7.28 Emission-Related Permanent Diagnostic Trouble Codes (DM28)

The purpose of this DM is to provide the list of permanent DTCs that are currently present. A permanent DTC is defined as a confirmed fault code and is stored in nonvolatile memory.

The lamp information (malfunction indicator lamp, red stop lamp, amber warning lamp, and protect lamp) should reflect the present state of the transmitting electronic component. The lamp information shall not convey temporary signals to provide for lamp test illumination, DTC numeric displays, or OBD readiness indication at key-on.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | Variable |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 253 |
| PDU Specific: | 128 |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |
| Parameter Group Number: | 64896 ($00FD80_h$) |

| | | | |
|---|---|---|---|
| Byte: 1 | bits 8-7 | Malfunction indicator lamp | see 5.7.1.4 |
| | bits 6-5 | Red stop lamp | see 5.7.1.5 |
| | bits 4-3 | Amber warning lamp | see 5.7.1.6 |
| | bits 2-1 | Protect lamp | see 5.7.1.7 |
| Byte: 2 | bits 8-7 | Flash malfunction indicator lamp | see 5.7.1.8 |
| | bits 6-5 | Flash red stop lamp | see 5.7.1.9 |
| | bits 4-3 | Flash amber warning lamp | see 5.7.1.10 |
| | bits 2-1 | Flash protect lamp | see 5.7.1.11 |
| Byte: 3 | bits 8-1 | SPN, 8 least significant bits of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: 4 | bits 8-1 | SPN, second byte of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: 5 | bits 8-6 | SPN, 3 most significant bits (most significant at bit 8) | see 5.7.1.12 |
| | bits 5-1 | FMI (most significant at bit 5) | see 5.7.1.13 |
| Byte: 6 | bit 8 | SPN conversion method | see 5.7.1.14 |
| | bits 7-1 | Occurrence count | see 5.7.1.15 |

NOTE: When the occurrence count is not available, it should be set to all ones, which is a value of 127.

NOTE: When there is zero or one DTC to report, then unused byte 7 and byte 8 of the CAN frame shall be set to 255 (as per SAE J1939-71).

NOTE: When there are no DTCs to report, each of bytes 3 through 6 shall be set to zero.

EXAMPLE 1: The following illustrates the message format for when there is more than one diagnostic trouble code.

Given:
a = lamp status (LS)
b = SPN
c = FMI
d = CM and OC

Message form will be as follows: a,b,c,d,b,c,d,b,c,d,b,c,d....etc. In this example, the transport protocol of SAE J1939-21 will have to be used to send the information because it requires more than 8 data bytes. Actually, any time there is more than one fault, the services of the transport protocol will have to be used.

EXAMPLE 2: The following illustrates the message format for when a request of the DM28 is made and there are zero previously active faults. The currently defined lamps (malfunction indicator lamp, red stop lamp, amber warning lamp, and protect lamp) should reflect the present state of the transmitting electronic component. In this example, the amber lamp is identified as being on.

The required setting for bytes 6 through 1 is shown below.

Given:

| | | |
|---|---|---|
| Byte: 1 | bits 8-7 = 00 (example of reporting off) | |
| | bits 6-5 = 00 (example of reporting off) | |
| | bits 4-3 = 01 (example of reporting on) | |
| | bits 2-1 = 00 (example of reporting off) | |
| Byte: 2 | bits 8-7 = 11 (example of reporting not available/don’t care) | |
| | bits 6-5 = 11 (example of reporting not available/don’t care) | |
| | bits 4-3 = 11 (example of reporting not available/don’t care) | |
| | bits 2-1 = 11 (example of reporting not available/don’t care) | |

Required Setting:

| | | |
|---|---|---|
| Bytes: 3-6 | SPN | = 0 (required setting for reporting no diagnostic trouble code) |
| | FMI | = 0 (required setting for reporting no diagnostic trouble code) |
| | OC | = 0 (required setting for reporting no diagnostic trouble code) |
| | CM | = 0 (required setting for reporting no diagnostic trouble code) |
| Byte: 7 | | = 255 |
| Byte: 8 | | = 255 |

5.7.29 DTC Counts (DM29)

The purpose of this DM is to identify the number of DTCs in each category. The categories are pending, all pending, MIL-On, previously MIL-On, and permanent DTCs.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | 8 bytes |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 158 |
| PDU Specific: | Destination address |
| Default Priority: | 6 |
| Parameter Group Number: | 40448 ($009E00_h$) |

| | | | |
|---|---|---|---|
| Byte: | 1 | Emission-related pending DTC count | see 5.7.29.1 |
| Byte: | 2 | All pending DTC count | see 5.7.29.2 |
| Byte: | 3 | Emission-related MIL-On DTC count | see 5.7.29.3 |
| Byte: | 4 | Emission-related previously MIL-On DTC count | see 5.7.29.4 |
| Byte: | 5 | Emission-related permanent DTC count | see 5.7.29.5 |
| Bytes: | 6-8 | Reserved for assignment by SAE | |

#### 5.7.29.1 Emission-Related Pending DTC Count

Identifies the current number of emission-related pending DTCs (DM6).

| | |
|---|---|
| Data Length: | 1 byte |
| Resolution: | 1 trouble code/bit |
| Data Range: | 0 to 250 |
| Type: | Measured |
| Suspect Parameter Number: | 4104 |
| Reference: | 5.7.29 |

#### 5.7.29.2 All Pending DTC Count

Identifies the current total number of pending DTCs, including emission-related ones (DM27).

| | |
|---|---|
| Data Length: | 1 byte |
| Resolution: | 1 trouble code/bit |
| Data Range: | 0 to 250 |
| Type: | Measured |
| Suspect Parameter Number: | 4105 |
| Reference: | 5.7.29 |

#### 5.7.29.3 Emission-Related MIL-On DTC Count

Identifies the current number of Emission-Related MIL-On DTCs (DM12).

| | |
|---|---|
| Data Length: | 1 byte |
| Resolution: | 1 trouble code/bit |
| Data Range: | 0 to 250 |
| Type: | Measured |
| Suspect Parameter Number: | 4106 |
| Reference: | 5.7.29 |

#### 5.7.29.4 Emission-Related Previously MIL-On Diagnostic Trouble Codes Count

Identifies the current number of emission-related previously MIL-On DTCs (DM23).

| | |
|---|---|
| Data Length: | 1 byte |
| Resolution: | 1 trouble code/bit |
| Data Range: | 0 to 250 |
| Type: | Measured |
| Suspect Parameter Number: | 4107 |
| Reference: | 5.7.29 |

#### 5.7.29.5 Emission-Related Permanent DTC Count

Identifies the current number of permanent DTCs (DM28) that are active and does not contain the count of any previously active permanent DTCs.

| | |
|---|---|
| Data Length: | 1 byte |
| Resolution: | 1 trouble code/bit |
| Data Range: | 0 to 250 |
| Type: | Measured |
| Suspect Parameter Number: | 4108 |
| Reference: | 5.7.29 |

5.7.30 Scaled Test Results (DM30)

This message conveys the scaled test results for the applicable test requested in DM7. The SPN and FMI convey the specific DTC for which the results are being reported. The test value, test limit maximum, and test limit minimum shall follow the conventions of Table 8.

| | |
|---|---|
| Transmission Rate: | Sent in response to DM7 (PGN 58112) when the results are available.<br>Acknowledgment control mode value of 1, 2, or 3 is required if the specific DM7 content is not supported or can't be performed.<br>(refer to SAE J1939-21 PGN 59392) |
| Data Length: | Variable |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 164 |
| PDU Specific: | Destination address |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |
| Parameter Group Number: | 41984 (00A400h) |

| | | | | |
|---|---|---|---|---|
| Byte: | 1 | | Test identifier | see 5.7.7.1 |
| Byte: | 2 | bits 8-1 | SPN, 8 least significant bits of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: | 3 | bits 8-1 | SPN, second byte of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: | 4 | bits 8-6 | SPN, 3 most significant bits (most significant at bit 8) | see 5.7.1.12 |
| | | bits 5-1 | FMI (most significant at bit 5) | see 5.7.1.13 |
| Bytes: | 5-6 | | SLOT identifier | see 5.7.30.1 |
| Bytes: | 7-8 | | Test value | see 5.7.30.2 |
| Bytes: | 9-10 | | Test limit maximum | see 5.7.30.3 |
| Bytes: | 11-12 | | Test limit minimum | see 5.7.30.4 |

EXAMPLE 1: The following illustrates the message format for when there is more than one test result to communicate. The DM30 response can contain more than one test result set (items a through g). Each set can use the same FMI or use different FMI values in the response.

Given:

a = test identifier
b = SPN
c = FMI
d = SLOT Identifier
e = test value
f = test limit maximum
g = test limit minimum

Message form will be as follows: a,b,c,d,e,f,g,a,b,c,d,e,f,g, a,b,c,d,e,f,g, a,b,c,d,e,f,g,....etc. The transport protocol of SAE J1939-21 has to be used to send the scaled test results message because it contains more than 8 data bytes.

#### 5.7.30.1 SLOT Identifier

The SLOT identifier is used to determine the multiplier and offset to be used to scale the test value, test limit maximum, and test limit minimum so that the results are displayable in engineering units. The SLOT identifiers are defined in SAE J1939DA, and SLOTs utilized shall define a 16 bit data value. For OBD products, check SAE J1939-84 for SLOT identifiers used in its testing.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | Not defined |
| Data Range: | 0 to 64255 |
| Type: | Measured |
| Suspect Parameter Number: | 4109 |
| Reference: | 5.7.30 |

#### 5.7.30.2 Test Value

The test value collected during the test. If the test performed does not have both a test limit minimum and maximum, then the appropriate limit value (maximum or minimum) shall be set to all ones. SAE J1939-71 defines this to mean not available. The test value shall be scaled according to the SLOT definition in SAE J1939DA. The test value shall follow the conventions of Table 8.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | Not defined |
| Data Range: | 0 to 64255 (see Table 8 for use of $FB00_h$, $FB01_h$, and $FE00_h$) |
| Type: | Measured |
| Suspect Parameter Number: | 4110 |
| Reference: | 5.7.30 |

#### 5.7.30.3 Test Limit Maximum

The test value must be less than or equal to test limit maximum in order for the test to pass. The test limit maximum shall be scaled according to the SLOT definition in SAE J1939DA. If the test performed does not have a test limit maximum, then the test limit maximum shall be set to all ones, which is defined by SAE J1939-71 to mean not available. The test limit maximum shall follow the conventions of Table 8.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | Not defined |
| Data Range: | 0 to 64255 (see Table 8 for use of $FFFF_h$) |
| Type: | Measured |
| Suspect Parameter Number: | 4111 |
| Reference: | 5.7.30 |

#### 5.7.30.4 Test Limit Minimum

The test value must be greater than or equal to test limit minimum in order for the test to pass. The test limit minimum shall be scaled according to the SLOT definition in SAE J1939DA. If the test performed does not have a test limit minimum, then the test limit minimum shall be set to all ones, which is defined by SAE J1939-71 to mean not available. The test limit minimum shall follow the conventions of Table 8.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | Not defined |
| Data Range: | 0 to 64255 (see Table 8 for use of $FFFF_h$) |
| Type: | Measured |
| Suspect Parameter Number: | 4112 |
| Reference: | 5.7.30 |

5.7.31 DTC to Lamp Association (DM31)

This message shall provide the applicable lamp(s) for each individual DTC. Only the lamp(s) associated with the specific DTC shall be reported. Those lamp(s) that are not relevant to the specific DTC should be reported as not available (i.e., report as 11). Each DTC reported shall indicate the relevance of each of the SAE-specified lamps in DM31. The use of the lamp fields are unique to this message. Other DMs, such as DM1 and DM2, always report the composite status of the lamps for the reporting system. In DM31, the lamp information reported is specific to each DTC and only that DTC. See Table 5 for additional specifications regarding the lamp information in DM31. The device(s) receiving this message can deduce the composite status of the DM31-specified lamps for the transmitting device by properly considering all lamp values for each of the specific DTCs.

The initial intent of DM31 was to include only the subset of currently active DTCs being reported in DM1 in order to help the service technician understand the current severity of each DTC. While there is no prohibition to DM31, including the entire list of supported or defined DTCs, it would use a lot of bandwidth and processing overhead. In the event that the entire list of supported or defined DTCs were included, the majority of the defined DTCs in the list would have no fault lamp data for the majority of the time.

The lamp information (malfunction indicator lamp, red stop lamp, amber warning lamp, and protect lamp) and the associated flash lamp information shall indicate the applicable lamp(s) for the specific DTC. The lamp information for each specific DTC shall not convey temporary signals to provide for lamp test illumination, DTC numeric displays, or OBD readiness indication at key-on.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | Variable |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 163 |
| PDU Specific: | Destination address |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |
| Parameter Group Number: | 41728 (00A300$_h$) |

| | | | | |
|---|---|---|---|---|
| Byte: | 1 | bits 8-1 | SPN, 8 least significant bits of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: | 2 | bits 8-1 | SPN, second byte of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: | 3 | bits 8-6 | SPN, 3 most significant bits (most significant at bit 8) | see 5.7.1.12 |
| | | bits 5-1 | FMI (most significant at bit 5) | see 5.7.1.13 |
| Byte: | 4 | bit 8 | SPN conversion method | see 5.7.1.14 |
| | | bits 7-1 | Occurrence count | see 5.7.1.15 |
| Byte: | 5 | bits 8-7 | DTCx Malfunction Indicator Lamp Support and Status | see 5.7.31.1 |
| | | bits 6-5 | DTCx Red Stop Lamp Support and Status | see 5.7.31.2 |
| | | bits 4-3 | DTCx Amber Warning Lamp Support and Status | see 5.7.31.3 |
| | | bits 2-1 | DTCx Protect Lamp Support and Status | see 5.7.31.4 |
| Byte: | 6 | bits 8-7 | DTCx Flash Malfunction Indicator Lamp Support and Status | see 5.7.31.5 |
| | | bits 6-5 | DTCx Flash Red Stop Lamp Support and Status | see 5.7.31.6 |
| | | bits 4-3 | DTCx Flash Amber Warning Lamp Support and Status | see 5.7.31.7 |
| | | bits 2-1 | DTCx Flash Protect Lamp Support and Status | see 5.7.31.8 |

NOTE: When the occurrence count is not available, it should be set to all ones, which is a value of 127.

NOTE: When there is zero or one DTC to report, then unused bytes 7 and 8 of the CAN frame shall be set to 255 (as per SAE J1939-71).

NOTE: When there are no DTCs to report, each of bytes 3 through 6 shall be set to zero.

EXAMPLE 1: The following illustrates the message format for when there is more than one diagnostic trouble code.

Given:

a = SPN
b = FMI
c = CM and OC
d = DTCx Malfunction Indicator Lamp Support and Status
e = DTCx Red Stop Lamp Support and Status
f = DTCx Amber Warning Lamp Support and Status
g = DTCx Protect Lamp Support and Status
h = DTCx Flash Malfunction Indicator Lamp Support and Status
i = DTCx Flash Red Stop Lamp Support and Status
j = DTCx Flash Amber Warning Lamp Support and Status
k = DTCx Flash Protect Lamp Support and Status

Message form will be as follows: a,b,c,d,e,f,g,h,i,j,k, a,b,c,d,e,f,g,h,i,j,k, a,b,c,d,e,f,g,h,i,j,k, a,b,c,d,e,f,g,h,i,j,k,....etc. In this example, the transport protocol of SAE J1939-21 will have to be used to send the information because it requires more than 8 data bytes.

EXAMPLE 2: The following illustrates the message format for when a request of the DM31 is made and there are zero DTCs to report. This example would represent a DTC free controller whose response to a DM31 request includes only DTCs found in DM1 and DM2.

Required Response:

| | | | |
|---|---|---|---|
| Bytes: 1-4 | | SPN | = 0 (required setting for reporting no diagnostic trouble code) |
| | | FMI | = 0 (required setting for reporting no diagnostic trouble code) |
| | | OC | = 0 (required setting for reporting no diagnostic trouble code) |
| | | CM | = 0 (required setting for reporting no diagnostic trouble code) |
| Byte: 5 | bits 8-7 | = 00 (example of reporting off) | |
| | bits 6-5 | = 00 (example of reporting off) | |
| | bits 4-3 | = 00 (example of reporting off) | |
| | bits 2-1 | = 00 (example of reporting off) | |
| Byte: 6 | = 255 | | |
| Byte: 7 | = 255 | | |
| Byte: 8 | = 255 | | |

5.7.31.1 DTCx Malfunction Indicator Lamp Support and Status

This parameter indicates the support and status of the malfunction indicator lamp as it pertains to the specific DTCx. This parameter in combination with DTCx flash MIL support and status provides two separate pieces of information. One is whether this parameter pertains to the specific DTC. If it does pertain to the specific DTC, then the value indicates whether the lamp operation that is applicable for the DTCx (i.e., currently off, on, on and slow flash, on and fast flash, short MIL active, short MIL not active, etc.).

If this lamp is applicable to the associated specific DTC, then the value for this parameter shall convey the current state for this DTC (i.e., 00, 01, or 10). If this lamp is not applicable to the currently active DTC, then the value shall be reported as unavailable (i.e., 11). None, one, or more than one of lamps may be associated with each specific DTC.

| | |
|---|---|
| 00 | Lamp off |
| 01 | Lamp on |
| 10 | Short MI |
| 11 | Unavailable |
| Type: | Status |
| Suspect Parameter Number: | 4113 |
| Reference: | 5.7.31 |

5.7.31.2 DTCx Red Stop Lamp Support and Status

This parameter indicates the support and status of the red stop lamp as it pertains to the specific DTCx. This parameter in combination with DTCx flash red stop lamp support and status provides two separate pieces of information. See Table 5. One is whether this parameter pertains to the specific DTC. If it does pertain to the specific DTC, then the value indicates whether the lamp operation that is applicable for the DTCx (i.e., currently off, on, on and slow flash, on and fast flash, etc.).

If this lamp is applicable to the associated specific DTC, then the value for this parameter shall convey the current state for this DTC (i.e., 00, 01, or 10). If this lamp is not applicable to the currently active DTC, then the value shall be reported as unavailable (i.e., 11). None, one, or more than one of lamps may be associated with each specific DTC.

| | |
|---|---|
| 00 | Lamp off |
| 01 | Lamp on |
| 10 | Reserved for assignment by SAE |
| 11 | Unavailable |
| Type: | Status |
| Suspect Parameter Number: | 4114 |
| Reference: | 5.7.31 |

5.7.31.3 DTCx Amber Warning Lamp Support and Status

This parameter indicates the support and status of the amber warning lamp as it pertains to the specific DTCx. This parameter in combination with DTCx flash amber warning lamp support and status provides two separate pieces of information. One is whether this parameter pertains to the specific DTC. If it does pertain to the specific DTC, then the value indicates whether the lamp operation that is applicable for the DTCx (i.e., currently off, on, on and slow flash, on and fast flash, etc.).

If this lamp is applicable to the associated specific DTC, then the value for this parameter shall convey the current state for this DTC (i.e., 00, 01, or 10). If this lamp is not applicable to the currently active DTC, then the value shall be reported as unavailable (i.e., 11). None, one, or more than one of lamps may be associated with each specific DTC.

| | |
|---|---|
| 00 | Lamp off |
| 01 | Lamp on |
| 10 | Reserved for assignment by SAE |
| 11 | Unavailable |
| Type: | Status |
| Suspect Parameter Number: | 4115 |
| Reference: | 5.7.31 |

5.7.31.4 DTCx Protect Lamp Support and Status

This parameter indicates the support and status of the protect lamp as it pertains to the specific DTCx. This parameter in combination with DTCx flash protect lamp support and status provides two separate pieces of information. One is whether this parameter pertains to the specific DTC. If it does pertain to the specific DTC, then the value indicates whether the lamp operation that is applicable for the DTCx (i.e., currently off, on, on and slow flash, on and fast flash, etc.).

If this lamp is applicable to the associated specific DTC, then the value for this parameter shall convey the current state for this DTC (i.e., 00, 01, or 10). If this lamp is not applicable to the currently active DTC, then the value shall be reported as unavailable (i.e., 11). None, one, or more than one of lamps may be associated with each specific DTC.

| | |
|---|---|
| 00 | Lamp off |
| 01 | Lamp on |
| 10 | Reserved for assignment by SAE |
| 11 | Unavailable |
| Type: | Status |
| Suspect Parameter Number: | 4116 |
| Reference: 5.7.31 | 5.7.31 |

5.7.31.5 DTCx Flash Malfunction Indicator Lamp Support and Status

This parameter indicates the support and status of the flash MIL lamp as it pertains to the specific DTCx. This parameter in combination with DTCx MIL command support and status provides two separate pieces of information. One is whether this parameter pertains to the specific DTC. If it does pertain to the specific DTC, then the value indicates whether the lamp operation that is applicable for the DTCx (i.e., currently off, on, on and slow flash, on and fast flash, short MIL active, short MIL not active, etc.).

If this lamp is applicable to the associated specific DTC, then the value for this parameter shall convey the current state for this DTC (i.e., 00, 01, or 10). If this lamp is not applicable to the currently active DTC, then the value shall be reported as unavailable (i.e., 11). None, one, or more than one of lamps may be associated with each specific DTC.

This parameter provides the capability to flash the MIL.

| | |
|---|---|
| 00 | Slow flash (1 Hz, 50% duty cycle) |
| 01 | Fast flash (2 Hz or faster, 50% duty cycle) |
| 10 | Class C DTC (for WWH OBD discriminatory display systems, not applicable for other OBD non-discriminatory display systems) |
| 11 | Unavailable/do not flash |
| Type: | Status |
| Suspect Parameter Number: | 4117 |
| Reference: | 5.7.31 |

5.7.31.6 DTCx Flash Red Stop Lamp Support and Status

This parameter indicates the support and status of the flash red stop lamp as it pertains to the specific DTCx. This parameter in combination with DTCx red stop lamp command support and status provides two separate pieces of information. One is whether this parameter pertains to the specific DTC. If it does pertain to the specific DTC, then the value indicates whether the lamp operation that is applicable for the DTCx (i.e., currently off, on, on and slow flash, on and fast flash, etc.).

If this lamp is applicable to the associated specific DTC, then the value for this parameter shall convey the current state for this DTC (i.e., 00 or 10). If this lamp is not applicable to the currently active DTC, then the value shall be reported as unavailable (i.e., 11). None, one, or more than one of lamps may be associated with each specific DTC.

| | |
|---|---|
| 00 | Slow flash (1 Hz, 50% duty cycle) |
| 01 | Fast flash (2 Hz or faster, 50% duty cycle) |
| 10 | Reserved for assignment by SAE |
| 11 | Unavailable/do not flash |
| Type: | Status |
| Suspect Parameter Number: | 4118 |
| Reference: | 5.7.31 |

5.7.31.7 DTCx Flash Amber Warning Lamp Support and Status

This parameter indicates the support and status of the flash amber warning lamp as it pertains to the specific DTCx. This parameter in combination with DTCx amber warning lamp command support and status provides two separate pieces of information. One is whether this parameter pertains to the specific DTC. If it does pertain to the specific DTC, then the value indicates whether the lamp operation that is applicable for the DTCx (i.e., currently off, on, on and slow flash, on and fast flash, etc.)

If this lamp is applicable to the associated specific DTC, then the value for this parameter shall convey the current state for this DTC (i.e., 00 or 10). If this lamp is not applicable to the currently active DTC, then the value shall be reported as unavailable (i.e., 11). None, one, or more than one of lamps may be associated with each specific DTC.

| | |
|---|---|
| 00 | Slow flash (1 Hz, 50% duty cycle) |
| 01 | Fast flash (2 Hz or faster, 50% duty cycle) |
| 10 | Reserved for assignment by SAE |
| 11 | Unavailable/do not flash |
| Type: | Status |
| Suspect Parameter Number: | 4119 |
| Reference: | 5.7.31 |

5.7.31.8 DTCx Flash Protect Lamp Support and Status

This parameter indicates the support and status of the flash protect lamp as it pertains to the specific DTCx. This parameter in combination with DTCx protect lamp command support and status provides two separate pieces of information. One is whether this parameter pertains to the specific DTC. If it does pertain to the specific DTC, then the value indicates whether the lamp operation that is applicable for the DTCx (i.e., currently off, on, on and slow flash, on and fast flash, etc.)

If this lamp is applicable to the associated specific DTC, then the value for this parameter shall convey the current state for this DTC (i.e., 00 or 10). If this lamp is not applicable to the currently active DTC, then the value shall be reported as unavailable (i.e., 11). None, one, or more than one of lamps may be associated with each specific DTC.

| | |
|---|---|
| 00 | Slow flash (1 Hz, 50% duty cycle) |
| 01 | Fast flash (2 Hz or faster, 50% duty cycle) |
| 10 | Reserved for assignment by SAE |
| 11 | Unavailable/do not flash |
| Type: | Status |
| Suspect Parameter Number: | 4120 |
| Reference: | 5.7.31 |

5.7.32 Regulated Exhaust Emission Level Exceedance (DM32)

DM32 provides the DTCs and associated timers related to a regulated exhaust emission level exceedance due to an emission control system malfunction. The DTCs that are reported may be active or previously active. For example, DM32 can be used to provide the DTCs and associated timers related to regulated exhaust NOx emission level exceedance due to an emission control system malfunction as is required with European heavy-duty OBD. Those reported shall include active or previously active DTCs.

Certain SPNs may be associated with other SPNs and be considered as “parent” SPN to them as well as a parent DTC to them. The set of SPNs associated with a given parent SPN are considered to be “child” SPNs. A parent SPN shall be declared active when any of its child SPNs are declared active. DM32 can be used to provide parent SPNs and associated timers due to certain emissions control system malfunctions. For Euro IV and V implementations, SPNs 4090, 4091, 4092, 4093, 4094, 4095, and 4096 are parent SPNs for NOx limit exceedances due to specific types of malfunctions. For Euro VI implementations, SPNs 5838, 5839, 5840, 5841, and 5842 are parent SPNs for specific types of NOx control system malfunctions. For China IV non-road implementations, SPNs 5838, 5840, 5841, and 5842 are parent SPNs for specific types of NOx control system malfunctions, and SPNs 6602, 6603, and 6604 are parent SPNs for particulate control malfunctions.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | Variable |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 162 |
| PDU Specific: | Destination address |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |

Parameter Group Number: 41472 (00A200$_h$)

| | | | | |
|---|---|---|---|---|
| Byte: | 1 | bits 8-1 | DTCx SPN, 8 least significant bits of SPN (most significant at bit 8) | see 5.7.32.1 |
| Byte: | 2 | bits 8-1 | DTCx SPN, second byte of SPN (most significant at bit 8) | see 5.7.32.1 |
| Byte: | 3 | bits 8-6 | DTCx SPN, 3 most significant bits (most significant at bit 8) | see 5.7.32.1 |
| | | bits 5-1 | DTCx FMI (most significant at bit 5) | see 5.7.32.1 |
| Bytes: | 4-5 | | DTCx total active time | see 5.7.32.3 |
| Bytes: | 6-7 | | DTCx total previously active time | see 5.7.32.4 |
| Byte: | 8 | | DTCx time until derate | see 5.7.32.5 |
| Byte: | 9-n | | Repeat pattern for bytes 1-8, see example. | |

EXAMPLE 1: The following illustrates the message format for multiple regulated exhaust emission exceedance DTCs with their timers.

Given:

a = DTCx SPN
b = DTCx FMI
c = DTCx total active time
d = DTCx total previously active time
e = DTCx time until derate

Bytes 1 through 8 in the DM32 definition establish the pattern for tokens a, b, c, d, and e. This pattern is repeated as many times as is necessary to report the NOx exceedance DTC and the associated timers that the system supports. Ten DTCs will result in a data length of 80 bytes and appear as abcde abcde abcde abcde abcde abcde abcde abcde abcde abcde in the reply. This message will be sent with SAE J1939-21 defined transport protocol for any system reporting two or more NOx exceedance DTCs.

EXAMPLE 2: The following illustrates the message format for when there are no DTCs and the timer values are zero and there is no derate or pending derate.

Given:

a = DTCx SPN
b = DTCx FMI
c = DTCx total active timed = DTCx total previously active time
e = DTCx time until derate

Bytes 1 through 8 in the DM32 definition establish the pattern for tokens a, b, c, d, and e. They shall be set as follows when there are no DTCs present or timer values to report.

| Field | Value to Report |
|---|---|
| a = SPN | 0 |
| b = FMI | 0 |
| c = DTCx Total Active Time | 0 |
| d = DTCx Total Previously Active Time | 0 |
| e = DTCx Time Until Derate | 0 |

5.7.32.1 DTCx SPN

The DTC(s) related to a regulated exhaust emission level exceedance due to an emission control system malfunction. These DTCs may be active or previously active.

For Euro IV and V implementations, SPNs 4090, 4091, 4092, 4093, 4094, 4095, and 4096 are parent SPNs for NOx limit exceedances due to specific types of malfunctions. Definitions of these SPNs are shown in Table 25.

For Euro VI implementations, SPNs 5838, 5839, 5840, 5841, and 5842 are parent SPNs for specific types of NOx control system malfunctions. Definitions of these SPNs are shown in Table 25.

For China IV non-road implementations, SPNs 5838, 5840, 5841, and 5842 are parent SPNs for specific types of NOx control system malfunctions, and SPNs 6602, 6603, and 6604 are parent SPNs for particulate control malfunctions. Definitions of these SPNs are shown in Table 25.

***Table 25 - Parent SPNs for certain system malfunctions***

| Regulation | Parent SPN | DTCx | Description |
|---|---|---|---|
| Euro IV and V | 4090 | NOx limits exceeded, root cause unknown | Indicates that on-board diagnostics has determined that the limits for NOx in the exhaust stream have been exceeded, but the root cause cannot be determined by the OBD system. |
| Euro IV and V | 4091 | NOx limits exceeded due to deactivation of EGR | Indicates that on-board diagnostics has determined that the limits for NOx in the exhaust stream have been exceeded due to deactivation of EGR. |
| Euro IV and V | 4092 | NOx limits exceeded due to incorrect EGR flow | Indicates that on-board diagnostics has determined that the limits for NOx in the exhaust stream have been exceeded due to incorrect EGR flow. |
| Euro IV and V | 4093 | NOx limits exceeded due to low diesel exhaust fluid consumption | Indicates that on-board diagnostics has determined that the limits for NOx in the exhaust stream have been exceeded due to low diesel exhaust fluid consumption. |
| Euro IV and V | 4094 | NOx limits exceeded due to insufficient diesel exhaust fluid quality | Indicates that on-board diagnostics has determined that the limits for NOx in the exhaust stream have been exceeded due to an insufficient diesel exhaust fluid quality. |
| Euro IV and V | 4095 | NOx limits exceeded due to interrupted diesel exhaust fluid dosing | Indicates that on-board diagnostics has determined that the limits for NOx in the exhaust stream have been exceeded due to an interruption in diesel exhaust fluid dosing activity. |
| Euro IV and V | 4096 | NOx limits exceeded due to empty diesel exhaust fluid tank | Indicates that on-board diagnostics has determined that the limits for NOx in the exhaust stream have been exceeded due to the diesel exhaust fluid tank being empty. |
| Euro VI and China IV non-road | 5838 | Engine exhaust gas recirculation valve malfunction | Indicates that the diagnostic system has determined that the engine has been operated with an exhaust gas recirculation (EGR) valve malfunction. This is a parent SPN that shall be active when any relevant EGR valve DTC is active, per the applicable regulatory document and section.(1) The FMI 31, total active time, total previously active time, and time until derate information associated with this SPN shall be reported through DM32. |
| Euro VI | 5839 | Diesel exhaust fluid consumption malfunction | Indicates that the diagnostic system has determined that the engine has been operated with a diesel exhaust fluid (reagent) consumption malfunction. This is a parent SPN that shall be active when any relevant diesel exhaust fluid (reagent) consumption DTC is potential or active, per the requirements of Annex XIII, Appendix 2, section 4 of the Euro VI implementing regulation. The FMI 31, total active time, total previously active time, and time until derate information associated with this SPN shall be reported through DM32. |
| Euro VI and China IV non-road | 5840 | Diesel exhaust fluid dosing malfunction | Indicates that the diagnostic system has determined that the engine has been operated with a diesel exhaust fluid (reagent) dosing malfunction. This is a parent SPN that shall be active when any relevant diesel exhaust fluid (reagent) dosing DTC is active, per the applicable regulatory document and section[1]. The FMI 31, total active time, total previously active time, and time until derate information associated with this SPN shall be reported through DM32. |
| Euro VI and China IV non-road | 5841 | Diesel exhaust fluid quality malfunction | Indicates that the diagnostic system has determined that the engine has been operated with a diesel exhaust fluid (reagent) quality malfunction. This is a parent SPN that shall be active when any relevant diesel exhaust fluid (reagent) quality DTC is active, per the applicable regulatory document and section.(1) The FMI 31, total active time, total previously active time, and time until derate information associated with this SPN shall be reported through DM32. |
| Euro VI and China IV non-road | 5842 | SCR monitoring system malfunction | Indicates that the diagnostic system has determined that the engine has been operated with an SCR monitoring system malfunction. This is a parent SPN that shall be active when any relevant SCR monitoring system DTC is active, per the requirements of Annex XIII, Appendix 2, section 4 of the Euro VI implementing regulation. The FMI 31, total active time, total previously active time, and time until derate information associated with this SPN shall be reported through DM32. |

| Regulation | Parent SPN | DTCx | Description |
|---|---|---|---|
| China IV non-road | 6602 | Exhaust Particulate Control System Removed or Blocked | Indicates that the diagnostic system has determined that the engine has been operated with a malfunction of the exhaust particulate control system relating to aftertreatment device removal or aftertreatment device blockage. This is a parent SPN that shall be active when any relevant particulate aftertreatment system malfunction DTC relating to aftertreatment device removal or aftertreatment device blockage is active, per China IV, HJ 1014-2020 Emissions control technical requirements of non-road diesel mobile machinery, Appendix D. The FMI 31, total active time, total previously active time, and time until derate information associated with this SPN shall be reported through DM32. |
| China IV non-road | 6603 | Exhaust Particulate Control System Loss of Function | Indicates that the diagnostic system has determined that the engine has been operated with a malfunction resulting in exhaust particulate control system loss of function. This is a parent SPN that shall be active when any exhaust particulate control system DTC is active, per China IV, HJ 1014-2020 Emissions control technical requirements of non-road diesel mobile machinery, Appendix D. The FMI 31, total active time, total previously active time, and time until derate information associated with this SPN shall be reported through DM32. |
| China IV non-road | 6604 | Exhaust Particulate Control System Diagnostics Failure | Indicates that the diagnostic system has determined that the engine has been operated with an exhaust particulate control system diagnostics malfunction. This is a parent SPN that shall be active when any relevant exhaust particulate control diagnostics system DTC is active, per China IV, HJ 1014-2020 Emissions control technical requirements of non-road diesel mobile machinery, Appendix D. The FMI 31, total active time, total previously active time, and time until derate information associated with this SPN shall be reported through DM32. |

(1) For Euro VI, Annex XIII, Appendix 2, section 4 of the Euro VI implementing regulation. For China IV, HJ 1014-2020 Emissions control technical requirements of non-road diesel mobile machinery, Appendix C.

5.7.32.2 DTCx FMI

The FMI associated with the emission control system malfunction detected in the sub-system identified by the SPN above.

5.7.32.3 DTCx Total Active Time

This timer provides the total number of hours the DTCx has been confirmed and active. This does not include the hours the DTC has been confirmed and previously active. This is the cumulative time, so if the DTCx goes inactive and then confirmed and active again, then it must continue to count from its previous value. The timer value is allowed to be erased after 400 days or 9600 hours of operation with the associated DTC being previously active during that period. Each timer is preceded in the data by a DTC (SPN + FMI) with which it is associated.

Euro VI implementations and China IV non-road: This timer provides the total number of hours the DTCx has been either potential or confirmed and active, as per regulatory requirements. This does not include the hours the DTC has been confirmed and previously active. This is the cumulative time, so if the DTCx goes inactive and then is potential or confirmed and active again, as per regulatory requirements, then it must continue to count from its previous value. The timer value is allowed to be erased after 36 engine operating hours of operation with the associated DTC being previously active during that period for Euro VI implementations and 40 hours for China IV non-road implementations. Each timer is preceded in the data by a DTC (SPN + FMI) with which it is associated.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | 0.2 h/bit, 0 offset |
| Data Range: | 0 to 12851 hours |
| Operational Range: | Same as data range |
| Type: | Measured |
| Suspect Parameter Number: | 4121 |
| Reference: | 5.7.32 |

5.7.32.4 DTCx Total Previously Active Time

Euro IV and V implementations: This timer provides the number of hours the NOx malfunction has been confirmed and previously active. This does not include the hours the DTC has been confirmed and active. The timer value is allowed to be erased after 400 days or 9600 hours of operation with the associated DTC being previously active during that period. This is the cumulative time during the 400 days or 9600 hours of operation. Each timer is preceded in the data by the DTC (SPN + FMI) with which it is associated.

Euro VI implementations and China IV non-road: This timer provides the total number of hours the DTCx has been confirmed and previously active. This does not include the hours the DTC has been potential or confirmed and previously active, as per the regulatory requirements. This is the cumulative time, so if the DTCx goes potential or confirmed and active and then confirmed and previously active again, as per regulatory requirements, then it must continue to count from its previous value. The timer value is allowed to be erased after 36 engine operating hours of operation with the associated DTC being previously active during that period for Euro VI implementations and 40 hours for China IV non-road implementations. Each timer is preceded in the data by a DTC (SPN + FMI) with which it is associated. See CB 4.2.1.4 for additional China IV non-road requirements for a frozen counter.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | 0.2 h/bit, 0 offset |
| Data Range: | 0 to 12851 hours |
| Operational Range: | Same as data range |
| Type: | Measured |
| Suspect Parameter Number: | 4122 |
| Reference: | 5.7.32 |

5.7.32.5 DTCx Time Until Derate

This timer provides the number of hours the malfunction has until the OBD required derate will occur. Each timer is preceded in the data by a DTC or parent DTC (SPN + FMI) with which it is associated. If the specific DTC is not required by the applicable OBD regulation to have a torque derate, then this parameter shall be sent as not available. For Euro IV and V implementations, if the DTC or parent DTC is confirmed and previously active, then the count shall be sent as 62.5 hours. For Euro VI and China IV non-road implementations, if the DTC or parent DTC is confirmed and previously active, the count shall be set per the regulatory requirements for that type of DTC or parent DTC. When the counter reaches zero, it shall remain at zero while the malfunction is active.

| | |
|---|---|
| Data Length: | 1 byte |
| Resolution: | 0.25 h/bit, 0 offset |
| Data Range: | 0 to 62.5 hours |
| Operational Range: | Same as data range |
| Type: | Measured |
| Suspect Parameter Number: | 4123 |
| Reference: | 5.7.32 |

5.7.33 Emission Increasing Auxiliary Emission Control Device Active Time (DM33)

The total engine run time while each of the emission increasing auxiliary emission control devices (EI-AECDs) is active. This service can support up to 198 EI-AECDs (due to TP data limits). EI-AECDs with variable actions or degrees of action, two separate engine run time totals shall be reported: one timer for total active time when commanding up to, but not including, 75% reduction of the maximum emissions control effectiveness, and one timer for total active time when commanding 75% or more reduction of the maximum emissions control effectiveness. EI-AECDs with single actions, only one timer shall be kept to total the active time.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | Variable |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 161 |
| PDU Specific: | Destination address |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |
| Parameter Group Number: | 41216 ($00A100_h$) |

| Start Position: | Length: | Parameter Name: | |
|---|---|---|---|
| 1 | 1 byte | EI-AECD Number | see 5.7.33.1 |
| 2-5 | 4 byte | EI-AECD Engine Hours Timer 1 | see 5.7.33.2 |
| 6-9 | 4 byte | EI-AECD Engine Hours Timer 2 | see 5.7.33.3 |

Note the value $FB_h$ in EI-AECD Number (see 5.7.33.1) facilitates a positive response for DM33 requests. The data length of 9 means that transport services (refer to SAE J1939-21) will be utilized in the communication of the information.

EXAMPLE 1: The following illustrates the message format when there are more than one EI-AEDCs supported by the ECU.

Given:
a = EI-AECD number
b = EI-AECD engine hours timer 1
c = EI-AECD engine hours timer 2

Message form is as follows: a,b,c, a,b,c, a,b,c, a,b,c....etc.

5.7.33.1 EI-AECD Number

The manufacturer-assigned number for the specific EI-AECD.

| | |
|---|---|
| Data Length: | 1 byte |
| Resolution: | Manufacturer-defined enumeration |
| Data Range: | 0 to 250 and 251 ($FB_h$)<br>A data value of 251 ($FB_h$) shall be reported to indicate the emission control system does not utilize EI-AECDs. Products prior to the 2012 publication may have used a data value of $FF_h$ to indicate the emission control system does not utilize EI-AECDs. |
| Type: | Status |
| Suspect Parameter Number: | 4124 |
| Reference: | 5.7.33 |

5.7.33.2 EI-AECD Engine Hours Timer 1

The total engine running hours recorded in the first timer for the EI-AECD. For EI-AECDs requiring only a single timer, timer 1 shall be used to report the total engine hours for the EI-AECD. For EI-AECDs requiring two timers, timer 1 shall report the total engine hours when the EI-AECD is commanding reduced emission control effectiveness up to, but not including, 75% of the maximum reduced emission control effectiveness.

NOTE: If the timer for any of the EI-AECDs has reached the maximum data range, the timer values for <u>all</u> EI-AECDs shall be divided by two to avoid overflow problems.

| | |
|---|---|
| Data Length: | 4 bytes |
| Resolution: | 1 min/bit, 0 minute offset |
| Data Range: | 0 to 4211081215 minutes |
| Type: | Measured |
| Suspect Parameter Number: | 4125 |
| Reference: | 5.7.33 |

5.7.33.3 EI-AECD Engine Hours Timer 2

The total engine running hours for the second timer for the EI-AECD. For EI-AECDs requiring only a single timer, timer 2 shall be reported as not available. For EI-AECDs requiring two timers, timer 2 shall report the total engine hours when the EI-AECD is commanding reduced emission control effectiveness of 75% or more of the maximum reduced emission control effectiveness.

NOTE: If the timer for any of the EI-AECDs has reached the maximum data range, the timer values for <u>all</u> EI-AECDs shall be divided by two to avoid overflow problems.

| | |
|---|---|
| Data Length: | 4 bytes |
| Resolution: | 1 min/bit, 0 minute offset |
| Data Range: | 0 to 4211081215 minutes |
| Type: | Measured |
| Suspect Parameter Number: | 4126 |
| Reference: | 5.7.33 |

5.7.34 NTE Status (DM34)

The status of engine operating in the NTE control areas for given pollutants, such as NOx and PM. The operating status includes outside the NTE control area, inside the NTE control area, inside the manufacturer-specific NTE limited testing region, and the deficiency active area.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | Variable |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 160 |
| PDU Specific: | Destination address |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |
| Parameter Group Number: | 40960 ($00A000_h$) |

| *Start Position* | *Length* | *Parameter Name* | |
|---|---|---|---|
| 1.7 | 2 bits | NOx NTE control area status | see 5.7.34.1 |
| 1.5 | 2 bits | Manufacturer-specific NOx NTE carve-out area status | see 5.7.34.2 |
| 1.3 | 2 bits | NOx NTE deficiency area status | see 5.7.34.3 |
| 1.1 | 2 bits | Reserved for assignment by SAE (set to 11) | |
| 2.7 | 2 bits | PM NTE control area status | see 5.7.34.4 |
| 2.5 | 2 bits | Manufacturer-specific PM NTE carve-out area status | see 5.7.34.5 |
| 2.3 | 2 bits | PM NTE deficiency area status | see 5.7.34.6 |
| 2.1 | 2 bits | Reserved for assignment by SAE (set to 11) | |
| 3.8 | 6 bytes | Reserved for assignment by SAE | |

5.7.34.1 NOx NTE Control Area Status

Status of engine operation within the bounded region of the engine's torque and speed map where emissions must not exceed a specific emission cap for NOx under the NTE requirement.

00 - Outside control area
01 - Inside control area
10 - Reserved for assignment by SAE
11 - Not available

| | |
|---|---|
| Data Length: | 2 bits |
| Resolution: | 4 states/2 bit, 0 offset |
| Data Range: | 0 to 3 |
| Operational Range: | Same as data range |
| Type: | Status |
| Suspect Parameter Number: | 4127 |
| Reference: | 5.7.34 |

5.7.34.2 Manufacturer-Specific NOx NTE Carve-Out Area Status

Status of engine operation within the manufacturer-specific NOx NTE carve-out area. The manufacturer-specific NOx NTE carve-out area is defined as bounded regions within the NTE control area for NOx where the manufacturer has limited NTE testing. If the application does not have a manufacturer specific NOx NTE carve-out area, then the application shall report not available. If supported and the engine is operating outside of the NOx NTE control area (SPN 4127), then this status shall be reported as outside area.

00 - Outside area
01 - Inside area
10 - Reserved for assignment by SAE
11 - Not available

Data Length: 2 bits
Resolution: 4 states/2 bit, 0 offset
Data Range: 0 to 3
Operational Range: Same as data range
Type: Status
Suspect Parameter Number: 4128
Reference: 5.7.34

5.7.34.3 NOx NTE Deficiency Area Status

Status of engine operation within the NOx NTE deficiency area. The NOx NTE deficiency area is defined as bounded regions or conditions within the NTE control area for NOx where the manufacturer has received a deficiency. If the application does not have, then the application shall report not available. If supported and the engine is operating outside of the NOx NTE control area (SPN 4127), then this status shall be reported as outside area.

00 - Outside area
01 - Inside area
10 - Reserved for assignment by SAE
11 - Not available

Data Length: 2 bits
Resolution: 4 states/2 bit, 0 offset
Data Range: 0 to 3
Operational Range: Same as data range
Type: Status
Suspect Parameter Number: 4129
Reference: 5.7.34

5.7.34.4 PM NTE Control Area Status

Status of engine operation within the bounded region of the engine's torque and speed map where emissions must not exceed a specific emission cap for PM under the NTE requirement.

00 - Outside control area
01 - Inside control area
10 - Reserved for assignment by SAE
11 - Not available

Data Length: 2 bits
Resolution: 4 states/2 bit, 0 offset
Data Range: 0 to 3
Operational Range: Same as data range
Type: Status
Suspect Parameter Number: 4130
Reference: 5.7.34

5.7.34.5 Manufacturer-Specific PM NTE Carve-Out Area Status

Status of engine operation within the manufacturer-specific PM NTE carve-out area. The manufacturer-specific PM NTE carve-out area is defined as bounded regions within the NTE control area for PM where the manufacturer has limited NTE testing. If the application does not have a manufacturer-specific PM NTE carve-out area, then the application shall report not available. If supported and the engine is operating outside of the PM NTE control area (SPN 4130), then this status shall be reported as outside area.

00 - Outside area
01 - Inside area
10 - Reserved for assignment by SAE
11 - Not available

| | |
|---|---|
| Data Length: | 2 bits |
| Resolution: | 4 states/2 bit, 0 offset |
| Data Range: | 0 to 3 |
| Operational Range: | Same as data range |
| Type: | Status |
| Suspect Parameter Number: | 4131 |
| Reference: | 5.7.34 |

#### 5.7.34.6 PM NTE Deficiency Area Status

Status of engine operation within the PM NTE deficiency area. The PM NTE deficiency area is defined as bounded regions or conditions within the NTE control area for PM where the manufacturer has received a deficiency. If the application does not have, then the application shall report not available. If supported and the engine is operating outside of the PM NTE control area (SPN 4130), then this status shall be reported as outside area.

00 - Outside area
01 - Inside area
10 - Reserved for assignment by SAE
11 - Not available

| | |
|---|---|
| Data Length: | 2 bits |
| Resolution: | 4 states/2 bit, 0 offset |
| Data Range: | 0 to 3 |
| Operational Range: | Same as data range |
| Type: | Status |
| Suspect Parameter Number: | 4132 |
| Reference: | 5.7.34 |

### 5.7.35 Immediate Fault Status (DM35)

The purpose of this DM is to enable the external test equipment to obtain the instantaneous status of diagnostic results. (This status is reported using the DTCs that are associated with each of the diagnostic algorithms.) DM35 is similar to DM1 and DM27. Whereas DM27 latches the pending state for two drive cycles and DM1 can latch the active state for three drive cycles with OBD, DM35 does not latch on and DTCs can be removed from the list as required.

The intended use of this data is for troubleshooting intermittent wiring problems. For example, it can be used to report the information from a “wiggle test” mode where the purpose is to find wiring hardness problems by pulling on wires and/or components.

Reporting the Immediate DTCs is done using the same format as is used to report active DTCs.

The lamp information (malfunction indicator lamp, red stop lamp, amber warning lamp, and protect lamp) should reflect the present state of the transmitting electronic component. The lamp information shall not convey temporary signals to provide for lamp test illumination, DTC numeric displays, or OBD readiness indication at key-on.

See Appendix H for how these diagnostic messages relate to other diagnostic messages that convey various kinds of DTCs.

Transmission Rate: On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)
A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392)

a) Unlike DM1, this service is by request only.
b) Once requested, this service should continue to transmit until key-off.
c) Once requested, this service should be transmitted whenever there is a change of status in this DTC list.
d) Message intervals should not be more frequent than 250 ms.
e) Optionally, this service may transmit a message every second in addition to or in lieu of transmitting on each status change.
f) Optionally, the service may begin or terminate transmission in response to a DM7 “command non-continuously monitored test” message.
g) In the case where an ECU only transmits on change of state, a service tool may optionally request the message every few seconds as required.

Data Length: Variable
Extended Data Page: 0
Data Page: 0
PDU Format: 159
PDU Specific: Destination address
Default Priority: 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used)
Parameter Group Number: 40704(009F00$_h$)

| | | | | |
|---|---|---|---|---|
| Byte: | 1 | bits 8-7 | Malfunction indicator lamp | see 5.7.1.4 |
| | | bits 6-5 | Red stop lamp | see 5.7.1.5 |
| | | bits 4-3 | Amber warning lamp | see 5.7.1.6 |
| | | bits 2-1 | Protect lamp | see 5.7.1.7 |
| Byte: | 2 | bits 8-7 | Flash malfunction indicator lamp | see 5.7.1.8 |
| | | bits 6-5 | Flash red stop lamp | see 5.7.1.9 |
| | | bits 4-3 | Flash amber warning lamp | see 5.7.1.10 |
| | | bits 2-1 | Flash protect lamp | see 5.7.1.11 |
| Byte: | 3 | bits 8-1 (most significant at bit 8) | SPN, 8 least significant bits of SPN | see 5.7.1.12 |
| Byte: | 4 | bits 8-1 (most significant at bit 8) | SPN, second byte of SPN | see 5.7.1.12 |
| Byte: | 5 | bits 8-6 (most significant at bit 8) | SPN, 3 most significant bits | see 5.7.1.12 |
| | | bits 5-1 | FMI (most significant at bit 5) | see 5.7.1.13 |
| Bytes: | 6 | bit 8 | SPN conversion method | see 5.7.1.14 |
| | | bits 7-1 | Occurrence count | see 5.7.1.15 |

NOTE: When the occurrence count is not available, it should be set to all ones, which is a value of 127.

NOTE: When there is zero or one DTC to report, then unused bytes 7 and 8 of the CAN frame shall be set to 255 (as per SAE J1939-71).

NOTE: When there are no DTCs to report, each of bytes 3 through 6 shall be set to zero.

NOTE: Unlike DM1, this service is by request only.

NOTE: The message format will follow that of DM1/DM2/DM6/DM27, etc., in that the lamps bits will be the current status of lamps (i.e., lamp bits in DM35 will be numerically the same value as that in DM1).

5.7.36 Harmonized Roadworthiness - Vehicle (DM36 - HRWV)

HRWV provides the aggregated roadworthiness from the VOBD (vehicle on-board diagnostics) to a scan tool or similar inquiry device. The VOBD function aggregates HRWS messages from individual sub-systems or components, summing the number of components or sub-systems that are not roadworthy and summing the number of incomplete critical diagnostics. Vehicle non-roadworthy component count counts the number of components that declare their (sub-) system to not be roadworthy. VOBD functions and processes are described in SAE J1939-03.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | 8 |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 253 |
| PDU Specific: | 100 |
| Default Priority: | 6 |
| Parameter Group Number: | 64868 (00FD64$_h$) |

| | | | | |
|---|---|---|---|---|
| Byte: | 1 | | Vehicle non-roadworthy component count | see 5.7.36.1 |
| Byte: | 2 | bits 8-7 | Vehicle continuous malfunction indicator | see 5.7.36.2 |
| | 2 | bits 6-5 | Vehicle malfunction indicator display strategy | see 5.7.36.3 |
| | 2 | bits 4-1 | Vehicle malfunction indicator activation mode | see 5.7.36.4 |
| Bytes: | 3-4 | | Vehicle incomplete monitor count | see 5.7.36.5 |
| Bytes: | 5-6 | | Vehicle Current Malfunction Indicator Accumulated Time | see 5.7.36.6 |
| Bytes | 6-8 | | Reserved for assignment by SAE (pad with FF$_h$) | |

NOTE: This construction for HRWV is intended to support future high-speed interrogation of the vehicle where timing constraints may not support the use of the transfer PG to provide a list of individual answers, without requiring the high-speed gateway device to act as a cache for all individual sub-systems and act as the VOBD function. This will likely require the VOBD function to retain a record of HRWS receipts in order to aggregate them correctly. Ongoing schemes to increment or decrement vehicle non-roadworthy component count or vehicle incomplete monitor count will be difficult to develop and would likely require a periodic re-initialization to ensure accurate counts.

5.7.36.1 Vehicle Non-Roadworthy Component Count

Vehicle non-roadworthy component count provides the sum of the (sub-) system or component non-roadworthiness counts. (See 5.7.37.1.) If the sum of all the counts is greater than 250, the value 250 shall be reported.

| | |
|---|---|
| Data Length: | 1 byte |
| Resolution: | 1 count/bit 0 offset |
| Data Range: | 0 to 250 counts |
| Operational Range: | Same as data range |
| Type: | Status |
| Suspect Parameter Number: | 4133 |
| Reference: | 5.7.36 |

5.7.36.2 Vehicle Continuous Malfunction Indicator

Vehicle continuous malfunction indicator indicates that one or more (sub-) systems or components requires that the malfunction indicator (MI) to be steady burning.

00 - Vehicle MI is not continuous
01 - Vehicle MI is continuous
10 - Reserved for assignment by SAE
11 - Not available/not required of this vehicle.

| | |
|---|---|
| Data Length: | 2 bits |
| Resolution: | 4 states/2 bit, 0 offset |
| Data Range: | 0 to 3 |
| Operational Range: | Same as data range |
| Type: | Status |
| Suspect Parameter Number: | 4134 |
| Reference: | 5.7.36 |

#### 5.7.36.3 Vehicle Malfunction Indicator Display Strategy

Vehicle malfunction indicator display strategy indicates if any system is configured to employ a discriminatory MI display. The value 00 indicates that all systems employ a non-discriminatory MI display.

00 - All systems employ a non-discriminatory MI display
01 - Some system employs a discriminatory MI display
10 - Reserved for assignment by SAE
11 - Not available/not required of this vehicle.

| | |
|---|---|
| Data Length: | 2 bits |
| Resolution: | 4 states/2 bit, 0 offset |
| Data Range: | 0 to 3 |
| Operational Range: | Same as data range |
| Type: | Status |
| Suspect Parameter Number: | 4135 |
| Reference: | 5.7.36 |

#### 5.7.36.4 Vehicle Malfunction Indicator Activation Mode

The vehicle malfunction indicator status provides the most severe form of MI display required by the failure status of any sub-system or component. For the enumeration shown, the MI activation mode is ordered from least severe to most severe. MI Activation Mode 1 indicates no malfunctions.

0000 - MI Activation Mode 1 (Off)
0001 - MI Activation Mode 2 (On Demand MI)
0010 - MI Activation Mode 3 (Short MI)
0011 - MI Activation Mode 4 (Continuous MI)
0100 - 1101 Reserved for assignment by SAE
1110 - Error
1111 - Not available/Not required of this system.

| | |
|---|---|
| Data Length: | 4 bits |
| Resolution: | 16 states/4 bit, 0 offset |
| Data Range: | 0 to 15 |
| Operational Range: | Same as data range |
| Type: | Status |
| Suspect Parameter Number: | 4136 |
| Reference: | 5.7.36 |

NOTE: MI Activation Mode 1 affirms that there are no malfunctions. No malfunction is consistent with the use of 0000. The term “Mode 1” is used to match the GTR regulation text.

#### 5.7.36.5 Vehicle Incomplete Monitor Count

Vehicle incomplete monitor count provides the number of incomplete diagnostic monitors for a given sub-system or component. A count of zero means that all monitors are complete and the vehicle is “ready” for inspection. If the sum exceeds 64255 counts, then the value 64255 shall be reported.

| | |
|---|---|
| Data Length: | 2 byte |
| Resolution: | 1 count/bit, 0 offset |
| Data Range: | 0 to 64 255 counts |
| Operational Range: | Same as data range |
| Type: | Status |
| Suspect Parameter Number: | 4137 |
| Reference: | 5.7.36 |

#### 5.7.36.6 Vehicle Current Malfunction Indicator Accumulated Time

Vehicle current MI accumulated time reports the accumulated count (in minutes) that the MIL is activated (on) for the current MI activation (or the last MI activation). Conditions include: Reset to 0000 when MIL state changes from deactivated to activated by a (sub-) system or component. Accumulate counts in minutes if MIL is activated (on). Do not change value while MIL is not activated (off); do not wrap to 0000 if value is 64255. This number should be the largest value of SPN 3295, minutes accumulated while MIL is activated, available from all applicable (sub-) systems or components, when no components demand the MI to light. SPN 3295 is contained in DM21.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | 1 minute, 0 minute offset |
| Data Range: | 0 to 64255 minutes |
| Operational Range: | Same as data range |
| Type: | Measured |
| Suspect Parameter Number: | 4138 |
| Reference: | 5.7.36 |

### 5.7.37 Harmonized Roadworthiness - System (DM37 - HRWS)

HRWS reports sub-system (or component) roadworthiness to the VOBD function to aggregate in the HRWV message (see 5.7.36). The VOBD function aggregates HRWS messages from individual components summing the number of components or sub-systems that are not roadworthy and summing the number of incomplete critical diagnostics. This collaboration process is discussed in SAE J1939-03.

| | |
|---|---|
| Transmission Rate: | 0.1 Hz or on change but no greater than 1 Hz<br>If requested, A NACK is required if PG is not supported<br>(refer to SAE J1939-21 PGNs 59904 and 59392) |
| Data Length: | 8 |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 253 |
| PDU Specific: | 99 |
| Default Priority: | 6 |
| Parameter Group Number: | 64867 (00FD63$_h$) |

| | | | | |
|---|---|---|---|---|
| Byte: | 1 | | System non-roadworthy component count | see 5.7.37.1 |
| Byte: | 2 | bits 8-7 | System continuous malfunction indicator | see 5.7.37.2 |
| | 2 | bits 6-5 | System malfunction indicator display strategy | see 5.7.37.3 |
| | 2 | bits 4-1 | System malfunction indicator activation mode | see 5.7.37.4 |
| Bytes: | 3-4 | | System incomplete monitor count | see 5.7.37.5 |
| Bytes: | 5-8 | | Reserved for assignment by SAE (pad with FF$_h$) | |

5.7.37.1 System Non-Roadworthy Component Count

System non-roadworthy component count provides the number of components (or sub-sub-systems) that a (sub-) system has determined are not roadworthy. Sub-systems or components that provide system continuous malfunction indicator status shall provide a minimum count of one; when they report their system continuous malfunction indicator (see 5.7.37.2) as 01, system MI is continuous. If the calculated count for a sub-system is greater than 250, the value 250 shall be reported.

| | |
|---|---|
| Data Length: | 1 byte |
| Resolution: | 1 count/bit 0 offset |
| Data Range: | 0 to 250 counts |
| Operational Range: | Same as data range |
| Type: | Status |
| Suspect Parameter Number: | 4139 |
| Reference: | 5.7.37 |

5.7.37.2 System Continuous Malfunction Indicator

System continuous malfunction indicator indicates that the system requires its MI (or the emissions malfunction indicator) to be steady burning.

00 - System MI is not continuous
01 - System MI is continuous
10 - Reserved for assignment by SAE
11 - Not available/Not required of this system.

| | |
|---|---|
| Data Length: | 2 bits |
| Resolution: | 4 states/2 bit, 0 offset |
| Data Range: | 0 to 3 |
| Operational Range: | Same as data range |
| Type: | Status |
| Suspect Parameter Number: | 4140 |
| Reference: | 5.7.37 |

5.7.37.3 System Malfunction Indicator Display Strategy

System malfunction indicator display strategy indicates whether the system uses a non-discriminatory MI display or a discriminatory MI display as permitted by local regulations.

00 - System employs a non-discriminatory MI display
01 - System employs a discriminatory MI display
10 - Reserved for assignment by SAE
11 - Not available/not required of this system.

| | |
|---|---|
| Data Length: | 2 bits |
| Resolution: | 4 states/2 bit, 0 offset |
| Data Range: | 0 to 3 |
| Operational Range: | Same as data range |
| Type: | Status |
| Suspect Parameter Number: | 4141 |
| Reference: | 5.7.37 |

5.7.37.4 System Malfunction Indicator Activation Mode

The system malfunction indicator status provides the form of MI display required by the failure status of the sub-system or component. For the enumeration shown, the MI activation mode is ordered from least severe to most severe. MI Activation Mode 1 indicates no malfunctions.

0000 - MI Activation Mode 1 (MI Off)
0001 - MI Activation Mode 2 (On Demand MI)
0010 - MI Activation Mode 3 (Short MI)
0011 - MI Activation Mode 4 (Continuous MI)
0100 - 1101 Reserved for assignment by SAE
1110 - Error
1111 - Not available/not required of this system.

| | |
|---|---|
| Data Length: | 4 bits |
| Resolution: | 16 states/4 bit, 0 offset |
| Data Range: | 0 to 15 |
| Operational Range: | Same as data range |
| Type: | Status |
| Suspect Parameter Number: | 4142 |
| Reference: | 5.7.37 |

NOTE: MI Activation Mode 1 affirms that there are no malfunctions. No malfunction is consistent with the use of 0000. The term Mode 1 is used to match the GTR regulation text.

5.7.37.5 System Incomplete Monitor Count

System incomplete monitor count provides the number of incomplete diagnostic monitors for a given sub-system or component. A count of zero means that all monitors are complete and the vehicle is “ready” for inspection. If the sum exceeds 64255 counts, then the value 64255 shall be reported.

| | |
|---|---|
| Data Length: | 2 byte |
| Resolution: | 1 count/bit 0 offset |
| Data Range: | 0 to 64255 counts |
| Type: | Status |
| Suspect Parameter Number: | 4143 |
| Reference: | 5.7.37 |

5.7.38 Harmonized Global Regulation Description (DM38 - HGRD)

HGRD provides a description of the UN/ECE WWH OBD global technical regulation (GTR) to which the sub-system or component complies. The description may include the identification of any local regulation amending or tailoring GTR content to the region.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | Variable |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 253 |
| PDU Specific: | 98 |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |
| Parameter Group Number: | 64866 ($00FD62_h$) |

| | | | |
|---|---|---|---|
| Byte: | 1-n | Global Technical Regulation Description | see 5.7.38.1 |

5.7.38.1 Global Technical Regulation Description

Global technical regulation description provides a textual description of the global technical regulations to which the sub-system or component complies. Individual components can comply with different regulations. Refer to SAE J1939-03 regarding reporting descriptions for multiple components.

| | |
|---|---|
| Data Length: | Variable - up to 200 characters (May be “*” delimited) |
| Resolution: | ASCII, 0 offset |
| Data Range: | 0 to 127 per byte |
| Operational Range: | Same as data range |
| Type: | Status |
| Suspect Parameter Number: | 4144 |
| Reference: | 5.7.38 |

NOTE: The ASCII character “*” is reserved as a delimiter in similar parameters; it shall only be used in descriptions to delimit specific regulation references. Data range is restricted to character codes 0 through 127 in harmony with the definition of character data in ISO PAS 27145-2. Refer to SAE J1939DA’s application of the ISO Latin 1 character set and message data byte order for ASCII data.

This information is not anticipated to be provided as a part of any high-speed sorting of vehicles for inspection and enforcement. It may be provided using the transfer PG through a gateway.

5.7.39 Harmonized Cumulative Continuous Malfunction Indicator - System (DM39 - HCMI)

Harmonized cumulative continuous malfunction indicator (HCMI) message provides the system specific cumulative information.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | 8 |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 253 |
| PDU Specific: | 97 |
| Default Priority: | 6 |
| Parameter Group Number: | 64865 ($00FD61_h$) |

| | | |
|---|---|---|
| Bytes: 1-4 | System cumulative continuous MI time | see 5.7.39.1 |
| Bytes: 5-6 | System greatest B1 counter | see 5.7.39.2 |
| Bytes: 7-8 | Reserved for assignment by SAE (pad with $FF_h$) | |

5.7.39.1 System Cumulative Continuous MI Time

System cumulative continuous MI time provides the total amount of time that the MI has been demanded to be illuminated during the life of the (sub-) system or component.

| | |
|---|---|
| Data Length: | 4 bytes |
| Resolution: | 0.05 h/bit, 0 offset |
| Data Range: | 0 to 210554060.75 hours |
| Operational Range: | Same as data range |
| Type: | Measured |
| Suspect Parameter Number: | 4145 |
| Reference: | 5.7.39 |

NOTE: A vehicle-centric view of this parameter is not indicated by GTR Module B, 4.7.1.2. Module B, 4.7.1 has been interpreted to indicate a vehicle-centric view for the current MI counter as provided by HRWV. DM21 defines the current MI counter.

5.7.39.2 System Greatest B1 Counter

System greatest B1 counter provides the total amount of time that one or more B1 DTCs have been active.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | 0.1 h/bit, 0 offset |
| Data Range: | 0 to 6 425.5 h |
| Operational Range: | Same as data range |
| Type: | Measured |
| Suspect Parameter Number: | 4146 |
| Reference: | 5.7.39 |

5.7.40 Harmonized B1 Failure Counts (DM40 - HB1C)

HB1C provides the system-specific individual B1 failure counters, when supported by the system.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | Variable |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 253 |
| PDU Specific: | 96 |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |
| Parameter Group Number: | 64864 (00FD60$_h$) |

| | | | | |
|---|---|---|---|---|
| Byte: | 1 | bits 8-1 | SPN, 8 least significant bits of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: | 2 | bits 8-1 | SPN, second byte of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: | 3 | bits 8-6 | SPN, 3 most significant bits (most significant at bit 8) | see 5.7.1.12 |
| | | bits 5-1 | FMI (most significant at bit 5) | see 5.7.1.13 |
| Bytes: | 4-5 | | Failure specific B1 counter | see 5.7.40.1 |
| Bytes: | 6-n | | repeat pattern for bytes 1 through 5; see example | |

EXAMPLE 1: The following illustrates the message format for multiple B1 counters.

Given:

a = SPN
b = FMI
c = Failure-specific B1 counter

Bytes 1 through 5 above establish the pattern for tokens a, b, and c. This pattern is repeated as many times as is necessary to report the failure-specific B1 counters that the system supports. Ten counters will result in a data length of 50 bytes and appear as abc <u>abc</u> abc <u>abc</u> abc <u>abc</u> abc <u>abc</u> abc <u>abc</u> in the reply. This message will be broadcast with transport protocol for any system supporting two or more B1 failure counters.

5.7.40.1 Failure-Specific B1 Counter

The failure-specific B1 counter provides an individual B1 counter, supported by the system. The counter provides the number of hours the B1 failure has been confirmed and active. Each counter is preceded in the data by a class B1 failure DTC (SPN + FMI) with which it is associated.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | 0.1 h/bit, 0 offset |
| Data Range: | 0 to 6 425.5 h |
| Operational Range: | same as data range |
| Type: | Measured |
| Suspect Parameter Number: | 4147 |
| Reference: | 5.7.40 |

#### 5.7.41 DTCs - A, Pending (DM41) (as part of Harmonized Failure Classification DTC Reporting)

Failures are partitioned by severity in harmonized regulations into classifications A, B1, B2, and C. Failures in each of these categories are further classified as pending, confirmed and active, and confirmed and previously active. This creates a cross product of 12 composite categories. Table 26 shows the messages defined used to communicate individual composite categories. Table 27 assigns the PGNs to be used. The PGNs for DM6, DM12, and DM23 are previously provided in other sections and are not repeated in Table 26. Each composite category uses the same structure for reporting a list of DTCs. This structure is shared with DM1, DM2, DM6, DM12, and DM23.

For backward compatibility faults reported using DMx1 - DMxC should also be reported using DM6, DM12, and DM23. DM6 would provide the pending DTCs for all classes A, B1, B2, and C. DM12 would provide all confirmed and active DTCs in classes A, B1, B2, and C. Finally, DM23 would provide all confirmed and previously active DTCs in classes A, B1, B2, and C. This construction assumes that an engine or vehicle will not simultaneously comply with both UN/ECE WWH OBD GTR and California Air Resources Board HD OBD or OBD II regulations.

##### 5.7.41.1 Harmonized Failure Classification DTC Reporting Messages

Table 26 displays the 12 messages used to convey DTCs by WWH OBD severity class and status.

***Table 26 - Fault reporting messages by status and severity class***

| **Status/Severity Class** | **Class A** | **Class B1** | **Class B2** | **Class C** | ***All*** |
|---|---|---|---|---|---|
| Pending | DM41 | DM44 | DM47 | DM50 | DM6 |
| Confirmed and Active | DM42 | DM45 | DM48 | DM51 | DM12 |
| Previously Active | DM43 | DM46 | DM49 | DM52 | DM23 |

Table 27 assigns PGNs to the messages displaying their PF and PS field values. All the PGNs are in data page 0 and extended data page 0 as described in SAE J1939-21.

***Table 27 - PGN assignments for harmonized failure reporting***

| **Message** | **PGN** | **$PGN_h$** | **PF** | **PS** | **Severity Class/Status** |
|---|---|---|---|---|---|
| DM41 | 64863 | 00FD5F | 253 | 95 | DTCs - A, Pending |
| DM42 | 64862 | 00FD5E | 253 | 94 | DTCs - A, Confirmed and Active |
| DM43 | 64861 | 00FD5D | 253 | 93 | DTCs - A, Previously Active |
| DM44 | 64860 | 00FD5C | 253 | 92 | DTCs - B1, Pending |
| DM45 | 64859 | 00FD5B | 253 | 91 | DTCs - B1, Confirmed and Active |
| DM46 | 64858 | 00FD5A | 253 | 90 | DTCs - B1, Previously Active |
| DM47 | 64857 | 00FD59 | 253 | 89 | DTCs - B2, Pending |
| DM48 | 64856 | 00FD58 | 253 | 88 | DTCs - B2, Confirmed and Active |
| DM49 | 64855 | 00FD57 | 253 | 87 | DTCs - B2, Previously Active |
| DM50 | 64854 | 00FD56 | 253 | 86 | DTCs - C, Pending |
| DM51 | 64853 | 00FD55 | 253 | 85 | DTCs - C, Confirmed and Active |
| DM52 | 64852 | 00FD54 | 253 | 84 | DTCs - C, Previously Active |

5.7.41.2 Harmonized Failure Classification DTC Reporting Message Format

Reporting the Harmonized Failure Classification DTC Reporting Messages is done using the same format as is used to report DM6, DM12, DM23, DM1, and DM2. The lamp information (malfunction indicator lamp, red stop lamp, amber warning lamp, and protect lamp) should reflect the present state of the transmitting electronic component. The lamp information shall not convey temporary signals to provide for lamp test illumination, DTC numeric displays, or OBD readiness indication at key-on.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21)<br>A NACK is required if PG is not supported<br>(refer to SAE J1939-21 PGN 59392) |
| Data Length: | Variable |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | See Table 27 |
| PDU Specific: | See Table 27 |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |
| Parameter Group Number: | See Table 27 |

| | | | |
|---|---|---|---|
| Byte: 1 | bits 8-7 | Malfunction indicator lamp | see 5.7.1.4 |
| | bits 6-5 | Red stop lamp | see 5.7.1.5 |
| | bits 4-3 | Amber warning lamp | see 5.7.1.6 |
| | bits 2-1 | Protect lamp | see 5.7.1.7 |
| Byte: 2 | bits 8-7 | Flash malfunction indicator | see 5.7.1.8 |
| | bits 6-5 | Flash red stop lamp | see 5.7.1.9 |
| | bits 4-3 | Flash amber warning lamp | see 5.7.1.10 |
| | bits 2-1 | Flash protect lamp | see 5.7.1.11 |
| Byte: 3 | bits 8-1 | SPN, 8 least significant bits of SPN<br>(most significant at bit 8) | see 5.7.1.12 |
| Byte: 4 | bits 8-1 | SPN, second byte of SPN<br>(most significant at bit 8) | see 5.7.1.12 |
| Byte: 5 | bits 8-6 | SPN, 3 most significant bits<br>(most significant at bit 8) | see 5.7.1.12 |
| | bits 5-1 | FMI (most significant at bit 5) | see 5.7.1.13 |
| Byte: 6 | bit 8 | SPN conversion method | see 5.7.1.14 |
| | bits 7-1 | Occurrence count | see 5.7.1.15 |

NOTE: When the occurrence count is not available, it should be set to all ones, which is a value of 127.

NOTE: When there is zero or one DTC to report, then unused bytes 7 and 8 of the CAN frame shall be set to 255 (as per SAE J1939-71).

NOTE: When there are no DTCs to report, each of bytes 3 through 6 shall be set to zero.

NOTE: The method for providing the SPN shown above matches the figure for version 4 in the definition for SPN 1706. The version 4 method is the only method to be used for reporting SPNs in DTCs for OBD II, HD OBD, and UN/ECE WWH OBD GTR.

The following two examples illustrate special cases for reporting DTCs.

EXAMPLE 1: The following illustrates the message format for when there is more than one diagnostic trouble code.

Given:
a = lamp status (LS)
b = SPN
c = FMI
d = CM and OC

Message form is as follows: a,b,c,d,b,c,d,b,c,d,b,c,d....etc. In this example, the transport protocol of SAE J1939-21 has to be used to send the information because it requires more than 8 data bytes. Transport protocol services shall be used any time there is more than one fault to be sent in a message defined in Table 27.

EXAMPLE 2: The following illustrates the message format for when a request of any DMx1-DMxC message is made and all test results indicate no trouble information. The currently defined lamps (malfunction indicator lamp, red stop lamp, amber warning lamp, and protect lamp) should reflect the present state of the transmitting electronic component. In this example, only the red stop lamp is identified as being on.

Bytes 3 through 6 shall be set as shown below. Previous drafts provided alternate settings that are obsolete for HD OBD, OBD II, and UN/ECE GTR compliant engines. The recommended setting for bytes 3 through 6 is shown below.

Given:

| | | |
|---|---|---|
| Byte: 1 | bits 8-7 = | 00 (example of reporting off) |
| | bits 6-5 = | 01 (example of reporting on) |
| | bits 4-3 = | 00 (example of reporting off) |
| | bits 2-1 = | 00 (example of reporting off) |
| Byte: 2 | bits 8-7 = | 11 (example of reporting not available/don't care) |
| | bits 6-5 = | 11 (example of reporting not available/don't care) |
| | bits 4-3 = | 11 (example of reporting not available/don't care) |
| | bits 2-1 = | 11 (example of reporting not available/don't care) |
| Bytes: 3-6 | SPN | = 0 (required setting for reporting no diagnostic trouble code) |
| | FMI | = 0 (required setting for reporting no diagnostic trouble code) |
| | OC | = 0 (required setting for reporting no diagnostic trouble code) |
| | CM | = 0 (required setting for reporting no diagnostic trouble code) |
| Byte: 7 | = 255 | |
| Byte: 8 | = 255 | |

#### 5.7.42 DTCs - A, Confirmed and Active (DM42)

See 5.7.41 for the message definition.

#### 5.7.43 DTCs - A, Previously Active (DM43)

See 5.7.41 for the message definition.

#### 5.7.44 DTCs - B1, Pending (DM44)

See 5.7.41 for the message definition.

#### 5.7.45 DTCs - B1, Confirmed and Active (DM45)

See 5.7.41 for the message definition.

#### 5.7.46 DTCs - B1, Previously Active (DM46)

See 5.7.41 for the message definition.

5.7.47 DTCs - B2, Pending (DM47)

See 5.7.41 for the message definition.

5.7.48 DTCs - B2, Confirmed and Active (DM48)

See 5.7.41 for the message definition.

5.7.49 DTCs - B2, Previously Active (DM49)

See 5.7.41 for the message definition.

5.7.50 DTCs - C, Pending (DM50)

See 5.7.41 for the message definition.

5.7.51 DTCs - C, Confirmed and Active (DM51)

See 5.7.41 for the message definition.

5.7.52 DTCs - C, Previously Active (DM52)

See 5.7.41 for the message definition.

5.7.53 Active Service Only DTCs (DM53)

The information communicated is limited to only the active service only diagnostic trouble codes that do not use an operator lamp and that are intended for use at the service location for the product. It is used to notify the service technician about information that can be helpful when troubleshooting the reported condition. These DTCs are not intended for vehicle operator display while the vehicle is in normal use as compared to when the vehicle is being repaired. The data contains a list of diagnostic codes and occurrence counts for active trouble codes. Note that this parameter group will be sent using the “multipacket transport” parameter group as specified in SAE J1939-21 when applicable.

The lamp information (malfunction indicator lamp, red stop lamp, amber warning lamp, and protect lamp) should reflect the present state of the transmitting electronic component. The lamp information shall not convey temporary signals to provide for lamp test illumination, DTC numeric displays, or OBD readiness indication at key-on.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | Variable |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 252 |
| PDU Specific: | 209 |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |
| Parameter Group Number: | 64721 (FCD1$_h$) |

| | | | |
|---|---|---|---|
| Byte: 1 | bits 8-7 | Malfunction indicator lamp | see 5.7.1.4 |
| | bits 6-5 | Red stop lamp | see 5.7.1.5 |
| | bits 4-3 | Amber warning lamp | see 5.7.1.6 |
| | bits 2-1 | Protect lamp | see 5.7.1.7 |
| Byte: 2 | bits 8-7 | Flash malfunction indicator lamp | see 5.7.1.8 |
| | bits 6-5 | Flash red stop lamp | see 5.7.1.9 |
| | bits 4-3 | Flash amber warning lamp | see 5.7.1.10 |
| | bits 2-1 | Flash protect lamp | see 5.7.1.11 |
| Byte: 3 | bits 8-1 | SPN, 8 least significant bits of SPN (most significant at bit 8) | see 5.7.1.12 |

| | | | | |
|---|---|---|---|---|
| Byte: | 4 | bits 8-1 | SPN, second byte of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: | 5 | bits 8-6 | SPN, 3 most significant bits (most significant at bit 8) | see 5.7.1.12 |
| | | bits 5-1 | FMI (most significant at bit 5) | see 5.7.1.13 |
| Byte: | 6 | bit 8 | SPN conversion method | see 5.7.1.14 |
| | | bits 7-1 | Occurrence count | see 5.7.1.15 |

NOTE: When the occurrence count is not available, it should be set to all ones, which is a value of 127.

NOTE: When there is zero or one DTC to report, then unused bytes 7 and 8 of the CAN frame shall be set to 255 (as per SAE J1939-71).

NOTE: When there are no DTCs to report, each of bytes 3 through 6 shall be set to zero.

EXAMPLE 1: The following illustrates the message format for when there is more than one diagnostic trouble code.

Given:
a = lamp status
b = SPN
c = FMI
d = CM and OC

Message form is as follows: a,b,c,d,b,c,d,b,c,d,b,c,d....etc. In this example, the transport protocol of SAE J1939-21 has to be used to send the information because it requires more than 8 data bytes. Actually, any time there is more than one fault, the services of the transport protocol have to be used.

EXAMPLE 2: The following illustrates the message format for when a request of the DM53 is made and there are zero active emissions faults. Note that the malfunction indicator lamp is off while any of the other three (red stop lamp, amber warning lamp, and protect lamp) could be on. In this example, all three are on.

The required setting for bytes 3 through 6 is shown below. The required setting shall be used for all products.

Given:

Byte: 1

bits 8-7 = 00 (example of reporting off)
bits 6-5 = 01 (example of reporting on)
bits 4-3 = 01 (example of reporting on)
bits 2-1 = 01 (example of reporting on)

Byte: 2

bits 8-7 = 11 (example of reporting not available/don’t care)
bits 6-5 = 11 (example of reporting not available/don’t care)
bits 4-3 = 11 (example of reporting not available/don’t care)
bits 2-1 = 11 (example of reporting not available/don’t care)

Bytes: 3-6 SPN= 0 (required setting for reporting no diagnostic trouble code)

FMI= 0 (required setting for reporting no diagnostic trouble code)

OC= 0 (required setting for reporting no diagnostic trouble code)

CM= 0 (required setting for reporting no diagnostic trouble code)

Byte: 7 = 255

Byte: 8 = 255

5.7.54 Previously Active Service Only DTCs (DM54)

The information communicated is limited to the previously active service only diagnostic trouble codes that are intended only for use at the service location for the product. It is used to notify the service technician about conditions that can be helpful when troubleshooting the reported condition. These DTCs are not intended for vehicle operator display while the vehicle is in normal use as compared to when the vehicle is being repaired. The data contains a list of diagnostic codes and occurrence counts for previously active trouble codes. Note that this parameter group will be sent using the “multipacket transport” parameter group as specified in SAE J1939-21 when applicable.

The lamp information (malfunction indicator lamp, red stop lamp, amber warning lamp, and protect lamp) should reflect the present state of the transmitting electronic component. The lamp information shall not convey temporary signals to provide for lamp test illumination, DTC numeric displays, or OBD readiness indication at key-on.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | Variable |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 252 |
| PDU Specific: | 210 |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |
| Parameter Group Number: | 64722 (FCD2$_h$) |

| | | | | |
|---|---|---|---|---|
| Byte: | 1 | bits 8-7 | Malfunction indicator lamp | see 5.7.1.4 |
| | | bits 6-5 | Red stop lamp | see 5.7.1.5 |
| | | bits 4-3 | Amber warning lamp | see 5.7.1.6 |
| | | bits 2-1 | Protect lamp | see 5.7.1.7 |
| Byte: | 2 | bits 8-7 | Flash malfunction indicator lamp | see 5.7.1.8 |
| | | bits 6-5 | Flash red stop lamp | see 5.7.1.9 |
| | | bits 4-3 | Flash amber warning lamp | see 5.7.1.10 |
| | | bits 2-1 | Flash protect lamp | see 5.7.1.11 |
| Byte: | 3 | bits 8-1 | SPN, 8 least significant bits of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: | 4 | bits 8-1 | SPN, second byte of SPN (most significant at bit 8) | see 5.7.1.12 |
| Byte: | 5 | bits 8-6 | SPN, 3 most significant bits (most significant at bit 8) | see 5.7.1.12 |
| | | bits 5-1 | FMI (most significant at bit 5) | see 5.7.1.13 |
| Byte: | 6 | bit 8 | SPN conversion method (shall be sent as a 0) | see 5.7.1.14 |
| | | bits 7-1 | Occurrence count | see 5.7.1.15 |

NOTE: When the occurrence count is not available, it should be set to all ones, which is a value of 127.

NOTE: When there is zero or one DTC to report, then unused bytes 7 and 8 of the CAN frame shall be set to 255 (as per SAE J1939-71).

NOTE: When there are no DTCs to report, each of bytes 3 through 6 shall be set to zero.

EXAMPLE 1: The following illustrates the message format for when there is more than one diagnostic trouble code.

Given:
a = lamp status
b = SPN
c = FMI
d = CM and OC

Message form will be as follows: a,b,c,d,<u>b,c,d</u>,b,c,d,<u>b,c,d</u>....etc. In this example, the transport protocol of SAE J1939-21 will have to be used to send the information because it requires more than 8 data bytes. Actually, any time there is more than one fault, the services of the transport protocol will have to be used.

EXAMPLE 2: The following illustrates the required message format for reporting DM54 when there are zero active faults. The currently defined lamps (malfunction indicator lamp, red stop lamp, amber warning lamp, and protect lamp) shall reflect the present state of the transmitting electronic component.

The required settings for bytes 3 through 6 for reporting no DTC information is shown below. Implementations are required to set bytes 3 through 6 to all zeros and bytes 7 and 8 to all ones when there are no trouble codes to report.

Given:

| | | |
|---|---|---|
| Byte: 1 | bits 8-7 | = 00 (example of reporting off) |
| | bits 6-5 | = 00 (example of reporting off) |
| | bits 4-3 | = 00 (example of reporting off) |
| | bits 2-1 | = 00 (example of reporting off) |
| Byte: 2 | bits 8-7 | = 11 (example of reporting not available/don't care) |
| | bits 6-5 | = 11 (example of reporting not available/don't care) |
| | bits 4-3 | = 11 (example of reporting not available/don't care) |
| | bits 2-1 | = 11 (example of reporting not available/don't care) |
| Bytes: 3-6 | SPN | = 0 (required setting for reporting no diagnostic trouble code) |
| | FMI | = 0 (required setting for reporting no diagnostic trouble code) |
| | OC | = 0 (required setting for reporting no diagnostic trouble code) |
| | CM | = 0 (required setting for reporting no diagnostic trouble code) |
| Byte: 7 | | = 255 |
| Byte: 8 | | = 255 |

### 5.7.55 Diagnostic Data Clear/Reset for All Service Only DTCs (DM55)

All of the diagnostic information pertaining to the active service only diagnostic trouble codes and previously active diagnostic trouble codes should be erased as well as the DTCs. Sent as a request whenever the service tool wishes to clear/reset diagnostic data relative to these DTCs. This is expected to occur once the problem has been corrected. Upon the completion of this operation or if there are no faults to clear, a positive acknowledgment shall be sent as required (refer to SAE J1939-21 PGN 59392). If for some reason a device cannot perform the requested action, then it is required to send a negative acknowledgment (refer to SAE J1939-21 PGN 59392). Implementers be aware that no positive or negative acknowledgment is sent when the request was sent to the global address.

All diagnostic information pertaining to the service only active DTCs and service only previously active DTCs includes:

- Number of DTCs
- Trouble code for non-emissions-related freeze frame data (can be read with DM4)
- Non-emissions-related freeze frame data (can be read with DM4)
- All non-emissions-related monitor test results (can be read with DM8 and DM30)
- Other manufacturer-specific "clearing/resetting" actions may also occur in response to this request message

All ECUs shall clear the DTCs and send a positive acknowledgment to this request message, when it is not directed to the global destination address, with ignition ON and with the engine not running.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data length: | 0 |
| Extended data page: | 0 |
| Data Page: | 0 |
| PDU Format: | 252 |
| PDU Specific: | 211 |
| Default priority: | 6 |
| Parameter group number: | 64723 ($00FCD3_h$) |

5.7.56 Model Year and Certification Engine Family (DM56)

This message provides access to the vehicle or engine model year and the engine emission family or vehicle test group. If for some reason a device cannot return the requested information, then it is required to send a negative acknowledgment (refer to SAE J1939-21 PGN 59392). Implementers be aware that no positive or negative acknowledgment is sent when the request was sent to the global address.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | Variable |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 252 |
| PDU Specific: | 199 |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |
| Parameter Group Number: | 64711 ($00FCC7_h$) |

| | | |
|---|---|---|
| Bytes: 1-8 | Model Year | see 5.7.56.1 |
| Bytes: 9-208 | Certification Engine Family Name | see 5.7.56.2 |

5.7.56.1 Model Year

The compliance model year of the OBD system of the engine and/or vehicle. For example, in the U.S., this is associated with either a vehicle model year or an engine model year, depending on how the emissions and OBD are certified. This message shall be 8 bytes and formatted as ASCII.

Bytes 1 through 4 = The applicable four-digit model year (i.e., 2011)

Byte 5 = The certification type; allowable values are either “E” for Engine, or “V” for Vehicle

Bytes 6 through 8 = “-MY,” to convey an abbreviation of “Model Year” in the data displayed to the end user

For example, to represent an OBD system certified for the 2011 engine model year, the value shall be 2011E-MY. To represent an OBD system that certifies the emissions on a vehicle dynamometer for the 2011 vehicle model year, the value shall be 2011V-MY. For vehicles in the U.S. above 14000 pounds, E shall be the only allowed certification type.

| | |
|---|---|
| Data Length: | 8 bytes |
| Resolution: | Not applicable |
| Data Range: | 00 00 00 00 00 00 00 $00_h$ to<br>FF FF FF FF FF FF FF $FF_h$ |
| Type: | ASCII and limited to printable characters only |
| Suspect Parameter Number: | 5844 |
| Reference: | 5.7.56 |

5.7.56.2 Certification Engine Family Name

The regulated family name per the applicable certification agency. For engines certified in the U.S., this shall begin with the 12-character engine family name as defined by EPA. If additional information is supplied, the individual fields shall be separated with the ASCII character "*" (2Ah). Up to 200 characters may be used. The characters are to be terminated with a 00h, which is the "null" character.

| | |
|---|---|
| Data Length: | Variable up to 200 bytes |
| Resolution: | Not applicable |
| Data Range: | $00_h$ to $FF_h$ |
| Type: | ASCII and limited to printable characters only |
| Suspect Parameter Number: | 5845 |
| Reference: | 5.7.56 |

5.7.57 OBD Information (DM57)

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | Variable |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 252 |
| PDU Specific: | 198 |
| Default Priority: | 6 (note that per SAE J1939-21 requirements, priority should be seven when transport protocol is used) |
| Parameter Group Number: | 64710 ($00FCC6_h$) |

| | | | |
|---|---|---|---|
| Byte: | 1.1 | Engine OBD warm-up sequence | see 5.7.57.1 |
| Bytes: | 1.3-8 | Reserved for assignment by SAE | |
| Bytes: | 2-3 | Plug-in Hybrid Engine Ignition Cycle Counter | see 5.7.57.2 |
| Bytes | 4-8 | Reserved for assignment by SAE | |

5.7.57.1 Engine OBD Warm-Up Sequence

Engine warm-up as defined by the applicable region's regulatory agency. In the U.S., "Engine OBD Warm-up" or "Warm-up cycle" means sufficient vehicle operation such that coolant temperature rises by at least 22.2 °C (40 °F) from engine starting and reaches a minimum temperature of 71.1 °C (160 °F) for gasoline engines and 60 °C (140 °F) for diesel engines.

00 - A warm-up cycle has not occurred on this engine start
01 - A warm-up cycle has occurred on this engine start
10 - SAE Reserved
11 - Not supported or not available.

| | |
|---|---|
| Data Length: | 2 bits |
| Resolution: | 4 states/2 bit, 0 offset |
| Data Range: | 0 to 3 |
| Operational Range: | Same as data range |
| Type: | Status |
| Suspect Parameter Number: | 5843 |
| Reference: | 5.7.57 |

5.7.57.2 Plug-In Hybrid Engine Ignition Cycle Counter

Plug-in Hybrid Engine ignition cycle counter counts trips that meet the fueled engine operation definition defined in 13 CCR 1971.1 (c) for plug-in Hybrid Vehicles. Ignition cycles are counted (only once), after the engine is started, and operates for at least 2 seconds plus or minus 1 second, as defined in (c) for fueled engine operation. Ignition cycles where the engine does not start are not counted. This counter shall be incremented only once per activation of the operator's vehicle ignition switch, tracked by SPN 3048 defined in 5.7.20.1.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | 1 cycle/bit, 0 offset |
| Data Range: | 0 to 64255 |
| Operational Range: | Same as data range |
| Type: | Measured |
| Suspect Parameter Number: | 3077 |
| Reference: | 5.7.57 |

### 5.7.58 Rationality Fault SP Data (DM58)

When it is not possible to provide an estimate for a measured SP, the SAE J1939-71 error indicator (e.g., FEh) is provided instead of "data." The rationality fault SP data value reported in a DM58 message provides a scaled value representing the measurement that was detected to be in error, when a rationality fault was detected for the measured SP. The rationality fault SP data value shall use the same scaling (gain and offset) as the measured SP.

DM58 provides the "OBD regulation defined data" of the parameter identified by its SPN requested using DM7 with TID 245, where the SP is subject to "rationality" fault detection (refer to 13 CCR 1971.1 (g)(3)) or to threshold fault detection (refer to, for example, 13 CCR 1971.1 (e)(9)), and the error indicator (e.g., FEh) is provided when such a rationality fault or threshold fault has been detected. Given the referenced definitions of "rationality" faults and "threshold monitors," only directly measured input signals are subject to reporting using DM58. DM58 does not alter the responses provided by DM30, as defined in any applicable OBD regulation.

OBD systems that do not support "Rationality Diagnostics" for a given input signal SP may exclude DM58 support for that SP. Support for DM58 is optional for non-OBD devices. All OBD devices that must support DM58 shall respond to all directly addressed requests using DM7 with an SAE J1939-21 acknowledgment message (PGN E800h) control byte value of 1 when the requested SPN is not supported with an "OBD regulation defined data" value. The SPNs supported for DM58 responses shall be identified as described in DM24.

Table 28 defines the expected response following a query using DM7 with TID 245 (F5h).

***Table 28 - Expected DM58 responses to a DM7 query***

| Condition(s) | DM7 Query Response: Rationality Fault SP Data Value | Comment |
|---|---|---|
| No faults detected for the requested SP. | Provide DM58: Use the "normal" process value defined by the SP's slot scaling (m) and offset (b) from SAE J1939DA for the Rationality Fault SP Data Value. [i.e., mX + b]. The most recently determined (measured) data shall be used. | This behavior is prescribed by SAE J1939-71, which assumes that the operational range is correctly specified for all expected conditions. |
| Rationality monitoring fault or threshold monitoring fault is detected for the requested SP, and the data value can be reliably extrapolated. | Provide DM58: Use the "normal" process value defined by the SP's slot scaling (m) and offset (b) from SAE J1939DA for the Rationality Fault SP Data Value. [i.e., mX + b]. The most recently determined (measured) data shall be used. Default or "limp" home values shall not be used. | This behavior is the exception to the behavior prescribed by SAE J1939-71 for data in the rationality fault data value. The data value display for the requested SP in its defined PG will be $FE_h$, $FE00_h$, or $FE000000_h$. The data in the DM30 response for the requested SP provides the value that lies behind the detected fault (when not superseded by a passing test result). The rationality fault SP data value provides the last measured value, regardless of its potential to be interpreted as error, for in-range data values. |
| Rationality monitoring fault or threshold monitoring fault is detected for the requested SP, and the data value cannot be reliably extrapolated. | Provide DM58: Use $FE_h$, $FE00_h$, or $FE000000_h$, or an enumerated type's error state for the rationality fault SP data value. | It is not possible to deduce the data. The data value display for the requested SP in its defined PG will be $FE_h$, $FE00_h$, or $FE000000_h$. This behavior is prescribed by SAE J1939-71.<br><br>NOTE: Signals that are detected outside the prescribed operational range for the SP are not typically categorized as rationality or threshold faults. |
| Special conditions exist for the requested SP [a no-fault special case]. | Provide DM58: Use $FB_h$, $FB00_h$, or $FB000000_h$, or an enumerated type's special conditions state for the rationality fault SP data value. | It is not possible to deduce the data. The data value display for the SP in its defined PG will be $FB_h$, $FB00_h$, or $FB000000_h$. This behavior is prescribed by SAE J1939-71 and is further defined for given SPs in SAE J1939DA. |
| The requested SPN is not supported for TID 245. | Do not provide DM58: NACK the DM7 request for TID 245 with the unsupported SPN. | This behavior is supported by SAE J1939-73, 5.7.7 and SAE J1939-73, 5.7.24. A negative acknowledgment response (control byte value of 1) shall be provided for SPNs that are not listed for DM58 responses in DM24. |
| TID 245 [DM58] is not supported for DM7 query. | Do not provide DM58: NACK the DM7 request for TID 245. | This behavior is prescribed by SAE J1939-73, 5.7.7. |
| DM7 is not supported. | Do not provide DM58: NACK the DM7 request. | This behavior is prescribed by SAE J1939-21. |

Notes:

- Rationality faults question the correctness of data that was scaled to a value in the defined operating range of the SP. Rationality faults most commonly use FMIs 20, 21, and 2, but may use other FMIs as described in Appendix A.
- Data reported by DM58 is independent of any value that may be reported in a DM30 freeze frame, as DM58 responses reflect the current operating conditions and not the conditions present when a freeze frame was recorded.
- Circuit faults most commonly use FMIs 3, 4, 5, and 6.
- Other faults can result in the ECU's inability to deduce a data value.
- Devices that do not support rationality fault or threshold fault detection are not expected to support TID 245 ($F5_h$) and may not support DM58.

| | | | | |
|---|---|---|---|---|
| Transmission Rate: | | | Sent in response to DM7 (PGN 58112 or 00E300h).<br>A NACK is required if SP or PG is not supported. | |
| Data Length: | | | 8 | |
| Extended Data Page: | | | 0 | |
| Data Page: | | | 0 | |
| PDU Format: | | | 251 | |
| PDU Specific: | | | 219 | |
| Default Priority: | | | 6 | |
| Parameter Group Number: | | | 64475 (00FBDBh) | |
| Byte: | 1 | | Test identifier | see 5.7.7.1 and see 5.7.58.1 |
| Byte: | 2 | bits 8-1 | Rationality Fault SPN, 8 least significant bits of SPN<br>(most significant at bit 8) | see 5.7.58.2 |
| Byte: | 3 | bits 8-1 | Rationality Fault SPN, second byte of SPN<br>(most significant at bit 8) | |
| Byte: | 4 | bits 8-6 | Rationality Fault SPN, 3 most significant bits<br>(most significant at bit 8) | |
| | | bits 5-1 | Reserved for assignment by SAE (set to binary ones) | |
| Bytes: | 5-8 | | Rationality fault SP data value | see 5.7.58.3 |

5.7.58.1 Test Identifier

The test identifier (SPN 1224) shall echo the value of TID 245 (F5h) that was provided in the directly addressed DM7 query. Additional TIDs for alternate uses of the DM58 format are not defined. The full range of test identifiers is described in 5.7.7.1. See Table 28 for all other response requirements.

5.7.58.2 Rationality Fault SPN

The rationality fault SPN (SPN 8555) identifies the parameter (by its SPN) associated with the data in the “Rationality Fault SP Data” value and determines the length of the data in “Rationality Fault SP Data” value. The rationality fault SPN value shall provide the data given in SPN 4148 of the DM7 query submitted to request rationality fault information. The bit ordering shall match the SPN bit ordering defined in 5.7.1, where the value of the SPN conversion method SPN is zero.

5.7.58.3 Rationality Fault SP Data

The rationality fault SP data value (SPN 8556) shall provide the data as defined in Table 28 for the parameter identified by its SPN in 5.7.58.2. The data will be the same value as the value reported for that parameter’s assigned PG when there are no fault conditions. When there are rationality fault conditions, the “normal process” data shall be provided using the SP’s defined scaling and offset to extrapolate the raw measurement. (An SP’s defined scaling and offset are defined by its SLOT assignment as documented by SAE J1939DA.) Where possible, the value measured for a threshold monitor is provided by DM58. When it is not possible to provide a value for a threshold monitor, the error indicator, FEh, is provided instead. When a value cannot be determined due to circuit failures (either externally or internally), an error indicator value (e.g., FEh) is provided. The values beginning with FBh may also be provided as described by the SP’s definition in SAE J1939DA and SAE J1939-71, Section 5.

The Rationality Fault SP Data value shall be reported in the lower order data bytes as required by the parameter data size starting in byte 5 and continuing in bytes 6, 7, and 8. Any Rationality Fault SP Data bytes not required for reporting the SP data value shall be filled with FFh. For example, if the parameter identified by the "Rationality Fault SPN" is a 2-byte parameter, then that parameter’s “data value” is reported in bytes 5 and 6 and bytes 7 and 8 are filled with FFh. The presentation layer concepts defined in SAE J1939-71, Section 5 (i.e., little endian ordering) shall be observed when filling bytes 5 through 8. Enumerated types shall be presented in the same manner as they are provided as freeze frame values in DM25 starting in byte 5 bit 8.

5.7.59 Diagnostic Readiness 4 (DM59)

Reports the diagnostic information relevant to a fourth PG conveying diagnostic readiness.

| | |
|---|---|
| Transmission Rate: | On request using PGN 59904 (refer to SAE J1939-21 PGN 59904)<br>A NACK is required if PG is not supported and it was a destination-specific request (refer to SAE J1939-21 PGN 59392) |
| Data Length: | 8 bytes |
| Extended Data Page: | 0 |
| Data Page: | 0 |
| PDU Format: | 193 |
| PDU Specific: | Destination address |
| Default Priority: | 6 |
| Parameter Group Number: | 64300 ($00FB2C_h$) |
| Bytes: 1-2 | Continuous-MI Time While Engine Operating |
| Bytes: 3-4 | Time Elapsed Since DTCs Were Erased |
| Bytes: 5-8 | Reserved for assignment by SAE |

5.7.59.1 Continuous-MI Time While Engine Operating

This is the same amount of time defined in DM21 SPN 3295 (see 5.7.21.3) but broadcast in units of hours.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | 1 hour, 0 hour offset |
| Data Range: | 0 to 64255 hours |
| Type: | Measured |
| Suspect Parameter Number: | 12319 |
| Reference: | 5.7.59.1 |

5.7.59.2 Time Elapsed Since DTCs Were Erased

This is the same amount of time defined in DM21 SPN 3296 (see 5.7.21.4) but broadcast in units of hours.

| | |
|---|---|
| Data Length: | 2 bytes |
| Resolution: | 1 hour, 0 hour offset |
| Data Range: | 0 to 64255 hours |
| Type: | Measured |
| Suspect Parameter Number: | 12320 |
| Reference: | 5.7.59.2 |

5.7.60 Extended DTC Information (DM60)

Extended DTC Information is used to provide additional information for reported DTCs. Extended DTC Information contains zero or more Extended DTC Information Records, where each Extended DTC Information Record provides additional information related to a reported DTC, such as network identity for controller applications (impaired and causal) associated with a DTC. See 5.7.60.4 for examples of uses for Extended DTC Information.

The SAE J1939-81 NAME, address, and network identifier belongs to either one or two of the following CA definitions:

- Impaired CA - a CA that is impaired as a result of another CA (e.g., essential heartbeat is missing).

- Causal CA - a CA that is causing impairment in one or more other CAs (e.g., heartbeat failed).

Further definitions are:

- Reporting CA - a CA that is reporting an impairment and has a presence on the network to which a tool is connected, either directly or via a bridge or network interconnect unit.

- DM60 Proxy - a CA that is able to communicate on the network to which the tool is connected on behalf of a reporting CA that is not on the same network. The method the DM60 proxy uses to gather and forward DTCs from secondary networks is OEM proprietary.

For CAs that are OBD regulated, the original DM communicating the DTCs shall include the DTC with the specific failure mode identifier in addition to the DTC with FMI 23, which indicates that DM60 information is available.

For controllers that are not OBD regulated, the DTCs should include the DTC with the specific failure mode identifier in addition to the DTC with FMI 23, which indicates more information is available by requesting DM60.

The information reported with DM60 may be cleared by means of DM3 or DM11 or DM22, whichever is applicable to the manufacturer's product.

| | |
|---|---|
| Transmission Rate | On request using PGN 59904<br>A NACK is required if PG is not supported (refer to SAE J1939-21 PGN 59392)<br>For OBD-regulated CAs, they shall only respond to DM60 requests that are directed to a specific address and not the global address of 255. |
| Data Length | Variable |
| Extended Data Page | 0 |
| Data Page | 0 |
| PDU Format | 250 |
| PDU Specific | 74 |
| Default Priority | 6 |
| Parameter Group Number | 64074 (00FA4Ah) |

| | | | | | |
|---|---|---|---|---|---|
| *Extended DTC Information Record* | | | | | see 5.7.60.1 |
| Byte | 1 | | | Record Length | see 5.7.60.1.1 |
| *DTC* | | | | | see 5.7.60.1.2 |
| Byte | 2 | | | SPN, 8 least significant bits of SPN | see 5.7.1.12 |
| Byte | 3 | | | SPN, second byte of SPN | see 5.7.1.12 |
| Byte | 4 | bits | 8-6 | SPN, 3 most significant bits | see 5.7.1.12 |
| | | bits | 5-1 | FMI | see 5.7.1.13 |
| Byte | 5 | bit | 8 | SPN conversion method | see 5.7.1.14 |
| | | bits | 7-1 | Occurrence count | see 5.7.1.15 |
| Byte | 6 | | | DTC Type | see 5.7.60.1.3 |
| *Controller Identity Header* | | | | | see 5.7.60.1.4 |
| Byte | 7 | bits | 8-6 | Controller Identity Type | see 5.7.60.1.4.1 |
| | | bits | 5-1 | Reserved for assignment by SAE | |
| Byte | 8 | | | Controller Identity Body Length | see 5.7.60.1.4.2 |
| Byte | 9 to k | | | Controller Identity Body | see 5.7.60.1.5 |
| Byte | k+1 to m | | | Future record data | see 5.7.60.3 |

It is possible that controllers will have more than one Extended DTC Information Record available. An Extended DTC Information Record is specific to one diagnostic trouble code and one diagnostic trouble code only has one Extended DTC Information Record. The maximum amount of extended DTC information data of 1785 bytes (refer to SAE J1939-21 transport protocol) limits the amount of Extended DTC Information Records that are included in this message.

NOTE: If there are no Extended DTC Information Records to report, then DM60 data in the response will be:

PGN = 64074

Byte: 1 = 0
8-2 = 0

When byte 1 is equal to zero, it identifies to the receiver that the other parameters in the message should not be interpreted. Also, notice that the values of the information put in bytes 1 through 8 are zero even though some of the parameters may have normally been set to all ones (binary) to indicate not available.

EXAMPLE: The following illustrates the message format when there are two or more Extended DTC Information Records.

Given:

a = Record Length
b = DTC
c = Controller Identity Header
d = Controller Identity Body
e = Future record data

Message form will be as follows: a,b,c,d,e,a,b,c,d,e,a,b,c,d,e,a,b,c,d,e,a,b,c,d,e....etc. Transport protocol will be used if this message will not fit into a single frame message.

5.7.60.1 Extended DTC Information Record

An Extended DTC Information Record provides additional information related to a reported DTC, such as network identity for controller applications (impaired and causal) associated with a DTC. An Extended DTC Information Record is specific to one diagnostic trouble code and one diagnostic trouble code only has one Extended DTC Information Record. The structure model of the Extended DTC Information data and the Extended DTC Information Record are illustrated in Figure 7.

5.7.60.1.1 Record Length

Record length specifies the number of data bytes associated with the Extended DTC Information Record. Record length value specifies the number of bytes after the Record Length byte that are associated with the current Extended DTC Information Record. The record length is the byte length of the DTC, DTC Type, Controller Identity Header, Controller Identity Body, and any other record data fields.

Data Length: 8 bit
Resolution: 1 byte/bit
Data Range: 0 to 255
Type: Status
Suspect Parameter Number: 21106
Reference: 5.7.60

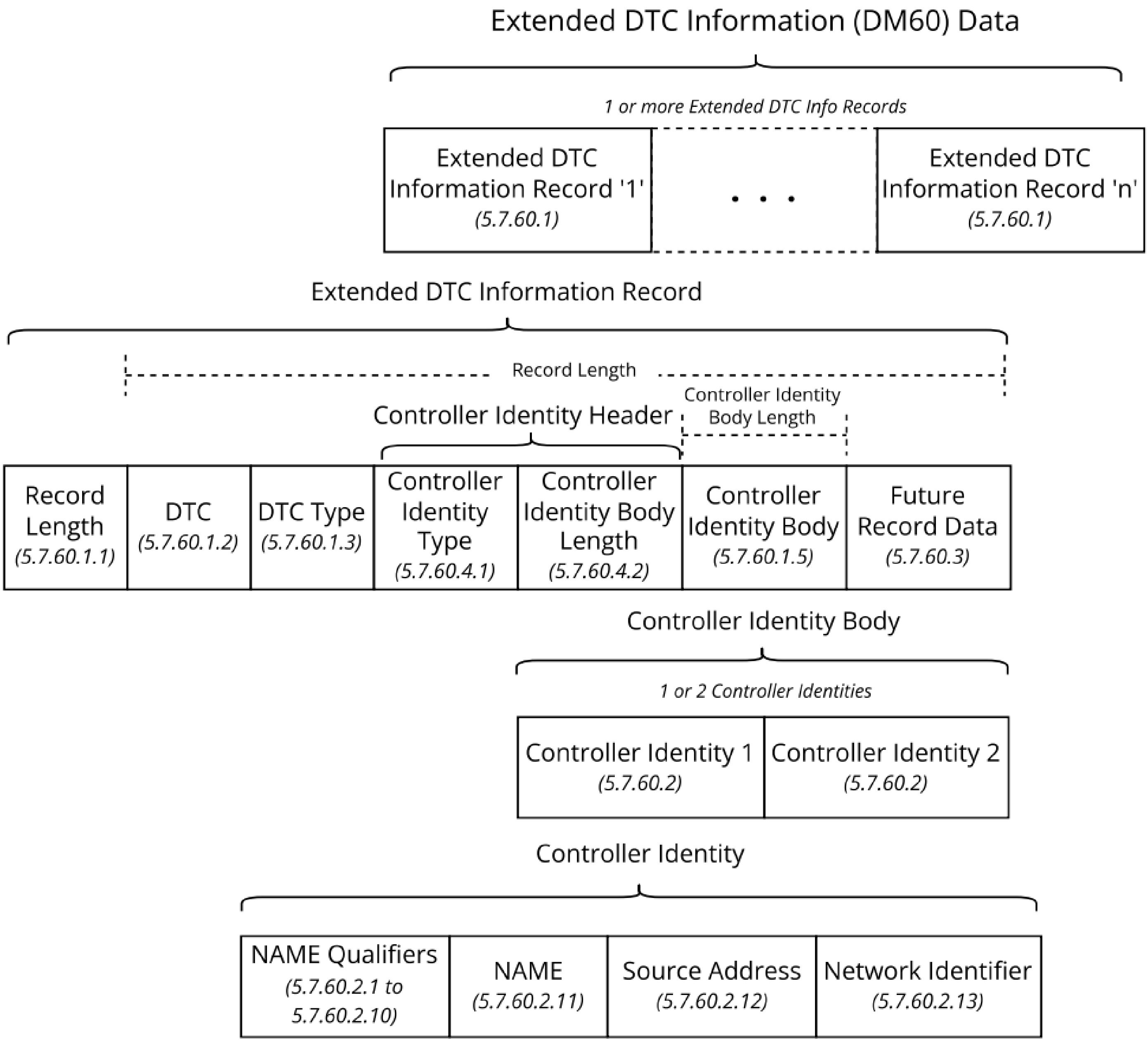


***Figure 7 - Extended DTC information data and record structures***

5.7.60.1.2 DTC

The DTC content specifies the DTC of the Extended DTC Information Record, and it is associated to the impaired CA. The SPN and occurrence count shall be equal to the SPN of the DM message reporting the FMI 23. The FMI shall provide the specific failure mode identifier.

5.7.60.1.3 DTC Type

The DTC Type value identifies the type of the specified DTC according to the DM message in which the DTC has been reported. Table 29 provides a list of all acceptable DMs and the corresponding value in column byte value. The most specific DTC type shall be used. For example, for U.S.-certified OBD product, a DTC is required to be reported in both DM1 and DM12; therefore, the DTC Type “2” (DM12) shall be specified for that DTC because DM12 is the more specific classification. ARB requires that both a MIL-ON (DM12) and Permanent fault (DM28) be stored prior to the end of an ignition cycle whenever an applicable malfunction is detected. Additionally, both faults are to remain active until the applicable monitor passes per ARBs required three driving cycles without a malfunction. In the event both DM12 and DM28 are active, DM12 shall be considered the priority and shall be reported.

| | |
|---|---|
| Data Length: | 8 bit |
| Resolution: | 1 DTC type/bit |
| Data Range: | 0 to 250 |
| Type: | Status |
| Suspect Parameter Number: | 21107 |
| Reference: | 5.7.60 |

***Table 29 - Mapping of DTC type to DM message***

| Byte Value | Diagnostic Message (DM) Name |
|---|---|
| 0 | DM1, Active Diagnostic Trouble Codes (DTCs) |
| 1 | DM2, Previously Active Diagnostic Trouble Codes (DTCs) |
| 2 | DM6, Emission-Related Pending Diagnostic Trouble Codes |
| 3 | DM12, Emission-Related MIL-On Diagnostic Trouble Codes |
| 4 | DM23, Emission-Related Previously MIL-On DTCs |
| 5 | DM27, All Pending DTCs |
| 6 | DM28, Emission-Related Permanent Diagnostic Trouble Codes |
| 7 | DM41, DTCs- A, Pending |
| 8 | DM42, DTCs- A, Confirmed and Active |
| 9 | DM43, DTCs- A, Previously Active |
| 10 | DM44, DTCs- B1, Pending |
| 11 | DM45, DTCs- B1, Confirmed and Active |
| 12 | DM46, DTCs- B1, Previously Active |
| 13 | DM47, DTCs- B2, Pending |
| 14 | DM48, DTCs- B2, Confirmed and Active |
| 15 | DM49, DTCs- B2, Previously Active |
| 16 | DM50, DTCs- C, Pending |
| 17 | DM51, DTCs- C, Confirmed and Active |
| 18 | DM52, DTCs- C, Previously Active |
| 19 | DM53, Active Service Only DTCs |
| 20 | DM54, Previously Active Service Only DTCs |
| 21 to 254 | Reserved for assignment by SAE |
| 255 | Not Available |

5.7.60.1.4 Controller Identity Header

The Controller Identity Header consists of the Controller Identity Type and the Controller Identity Body Length.

5.7.60.1.4.1 Controller Identity Type

Controller Identity Type specifies the content of the Controller Identity Body.

$000_b$ = Controller Identity Body contains only the causal CA identity
$001_b$ = Controller Identity Body contains only the impaired CA identity
$010_b$ = Controller Identity Body contains the causal CA followed by the impaired CA identity
$011_b$ to $111_b$ = Reserved for assignment by SAE

If the Controller Identity Type is $000_b$, then the Controller Identity Body Length shall be 13 ($0D_h$) and the Controller Identity Body contains:

NAME qualifiers for NAME of causal CA
SAE J1939 NAME of the causal CA
Source address - address of the causal CA
Network identifier - network ID of the causal CA

If the Controller Identity Type is $001_b$, then the Controller Identity Body Length shall be 13 ($0D_h$) and the Controller Identity Body contains:

NAME qualifiers for NAME of impaired CA
SAE J1939 NAME of the impaired CA
Source address - address of the impaired CA
Network identifier - network ID of the impaired CA

If the Controller Identity Type is $010_b$, then the Controller Identity Body Length shall be 26 ($1A_h$) and the Controller Identity Body contains:

NAME qualifiers for NAME of causal CA
SAE J1939 NAME of the causal CA
Source address - address of the causal CA
Network identifier - network ID of the causal CA
NAME qualifiers for NAME of impaired CA
SAE J1939 NAME of the impaired CA
Source address - address of the impaired CA
Network identifier - network ID of the impaired CA

| | |
|---|---|
| Data Length: | 3 bit |
| Data Range: | 0 to 7 |
| Type: | Status |
| Suspect Parameter Number: | 21108 |

5.7.60.1.4.2 Controller Identity Body Length

The number of data bytes associated with the Controller Identity Body. When the Controller Identity Body contains one controller identity, the Controller Identity Body Length value shall be 13 ($0D_h$). When the Controller Identity Body contains two controller identities, the Controller Identity Body Length value shall be 26 ($1A_h$).

| | |
|---|---|
| Data Length: | 8 bit |
| Data Range: | 0 to 255 |
| Type: | Status |
| Suspect Parameter Number: | 21109 |

5.7.60.1.5 Controller Identity Body

The Controller Identity Body consists of one or two Controller Identities as defined by the Controller Identity Type. The first controller identity in the Controller Identity Body occupies bytes 9 to 21 of the Extended DTC Information Record, as shown in Figure 8. If there is a second Controller Identity in the Controller Identity Body, then the second Controller Identity occupies bytes 22 to 34 of the Extended DTC Information Record, as shown in Figure 9.

See 5.7.60.2 for the data content and structure of an individual Controller Identity object.

5.7.60.2 Controller Identity

A single Controller Identity (CI) consists of the information described in this section. The Controller Identity structure is shown in Figures 8 and 9. The data structure shown in Figures 8 and 9 are the same. The only difference between Figures 8 and 9 is Figure 8 shows the Extended DTC Information Record data bytes occupied by the first CI in the Controller Identity Body, and Figure 9 shows the Extended DTC Information Record data bytes occupied by a conditional second CI in the Controller Identity Body. The first Controller Identity in the Controller Identity Body occupies bytes 9 to 21 of the Extended DTC Information Record. If there is a second Controller Identity in the Controller Identity Body, then the second Controller Identity occupies bytes 22 to 34 of the Extended DTC Information Record. The CA described by the Controller Identity, as either the identity of the causal CA or the impaired CA, is specified by the Controller Identity Type.

| | | | | | |
|---|---|---|---|---|---|
| *SAE J1939 NAME Qualifiers* | | | | | |
| Byte | 9 | bit | 8 | CI Manufacturer Code Qualifier Flag | see 5.7.60.2.9 |
| | | bit | 7 | CI ECU Instance Qualifier Flag | see 5.7.60.2.8 |
| | | bit | 6 | CI Function Instance Qualifier Flag | see 5.7.60.2.7 |
| | | bit | 5 | CI Function Qualifier Flag | see 5.7.60.2.6 |
| | | bit | 4 | CI Reserved Qualifier Flag | see 5.7.60.2.5 |
| | | bit | 3 | CI Vehicle System Qualifier Flag | see 5.7.60.2.4 |
| | | bit | 2 | CI Vehicle System Instance Qualifier Flag | see 5.7.60.2.3 |
| | | bit | 1 | CI Industry Group Qualifier Flag | see 5.7.60.2.2 |
| Byte | 10 | bit | 8 | CI Identity Number Qualifier Flag | see 5.7.60.2.10 |
| | | bit | 7 | CI Arbitrary Address Capable Qualifier Flag | see 5.7.60.2.1 |
| | | bits | 6-1 | Reserved for assignment by SAE (all bits set to 1's) | |
| *SAE J1939 NAME of Controller Identity CA* | | | | | *see 5.7.60.2.11* |
| Byte | 11 | | | Identity Number (LSB) (SPN 2837) | Refer to SAE J1939-81 |
| Byte | 12 | | | Identity Number | Refer to SAE J1939-81 |
| Byte | 13 | bits | 5-1 | Identity Number (MSB) | |
| | | bits | 8-6 | Manufacturer code (LSB) (SPN 2838) | Refer to SAE J1939-81 |
| Byte | 14 | | | Manufacturer code (MSB) | |
| Byte | 15 | bits | 3-1 | ECU Instance (SPN 2840) | Refer to SAE J1939-81 |
| | | bits | 8-4 | Function Instance (SPN 2839) | Refer to SAE J1939-81 |
| Byte | 16 | | | Function (SPN 2841) | Refer to SAE J1939-81 |
| Byte | 17 | bit | 1 | Reserved | |
| | | bits | 8-2 | Vehicle System (SPN 2842) | Refer to SAE J1939-81 |
| Byte | 18 | bits | 4-1 | Vehicle System Instance (SPN 2843) | Refer to SAE J1939-81 |
| | | bits | 7-5 | Industry Group (SPN 2846) | Refer to SAE J1939-81 |
| | | bit | 8 | Arbitrary Address Capable (SPN 2844) | Refer to SAE J1939-81 |
| Byte | 19 | | | Source Address of CA | see 5.7.60.2.12 |
| Byte | 20 to 21 | | | Network Identifier for CA | see 5.7.60.2.13 |

***Figure 8 - First controller identity in controller identity body***

| | | | | | |
|---|---|---|---|---|---|
| *SAE J1939 NAME Qualifiers* | | | | | |
| Byte | 22 | bit | 8 | CI Manufacturer Code Qualifier Flag | see 5.7.60.2.9 |
| | | bit | 7 | CI ECU Instance Qualifier Flag | see 5.7.60.2.8 |
| | | bit | 6 | CI Function Instance Qualifier Flag | see 5.7.60.2.7 |
| | | bit | 5 | CI Function Qualifier Flag | see 5.7.60.2.6 |
| | | bit | 4 | CI Reserved Qualifier Flag | see 5.7.60.2.5 |
| | | bit | 3 | CI Vehicle System Qualifier Flag | see 5.7.60.2.4 |
| | | bit | 2 | CI Vehicle System Instance Qualifier Flag | see 5.7.60.2.3 |
| | | bit | 1 | CI Industry Group Qualifier Flag | see 5.7.60.2.2 |
| Byte | 23 | bit | 8 | CI Identity Number Qualifier Flag | see 5.7.60.2.10 |
| | | bit | 7 | CI Arbitrary Address Capable Qualifier Flag | see 5.7.60.2.1 |
| | | bits | 6-1 | Reserved for assignment by SAE (all bits set to 1's) | |
| *SAE J1939 NAME of Controller Identity CA* | | | | | *see 5.7.60.2.11* |
| Byte | 24 | | | Identity Number (LSB) (SPN 2837) | Refer to SAE J1939-81 |
| Byte | 25 | | | Identity Number | Refer to SAE J1939-81 |
| Byte | 26 | bits | 5-1 | Identity Number (MSB) | |
| | | bits | 8-6 | Manufacturer code (LSB) (SPN 2838) | Refer to SAE J1939-81 |
| Byte | 27 | | | Manufacturer code (MSB) | |
| Byte | 28 | bits | 3-1 | ECU Instance (SPN 2840) | Refer to SAE J1939-81 |
| | | bits | 8-4 | Function Instance (SPN 2839) | Refer to SAE J1939-81 |
| Byte | 29 | | | Function (SPN 2841) | Refer to SAE J1939-81 |
| Byte | 30 | bit | 1 | Reserved | |
| | | bits | 8-2 | Vehicle System (SPN 2842) | Refer to SAE J1939-81 |
| Byte | 31 | bits | 4-1 | Vehicle System Instance (SPN 2843) | Refer to SAE J1939-81 |
| | | bits | 7-5 | Industry Group (SPN 2846) | Refer to SAE J1939-81 |
| | | bit | 8 | Arbitrary Address Capable (SPN 2844) | Refer to SAE J1939-81 |
| Byte | 32 | | | Source Address of CA | see 5.7.60.2.12 |
| Byte | 33 to 34 | | | Network Identifier for CA | see 5.7.60.2.13 |

***Figure 9 - Second controller identity in controller identity body***

5.7.60.2.1 CI Arbitrary Address Capable Qualifier Flag

The Arbitrary Address Capable qualifier flag indicates if the Arbitrary Address Capable component of the Controller Identity NAME contains data for that part of the NAME or is not available. The type of CA (causal CA or impaired CA) is based upon the Controller Identity Type (see 5.7.60.1.4.1) and the instance of the Controller Identity (first or second).

0 = Arbitrary Address Capable of the Controller Identity NAME represents valid NAME information
1 = Arbitrary Address Capable of the Controller Identity NAME is not available

Data Length: 1 bit
Type: Status
Suspect Parameter Number: 21110 (Causal Arbitrary Address Capable Qualifier Flag)
21206 (Impaired Arbitrary Address Capable Qualifier Flag)

5.7.60.2.2 CI Industry Group Qualifier Flag

The Industry Group qualifier flag indicates if the Industry Group component of the Controller Identity NAME contains data for that part of the NAME or is not available. The type of CA (causal CA or impaired CA) is based upon the Controller Identity Type (see 5.7.60.1.4.1) and the instance of the Controller Identity (first or second).

0 = Industry Group of the Controller Identity NAME represents valid NAME information
1 = Industry Group of the Controller Identity NAME is not available

Data Length: 1 bit
Type: Status
Suspect Parameter Number: 21111 (Causal Industry Group Qualifier Flag)
21207 (Impaired Industry Group Qualifier Flag)

5.7.60.2.3 CI Vehicle System Instance Qualifier Flag

The Vehicle System Instance qualifier flag indicates if the Vehicle System Instance component of the Controller Identity NAME contains data for that part of the NAME or is not available. The type of CA (causal CA or impaired CA) is based upon the Controller Identity Type (see 5.7.60.1.4.1) and the instance of the Controller Identity (first or second).

0 = Vehicle System Instance of the Controller Identity NAME represents valid NAME information
1 = Vehicle System Instance of the Controller Identity NAME is not available

Data Length: 1 bit
Type: Status
Suspect Parameter Number: 21112 (Causal Vehicle System Instance Qualifier Flag)
21208 (Impaired Vehicle System Instance Qualifier Flag)

5.7.60.2.4 CI Vehicle System Qualifier Flag

The Vehicle System qualifier flag indicates if the Vehicle System component of the Controller Identity NAME contains data for that part of the NAME or is not available. The type of CA (causal CA or impaired CA) is based upon the Controller Identity Type (see 5.7.60.1.4.1) and the instance of the Controller Identity (first or second).

0 = Vehicle System of the Controller Identity NAME represents valid NAME information
1 = Vehicle System of the Controller Identity NAME is not available

Data Length: 1 bit
Type: Status
Suspect Parameter Number: 21113 (Causal Vehicle System Qualifier Flag)
21209 (Impaired Vehicle System Qualifier Flag)

5.7.60.2.5 CI Reserved Qualifier Flag

The Reserved qualifier flag indicates if the Reserved component of the Controller Identity NAME contains data for that part of the NAME or is not available. The type of CA (causal CA or impaired CA) is based upon the Controller Identity Type (see 5.7.60.1.4.1) and the instance of the Controller Identity (first or second).

0 = Reserved of the Controller Identity NAME represents valid NAME information
1 = Reserved of the Controller Identity NAME is not available

Data Length: 1 bit
Type: Status
Suspect Parameter Number: 21114 (Causal Reserved Qualifier Flag)
21210 (Impaired Reserved Qualifier Flag)

#### 5.7.60.2.6 CI Function Qualifier Flag

The Function qualifier flag indicates if the Function component of the Controller Identity NAME contains data for that part of the NAME or is not available. The type of CA (causal CA or impaired CA) is based upon the Controller Identity Type (see 5.7.60.1.4.1) and the instance of the Controller Identity (first or second).

0 = Function of the Controller Identity NAME represents valid NAME information
1 = Function of the Controller Identity NAME is not available

| | |
|---|---|
| Data Length: | 1 bit |
| Type: | Status |
| Suspect Parameter Number: | 21115 (Causal Function Qualifier Flag)<br>21211 (Impaired Function Qualifier Flag) |

#### 5.7.60.2.7 CI Function Instance Qualifier Flag

The Function Instance qualifier flag indicates if the Function Instance component of the Controller Identity NAME contains data for that part of the NAME or is not available. The type of CA (causal CA or impaired CA) is based upon the Controller Identity Type (see 5.7.60.1.4.1) and the instance of the Controller Identity (first or second).

0 = Function Instance of the Controller Identity NAME represents valid NAME information
1 = Function Instance of the Controller Identity NAME is not available

| | |
|---|---|
| Data Length: | 1 bit |
| Type: | Status |
| Suspect Parameter Number: | 21116 (Causal Function Instance Qualifier Flag)<br>21212 (Impaired Function Instance Qualifier Flag) |

#### 5.7.60.2.8 CI ECU Instance Qualifier Flag

The ECU Instance qualifier flag indicates if the ECU Instance component of the Controller Identity NAME contains data for that part of the NAME or is not available. The type of CA (causal CA or impaired CA) is based upon the Controller Identity Type (see 5.7.60.1.4.1) and the instance of the Controller Identity (first or second).

0 = ECU Instance of the Controller Identity NAME represents valid NAME information
1 = ECU Instance of the Controller Identity NAME is not available

| | |
|---|---|
| Data Length: | 1 bit |
| Type: | Status |
| Suspect Parameter Number: | 21117 (Causal ECU Instance Qualifier Flag)<br>21213 (Impaired ECU Instance Qualifier Flag) |

#### 5.7.60.2.9 CI Manufacturer Code Qualifier Flag

The Manufacturer code qualifier flag indicates if the Manufacturer component of the Controller Identity NAME contains data for that part of the NAME or is not available. The type of CA (causal CA or impaired CA) is based upon the Controller Identity Type (see 5.7.60.1.4.1) and the instance of the Controller Identity (first or second).

0 = Manufacturer of the Controller Identity NAME represents valid NAME information
1 = Manufacturer of the Controller Identity NAME is not available

| | |
|---|---|
| Data Length: | 1 bit |
| Type: | Status |
| Suspect Parameter Number: | 21118 (Causal Manufacturer Qualifier Flag)<br>21214 (Impaired Manufacturer Qualifier Flag) |

5.7.60.2.10 CI Identity Number Qualifier Flag

The Identity Number qualifier flag indicates if the Identity Number component of the Controller Identity NAME contains data for that part of the NAME or is not available. The type of CA (causal CA or impaired CA) is based upon the Controller Identity Type (see 5.7.60.1.4.1) and the instance of the Controller Identity (first or second).

0 = Identity Number of the Controller Identity NAME represents valid NAME information
1 = Identity Number of the Controller Identity NAME is not available

| | |
|---|---|
| Data Length: | 1 bit |
| Type: | Status |
| Suspect Parameter Number: | 21119 (Causal Identity Number Qualifier Flag) |
| | 21215 (Impaired Identity Number Qualifier Flag) |

5.7.60.2.11 SAE J1939 NAME of Controller Identity CA

Identifies the SAE J1939 NAME for the CA associated to the Controller Identity. Refer to SAE J1939-81 for definition of the SAE J1939 NAME object. The type of CA (causal CA or impaired CA) associated to the SAE J1939 NAME is based upon the Controller Identity Type (see 5.7.60.1.4.1) and the instance of the Controller Identity (first or second).

| | |
|---|---|
| Data Length: | 64 bit |
| Type: | Measured |
| Suspect Parameter Number: | 21123 (SAE J1939 NAME of Causal CA) |
| | 21122 (SAE J1939 NAME of Impaired CA) |

5.7.60.2.12 Source Address of Controller Identity CA

A value from 0 to 253 represents the source address associated to the Controller Identity CA at the point in time the DTC has been created. A value of 254 means the Controller Identity CA with the NAME could not successfully claim an address. A value of 255 (Global address) means no address information available.

In systems using the dynamic address claim capabilities, the source address might be incorrect since the reported address might be used by another CA. Using the NAME provides a stable link to the Controller Identity CA.

The type of CA (causal CA or impaired CA) associated to the source address is based upon the Controller Identity Type (see 5.7.60.1.4.1) and the instance of the Controller Identity (first or second).

| | |
|---|---|
| Data Length: | 8 bit |
| Resolution: | one address per bit |
| Data Range: | 0 to 255 |
| Type: | Status |
| Suspect Parameter Number: | 21120 (Source Address of Causal CA) |
| | 21216 (Source Address of Impaired CA) |

5.7.60.2.13 Network Identifier for Controller Identity CA

The network identifier contains an indication of the communication network associated with the described NAME and source address for the Controller Identity CA. The network identity is manufacturer specific. The value assignments are shown in Table 30. The type of CA (causal CA or impaired CA) associated to the network identification is based upon the Controller Identity Type (see 5.7.60.1.4.1) and the instance of the Controller Identity (first or second).

| | |
|---|---|
| Data Length: | 16 bit |
| Resolution: | 1 network per bit |
| Data Range: | 0 to $FDFF_h$ |
| Type: | Status |
| Suspect Parameter Number: | 21121 (Network Identifier for Causal CA) |
| | 21217 (Network Identifier for Impaired CA) |

*Table 30 - Network identifier value assignments*

| Network Identifier Value | Definition |
|---|---|
| 0 | Current data link |
| 1 to FAFF$_h$ | Manufacturer-specified network number (see 5.7.60.2.13.1) |
| FB00$_h$ | Reserved for assignment by SAE |
| FB01$_h$ to FB12$_h$ | SAE Network Enumeration (see 5.7.60.2.13.2) |
| FB13$_h$ to FDFF$_h$ | Reserved for assignment by SAE |
| FE00$_h$ to FEFF$_h$ | SAE Reserved / Error Indicator |
| FF00$_h$ to FFFF$_h$ | Data Not Available / Network cannot be identified |

In a typical usage, the Network Identifier may reflect the current network as shown in Figure 10.

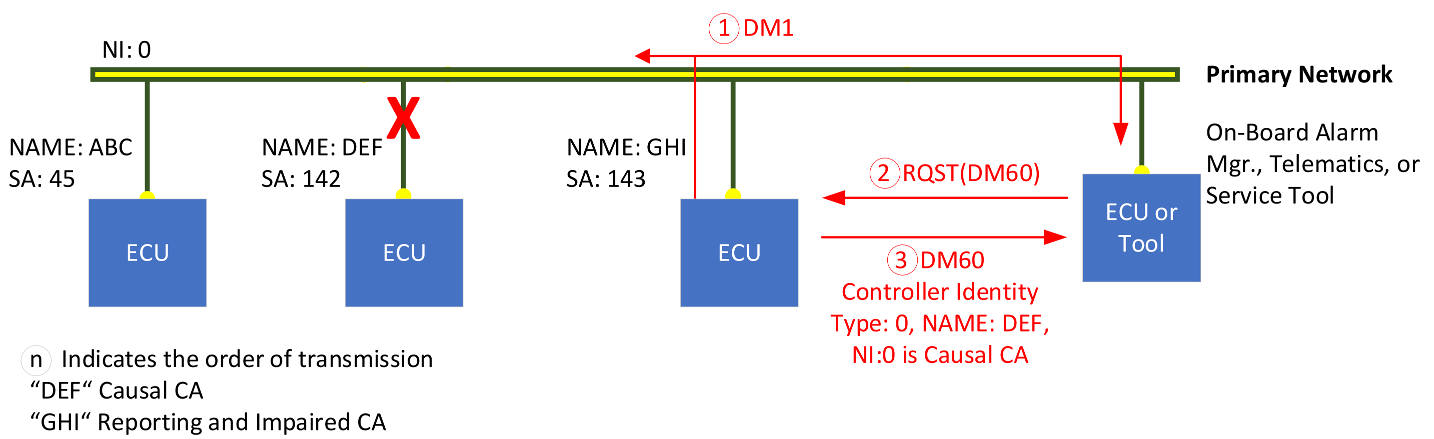


*Figure 10 - DM60 network identification on a single network*

Systems commonly have additional networks, which may or may not be bridged to the primary network. The network identifier is used to indicate on which of the additional networks the device originating the fault resides, as shown in Figure 11.

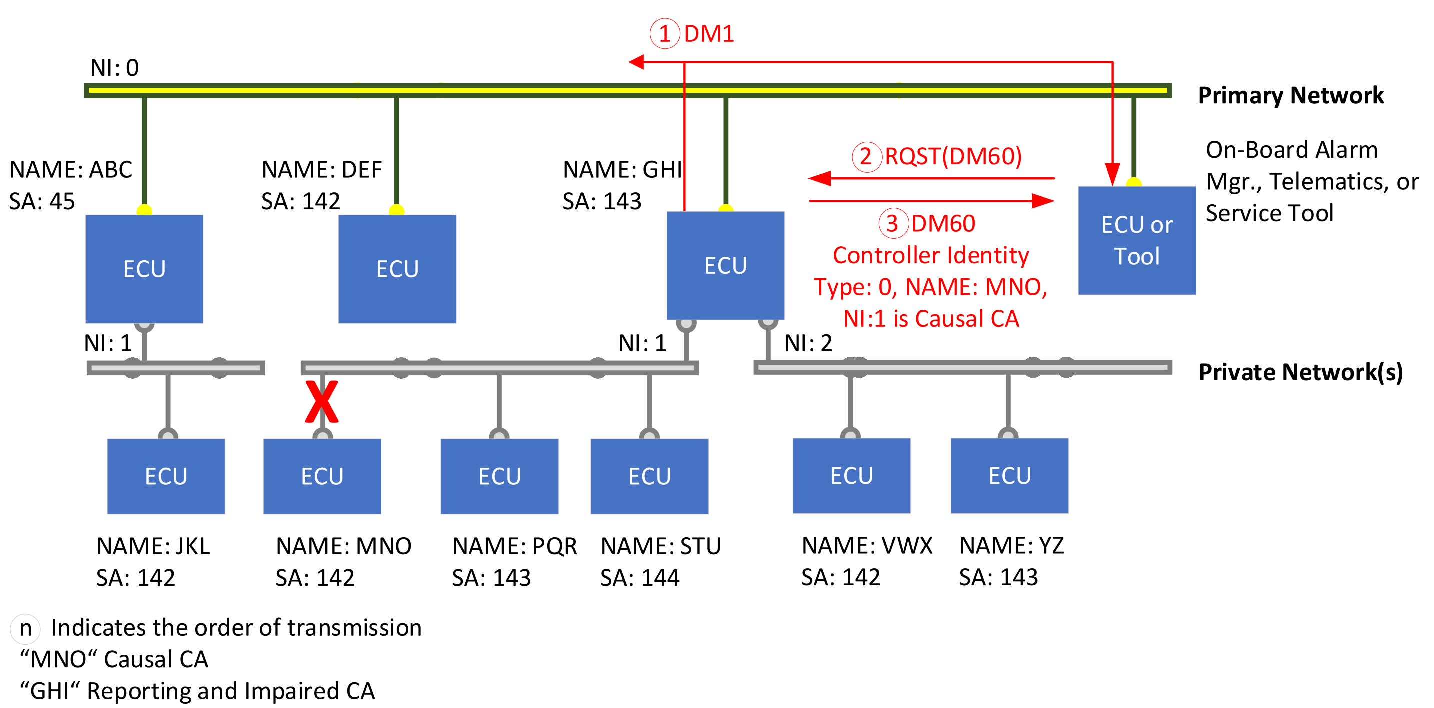


*Figure 11 - DM60 network identification in a multiple network environment*

The network identifier, associated with a specific DTC in the DM60 message, permits the reporting CA to identify a specific network on which the associated fault originates. By communicating the combination of DTC, NAME, and network identifier, it is possible not only for a CA to report faults originating on the directly attached network but also to act as a proxy and report faults that originate on an otherwise inaccessible network.

5.7.60.2.13.1 Manufacturer-Specified Network Number

The manufacturer-specified network number range provides a reference to a specific network, in the context of that system or ECU. The manufacturer of the ECU/system may assign the numbers referenced to the individual ECU (e.g., different ECUs could each have a unique network #1) or system referenced (e.g., all ECUs “network #1” reference the same network).

There is no restriction on the use of values within this range, permitting the flexibility to enumerate many networks, or by splitting the range or value into proprietary bit-fields, it could represent a hierarchy of networks.

5.7.60.2.13.2 SAE Network Enumeration

The SAE network enumeration range permits the ECU to reference a specific network using the same list of networks as defined in DM13. Note that the current data link is excluded from this range and is identified by the value 0. The network assignments are shown in Table 31.

***Table 31 - SAE network enumeration***

| SAE Network Enumeration | Network Name | Reference |
|---|---|---|
| $FB01_h$ | SAE J1587 network | 5.7.13.5 |
| $FB02_h$ | SAE J1922 network | 5.7.13.6 |
| $FB03_h$ | SAE J1939 network #1, primary vehicle network | 5.7.13.7 |
| $FB04_h$ | SAE J1939 network #2 | 5.7.13.8 |
| $FB05_h$ | ISO 9141 network | 5.7.13.9 |
| $FB06_h$ | SAE J1850 network | 5.7.13.10 |
| $FB07_h$ | Other, manufacturer-specified port | 5.7.13.11 |
| $FB08_h$ | SAE J1939 network #3 | 5.7.13.12 |
| $FB09_h$ | Proprietary network #1 | 5.7.13.16 |
| $FB0A_h$ | Proprietary network #2 | 5.7.13.17 |
| $FB0B_h$ | SAE J1939 network #4 | 5.7.13.18 |
| $FB0C_h$ | SAE J1939 network #5 | 5.7.13.19 |
| $FB0D_h$ | SAE J1939 network #6 | 5.7.13.20 |
| $FB0E_h$ | SAE J1939 network #7 | 5.7.13.21 |
| $FB0F_h$ | SAE J1939 network #8 | 5.7.13.22 |
| $FB10_h$ | SAE J1939 network #9 | 5.7.13.23 |
| $FB11_h$ | SAE J1939 network #10 | 5.7.13.24 |
| $FB12_h$ | SAE J1939 network #11 | 5.7.13.25 |

5.7.60.3 Future Record Data

Reserved for assignment by SAE.

5.7.60.4 Usage Examples

5.7.60.4.1 Only Causal CA Provided for DTC

One form of Extended DTC Information Record has a Controller Identity Body with a single controller identity that describes the causal CA associated with a DTC. This form indicates that the DTC has been detected by the impaired CA (reporting CA) as a result of an issue with the causal CA. In this form, the impaired CA identity is inferred as the source of the DM60 message, i.e., the reporting CA. One use of this form of the Extended DTC Information Record is when a controller fails to receive an SP from another controller connected on the same network (see Figure 12) or connected on a subnetwork (see Figure 13). The impaired CA can use this Extended DTC Information Record form to provide the identity of the CA that is the expected source of the SP (causal CA).

Figure 12 illustrates a situation where CA "GHI" fails to receive an SP from CA "DEF" via communications on the same network as the DM60 reporting. CA "GHI" is the impaired CA since its operations are impaired as a result of not receiving the SP, and CA "DEF" is the causal CA since it is the expected source of that SP. The Controller Identity Type is reported as $000_b$, indicating the causal CA identity is the only identity provided in the Controller Identity Body and the Controller Identity Body Length shall be 13 bytes ($0D_h$). The Controller Identity Body contains 13 bytes describing the NAME qualifiers, NAME, source address, and network identifier for CA "DEF." The CA "DEF" network identifier value indicates the CA "DEF" network connection where SP is communicated to CA "GHI."

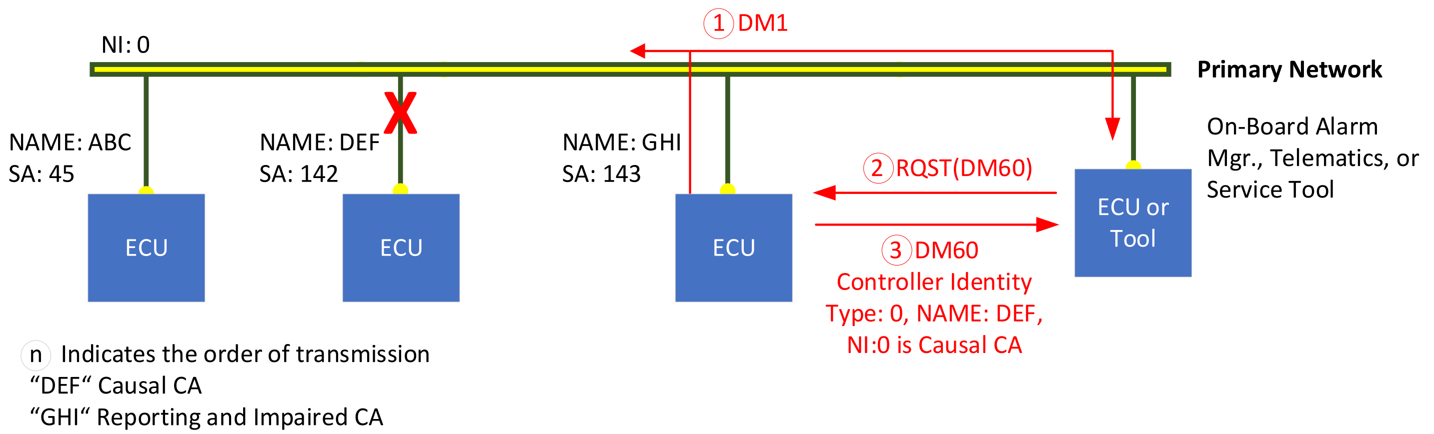


*Figure 12 - Causal CA on same network*

Figure 13 illustrates a situation where CA "GHI" fails to receive an SP from CA "MNO" via communications on a subnetwork. This example is the same as the Figure 12 example except that CA "MNO" is the causal CA. The Controller Identity Body for this example contains 10 bytes describing the NAME qualifiers, NAME, source address, and network identifier for CA "MNO," and the CA "MNO" network identifier shall indicate the subnetwork connection where the missing SP is communicated to CA "GHI."

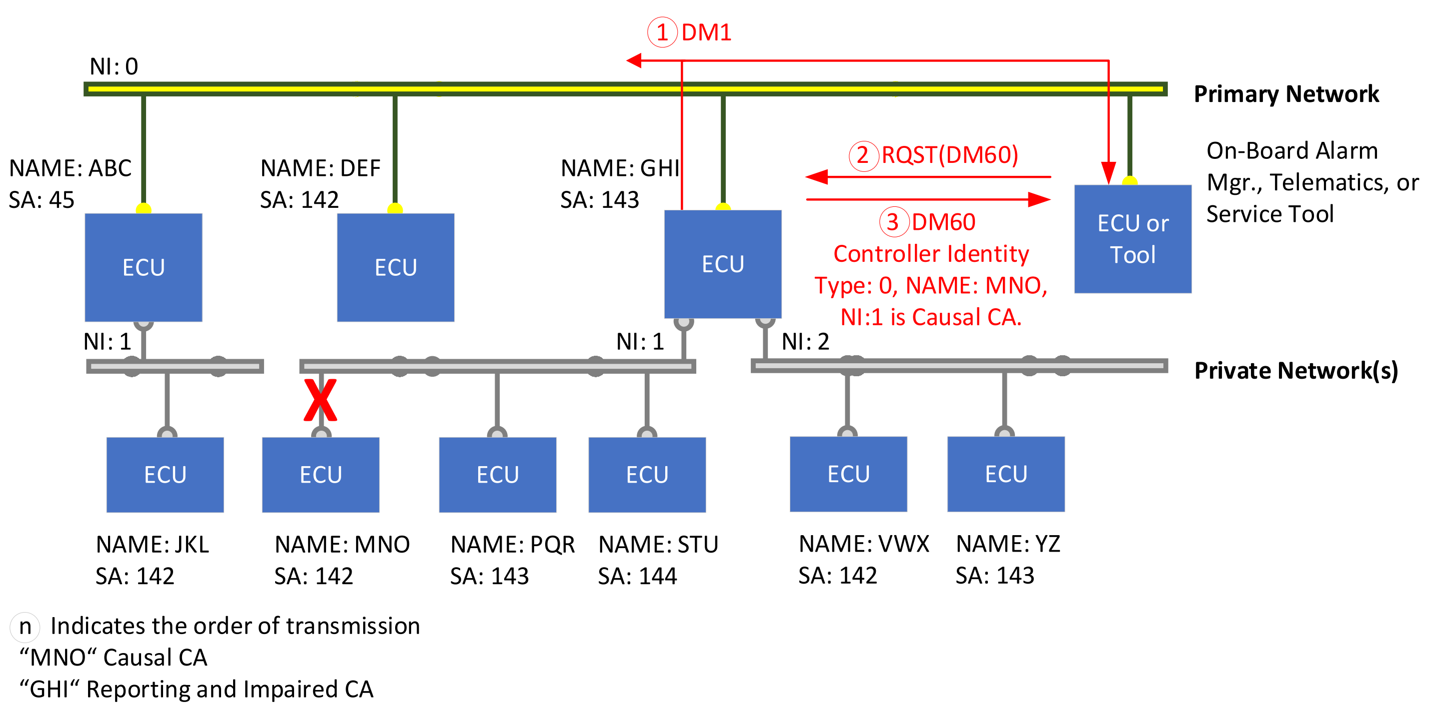


*Figure 13 - Causal CA on subnetwork*

5.7.60.4.2 Only Impaired CA Provided for DTC

One form of Extended DTC Information Record has a Controller Identity Body with a single controller identity that explicitly describes the impaired CA associated with the DTC. This form indicates the DTC was detected and originally reported by the impaired CA, the impaired CA is the causal CA, and the DTC is not associated to the reporting CA. This form is appropriate when a DM60 proxy is reporting the DTCs and DM60 Extended DTC Information Records on behalf of a reporting CA on a subnetwork where the DTC is attributed to the impaired CA, such as an issue with its I/O or a rationality fault.

Figure 14 illustrates a situation where CA “XYZ” is reporting (on a private subnetwork) a DTC for a detected I/O or rationality fault and CA “GHI” is reporting that DTC on behalf of CA “XYZ” on the primary network. CA “GHI” is the reporting CA on the primary network and the DM60 proxy CA for CA “XYZ” on the subnetwork “NI:2.” CA “XYZ” is the impaired CA and reporting CA on subnetwork “NI:2.” CA “GHI” reports the DTC reported by CA “XYZ” and includes the DTC with FMI 23 in its DM1 message. In the Extended DTC Information Record for the CA “XYZ” DTC, the Controller Identity Type is reported as $001_b$, indicating the impaired CA identity is the only identity provided in the Controller Identity Body and the Controller Identity Body Length shall be 13 bytes ($0D_h$). The Controller Identity Body contains 13 bytes describing the NAME qualifiers, NAME, source address, and network identifier for CA “XYZ.” This Extended DTC Information Record indicates the CA impaired by the DTC is CA “XYZ” and is not the DM60 reporting CA “GHI.”

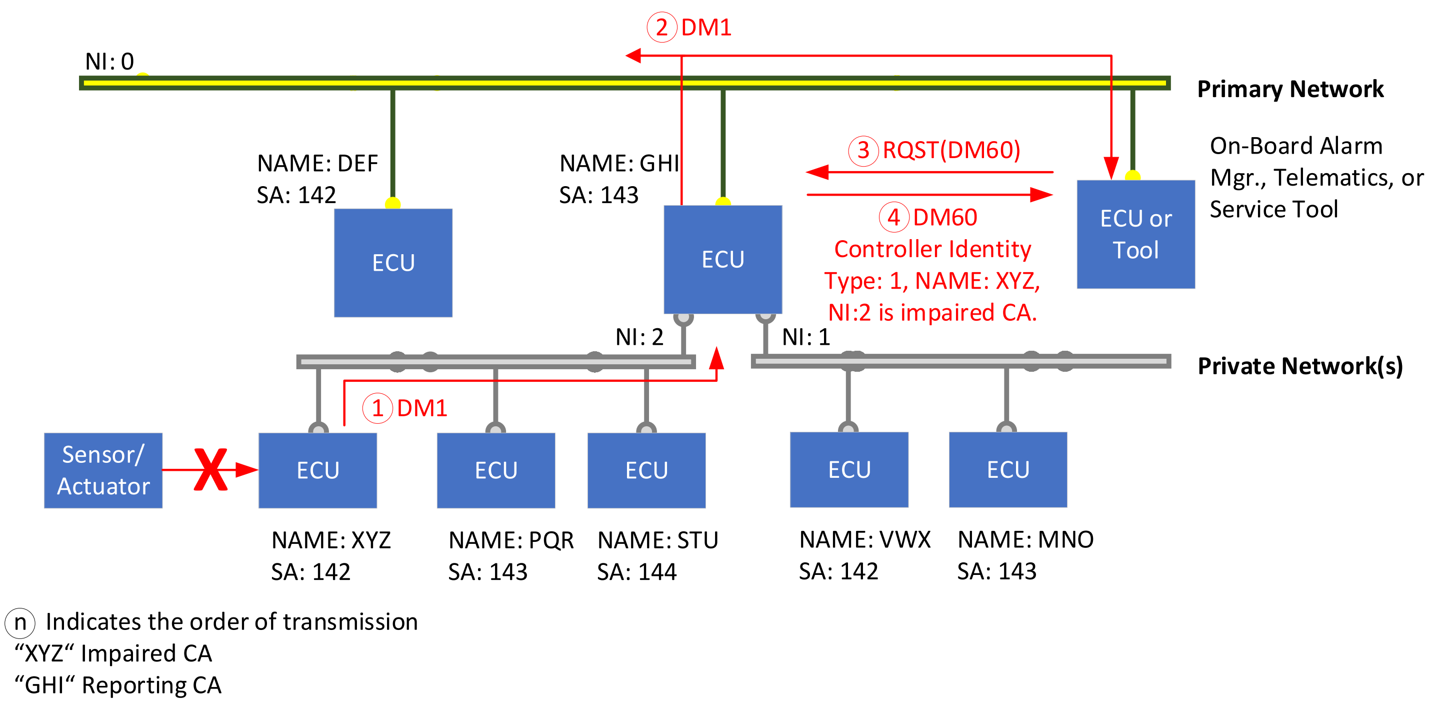


***Figure 14 - Impaired CA on subnetwork***

5.7.60.4.3 Causal CA and Impaired CA Provided for DTC

One form of Extended DTC Information Record has a Controller Identity Body with two controller identities: one that describes the causal CA associated with the DTC and one that describes the impaired CA associated with the DTC. This form indicates the DTC was detected and originally reported by the impaired CA and the DTC is a result of an issue with the causal CA. This form is appropriate when a DM60 proxy is reporting the DTCs and DM60 Extended DTC Information Records on behalf of a reporting CA on a subnetwork and DTC by the impaired CA is because it fails to receive an SP from another controller, either connected on the same network or a subnetwork.

Figure 15 illustrates a situation where CA "XYZ" fails to receive an SP from CA "PQR" via communications on the same subnetwork, and CA "GHI" is the DM60 proxy on the primary network reporting the DTC on behalf of CA "XYZ." CA "XYZ" is the impaired CA since its operations are impaired as a result of not receiving the SP, and CA "PQR" is the causal CA since it is the expected source of that SP. In the Extended DTC Information Record sent by CA "GHI" for the CA "XYZ" DTC, the Controller Identity Type is reported as $010_b$, indicating the Controller Identity Body contains two controller identities: the first controller identity describes the causal CA and the second controller identity describes the impaired CA. The Controller Identity Body Length shall be 26 bytes ($1A_h$). This Extended DTC Information Record indicates the impaired CA for the DTC is CA "XYZ," and CA "PQR" is the causal CA for that DTC; the DTC is not associated with the DM60 reporting CA "GHI."

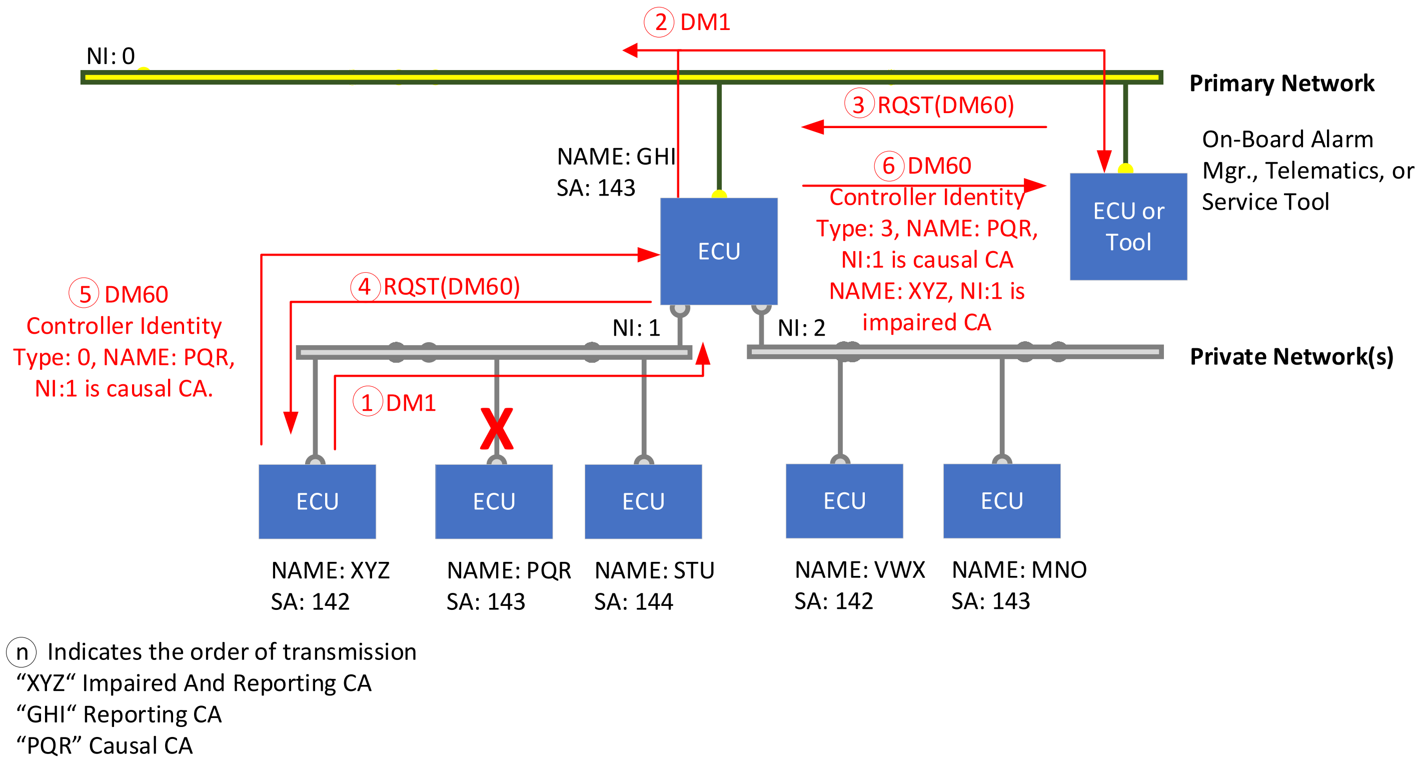


***Figure 15 - Impaired and causal CA on subnetwork***

## 6. NOTES

### 6.1 Revision Indicator

A change bar (l) located in the left margin is for the convenience of the user in locating areas where technical revisions, not editorial changes, have been made to the previous issue of this document. An (R) symbol to the left of the document title indicates a complete revision of the document, including technical revisions. Change bars and (R) are not used in original publications, nor in documents that contain editorial changes only.

PREPARED BY SAE TRUCK BUS CONTROL AND COMMUNICATIONS NETWORK COMMITTEE

# APPENDIX A - FAILURE MODE IDENTIFICATION CODES

## A.1 GENERAL RULES

The following definitions shall be applicable when using FMIs. Examples have been included to help achieve consistent usage of the failure mode identifiers. Not all FMIs are applicable to a given SPN. For example, a controller diagnosing a particular input, such as SPN 91 (Accelerator Pedal Position 1), may use FMIs 3 and 4 and, therefore, would not use FMIs 5 and 6.

### A.1.1 Assumptions and Definitions Used for the FMI Definitions

Data - Any information pertaining to physical conditions that is communicated to an electronic module in the form of voltage, current, PWM signals, or data streams.

Real World - Mechanical parameters or operating conditions that can be measured in the form of voltage, current, PWM signals, data streams, etc.

Signal Range - Definitions are shown in Figure A1, which also contains the definitions for regions a through k.

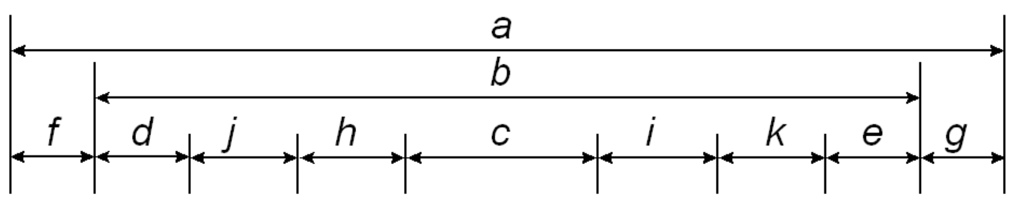


***Figure A1 - Signal ranges***

| | |
|---|---|
| Region a | Total signal input range possible that can be seen by an electronic module. |
| Region b | Total signal range physically possible as is defined by an application. The CARB-defined rationality fault diagnostic condition is applicable anywhere in this region. |
| Region c | Range defined as normal for a given real-world measurement. |
| Region d | Range defined as below normal, most severe level, of what is considered normal for the given real-world measurement. |
| Region e | Range defined as above normal, most severe level, of what is considered normal for the given real-world measurement. |
| Region f | Range that is low outside the range of what is considered physically possible for a given system, indicating a short to a low source has occurred. |
| Region g | Range that is high outside the range of what is considered physically possible for a given system, indicating a short to a high source has occurred. |
| Region h | Range defined as below normal, least severe level, of what is considered normal for a given real-world measurement. |
| Region i | Range defined as above normal, least severe level, of what is considered normal for a given real-world measurement. |

Region j Range defined as below normal, moderately severe level, of what is considered normal for a given real-world measurement.

Region k Range defined as above normal, moderately severe level, of what is considered normal for a given real-world measurement.

### A.1.2 FMI and Description

FMIs 0 to 31 are used to identify the failure mode relative to the DTC being reported. They should be assigned considering the range model definitions and according to the guidance given here and in each of the definitions.

FMIs 16 or 18 are recommended for use to indicate an OBD emission threshold monitor is exceeding the OBD-defined emission threshold.

FMIs 10, 13, and 7 are to be used to identify the malfunctions for OBD-defined feedback control monitoring requirements. The California Code of Regulation 1971.1 defines that the malfunctions of the feedback control shall be monitored for the following three conditions: (A) if the system fails to begin feedback control within a manufacturer-specified time interval; (B) if a failure or deterioration causes open loop or default operation; or (C) if feedback control has used up all of the adjustment allowed by the manufacturer. FMI 10 will be used to declare malfunction A, FMI 13 for malfunction B, and FMI 7 for malfunction C. If a manufacturer does not isolate these failure modes, then FMI 7 shall be used to indicate one or more of the three feedback control malfunctions have been detected.

FMIs 2, 20, and 21 provide some rationality coverage. If the rationality diagnostics do not distinguish between high and low, then FMI 2 shall be used. If high and low are isolated because unique troubleshooting procedures are required, then FMIs 20 and 21 shall be used. FMI 2 may also be used if there are additional malfunctions detected that are not relevant to FMIs 20 and 21.

#### A.1.2.1 FMI = 0 - Data Valid but Above Normal Operational Range - Most Severe Level

The signal communicating information is within a defined acceptable and valid range, but the real-world condition is above what would be considered normal as determined by the predefined most severe level limits for that particular measure of the real-world condition (region e of the signal range definition). Broadcast of data values is continued as normal.

#### A.1.2.2 FMI = 1 - Data Valid but Below Normal Operational Range - Most Severe Level

The signal communicating information is within a defined acceptable and valid range, but the real-world condition is below what would be considered normal as determined by the predefined least severe level limits for that particular measure of the real-world condition (region d of signal range definition). Broadcast of data values is continued as normal.

#### A.1.2.3 FMI = 2 - Data Erratic, Intermittent, or Incorrect

Erratic or intermittent data includes all measurements that change at a rate that is not considered possible in the real-world condition and must be caused by improper operation of the measuring device or its connection to the module. If the malfunction can be pinpointed, the broadcast of data is to be substituted with the “error indicator” value. This is to avoid broadcasting known bad data. If the malfunction cannot be isolated to a single component, the measured value is to be broadcast since it is not known which data may be in error.

Incorrect data includes any data not received and any data that is exclusive of the situations covered by FMIs 3, 4, 5, and 6 as follows in A.1.2.4 through A.1.2.7. Data can also be considered incorrect if it is inconsistent with other information collected or known about the system. See FMI 20 and FMI 21 for systems which desire to have separate DTCs for a rationality check for data drifted high and another DTC for a rationality check for data drifted low for the same component.

FMI 2 is applicable for rationality-type failures (see 3.23).

A.1.2.4 FMI = 3 - Voltage Above Normal or Shorted to High Source

a. A voltage signal, data or otherwise, is above the predefined limits that bound the range (region g of the signal range definition). Broadcast of data value is substituted with the “error indicator” value.

b. Any signal external to an electronic control module whose voltage remains at a high level when the ECM commands it to low. Broadcast of data value is substituted with the “error indicator” value.

A.1.2.5 FMI = 4 - Voltage Below Normal or Shorted to Low Source

a. A voltage signal, data or otherwise, is below the predefined limits that bound the range (region f of the signal range definition). Broadcast of data value is substituted with the “error indicator” value.

b. Any signal external to an electronic control module whose voltage remains at a low level when the ECM commands it to high. Broadcast of data value is substituted with the “error indicator” value.

A.1.2.6 FMI = 5 - Current Below Normal or Open Circuit

a. A current signal, data or otherwise, is below the predefined limits that bound the range (region f of the signal range definition). Broadcast of data value is substituted with the “error indicator” value.

b. Any signal external to an electronic control module whose current remains off when the ECM commands it on. Broadcast of data value is substituted with the “error indicator” value.

A.1.2.7 FMI = 6 - Current Above Normal or Grounded Circuit

a. A current signal, data or otherwise, is above the predefined limits that bound the range (region g of the signal range definition). Broadcast of data value is substituted with the “error indicator” value.

b. Any signal external to an electronic control module whose current remains on when the ECM commands it off. Broadcast of data value is substituted with the “error indicator” value.

A.1.2.8 FMI = 7 - Mechanical System Not Responding or Out of Adjustment

Any fault detected as the result of an improper mechanical adjustment or an improper response or action of a mechanical system that, with a reasonable confidence level, is not caused by an electronic or electrical system failure. This type of fault may or may not be directly associated with the value of general broadcast information.

This FMI is applicable for rationality-type failures (see 3.23).

A.1.2.9 FMI = 8 - Abnormal Frequency or Pulse Width or Period

To be considered in cases of FMIs 4 and 5. Any frequency or PWM signal that is outside the predefined limits which bound the signal range for frequency or duty cycle (outside region b of the signal definition). Also, if the signal is an ECM output, any signal whose frequency or duty cycle is not consistent with the signal that is emitted. Broadcast of data value is substituted with the “error indicator” value. Any failure detected as the result of expected system response occurring either too frequently or too infrequently (e.g., DPF regeneration frequency).

A.1.2.10 FMI = 9 - Abnormal Update Rate

Any failure detected when receipt of data via the data link or as input from a smart actuator or smart sensor is not at the update rate expected or required by the ECM (outside region c of the signal range definition). Also, any error detected causing the ECM not to send information at the rate required by the system. This type of fault may or may not be directly associated with the value of general broadcast information.

This FMI is applicable for rationality-type failures (see 3.23).

A.1.2.11 FMI = 10 - Abnormal Rate of Change

Any data, exclusive of the abnormalities covered by FMI 2, that is considered valid but whose data is changing at a rate that is outside the predefined limits that bound the rate of change for a properly functioning system (outside region c of the signal range definition). Broadcast of data values is continued as normal.

This FMI is applicable for rationality-type failures (see 3.23).

A.1.2.12 FMI = 11 - Root Cause Not Known

It has been detected that a failure has occurred in a particular sub-system, but the exact nature of the fault is not known. Broadcast of data value is substituted with the “error indicator” value.

A.1.2.13 FMI = 12 - Bad Intelligent Device or Component

Internal diagnostic procedures have determined that the failure is one which requires the replacement of the ECU, used here to mean the packaged unit that includes some microprocessor and its associated components and circuits. It can be assumed that the communications sub-system is not the part that has failed and that the manufacturer has determined that there is no serviceable component smaller than the ECU involved in the failure. Broadcast of data value is substituted with the “error indicator” value if appropriate, as there may or may not be any broadcast data involved. This error is to include all internal controller trouble codes that cannot be caused by connections or systems external to the controller.

This FMI is applicable for rationality-type failures (see 3.23).

A.1.2.14 FMI = 13 - Out of Calibration

A failure detected that can be identified to be the result of not being properly calibrated. This can be the case for a sub-system that can identify that the calibration attempting to be used by the controller is out of date. Or it can be the case that the mechanical sub-system is determined to be out of calibration. FMI 13 can also be used to indicate missing network data that is received with the not available indicator (i.e., $FF_h$; refer to SAE J1939-71). This failure mode does not relate to the signal range definition as do many of the FMIs.

This FMI is applicable for rationality-type failures (see 3.23).

A.1.2.15 FMI = 14 - Special Instructions #1

“Special Instructions #1” is the FMI to be used by manufacturers to associate an SPN to a nonstandard failure mode or when the on-board system can isolate the failure to a small number of choices but not to a single point of failure. When this FMI is used, there is a clear necessity for the service technician to take some action to complete the specific diagnosis, and the manufacturer has provided instructions for the completion of that diagnosis.

There are two legacy cases where this FMI could also be used: (1) for emission-related diagnostics where the particular failure cannot be separated between a sensor out of range and the case where the actual value is at the edge of a diagnostic region, and (2) for the older SPNs 611 to 615 where the problem is in determining which of two or more circuits (that may interact) is the one that needs repair.

SPNs 611 through 615 are defined as “System Diagnostic Codes” and are used to identify failures that cannot be tied to a specific field replaceable component. Specific sub-system fault isolation is the goal of any diagnostic system, but for various reasons this cannot always be accomplished. These SPNs allow the manufacturer some flexibility to communicate “non-specific component” diagnostic information. Since SPNs 611 through 615 use the standard SPN/FMI format, it allows the use of standard diagnostic tools, electronic dashboards, satellite systems, and other advanced devices that scan parameter groups containing the SPN/FMI formats. Because manufacturer-defined codes are not desirable in terms of standardization, the use of these codes should only occur when diagnostic information cannot be communicated as a specific component and failure mode.

Possible reasons for using a system diagnostic code include:

1. Cost of specific component fault isolation is not justified.

2. New concepts in total vehicle diagnostics are being developed.

3. New diagnostic strategies that are not component specific are being developed.

Due to the fact that SPNs 611 through 615 are manufacturer defined and are not component specific, FMIs 0 through 13 and FMIs 15 through 31 have little meaning. Therefore, FMI 14, “Special Instructions,” is usually used. The goal is to refer the service personnel to the manufacturer’s troubleshooting manual for more information on the particular diagnostic code. This failure mode does not relate to the signal range definition as do many of the FMIs. This type of fault may or may not be directly associated with the value of general broadcast information.

This FMI is applicable for rationality-type failures (see 3.23).

A.1.2.16 FMI = 15 - Data Valid but Above Normal Operating Range - Least Severe Level

The signal communicating information is within a defined acceptable and valid range, but the real-world condition is above what would be considered normal as determined by the predefined least severe level limits for that particular measure of the real-world condition (region i of signal range definition). Broadcast of data values is continued as normal.

A.1.2.17 FMI = 16 - Data Valid but Above Normal Operating Range - Moderately Severe Level

The signal communicating information is within a defined acceptable and valid range, but the real-world condition is above what would be considered normal as determined by the predefined moderately severe level limits for that particular measure of the real-world condition (region k of signal range definition). Broadcast of data values is continued as normal.

*Note that when FMIs 16 and 18 are used for threshold monitors as described in 13 CCR 1971.1 (e)(9), it can be the case that the error indicator (e.g., $FE_h$) shall be sent instead of a data value. In such cases, an extrapolated data value will be available using DM58. Where there is a need to pinpoint disparate root causes for threshold monitors, FMIs 0 and 1 may be appropriated to distinguish the failures from 16 and 18 (respectively).*

A.1.2.18 FMI = 17 - Data Valid but Below Normal Operating Range - Least Severe Level

The signal communicating information is within a defined acceptable and valid range, but the real-world condition is below what would be considered normal as determined by the predefined least severe level limits for that particular measure of the real-world condition (region h of signal range definition). Broadcast of data values is continued as normal.

*Note that when FMIs 16 and 18 are used for threshold monitors as described in 13 CCR 1971.1 (e)(9), it can be the case that the error indicator (e.g., $FE_h$) shall be sent instead of a data value. In such cases, an extrapolated data value will be available using DM58. Where there is a need to pinpoint disparate root causes for threshold monitors, FMIs 0 and 1 may be appropriated to distinguish the failures from 16 and 18 (respectively).*

A.1.2.19 FMI = 18 - Data Valid but Below Normal Operating Range - Moderately Severe Level

The signal communicating information is within a defined acceptable and valid range, but the real-world condition is below what would be considered normal as determined by the predefined moderately severe level limits for that particular measure of the real-world condition (region j of signal range definition). Broadcast of data values is continued as normal.

A.1.2.20 FMI = 19 - Received Network Data in Error

Any failure that is detected when the data received via the network is found substituted with the “error indicator” value (i.e., $FE_h$; refer to SAE J1939-71). This type of failure is associated with received network data. The component used to measure the real-world signal is wired directly to the module sourcing the data to the network and not to the module receiving the data via the network. This FMI is applicable to regions f and g of the signal range definition. This type of fault may or may not be directly associated with the value of general broadcast information.

A.1.2.21 FMI = 20 - Data Drifted High

Systems that use one DTC to report data drifted high and data drifted low, rationality failures for a component shall use FMI 2. When a product has separate DTCs for a rationality check for data drifted high and another DTC for a rationality check for data drifted low for the same component, it shall then use FMI 20 and FMI 21 accordingly.

The signal communicating information is within a defined acceptable and valid range, but the real-world condition is above what would be considered normal when compared to other measurements. This can include sensor drifts, measurements that do not seem possible when compared with other data, and measurements that change at a rate that is not considered possible in the real world or whose values themselves do not seem possible in the real world. It is understood that it is not feasible to always differentiate the cause of the data drifted high (e.g., Is the INTAKE MANIFOLD PRESSURE high because the sensor has drifted, or is there a mechanical problem with either the turbocharger or the hose connections?). This FMI is applicable to region b of the signal range definition. If the malfunction can be pinpointed, the broadcast of data is to be substituted with the “error indicator” value. This is to avoid broadcasting known bad data. If the malfunction cannot be isolated to a single component, the measured value is to be broadcast since it is not known which data may be in error.

This FMI is applicable for rationality-type failures (see 3.23).

A.1.2.22 FMI = 21 - Data Drifted Low

Systems that use one DTC to report data drifted high and data drifted low, rationality failures for a component shall use FMI 2. When a product has separate DTCs for a rationality check for data drifted high and another DTC for a rationality check for data drifted low for the same component, it shall then use FMI 20 and FMI 21 accordingly.

The signal communicating information is within a defined acceptable and valid range, but the real-world condition is below what would be considered normal when compared to other measurements. This can include sensor drifts, measurements that do not seem possible when compared with other data, and measurements that change at a rate that is not considered possible in the real world or whose values themselves do not seem possible in the real world. It is understood that it is not feasible to always differentiate the cause of the data drifted low (e.g., Is the INTAKE MANIFOLD PRESSURE low because the sensor has drifted, or is there a mechanical problem with either the turbocharger or the hose connections?). If the malfunction can be pinpointed, the broadcast of data is to be substituted with the “error indicator” value. This is to avoid broadcasting known bad data. If the malfunction cannot be isolated to a single component, the measured value is to be broadcast since it is not known which data may be in error.

This FMI is applicable for rationality-type failures (see 3.23).

A.1.2.23 FMI = 22 - Special Instructions #2

“Special Instructions #2” is the FMI to be used by manufacturers to associate an SPN to a nonstandard failure mode or when the on-board system can isolate the failure to a small number of choices but not to a single point of failure. When this FMI is used, there is a clear necessity for the service technician to take some action to complete the specific diagnosis, and the manufacturer has provided instructions for the completion of that diagnosis.

This failure mode does not relate to the signal range definition as do many of the FMIs.

This type of fault may or may not be directly associated with the value of general broadcast information.

This FMI is applicable for rationality-type failures (see 3.23).

A.1.2.24 FMI = 23 - Request DM60 for More Information

Additional information, such as the causal and impaired CA’s NAME, addresses, network identifiers, DTC type, DTC, etc., about this issue can be acquired by requesting DM60 (extended DTC information). This FMI can be used in open systems where CAs are connected on a plug and play basis without a qualified system integration phase and some SPNs are just optional.

FMI 23 is not specific to any particular region of the range model.

Broadcast of data values may or may not be impacted.

FMI 23 is not applicable for rationality-type failures (see 3.23). It may accompany a specific failure mode DTC, which is a rationality malfunction.

Products shall not use a DTC with FMI 23 in freeze frame messages DM4 or DM25.

A.1.2.25 FMI = 24 - Assurance Data Error

Assurance Data pertains to any data used to check the integrity and/or authenticity of a message. Examples include a Counter and CRC for integrity or a CMAC for authenticity. The assurance data can be conveyed in different manners, including:

a. Parameters that are embedded in a message's data field.

b. The SAE J1939-76 Functional Safety Communications Protocol.

c. Various SAE J1939-22 assurance data trailer content.

Received Assurance Data is in Error when any verification method fails its evaluation, such as a counter being stuck or a mismatch between received and calculated CRC values. When this FMI is reported, all data in the received message shall be considered invalid as it is unknown if one or more parameters were affected.

For integrity parameters embedded in a message's data field, this FMI can be reported against the received SPN(s) of interest or the SPN(s) of the specific integrity parameter(s) involved (e.g., a CRC SPN).

In SAE J1939-76 applications where more than one Safety Header Message (SHM) instance is being received (i.e., SHMs are used to protect more than one PG), it can be more beneficial to report this FMI against the Safety Data Message (SDM) parameter(s) of interest. Reporting this FMI against an SHM parameter will not indicate which SDM the fault is associated with.

This FMI may also be used to report a lack of received assurance data when the receiver has been configured to expect assurance data. For example, a certain SAE J1939-22 C-PG is received without an assurance data trailer despite the receiver being configured to expect one. Failure to receive an entire expected message, regardless of the inclusion of Assurance Data, should be reported via FMI 9.

A.1.2.26 FMIs = 25 through 30 - Reserved for Assignment by SAE

A.1.2.27 FMI = 31 - Condition Exists

This FMI is used to indicate that the condition identified by the SPN exists when no other applicable FMI exists or in cases when the reported SPN name spells out the component and a nonstandard failure mode. This type of fault may or may not be directly associated with the value of general broadcast information. This FMI will mean not available when the associated SPN is also not available as when the remainder of a packet is filled with binary ones after all data has been transmitted.

This FMI is applicable for rationality-type failures (see 3.23).

# APPENDIX B - ASSUMPTIONS USED TO DESIGN MEMORY ACCESS

## B.1 ASSUMPTIONS USED IN THE DESIGN OF MEMORY ACCESS

B.1.1 Memory data is transferred in byte pieces and if the memory width is other than an integer number of bytes, an extra full byte is used to contain the remaining bits.

B.1.2 It would be useful to have a direct address into memory, as well as a spatial (object or symbolic) referencing address. (As an example, a single 24-bit address would suffice for the direct address while 256 16-bit addresses could divide space and the standard could predefine the meaning of the first 128 spaces, while allowing the users to define and use the other 128 spaces proprietarily. It appears a 5-bit space identifier and a 19-bit object identifier would work since it would allow referencing SPNs directly - although a different length may be ultimately chosen. In fact, the pointer is presently 24 bits, while the pointer extension is 8 bits.)

B.1.3 It is desired to generate a memory access function without adding another transport protocol capable of handling more than 1785 bytes to the standard; thereby, data transfers are limited to lengths under 1785 bytes (refer to SAE J1939-21, 3.10.1.1).

B.1.4 Several security types shall be handled to satisfy all users. They are:

B.1.4.1 No security.

B.1.4.2 Password form of security.

B.1.4.3 Re-entrant security, which the manufacturer may optionally chose to implement, where the device allows multiple operations after a security level has been established.

B.1.4.4 Some more elaborate scheme similar to seed/key.

B.1.4.5 A User_Level request, which controls the user's privileges with the option for further security.

B.1.4.6 A means of increasing the effective seed/key size by requiring multiple iterations and/or mathematically combining the seeds and keys.

B.1.5 Minimum number of new PGs would be preferred (so that filtering and software overhead are minimized), so items that are time/message independent are combined (overlaid) to reduce the message set. (Obviously, the message set can be extended if the overlaying appears too complex or is desired for any other reason.)

B.1.6 Prefer single packet messages for the memory access invocation and control to reduce software overhead and improve speed of interchange, while need multipacketed messages for data transfer to provide reasonable lengths and improve transfer efficiency.

B.1.7 Reprogramming of "program memory" could be handled by any one of three general choices:

B.1.7.1 Use of a write operation in combination with some form of execution control table for enabling/disabling execution within sections of the program memory that are being modified in combination with a hardware configuration such that writing to these sections of program does not interfere with operation of other sections of program.

B.1.7.2 A boot loader approach where a proprietary program for reloading executable memory is loaded using the memory access operation of the standard and execution is then transferred to this proprietary reloading program. There is no need to standardize the data transfer utilized by the operation of this proprietary reloading program, but only the memory access operation loading said program and transferring control to it.

B.1.7.3 A completely proprietary technique, which is already possible using other features of this network standard.

B.1.8 Memory need only be addressed in one direction. Assume start at the lowest address and operate toward a higher address for this proposal (i.e., only an incrementing pointer is provided).

B.1.9 Also assume that for multipacket data sets the transport packet number shall be combined with the pointer provided in the original memory access to decode the address(es) for each packet.

B.1.10 All memory access requests originate at a tool and are considered commands to the device. The device, however, controls whether the request is handled.

B.1.11 Design to provide access for a single “tool” to access a single “device.” Then later, if it is desired, one can allow any node to function as a “tool” communicating with any other node functioning as a “device.” Also, if an OEM desires to allow more than one tool to access their device simultaneously, all they need additional is software to handle the different accesses.

## B.2 ASSUMPTIONS FOR DATA SECURITY

B.2.1 More of the committee members desired to use two messages over a single message, which at times was single-frame and at other times multi-frame, necessitating transport session.

B.2.2 A single message containing either a seed or a key is better than a separate message for seed and another for key since it uses fewer PGs.

B.2.3 A length parameter, while not inherently required, simplifies software handling enough to warrant inclusion.

B.2.4 No need to pack these parameters as it still takes a minimum of five frames to send any seed or key with a length between 8 and 13 bytes, so leave separate for ease of parsing.

# APPENDIX C - APPLICATION RULES REGARDING MEMORY ACCESS PGS

## Memory Access State Transition Diagram

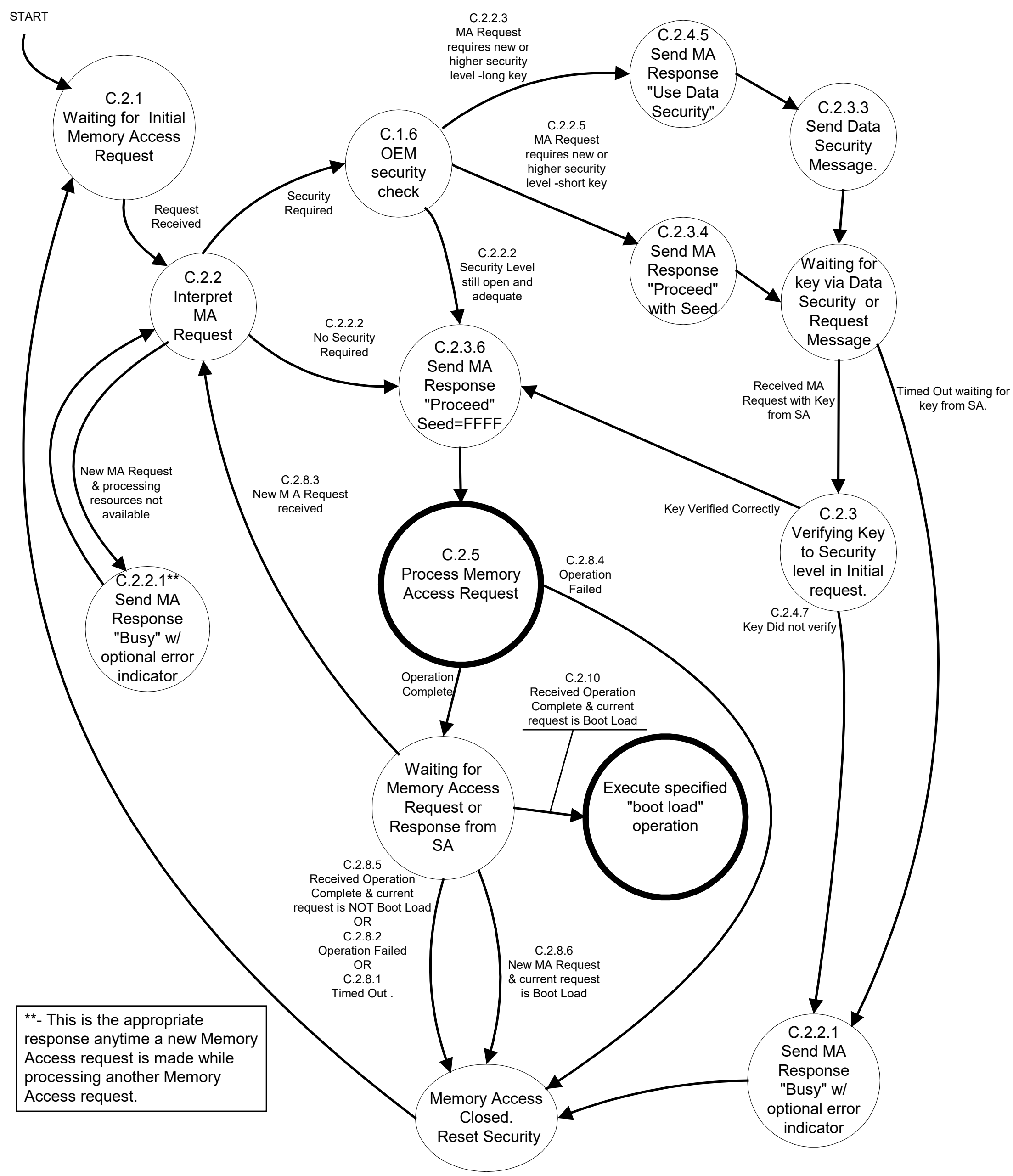


***Figure C1 - Memory access state transition diagram***

## C.1 GENERAL RULES

The following general rules shall be considered:

C.1.1 Only memory access operations initiated by a tool (using memory access request) are required to be honored. However, the manufacturer may choose to allow any network node to operate as either a tool or device, as long as it meets the functions presented in Appendix C for tool or device.

C.1.2 A device is required to support only one session of memory access at any one time (it may therefore reject all other requests with status of busy).

C.1.3 A tool may be designed to initiate memory access operations with more than one device at any given time.

C.1.4 There will be no specific messages to:

C.1.4.1 “Undo” a write request.

C.1.4.2 Abort an operation (obviously, failure to transfer data, etc., will cause a failure, which could be construed an abort).

C.1.5 A device may impose any number of additional constraints on when memory access requests are honored (see 5.7.14.4).

C.1.6 A manufacturer may choose to allow their device(s) to allow re-entrant security, wherein a tool that has already made a memory access request and established a security level may send additional memory access requests following the successful completion of the present operation, using the established security.

C.1.7 The device needs time-out functions for:

C.1.7.1 Failure to receive further security from a tool when the device has required same.

C.1.7.2 Failure to receive a complete transfer of the data set once an operation was allowed.

C.1.7.3 Failure to hear a close from a tool.

C.1.8 A tool needs time-out functions for:

C.1.8.1 Failure to receive a memory access response from a device to which it has sent a request.

C.1.8.2 Failure to receive a complete transfer of the data set once a read operation was allowed.

C.1.8.3 Failure to hear a close from a device.

## C.2 THE FOLLOWING IS A NARRATIVE OF A TYPICAL APPLICATION OF THIS PROTOCOL

It is only required that memory access operations be available once a node has become operational upon the network and satisfied any manufacturer-specific interlock requirements. Software functions that will need to be finished before memory access becomes available include address claiming, updating of instance fields within the NAME, and any other configuration matters that the manufacturer deems necessary as a precursor to allowing operation of the memory access software. A diagram showing the memory access state transitions for a device has been included along with message transmission diagrams for several cases (see Appendix E). These diagrams should be used along with the following text to generate the software modules for a device. There is presently no diagram for a tool, and the text and message transition diagrams in Appendix E should be used as the reference in designing the tool’s software.

C.2.1 Initial Memory Access Request

The tool sends a memory access request to the device. This consists of the address of the memory within the device to be accessed (pointer, pointer extension, and pointer type), the length of the memory the tool desires to operate upon (length/number requested), the operation requested (command = erase, read, write, boot load, or EDCP generation), and, if utilized by the device, any necessary User_Level or password information within the Key/User_Level parameter. If needed, based upon the device's particular requirements, it extracts from the message identifier (refer to SAE J1939-21, 3.1), the source (refer to SAE J1939-21, 3.2.6), and destination (refer to SAE J1939-21, 3.2.4, 3.2.5, and 3.2.5.1).

C.2.2 Device Response to Initial Memory Access Request

The device responds to this request with a memory access response as follows:

C.2.2.1 If the device is busy or has identified an error within the request (such as the pointer is not on a memory boundary for the memory being selected, the space being undefined, etc.), the device transmits a seed of all ones ($FFFF_h$) and a status of busy with the EDCP extension set to either no error indicator/EDC parameter available ($FF_h$) or data in error indicator/EDC parameter is an error indicator ($06_h$ or $07_h$) as desired by the manufacturer, the error indicator/EDC parameter as appropriate for the EDCP extension, and the length/number allowed to be zero. The number allowed needs not be interpreted by the tool as it has no specific meaning in the context of this message. The tool has a choice with handling of the EDCP extension and error indicator/EDC parameter. The tool is NOT required to assign any meaning to these items but may optionally (at the manufacturer's discretion) use the assigned values and definitions to imply specific meanings. The tool needs to try again later. See C.2.3.1.

C.2.2.2 If the device is not busy, and no security is required or was established in a previous operation (as would occur when the manufacturer has allowed re-entry to the memory access operation for a tool that has as yet not issued a "close"), or the password transmitted has been accepted, the device transmits the allowed number of objects or memory length within length/number allowed, a seed of all ones ($FFFF_h$) to indicate no further key is required (see 5.7.15.4 and Table 14), and a status of proceed with the EDCP extension set to either no error indicator/EDC parameter available ($FF_h$) or data in error indicator/EDC parameter is an error indicator ($06_h$ or $07_h$) as desired by the manufacturer, the error indicator/EDC parameter (most likely $00_h$ since no error) as appropriate for the EDCP extension. The tool has a choice with handling of the EDCP extension and error indicator/EDC parameter. The tool is NOT required to assign any meaning to these items but may optionally (at the manufacturer's discretion) use the assigned values and definitions to imply specific meanings. The requested operation can begin. (Remember that for the optional manufacturer re-entry, the device may have other established rules regarding whether the re-entrant operation was truly at the security level previously established.) See C.2.5.

C.2.2.3 If the device is not busy, and "long" seed/key security is required and if a valid User_Level was provided (when utilized by the device), the device transmits a length/number allowed of zero, a seed equal to one ($0001_h$), and a status of proceed with the EDCP extension set to either no error indicator/EDC parameter available ($FF_h$) or data in error indicator/EDC parameter is an error indicator ($06_h$ or $07_h$) as desired by the manufacturer, the error indicator/EDC parameter (most likely $00_h$ since no error) as appropriate for the EDCP extension. The tool has a choice with handling of the EDCP extension and error indicator/EDC parameter. The tool is NOT required to assign any meaning to these items but may optionally (at the manufacturer's discretion) use the assigned values and definitions to imply specific meanings. Memory access may continue. See C.2.3.3.

C.2.2.4 If the device is not busy, but security was required and some security violation or error has occurred (such as an invalid User_Level or password), the device transmits a seed of all ones ($FFFF_h$) and a status of busy, with the EDCP extension set to either no error indicator/EDC parameter available ($FF_h$) or data in error indicator/EDC parameter is an error indicator ($06_h$ or $07_h$) as desired by the manufacturer, the error indicator/EDC parameter as appropriate, and the length/number allowed to be zero. The number allowed needs not be interpreted by the tool as it has no specific meaning in the context of this message. The tool has a choice with handling of the EDCP extension and error indicator/EDC parameter. The tool is NOT required to assign any meaning to these items but may optionally (at the manufacturer's discretion) use the assigned values and definitions to imply specific meanings. The tool needs to try again later. See C.2.3.1.

C.2.3 Tools Action on Security Response

The tool responds to the memory access response message(s) controlling security of a memory access operation in one of several ways. (Remember that the tool always has a choice of how to handle the EDCP extension and error indicator/EDC parameter.) While the tool is NOT required to assign any meaning to these items, it may optionally (at the manufacturer's discretion) make use of the assigned values and definitions (see 5.7.15.3) to imply specific meanings. This supposes that the device being communicated with has also chosen to use the EDCP extension and error indicator/EDC parameter to indicate error conditions. The responses are as follows:

C.2.3.1 If the tool receives a memory access response with a status of busy, it needs to try the request again later, unless the busy was really indicating an error in the request. If the manufacturer has provided diagnostics of such errors, this will be indicated by the EDCP extension and the error identification will be within the error indicator/EDC parameter. If there was an identified error, the tool could then choose to correct the "problem" and issue another request. (It is felt that some manufacturers will wish to provide no further indication of invalid security, as this would only aide those trying to defeat the security. This is their choice. See 5.7.15.3.) See C.2.1.

C.2.3.2 If the tool sees a memory access response with a status of proceed, a length/number allowed of zeros, and a seed equal to all zeros ($0000_h$), then the seed (see 5.7.15.4) has been sent previously by the device and the device is expecting the tool to begin sending the key corresponding to the seed (using another memory access request message). This request should contain the key based upon the received seed, plus all of the memory access request parameters (pointer type, pointer extension, pointer, length/number requested, and command) that were in the initial request. See C.2.4.

C.2.3.3 If the tool sees a memory access response with a status of proceed, a length/number allowed of zero, and a seed equal to one ($0001_h$), then a "long" seed and key are to be used (refer to the data security message document). The tool should now expect a data security message (with a long seed). Following the receipt of a long seed from a data security message, the tool should reply with the corresponding long key using another data security message. The device then answers the tool with another of the messages identified here in C.2.3. See C.2.3.

C.2.3.4 If the tool sees a memory access response with a status of proceed, a length/number allowed of zero, and a seed not equal to zero, one, or all ones ($0000_h$, $0001_h$, or $FFFF_h$), then this is the seed from the device. The tool may now begin sending the key corresponding to the seed using another memory access request message. This request should contain the key based upon the received seed, plus all of the other memory access request parameters (pointer type, pointer extension, pointer, length/number requested, and command) that were in the initial request. See C.2.4.

C.2.3.5 If the tool sees a memory access response with a status of proceed, a length/number allowed of zero, and a seed equal to all ones ($FFFF_h$), then the device feels the key transfer has been completed but the key verification is not completed (or some other similar delay) and the operation cannot yet begin. There may have been an error indicator in the error indicator/EDC parameter at the manufacturer's choice (see 5.7.15.3). The tool shall not begin data transfer yet (if there is to be one). The tool should in general send another memory access request to the device, with a key of all ones ($FFFF_h$) plus all of the other memory access request parameters (pointer type, pointer extension, pointer, length/number requested, and command) that were in the initial request. See C.2.4. However, if the tool is waiting for data from the device, it may choose simply to continue waiting instead of sending another request. See C.2.5.

C.2.3.6 If the tool sees a memory access response with a status of proceed, a non-zero length/number allowed, and a seed equal to all ones ($FFFF_h$), then the device feels the data transfer may begin. The tool should consider the device is now ready to begin the requested operation. See C.2.5.

C.2.4 If the device has not previously signaled that it was busy, it responds to the next memory access request with a memory access response as follows:

C.2.4.1 If the device has become busy, the device transmits a seed of all ones ($FFFF_h$) and a status of busy with the EDCP extension set to either no error indicator/EDC parameter available ($FF_h$) or data in error indicator/EDC parameter is an error indicator ($06_h$ or $07_h$) as desired by the manufacturer, the error indicator/EDC parameter as appropriate for the EDCP extension, and the length/number allowed to be zero. The number allowed needs not be interpreted by the tool, as it has no specific meaning in the context of this message. The tool has a choice with handling of the EDCP extension and error indicator/EDC parameter. The tool is NOT required to assign any meaning to these items but may optionally (at the manufacturer's discretion) use the assigned values and definitions to imply specific meanings. The tool needs to try again later. See C.2.3.1 to see tool's action.

C.2.4.2 If the device is still not busy, and security was required (including receipt of a valid User_Level, if it was required), and the device feels a complete key has been received, thus requiring no additional seed/key combinations, but the device has as yet been unable to complete the verification of the key, the device transmits a zero for length/number allowed, a seed of all ones($FFFF_h$), and a status of proceed with the EDCP extension set to either no error indicator/EDC parameter available ($FF_h$) or data in error indicator/EDC parameter is an error indicator ($06_h$ or $07_h$) as desired by the manufacturer, the error indicator/EDC parameter as appropriate for the EDCP extension. The tool has a choice with handling of the EDCP extension and error indicator/EDC parameter. The tool is NOT required to assign any meaning to these items but may optionally (at the manufacturer's discretion) use the assigned values and definitions to imply specific meanings. See C.2.3 to see tool's action.

C.2.4.3 If the device is still not busy, and security was required (including receipt of a valid User_Level, if it was required), and the device feels a complete key has been received, thus requiring no additional seed/key combinations, and the device has validated (accepted) the key, the device transmits a non-zero length/number allowed (with the value representing the actual length the device is willing to allow the tool to operate upon), a seed of all ones ($FFFF_h$), and a status of proceed with the EDCP extension set to either no error indicator/EDC parameter available ($FF_h$) or data in error indicator/EDC parameter is an error indicator ($06_h$ or $07_h$) as desired by the manufacturer, the error indicator/EDC parameter as appropriate for the EDCP extension. The tool has a choice with handling of the EDCP extension and error indicator/EDC parameter. The tool is NOT required to assign any meaning to these items but may optionally (at the manufacturer's discretion) use the assigned values and definitions to imply specific meanings. The requested operation can begin. See C.2.4.3 to see tool's action.

C.2.4.4 If the device is still not busy, and security was required (including receipt of a valid User_Level, if it was required), but the seed has NOT been sent yet and the use of a long seed/key (see 5.7.18) is NOT required, the device transmits a length/number allowed of zero, a seed not equal to either all zeros or all ones ($0000_h$ or $FFFF_h$) or one (implying use long seed; see Table 14), and a status of proceed with the EDCP extension set to either no error indicator/EDC parameter available ($FF_h$) or data in error indicator/EDC parameter is an error indicator ($06_h$ or $07_h$) as desired by the manufacturer, the error indicator/EDC parameter as appropriate for the EDCP extension. The tool has a choice with handling of the EDCP extension and error indicator/EDC parameter. The tool is NOT required to assign any meaning to these items but may optionally (at the manufacturer's discretion) use the assigned values and definitions to imply specific meanings. Memory access may continue. See C.2.3 to see tool's action.

C.2.4.5 If the device is still not busy, and security was required (including receipt of a valid User_Level, if it was required), but the use of a long seed/key (see 5.7.18) is required, the device transmits a length/number allowed of zero, a seed equal to one ($0001_h$), and a status of proceed with the EDCP extension set to either no error indicator/EDC parameter available ($FF_h$) or data in error indicator/EDC parameter is an error indicator ($06_h$ or $07_h$) as desired by the manufacturer, the error indicator/EDC parameter as appropriate for the EDCP extension. The tool has a choice with handling of the EDCP extension and error indicator/EDC parameter. The tool is NOT required to assign any meaning to these items but may optionally (at the manufacturer's discretion) use the assigned values and definitions to imply specific meanings. The device should also transmit a data security message with an appropriate seed (outlined within the data security message documentation). Memory access may continue. See C.2.3.1 to see tool's action.

C.2.4.6 If the device is still not busy, and security was required (including receipt of a valid User_Level if it was required), and the seed has been sent but reception of the key has NOT occurred, and the device has timed-out waiting for the tool, the device may transmit another memory access response message with a length/number allowed of zero, a seed equal to all zeros ($0000_h$), and a status of proceed with the EDCP extension set to either no error indicator/EDC parameter available ($FF_h$) or data in error indicator/EDC parameter is an error indicator ($06_h$ or $07_h$) as desired by the manufacturer, the error indicator/EDC parameter as appropriate for the EDCP extension. The tool has a choice with handling of the EDCP extension and error indicator/EDC parameter. The tool is NOT required to assign any meaning to these items but may optionally (at the manufacturer's discretion) use the assigned values and definitions to imply specific meanings. Memory access may continue. See C.2.3 to see tool's action. Alternately, the device may choose to discontinue the operation. See C.2.10.

C.2.4.7 If the device is still not busy, and security was required, and an invalid key was received, the device transmits a seed of all ones ($FFFF_h$) and a status of busy with the EDCP extension set to either no error indicator/EDC parameter available ($FF_h$) or data in error indicator/EDC parameter is an error indicator ($06_h$ or $07_h$) as desired by the manufacturer, the error indicator/EDC parameter as appropriate for the EDCP extension, and the length/number allowed to be zero. The number allowed needs not be interpreted by the tool, as it has no specific meaning in the context of this message. The tool has a choice with handling of the EDCP extension and error indicator/EDC parameter. The tool is NOT required to assign any meaning to these items but may optionally (at the manufacturer's discretion) use the assigned values. See C.2.3.1 to see tool's action.

C.2.5 Begin a requested memory access operation when the tool has seen a status of proceed and a seed equal to all ones ($FFFF_h$), then it recognizes that the device is willing to allow the requested memory access operation to begin. The device should have retained any internal state information indicating that it has signaled the tool of its own readiness to allow said operation. The next step depends upon the type of operation initiated with the command parameter of the initial memory access request.

NOTE: The command and the length/number requested within the memory access request message(s) should have been either constant or changed to what the device was willing to allow (see C.2.4.3). Any other alteration during the sequence should cause the device to reject the operation. See 5.7.14.4.6.

C.2.5.1 If the memory access request command was an erase, the device should process the erase command that it allowed and, when completed, initiate the close sequence. See C.2.6.

C.2.5.2 If the memory access request command was a read, the tool allows the device to initiate a transfer using the binary data transfer PG, either as a single packet or as a multipacketed message within a transport session depending upon the length involved. (If a transport session is required, it follows the rules in SAE J1939-21.) When the transfer is completed, the device initiates the close sequence. See C.2.6.

C.2.5.3 If the memory access request command was a write or a boot load with data (non-zero length/number requested), the device allows the tool to initiate a transfer using the binary data transfer PG, either as a single packet or as a multipacketed message within a transport session depending upon the length involved. (If a transport session is required, it follows the rules in SAE J1939-21.) When the transfer is completed and when the write operation has finished (successfully or not), the device initiates the close sequence. If the command was a boot load without data (zero length/number requested), the device should initiate the close sequence exactly as when a data transfer had completed. See C.2.6.

C.2.5.4 If the memory access request command was an EDCP generation, the device reads the data from the length of memory at the address it has allowed access to and then generates the requested checksum (or CRC, etc.) initiated for these locations. When the checksum is generated, the device initiates the close sequence. See C.2.6.

C.2.6 Memory access close sequence is upon completion of a memory access operation, as follows:

C.2.6.1 If the memory access request command was an erase, write, boot load, or EDCP generation, the device transmits a memory access response with a status of operation completed or operation failed depending upon the success/failure of the requested operation. The EDCP extension identifies whether the error indicator/EDCP is used (remember this is at the manufacturer's discretion). It also identifies how to interpret itself and said error indicator/EDC parameter. The length/number allowed should be zero, the seed should be equal to all zeros ($0000_h$), and the error indicator/EDC parameter as appropriate for the EDCP extension. The tool has a choice with handling of the EDCP extension and error indicator/EDC parameter. The tool is NOT required to assign any meaning to these items but may optionally (at the manufacturer's discretion) use the assigned values and definitions to imply specific meanings. The tool may interpret the length/number allowed and seed parameters as having no meaning. This memory access response is to be transmitted only after any internal processes invoked by the memory access have completed. See C.2.7.

C.2.6.2 If the memory access request command was a read, the device transmits a memory access response with a status of operation completed or operation failed depending upon the success/failure of the requested operation. The EDCP extension identifies whether the error indicator/EDCP is used and/or how to handle itself and said error indicator/EDC parameter. The length/number allowed should be zero, the seed should be equal to all zeros ($0000_h$), and the error indicator/EDC parameter as appropriate for the EDCP extension. The tool has a choice with handling of the EDCP extension and error indicator/EDC parameter. The tool is NOT required to assign any meaning to the EDCP extension and error indicator/EDC parameter but may optionally (at the manufacturer's discretion) use the assigned values and definitions to imply specific meanings. The tool may interpret the length/number allowed and seed parameters as having no meaning (it may also choose to simply dispose of them). This memory access response is transmitted immediately after the binary data transfer has completed (transport session has closed if one was required for the data transfer). See C.2.7.

C.2.7 When the tool receives the memory access response from the device indicating operation completed or failed, and the tool wishes to end the memory access connection, it transmits a memory access request indicating a status of either operation completed or operation failed from its perspective. (The tool should have checked the EDCP extension and error indicator/EDC parameter as a part of its decision process.) The tool should send another memory access request to the device, with a key of all ones ($FFFF_h$) plus all of the other memory access request parameters (pointer type, pointer extension, pointer, command, and length/number allowed) that were in the initial request (except that the length/number requested may also be the value it changed to when the device indicated it was willing to allow the operation). See C.2.4.3. The device may treat all of these other parameters as having no meaning. In the case of boot load, a memory access request of operation failed from the tool shall prevent the device from transferring execution to the address specified within the original boot load request.

C.2.8 After the device initiates the close sequence, it waits for a memory access request from the tool. The action taken by the device depends upon the original request, as well as the response from the tool. The following are the possible device actions.

NOTE: It is expected that only a successful completion of the execution of the boot load command prevents the system from returning to the same operational mode it was in prior to the memory access request which initiated a boot load operation. See C.2.9.

C.2.8.1 If there is no response from the tool within 100 ms (tolerance of 25 ms), plus any additional delay needed to account for the delay of any bridges within the system, of the device transmission, the device shall reset any optional re-entrant security levels and return to the initial state for memory access and to whatever operation mode it was in prior to the original memory access request and may optionally consider this operation failed.

C.2.8.2 If the memory access response from the device to initiate the close sequence was operation failed, then regardless of the response from the tool, the device shall reset any optional re-entrant security levels and return to the initial state for memory access and to whatever operation mode it was in prior to the original memory access request and shall consider this operation failed.

C.2.8.3 If the response from the tool (i.e., the tool with the source address from which the initial memory access operation came) is another memory access request, the manufacturer has allowed the optional re-entrant security, and the initial request was other than boot load, the device shall consider this operation completed and shall return to the internal state where it processes the memory access requests with re-entrant security.

C.2.8.4 If the response from the tool is operation failed, the device shall reset any optional re-entrant security levels and return to the initial state for memory access and to whatever operation mode it was in prior to the original memory access request and shall consider this operation failed.

C.2.8.5 If the memory access response from the device to initiate the close sequence was operation completed and the memory access request from the tool to complete the close sequence was operation completed, then the device resets any optional re-entrant security levels and returns to the operational mode it was in prior to the memory access request that initiated this sequence, unless the request was a boot load command. When the request has been a boot load command, the device should transfer execution. See C.2.9.

C.2.8.6 If the response from the tool is another memory access request, and the initial operation was boot load, the device shall reset any optional re-entrant security levels and return to the initial state for memory access and to whatever operation mode it was in prior to the original memory access request, thus effectively considering the boot load operation failed.

C.2.9 If a boot load command is successfully completed (both the tool and the device sent operation completed), then the device transfers execution to the address that was determined from the pointer, pointer extension, and pointer type of the initial memory access request. At such time several results are possible, they are:

C.2.9.1 If there was no data to be sent (a zero length/number requested in original request), the device will simply transfer execution to another location (may be used simply as a means to invoke a new mode of program operation within the device, such as “reset,” switch to an internal “loader,” etc.).

C.2.9.2 If there was new data transferred, it may have been a new program that the device is simply to begin operating from. One possibility for this new program is that it is a reprogramming program designed to provide a more time-efficient means of reprogramming the device’s executable memory. In such a case, the tool and the device may then intercommunicate by means of the boot load data PG. This boot load data PG can be transmitted from the tool to the device with the parameters in any format that meets the needs of the device being programmed. The boot load data PG can be transmitted from the device to the tool as an ACK/NAK sequence or to control timing in any way deemed necessary to achieve the transfer between the tool and the device. It is expected that the device will no longer respond to any other PGs transmitted to it; however, the tool will be required to maintain communications to the remainder of the network devices. It will also become the tool’s further responsibility to act on the device’s behalf in any network management functions, in particular to prevent an address claim by any other node of the address being used by the device being programmed.

C.2.9.3 If there was new data transferred, it may simply have been an addition to the existing program. This possibility would have required the manufacturer to have left space available for such an addition and used a memory type that could have new data added without damage to the old. Should this have been the case, execution would simply transfer to the new address just as when no new data had been added. Whether or not a reset or other operation would be necessary would be at the manufacturer’s discretion.

C.2.10 If a tool fails to hear the memory access response message with operation completed or operation failed from a device within what it considers to be appropriate time, it may send a memory access request of status request to the device. If the tool receives no response within 0.25 seconds, it shall consider the device is not going to respond and return to a mode of operation appropriate for this “failure” (i.e., try to determine if the device is still operating, if data has been damaged, etc.). If the tool receives a memory access response of proceed from the device, the tool should recognize that the device has already returned to the waiting for request state and considers the previous request completed and whether it was successful or failed can no longer be determined. The tool may wish to attempt to determine why the device completed without it hearing the response (possible reasons are the response was not sent, bus communication is impaired, device had timed-out hearing the tool at one of the interchange points, etc.).

# APPENDIX D - APPLICATION RULES REGARDING DATA SECURITY MESSAGE (DM18)

## D.1 GENERAL RULES

The following general rules shall be adhered to:

D.1.1 The message should only be sent to a specific destination, never to the global address or the unavailable address.

D.1.2 A tool or a device shall have set the key parameter in the memory access request message or the seed parameter in the memory access response message (as appropriate) to identify that the long seed/key is being used prior to transmission of the data security message by either. This enables the software in both to have a basis upon which to “flag” that the data security message is going to be used to provide long seed/key.

D.1.3 A tool or a device upon seeing either a memory access request message or a memory access response message identifying that the long seed/key is being used shall set the appropriate “flags” within the respective software such that they look for the data security message and process it.

## D.2 MESSAGE EXCHANGE RULES

The following outlines the procedure for using the data security message to send/receive long seed/key within a memory access sequence.

D.2.1 A device that has received a memory access request message (see memory access request [5.7.14] and response [5.7.15] documentation) for which it is going to send a seed using the data security message should set the seed parameter within the memory access response message to indicate that the seed will actually be sent using the Data_Security message. The data security message with the long seed should then be sent within 0.25 seconds of the memory access response message. (A tool may use a time-out of twice this value plus whatever bridge delays it has determined is appropriate for the particular system. This requires the tool to have identified the system prior to this operation.)

D.2.2 A tool that has received a Data_Security message containing a long seed from a device should send the long key of that long seed back to the device with the data security message within 0.25 seconds. (Note that bridges, when utilized, need to be accounted for in the time-out number. A device may use a time-out of twice this value plus whatever bridge delays it has determined is appropriate for the particular system. This, of course, requires the device to have identified the system prior to this operation.)

D.2.3 A device that has received a data security message containing a long key (presumably of a long seed it had previously sent) from a tool should verify the seed and then continue with the memory access as outlined in Appendix C of the memory access request and response operation. (The two cases of seed verified and failed are outlined separately. See Figures E9 and E10.)

APPENDIX E - MEMORY ACCESS INFORMATION

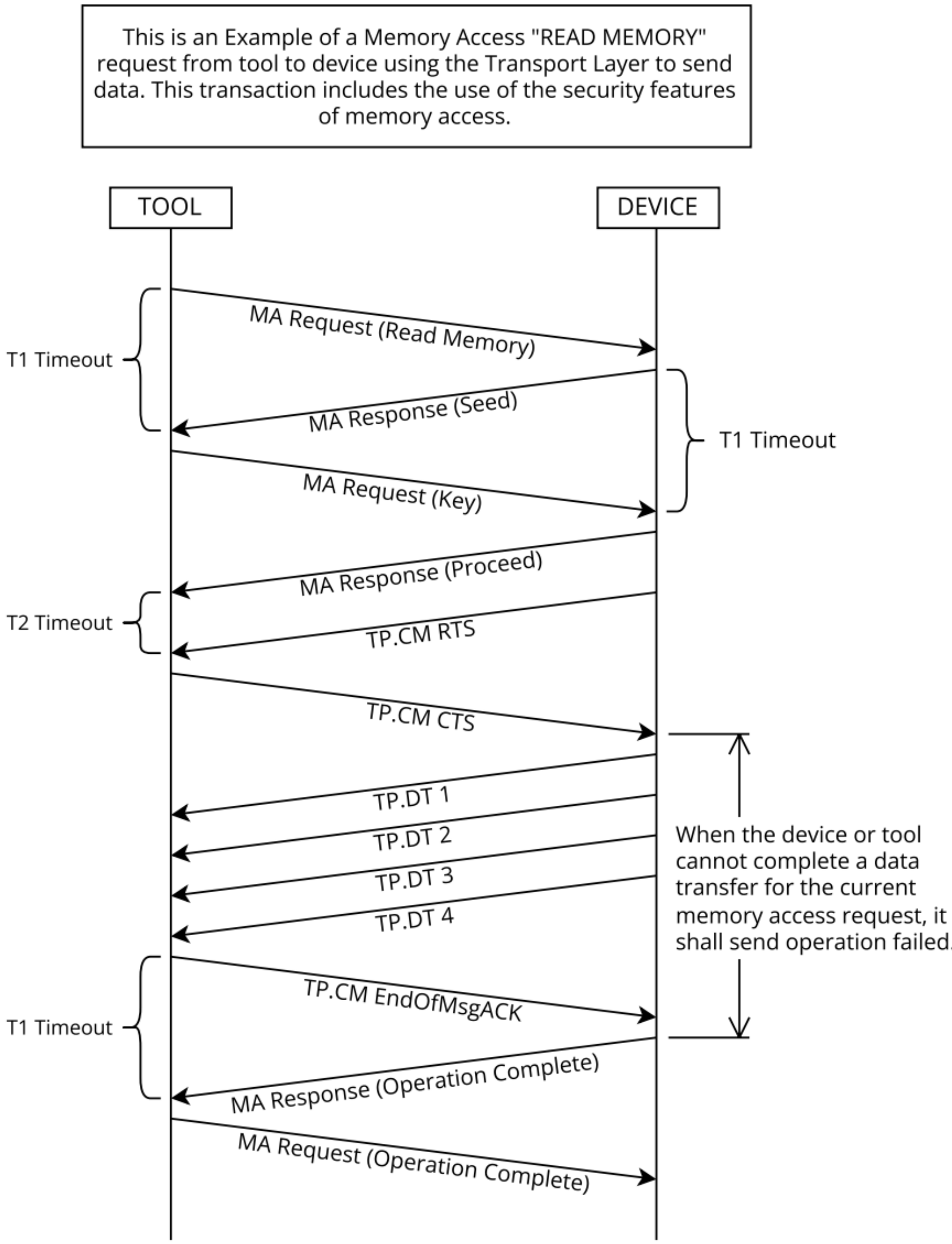


***Figure E1 - Example - message sequence to accomplish memory read operation with security (short form of security)***

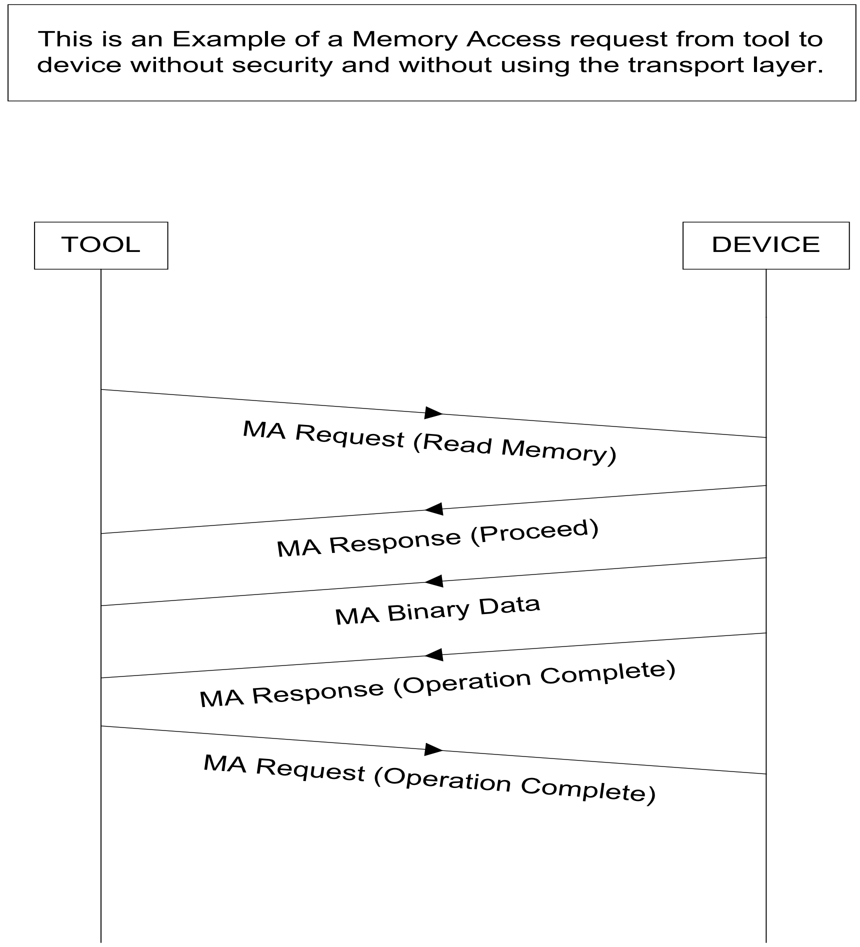


***Figure E2 - Example - message sequence to accomplish memory read operation without security***

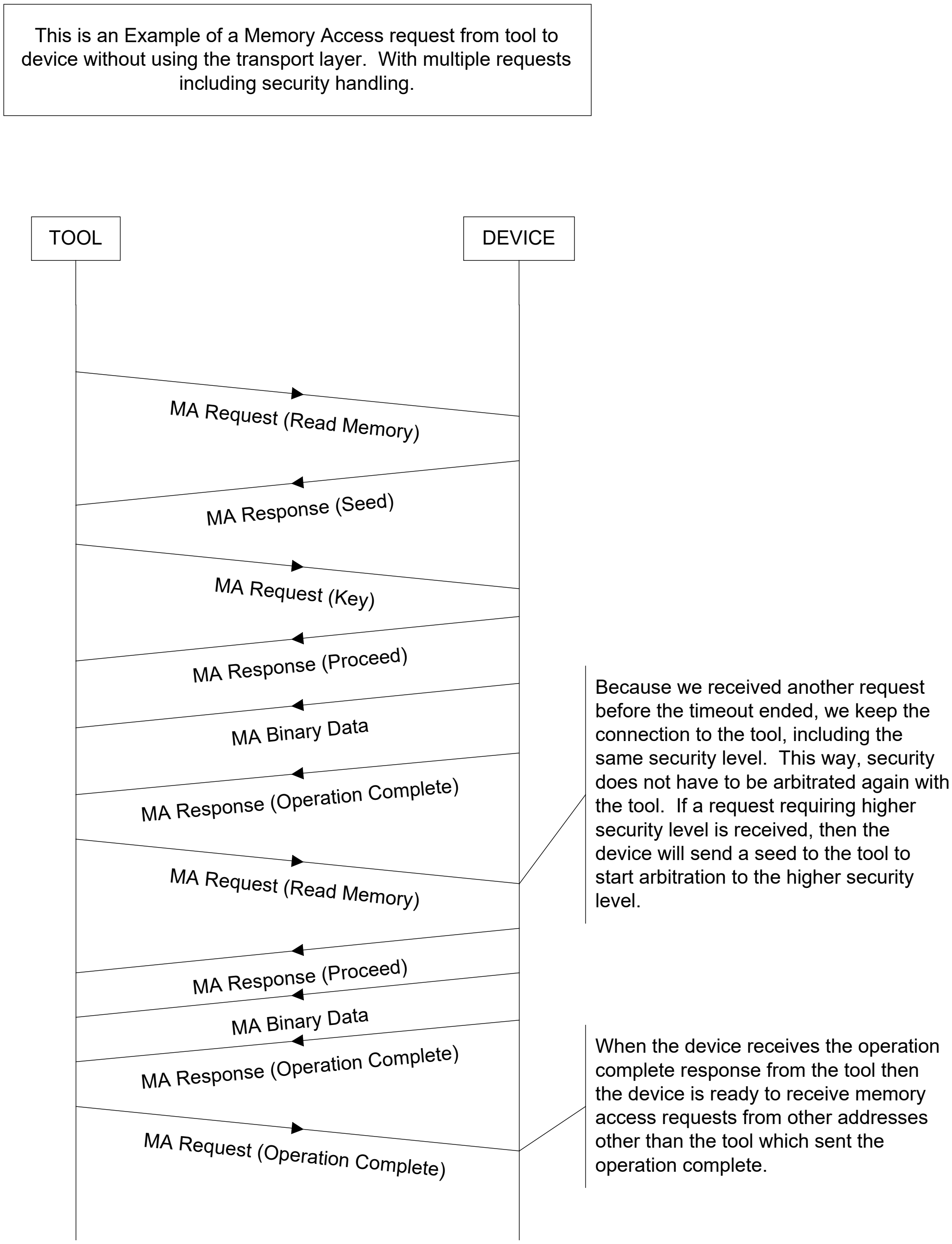


***Figure E3 - Example - message sequence to accomplish multiple memory read operation with security (short form of security)***

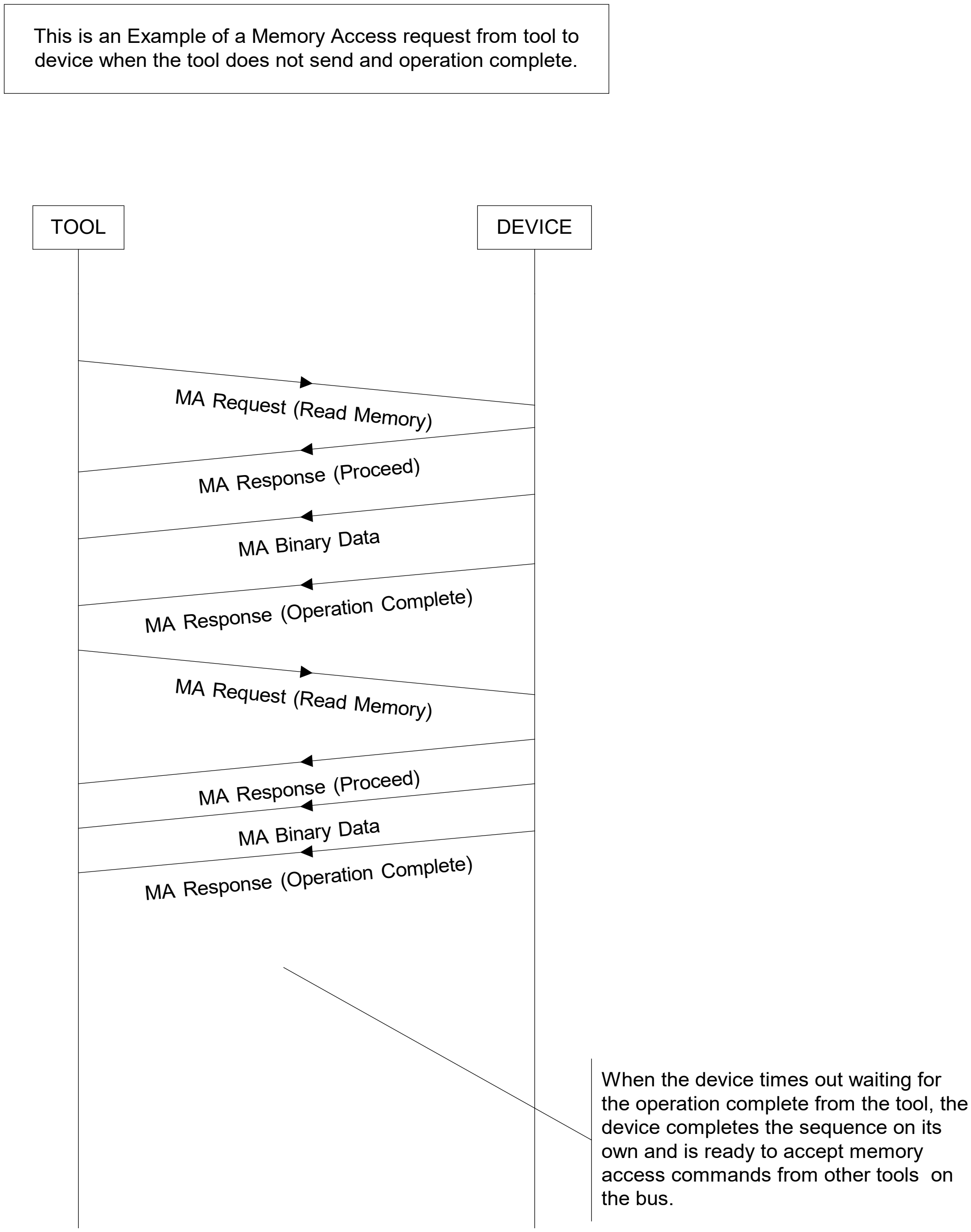


***Figure E4 - Example - tool does not send an operation complete to conclude the memory access session***

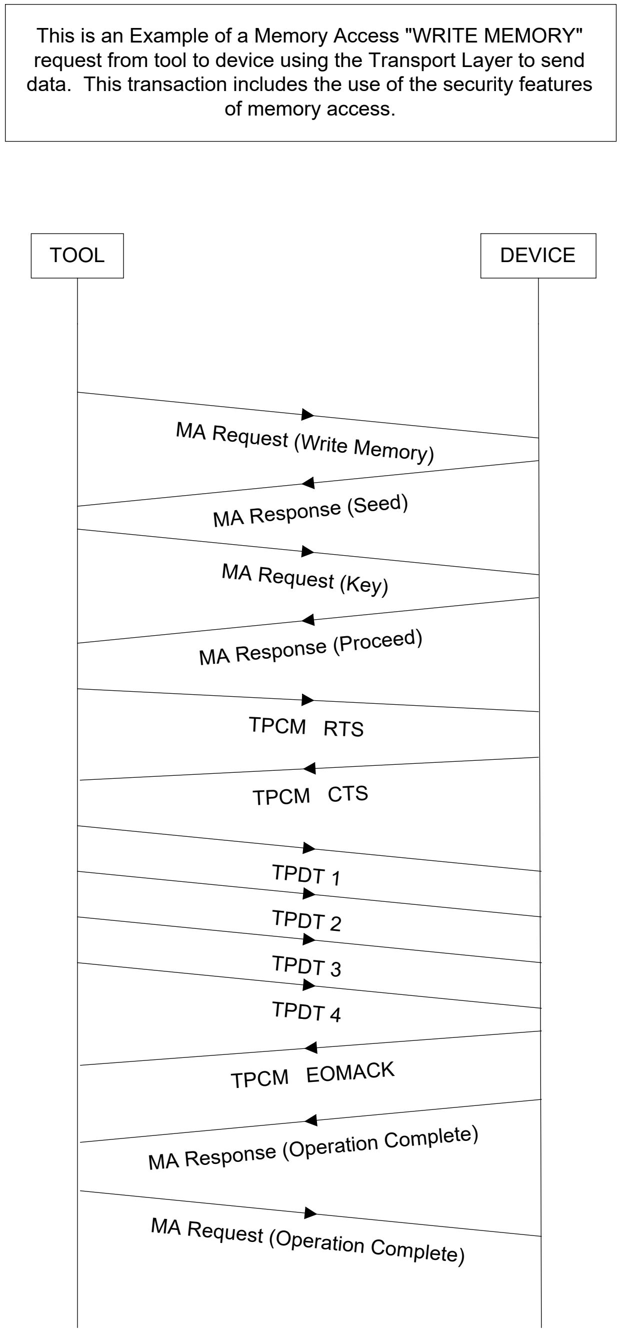


***Figure E5 - Example - write memory using transport protocol to send the data; also uses the short form of security***

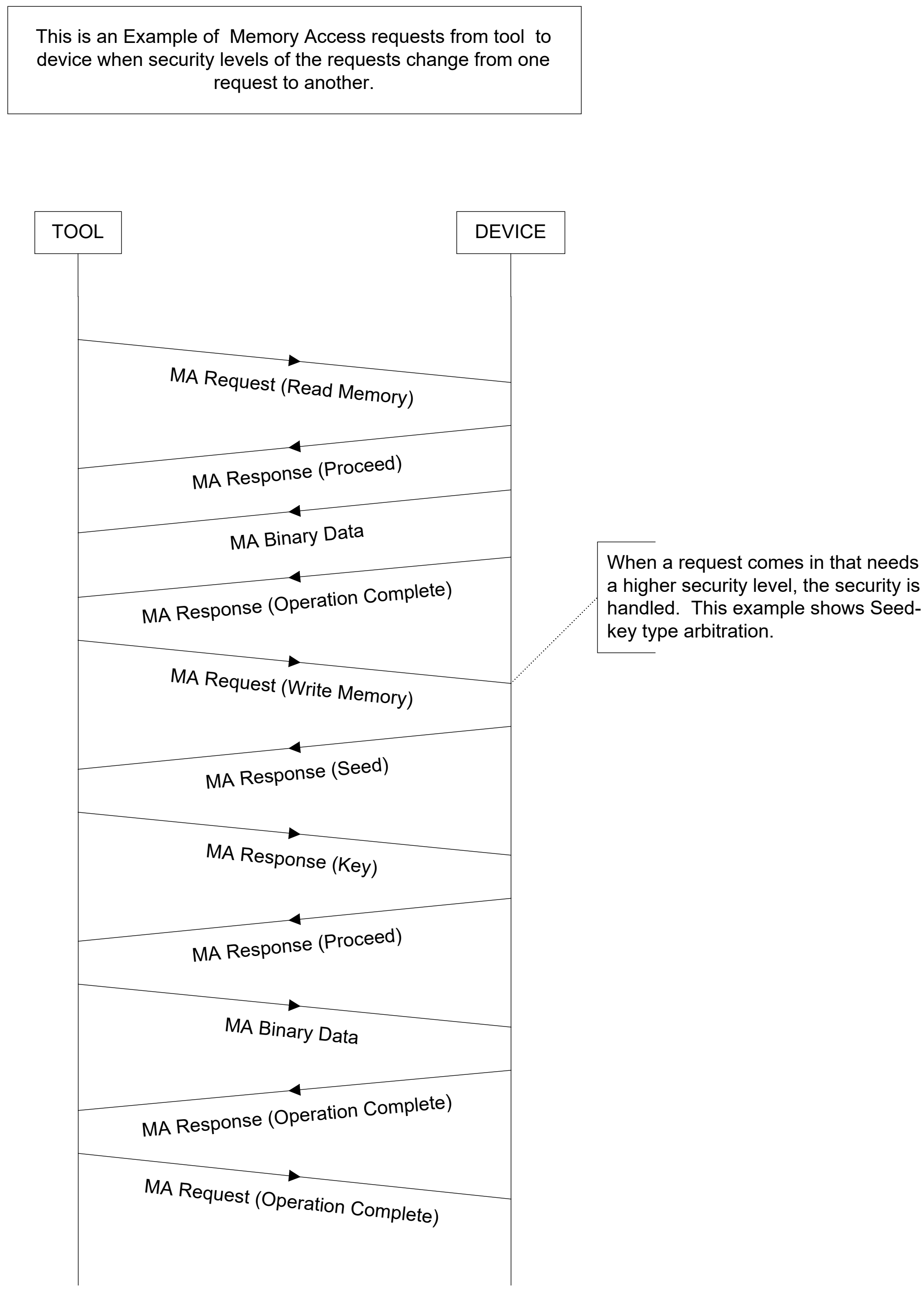


*Figure E6 - Example - memory access tool to device operations requiring different security levels*

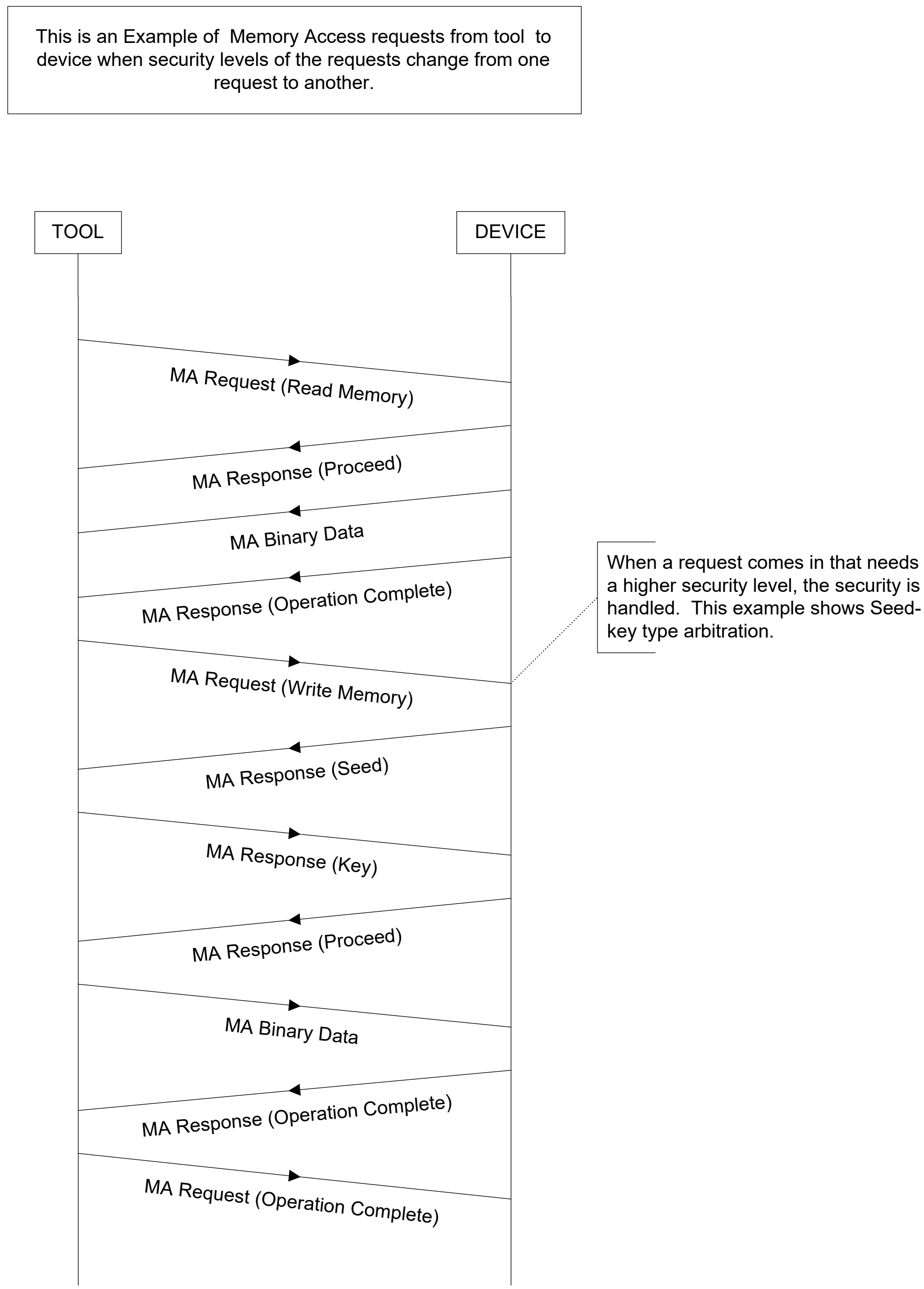


***Figure E7 - Example - memory access operation failed due to transport protocol session failure***

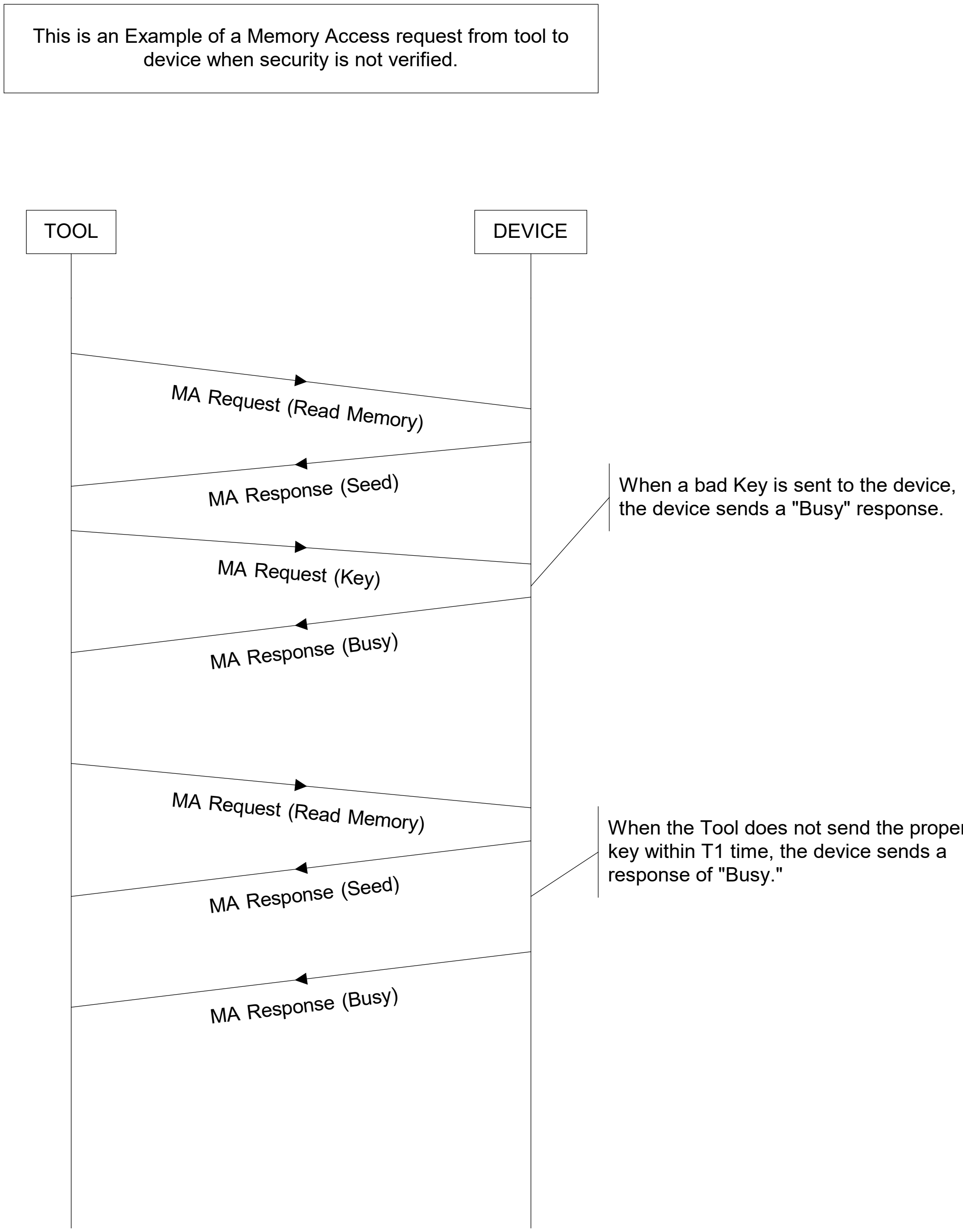


*Figure E8 - Example - memory access operation where security is not verified*

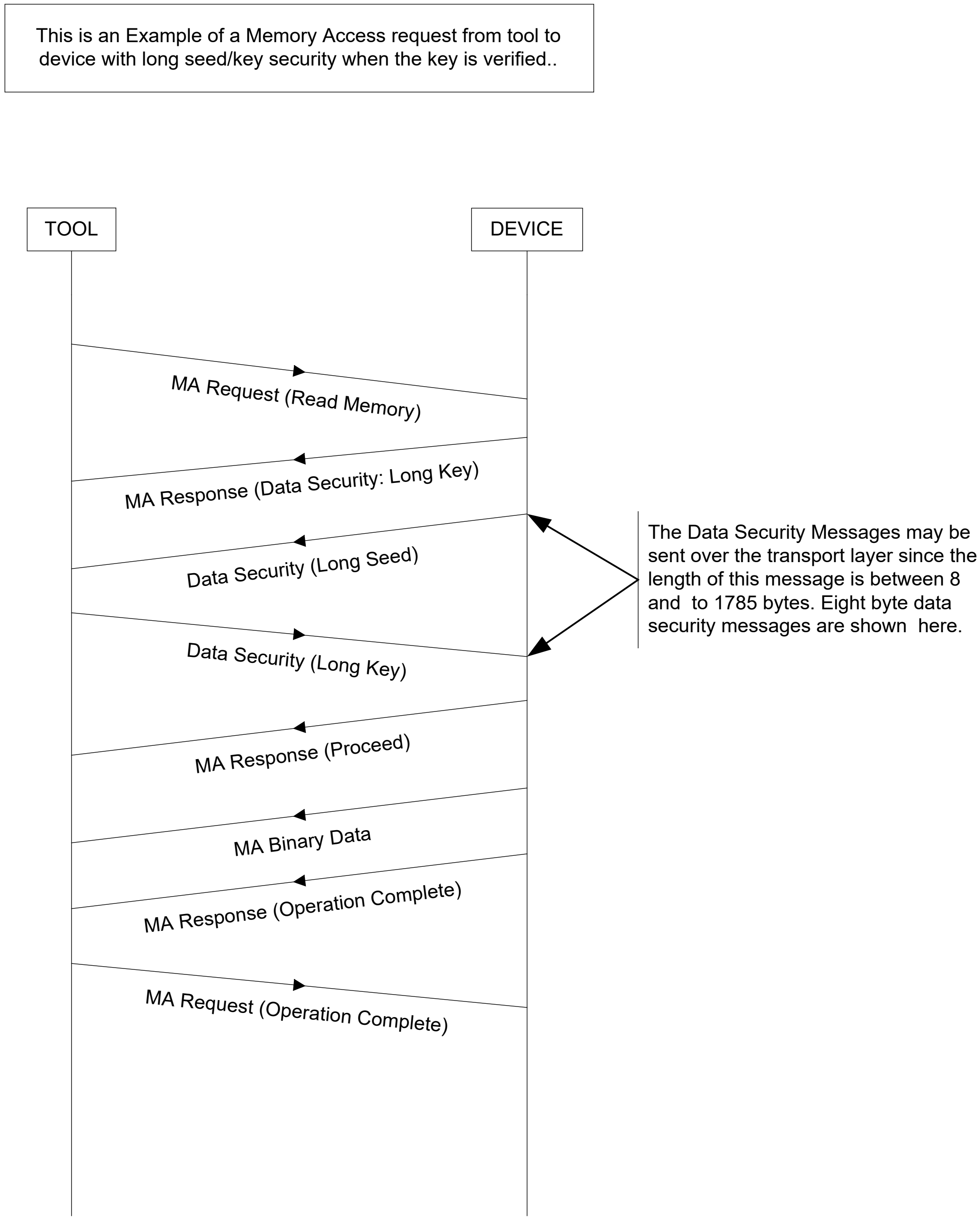


*Figure E9 - Example - memory access operation using the long seed and key*

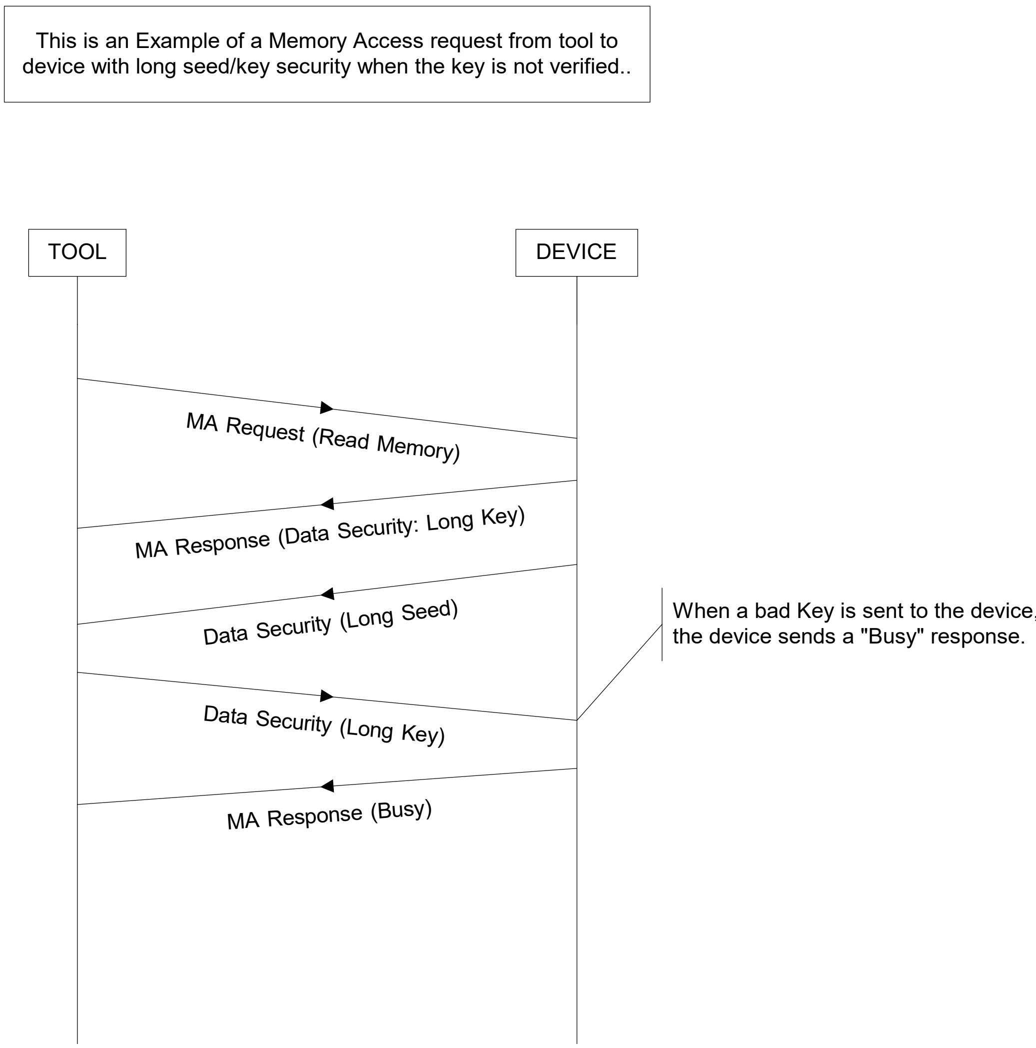


***Figure E10 - Example - tool does not send valid key so device responds with "busy"***

# APPENDIX F - RESTRICTIONS ASSOCIATED WITH PROPRIETARY SPNS

1. When suspect parameter number assignment is contemplated, SAE J1939-defined SPNs (SPNs 0 to 516095) should be considered along with proprietary SPNs (516096 [$7E000_h$] through 524287 [$7FFFF_h$]). If the information to be diagnosed is of general interest, then an SAE J1939-defined SPN (SPNs 0 to 516095) should be sought through the SAE J1939 Truck and Bus Control and Communications Network Committee. If the diagnosed information is emissions related, then an SAE J1939-defined SPN (SPNs 0 to 516095) shall be sought through the SAE J1939 Truck and Bus Control and Communications Network Committee.

2. The suspect parameter numbers for proprietary diagnostics shall not be used for communicating emissions-related diagnostics. If the diagnosed information is emissions related, then an SAE J1939-defined SPN (SPNs 0 to 516095) should be sought through the SAE J1939 Truck and Bus Control and Communications Network Committee.

3. The interpretation of the diagnostic trouble codes using proprietary SPNs varies by manufacturer. For example, even though two different implements may use the same proprietary SPN for their diagnostics, manufacturer “A’s” reported diagnostic using a proprietary SPN is more likely to be different from manufacturer “B’s” diagnostic using the same proprietary SPN. The interpretation of the diagnostic trouble codes with proprietary SPNs is dependent on the source address of the diagnostic message. The source address and its associated manufacturer ID from its SAE J1939 NAME should be used if a device is to translate proprietary SPN diagnostic trouble codes to manufacturer-specific text descriptions.

4. Generic scan tools, service tools, and other ECUs should be capable of presenting DTCs with proprietary SPNs in its numerical representation (i.e., SPN-FMI display format). However, this SAE Standard does not require the translation of these DTCs into any textual representation by generic scan tools, service tools, and other ECUs. Generic scan tools, service tools, and other ECUs are encouraged to present a generic phrase to the display, such as “Manufacturer Defined SPN-FMI. See Manufacturer Service Literature,” or interpret the proprietary SPN through use of source address and associated manufacturer code from the SAE J1939 NAME to determine applicable display DTC text when encountering DTCs with proprietary SPNs.

5. This SAE Standard imposes no restrictions upon performing textual translations of DTCs with proprietary SPNs by scan tools, service tools, and other ECUs designed by or designed for a specific manufacturer. Any devices that perform textual translations of DTCs with proprietary SPNs shall use the appropriate information, including source address and the associated SAE J1939 NAME, when performing these translations.

6. Each ECU manufacturer is responsible for the appropriate management of their assignments and usage of the proprietary diagnostic SPNs if these are utilized by their products.

7. The SAE J1939 Truck and Bus Control and Communications Network Committee does not have any intentions to expand the number of proprietary or manufacturer-specific SPNs beyond the current range of 8192 proprietary or manufacturer-specific SPNs.

# APPENDIX G - FAULT MANAGEMENT NARRATIVE

Appendix G provides examples that illustrate potential operating sequences for managing faults and managing which messages are used to communicate them. Section G.1 introduces Appendix G. Section G.2 reviews the active and previously active concepts for non-OBD-regulated systems and components. Section G.3 narrates OBD II concepts as defined by 13 CCR 1968.2 for engines and vehicles prior to the 2009 model year.

## G.1 FAULT MANAGEMENT NARRATIVE - INTRODUCTION

Section G.1 defines the scope of Appendix G and identifies the purpose of the models given in Figures G1 and G2.

### G.1.1 Fault Management Narrative Models

Figures G1 and G2 illustrate the relationships between diagnostic messages used to provide diagnostic conditions or diagnostic trouble codes (DTCs) from the vehicle’s electronic components to a diagnostic service tool. The figures focus upon defining when particular SAE J1939 services are used, based on when the diagnostic condition was detected. The figures “sort” DTCs, showing when DM1, DM2, DM6, DM12, and DM23 are used to convey the status of the condition. DM6 (pending), DM12 (confirmed, MIL-On), and DM23 (confirmed, MIL-Off) are required by governmental regulations of emissions-related, OBD-compliant components or systems. The governmental regulations related to 2004 to 2008 model year passenger cars identified by entry B in Table 2.

### G.1.2 Fault Narrative Model Limitations

These figures are not exhaustive. In Figure G2, many additional requirements defined or implied by 13 CCR 1968.2 are not modeled. For example, the effects of a diagnostic clear request (DM3/DM11) are not modeled. Requirements to store and manage freeze frame information (provided by DM24/DM25) when conditions are detected are also not discussed.

### G.1.3 Fault Management Narrative Organization

Figure G1 models fault reporting for components and systems that are not emissions related. Figure G2 models fault reporting for OBD-compliant components and systems meeting 13 CCR 1968.2. The figures show how the status of a fault or DTC is managed. The reporting requirements are then noted as semantic actions for individual states. The narrative for Figure G1 is given in G.2. Figure G2 is discussed in G.3.

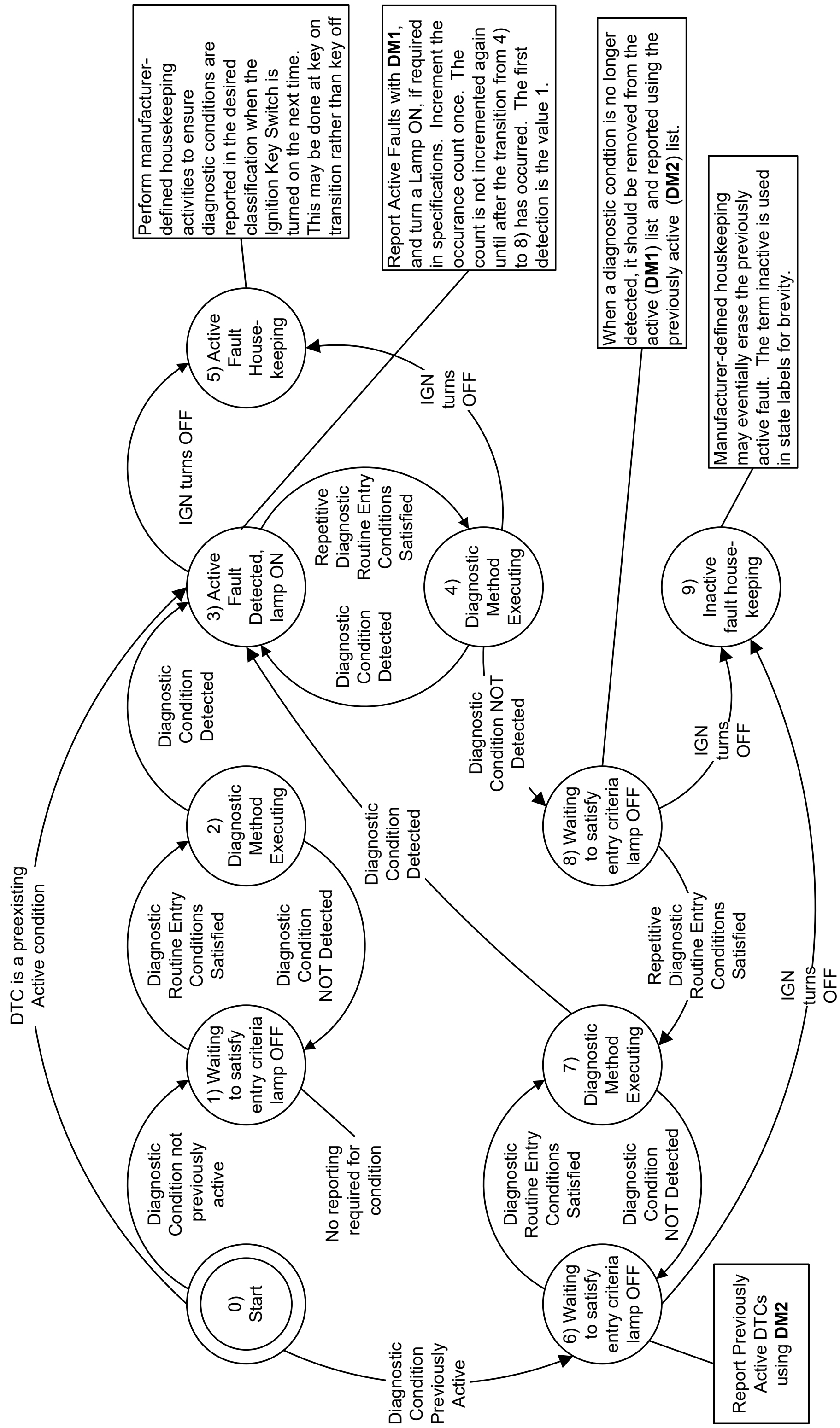


***Figure G1 - Diagnostic reporting services before HD OBD***

## G.2 FAULT MANAGEMENT NARRATIVE FOR NON-OBD-RELATED COMPONENTS/SYSTEMS

G.2.1 through G.2.4 discuss Figure G1. Together, they form a narrative for components that are not emissions related and have not been regulated.

### G.2.1 Fault Management Narrative for Non-OBD-Related Components Introduction

Figure G1 shows the relationship between DM1 and DM2 for components and systems that are not required to comply with OBD provisions. These components will not support DM6, DM12, and DM23. They will use DM1 to indicate active diagnostic conditions and DM2 to provide previously active diagnostic conditions. Figure G1 shows an abstract model of events. Diagnostic method details are not shown to focus upon illustrating the relationship between DM1 and DM2.

Figure G1 presents a state transition diagram that treats each fault or DTC as a separate token. By placing the set of DTCs supported by the system as tokens into state 0, the diagram sorts the DTCs, identifying the subsets of active and previously active DTCs. Figure G1 also discusses how the occurrence count is incremented, when the “DTC” is detected. State labels use the term “inactive” to describe the previously active faults reported by DM2 to better fit the label into the circles in Figure G1.

### G.2.2 Fault Management Narrative for Non-OBD-Related Components Active DTCs

The state transition diagram begins at state 0. In state 0, each DTC waits for the diagnostic process to begin. In Figure G1, the process uses a transition of the ignition key from off to on. A DTC may transition to one of three states from state 0. If the DTC was not detected previously, it transitions to state 1 and waits for its entry conditions to be satisfied. When entry conditions are satisfied, state 2 models its evaluation process.

After a decision is reached in state 2, the DTC transitions from state 2 back to state 1 or forward to state 3. State 3 captures the active faults for detected diagnostic conditions. Transitions into state 3 from state 2 or into state 3 from state 7 increment the occurrence count for the DTC. The occurrence count is not incremented when state 3 is entered from state 4. Depending upon the severity of the condition, a lamp can be lit when the condition has been detected. MIL-status will not be indicated as “on” for non-OBD-related components.

Diagnostics are usually evaluated iteratively. In Figure G1, this is modeled by the cycles from state 2 to state 1, from state 7 to state 6, and from state 4 back to state 3. These cycles maintain the initial sorting from state 0, keeping the active and previously active subsets separate from each other and the rest of the DTCs. Repeated evaluation of a diagnostic method, after it has been detected, can lead to the conclusion that the failure condition detected is no longer present. This is modeled by the transition from state 4 to state 8. During this transition, any lamp that was illuminated for the DTC is extinguished. G.2.3 discusses the previously active states in the model.

The DTC can be recognized as a previously existing active fault. In this case, the DTC transitions from state 0 to state 3. Upon a transition from state 0 to state 3, a “trouble lamp” may be turned on, but the occurrence count is not incremented.

### G.2.3 Fault Management Narrative for Non-OBD-Related Components Previously Active DTCs

The state transition diagram begins at state 0. In state 0, each DTC waits for the diagnostic process to begin. In Figure G1, the process uses a transition of the ignition key from off to on. A DTC can transition to one of three states from state 0. If the DTC was not detected previously, it transitions to state 1 and waits for its entry conditions to be satisfied.

Previously Active DTCs transition from state 0 to state 6 in Figure G1. Since they are not active, they do not require any lamp to illuminate. After the entry conditions are satisfied, the DTC token transitions from state 6 to state 7. The DTC transitions from state 7 back to state 6 or forward to state 3 after a diagnostic decision has been reached in state 7. State 3 captures the Active DTCs for detected conditions. G.2.2 discusses Figure G1 for active faults.

### G.2.4 Fault Management Narrative for Non-OBD-Related Components Housekeeping

Housekeeping across key-on/key-off cycles is modeled in states 5 and 9. The transition from state 0 to state 3 shows one effect of “housekeeping.” For this transition, the existence of a DTC was recalled from the prior key cycle, and is now provided using DM1. The transition from state 0 to state 3 does not increment the occurrence count.

Information can be saved to administer the transitions from state 0 to states 1, 3, and 6, depending upon the memory capabilities of the component. Clearly, components with only volatile memory (ROM and RAM memory) will not be capable of transitioning from state 0 to states 6 or 3. They will not provide occurrence counts across key cycles.

In this example, ignition key cycles are discussed as the defining trip events. Other endpoints are possible. For example, the endpoints may be defined by engine-start to engine start. Definitions in OBD regulations include engine start endpoints. The SAE J1939-73 standard does not require a specific endpoint definition for DM1 and DM2 for all vehicle components.

## G.3 SAE J1939-73 FAULT MANAGEMENT FOR 1968.2 NARRATIVE

G.3.1 and G.3.2 provide a narrative for Figure G2, which models the relationships among DM1, DM2, DM6, DM12, and DM23. Figure G.2 was modeled from the 2002 regulation text for 13 CCR 1968.2, which applies to model year 2004- to 2008-era passenger cars.

### G.3.1 SAE J1939-73 Fault Management for 1968.2 Narrative Introduction

Figure G2 shows a state transition diagram that illustrates the use of SAE J1939 diagnostic services for reporting diagnostic trouble codes (DTCs) under the requirements of 13 CCR 1968.2. The construction of DM12 and DM23 definitions require Figure G2 to illustrate the rules for MIL illumination. Since the regulation discusses fault recording in terms of drive cycles, Figure G2 shows the consequences of drive cycles on the diagnostics results and does not discuss ignition key state transitions.

Like Figure G1, Figure G2 sorts DTCs by treating them as tokens in the state transition diagram. The state transitions sort the DTC tokens into pending (DM6) and confirmed faults (DM12, DM23). Confirmed faults are defined in SAE J1939-73 to be further distinguished by whether they command the MIL to light. DM12 conveys confirmed faults that require the MIL to be on. DM23 conveys confirmed faults after the MIL is permitted to be turned off.

To appropriately discuss MIL illumination and fault deletion from the confirmed fault list, two variables are provided for each DTC token that the semantic actions of the state transition diagram maintain. The MIL countdown counter manages the three sequential “pass” results needed to turn the MIL-Off. MIL warm-up countdown tracks the 40 warm-up cycles required before a confirmed fault may be erased.

Reporting OBD DTCs using the existing active (DM1) and previously active (DM2) services can further enhance service. Figure G2 shows the linkage between pending and confirmed concepts for OBD and the active/previously active concepts that SAE J1939-73 originally provided. Through this linkage, the OBD faults can be reported in a way that is backward compatible with prior SAE J1939-73 versions. Thus, this construction creates two separate sets of services to report a DTC. Figure G2 shows how the sets interrelate.

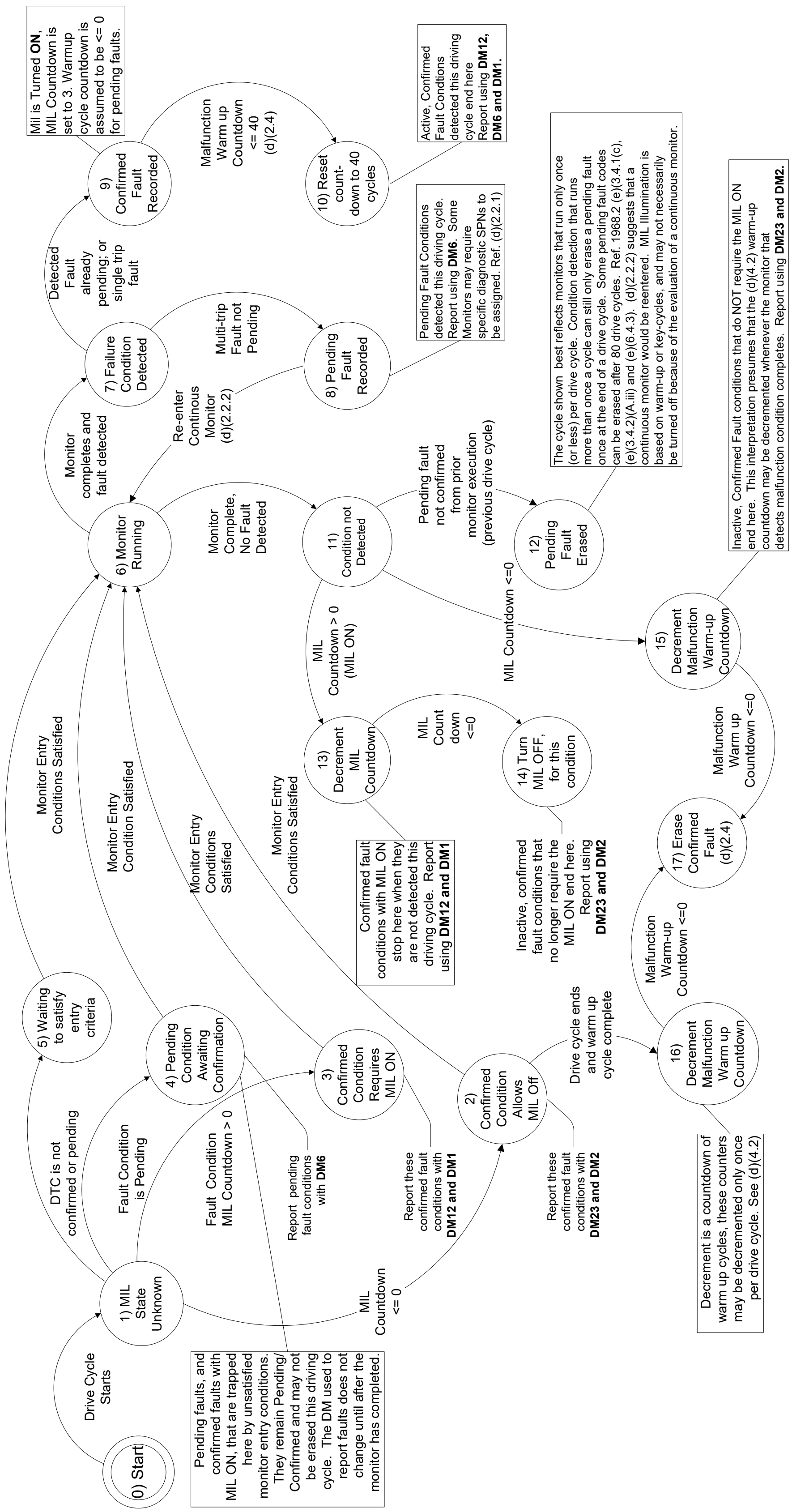


*Figure G2 - SAE J1939-73 fault management for 1968.2 [MY 2004-2008]*

G.3.2 SAE J1939-73 Fault Management for 1968.2 - Initial State

Like Figure G1, Figure G2 sorts DTCs by treating them as tokens in the state transition diagram. Each of the states on Figure G2 is numbered starting with the initial state, 0. Individual diagnostic conditions transition among the states.

All diagnostic conditions are treated as undetected, and untested, for the current drive cycle when the drive cycle starts. Then the diagnostic conditions or DTCs progress through the states based on past detection history and detection during the current drive cycle. Because the ignition key can be turned off at any time, nearly all states shown are practically final states as well. Some diagnostic methods may not meet their enable or entry criteria in a given drive cycle. Figure G2 models entry criteria concepts to show the consequences of drive cycles on the services used to report faults.

State 0 transitions to state 1 when a drive cycle starts (immediately after the engine starts). The transitions from state 1 determine whether the MIL shall be illuminated from a previously reported, confirmed fault code (shown in state 3) or will remain unlit (after the bulb check). DTCs that have an MIL countdown greater than 0 driving cycles progress to state 3 and cause the MIL to illuminate. Refer to 13 CCR 1968.2 (d)(2.3). All other DTCs progress to states 2, 4, or 5, depending on whether they are recorded as confirmed (state 2), pending (state 4), or not pending or confirmed (state 5).

G.3.3 SAE J1939-73 Fault Management for 1968.2 Narrative - Detected Conditions

Some diagnostics algorithms require specific operating conditions before they can be evaluated. Continuous diagnostics monitor DTCs transition to state 6 from states 2, 3, 4, and 5 without delay. Monitors with entry conditions wait until their entry conditions are satisfied. When the monitor completes, it makes a single decision whether the fault condition (DTC) is detected or not detected. Detected faults transition to state 7. Faults that were not detected transition to state 11.

A detected condition can be previously pending. This is modeled by the transition from state 7 to state 9. Detection of a pending fault makes it confirmed. The MIL must be illuminated. Here, a countdown counter is shown being set to 3 to accommodate the three subsequent trip illuminations required of confirmed faults. Refer to 13 CCR 1968.2 (d)(2.3). The transition from state 9 to state 10 ensures that the warm-up cycle countdown to erase the confirmed fault is set to 40 warm-up cycles after the MIL is turned on. Refer to 13 CCR 1968.2 (d)(2.4). Fault conditions that dwell in state 10 are reported as confirmed faults using DM12. They are also reported as pending faults using DM6 because they were detected during the current drive cycle. Refer to 13 CCR 1968.2 (d)(2.2.1). Finally, they shall be reported as active faults using DM1.

The transition from state 7 to state 8 illustrates the requirements for detected conditions that were not previously pending. When pending faults are recorded (refer to 13 CCR 1968.2 [d][2.2.1]), they are reported with DM6. They can become confirmed with an iterative evaluation of the test method shown by the transition from state 8 to state 6. Refer to 13 CCR 1968.2 (d)(2.2.2). Regardless, they transition from state 1 to state 4 after the next cycle restarts the model. Refer to 13 CCR 1968.2 (d)(2).

G.3.4 SAE J1939-73 Fault Management for 1968.2 Narrative - Undetected Conditions

State 11 begins the processes for diagnostic conditions that were not detected this driving cycle. State 12 illustrates the erasure of pending faults when they are not detected in the succeeding drive cycle. State 13 signals the illumination of the MIL. DTCs remaining in state 13 require the MIL to be illuminated. The transition from state 13 to state 14 signals that the MIL may be extinguished for the condition. Conditions in state 13 are reported using DM12 and DM1 because they are commanding the MIL-On. Conditions in State 14 are reported with DM23 and DM2.

State 15 begins the process for erasing confirmed fault conditions. If a diagnostic condition no longer commands the MIL-On and passes its diagnostics, it begins the process of counting down its cycles to erasure, shown in state 17. The warm-up cycle countdown permitted in 13 CCR 1968.2 (d)(4.2) does not require the diagnostic to have completed as long as a qualified warm-up cycle was achieved. This is shown by the transitions from state 2 to state 16 and from state 16 to state 17.

Transitions from states 10, 12, 13, 14, 15, 16, and 17 back to state 6 for continuous diagnostics are not shown. They do not improve the illustration of the relationships of SAE J1939-73 DM1, DM2, DM6, DM12, and DM23 services. The explicit modeling of cycling would have to also include provisions to ensure that counters based on warm-up cycles are not decremented more than once a warm-up cycle.

## APPENDIX H - DIAGNOSTIC TROUBLE CODE MESSAGE USAGE AND ASSOCIATION

This appendix provides examples that illustrate potential operating sequences for managing faults and managing which messages are used to communicate them. Figure H1 shows a possible flow and progression from first detection to an OBD DTC being logged. Figures H2 through H5 display signal charts illustrating the mutual support for DTC displays for HD OBD and Euro VI regulations. Figures H4 and H5 were drawn based on Euro VI as given in UN ECE R49-05 Annex 9B and remain correct for UN/ECE R49-07. Figure 2 of SAE J1939-84 provides a similar signal chart for the test process used for HD OBD certified engines. Section H.1 discusses distributed system effects for HD OBD implementations.

As can be seen, SAE J1939-84 uses two distinct faults to exercise many of the messages defined in SAE J1939-73 for HD OBDcompliant engines. In addition to the status of DTCs in DM1, DM2, DM12, DM23, and DM28, Section 6 of SAE J1939-84 evaluates required data displays in DM5, DM7, DM11, DM19, DM20, DM21, DM22, DM24, DM25, DM29, DM30, DM33, and other DM messages. Evaluation criteria in many SAE J1939-84 tests rely upon the directives given in the notes for DM messages.

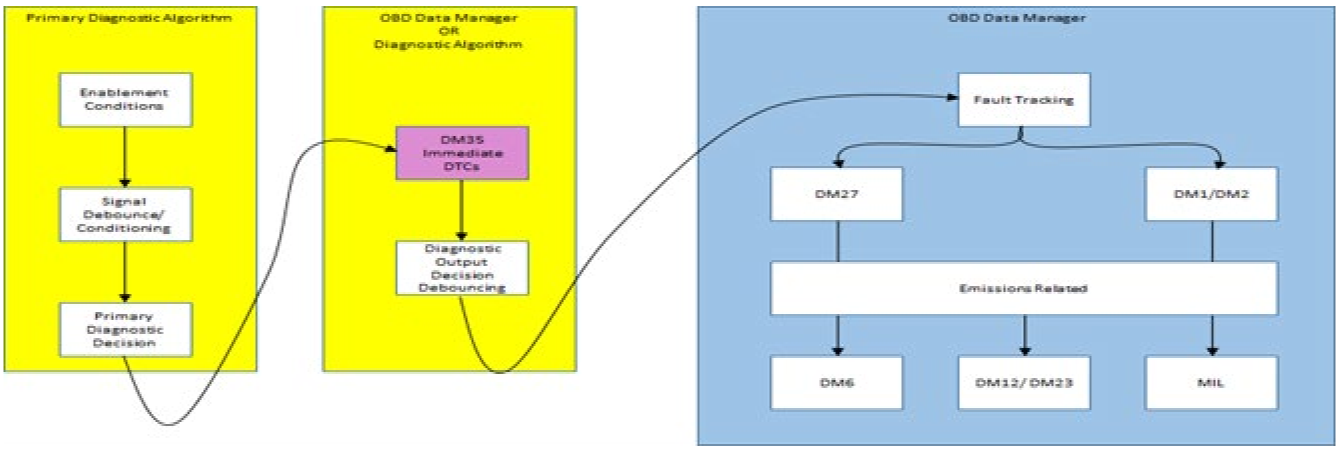


*Figure H1 - Malfunction detection to confirmed DTC*

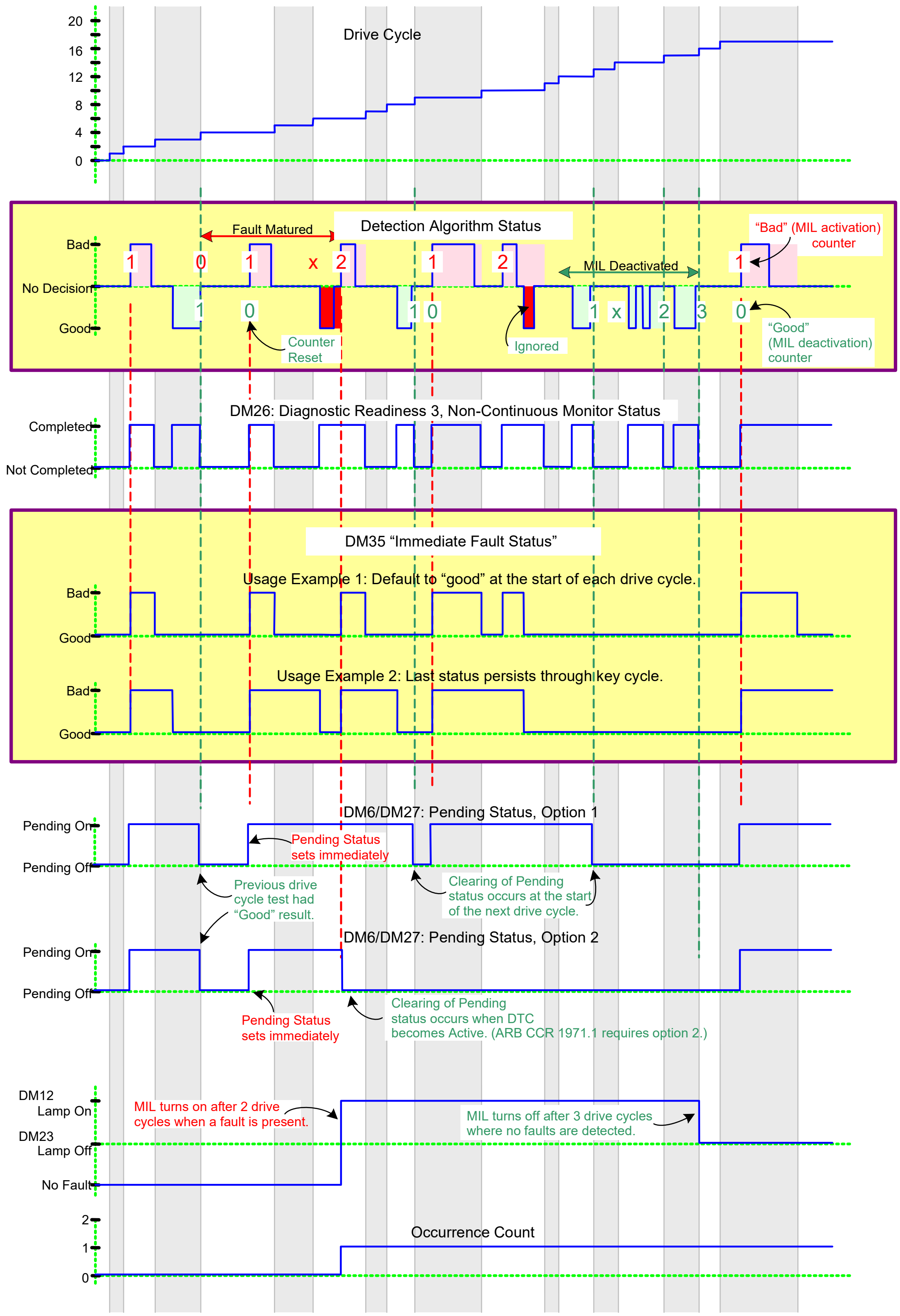


***Figure H2 - OBD drive cycle logic***

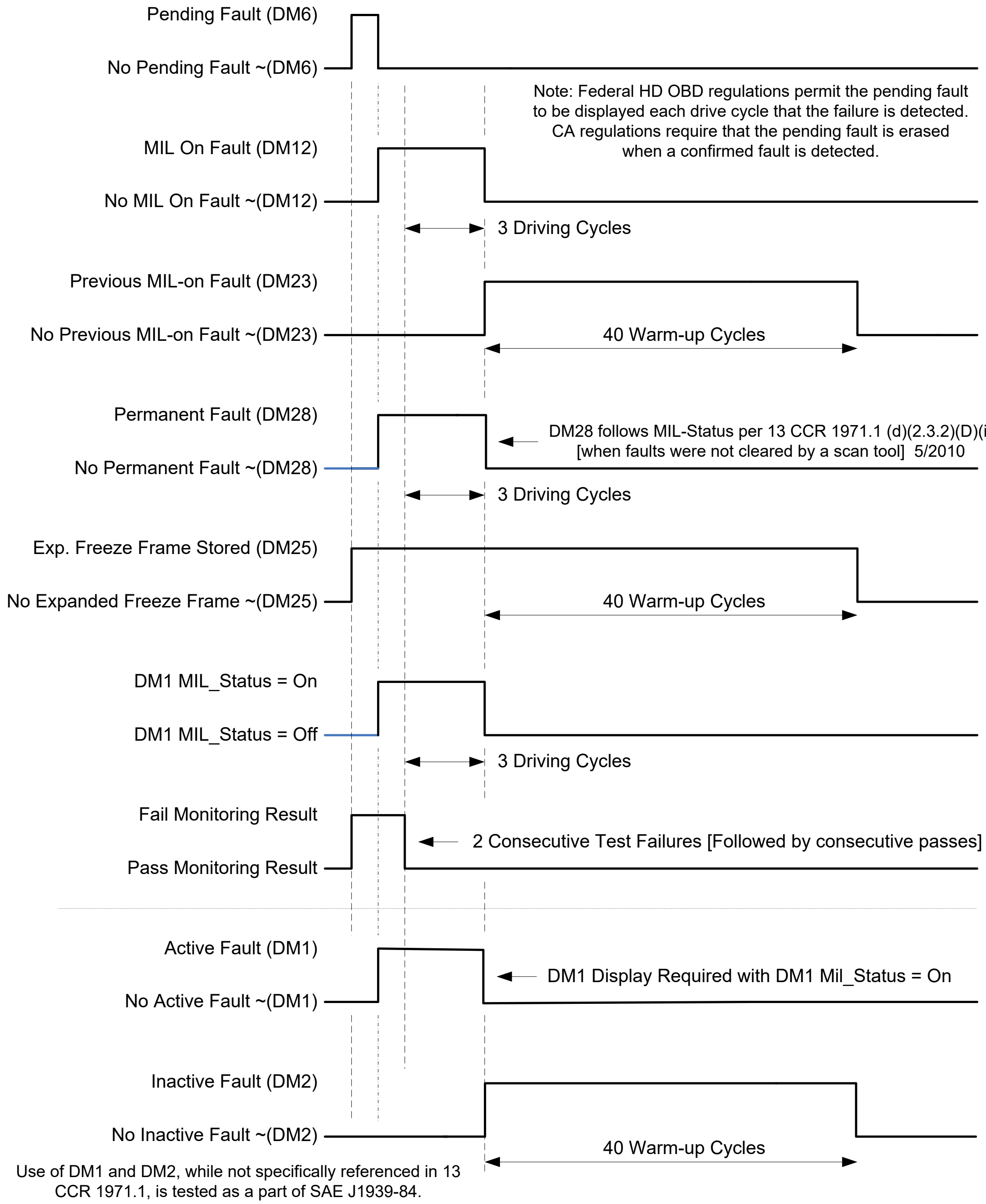


*Figure H3 - 2013 ARB 13 CCR 1971.1 SAE J1939 diagnostic message signal chart*

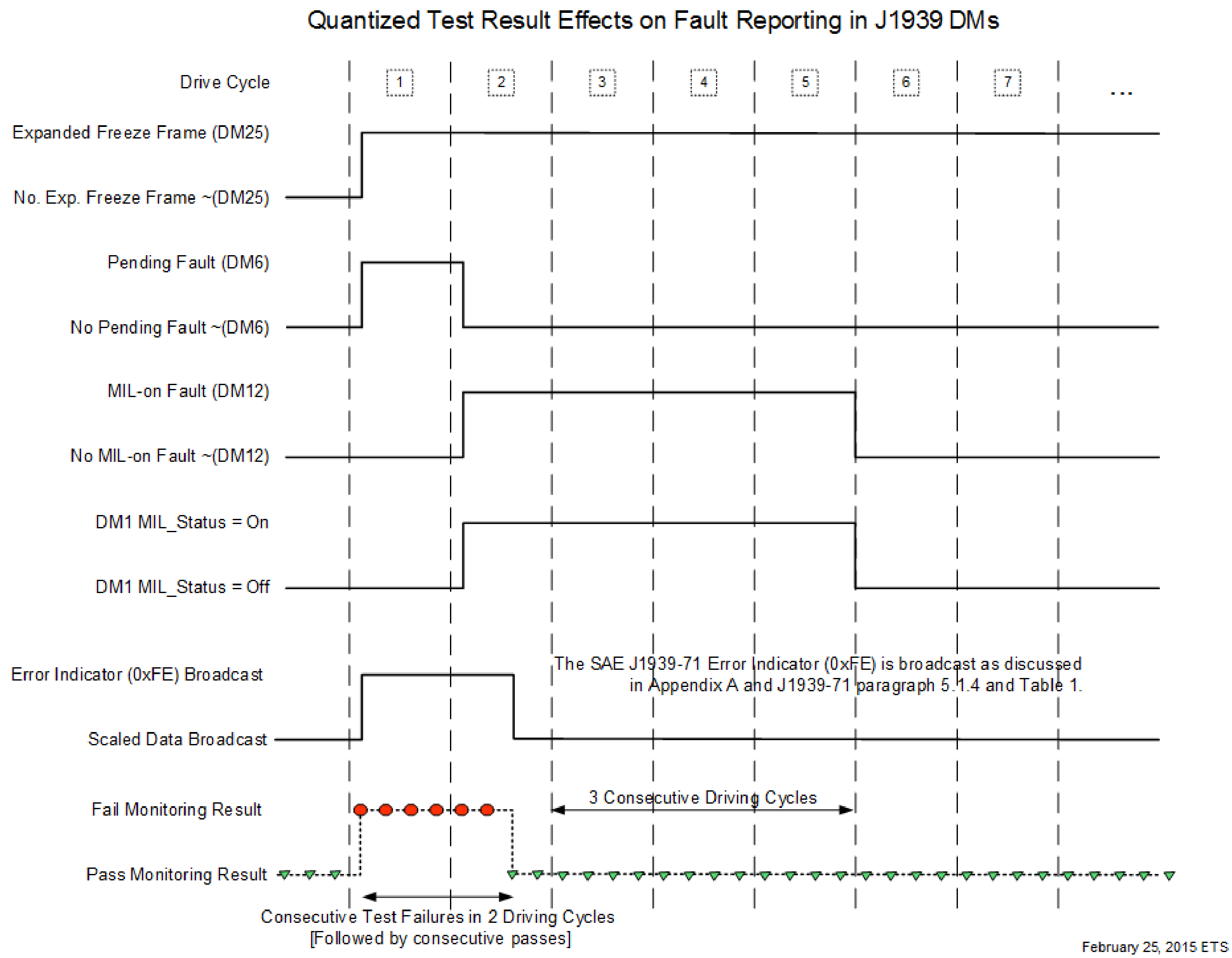


*Figure H4 - Quantized test result effects on fault reporting in SAE J1939 DM(s)*

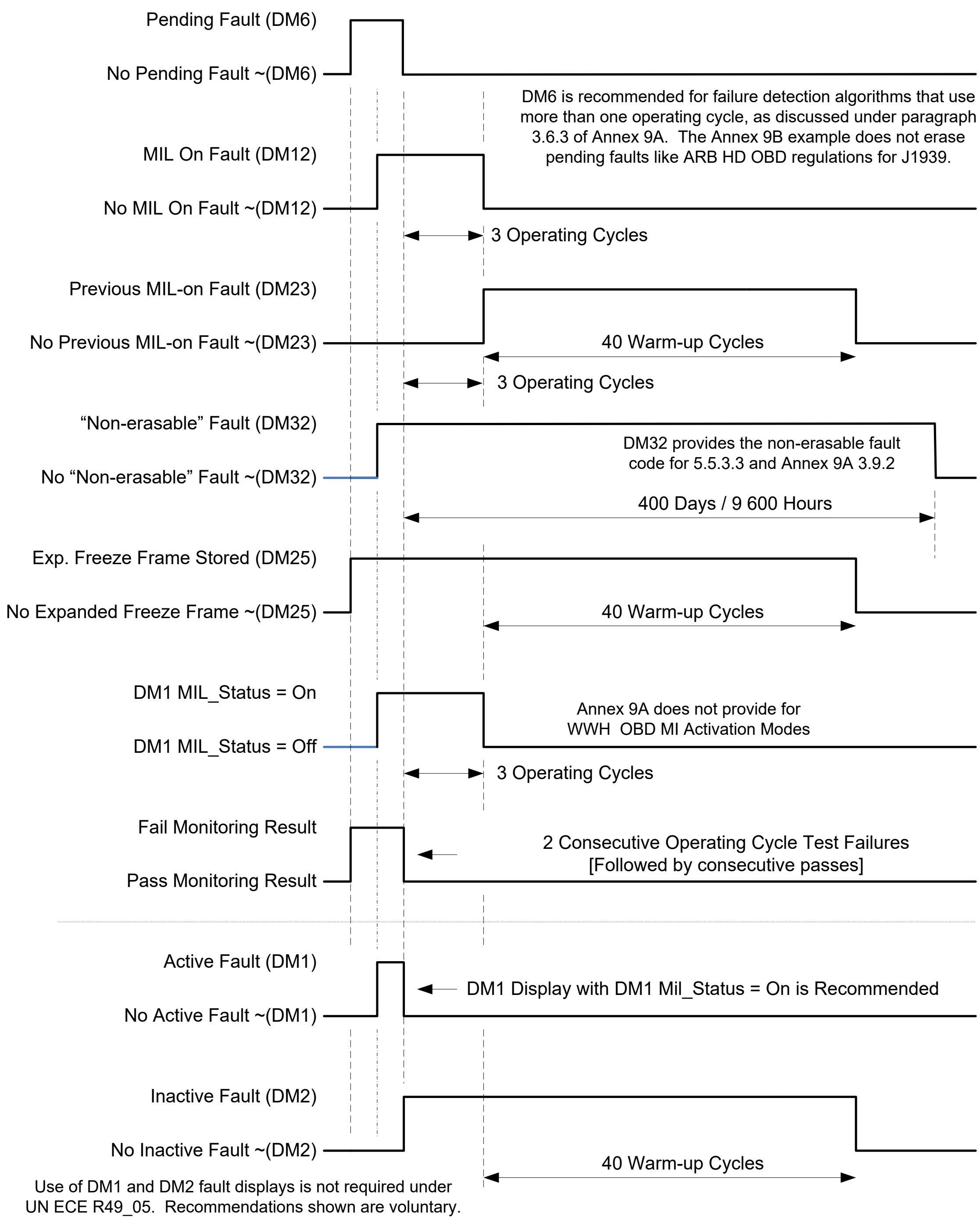


***Figure H5 - UN/ECE R49-05 SAE J1939 diagnostic message signal chart - two trip example***

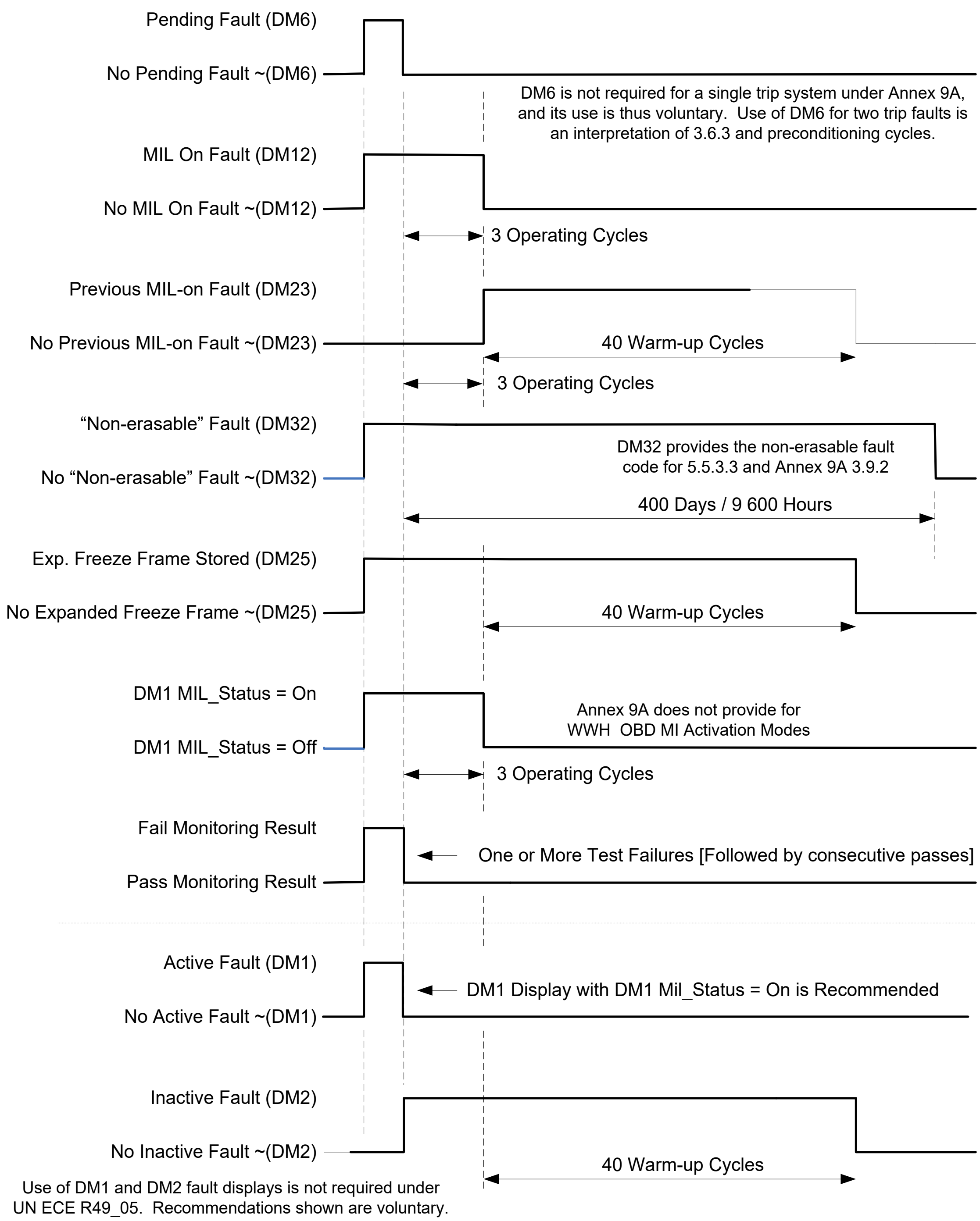


***Figure H6 - UN/ECE R49-05 SAE J1939 diagnostic message signal chart - single trip example***

## H.1 DISTRIBUTED SYSTEM EFFECTS

This section discusses the effects from distributed allocation of HD OBD functional elements across multiple devices. A monolithic implementation has but one HD OBD device. Distributed implementations have more than one device. Three categories of devices are defined here:

| | |
|---|---|
| Primary OBD Device | Device that provides the majority of support for SAE J1939-73 diagnostic messages. |
| Secondary OBD Device | Device that provides a minority of support for SAE J1939-73 diagnostic messages. |
| SAE J1939 Contributor | Devices that contribute data stream parameters relative to sensors and actuators. |

**Tabular DM Support**

The support provided by each of the three categories of devices that were defined above is shown for each message required by an HD OBD system. The table “DM Support Key” below defines most of the support comments. Some parameters or messages are only allowed from a single device (e.g., VIN).

| Tabular DM Support Key | |
|---|---|
| Yes | Yes, support is required. There may be cases where it is not from the applicable device or contributor. |
| May | Possible to come from this device. |
| No | No, support is not typical and may require agency approval. |
| Option | Optional; OBD does not require this support. |

***Table H1 - Message support by OBD-compliant devices and SAE J1939 network contributors***

| Row # | Function | PGN | Acronym | SPN # | Primary OBD Device | Secondary OBD Device | SAE J1939 Contributor | Description |
|---|---|---|---|---|---|---|---|---|
| Diagnostic Services | | | | | | | | |
| | Read DTCs and Lamps (MIL, RSL, AWL, Protect) | 65226 | DM1 | | Yes - MIL Status | Yes - MIL Status | No - for OBD DTCs and MIL_Status | All active DTCs and some lamps (MIL, RSL, AWL, Protect) |
| 2 | Read DTCs | 65236 | DM12 | | Yes | May | No | Emission-related MIL-On diagnostic trouble codes |
| 3 | Read DTCs | 64949 | DM23 | | Yes | May | No | Emission-related previously MIL-On diagnostic trouble codes |
| 4 | Read Pending DTCs | 65231 | DM6 | | Yes | May | No | Emission-related pending DTCs |
| 5 | Read All Pending DTCs | 64898 | DM27 | | Option | Option | No | All pending DTCs inclusive of emissions and non-emissions related |
| 6 | Read Permanent DTCs | 64896 | DM28 | | Yes | May | No | Emission-related permanent DTCs |
| 7 | Read DTC Counts | 40448 | DM29 | | Yes | May | No | Number of regulated DTC counts (pending, permanent, MIL-On, PMIL-On) |
| 8 | Read DTCs-Immediate | 40704 | DM35 | | Option | Option | No | Instantaneous status of diagnostic results |
| 9 | Command Test | 58112 | DM7 | | Yes | May | No - for OBD DTCs | Commanded test; system, device, or component |
| 10 | Scaled Test Results | 41984 | DM30 | | Yes | May | No - for OBD DTCs | Test results scaled |
| 11 | Clear DTCs | 65235 | DM11 | | Yes | Yes | Yes | Clear diagnostic information and active DTCs |

| Row # | Function | PGN | Acronym | SPN # | Primary OBD Device | Secondary OBD Device | SAE J1939 Contributor | Description |
|---|---|---|---|---|---|---|---|---|
| 12 | Clear DTCs | 65228 | DM3 | | No for ARB OBD | No for ARB OBD | No for ARB OBD | Clear diagnostic information and previously active DTCs |
| 13 | SPN Support | 64950 | DM24 | | Yes | Yes | No | SPN support for data stream and expanded freeze frame |
| 14 | Expanded Freeze Frame | 64951 | DM25 | | Yes | May | No | Expanded freeze frame (format allows DTC and manufacturer-specified number parameters) |
| 15 | Diagnostic Readiness | 65230 | DM5 | | Yes | May | No | OBD compliance |
| 16 | Diagnostic Readiness | 65230 | DM5 | | Yes | May | No | Monitors supported and their status (diagnostic readiness) |
| 17 | Diagnostic Readiness for This Trip | 64952 | DM26 | | Yes | May | No | Monitors supported and their status for this trip |
| 18 | Monitor Performance Ratio | 49664 | DM20 | | Yes | May | No | Indicates how often monitors complete compared to vehicle operation |
| 19 | Emission Increasing-AECD Active Time | 41216 | DM33 | | Yes | May | No | Engine emissions increasing AECDs and associated timers (not required HD EPA) |
| 20 | NTE Status | 40960 | DM34 | | Yes | May | No | Engine emissions not-to-exceed status |
| 21 | Rationality Fault SP Data | 64475 | DM58 | | Yes | May | No | SP data value for an SP with a rationality diagnostics fault |
| | | | | | | | | |
| Communication Services | | | | | | | | |
| 22 | Communication | 59904 | RQST | | Yes | Yes | Yes - Not supported by single PGN sensors | Request PG (Devices that do not support RQST will not support any SAE J1939-73-defined DM or request for any SAE J1939DA PGN/SPN) |
| 23 | Communication | 59392 | ACKM | | Yes | Yes | Yes - Not supported by single PGN sensors | Acknowledgment message (Same PGN is used for ACK and NACK. NACK includes reason codes, e.g., "busy.") |
| 24 | Communication | 60416 | TP.CMxx | | Yes | Yes | May | Transport protocol connection management (Manages TP.DT for destination-specific requests) |
| 25 | Communication | 60160 | TP.DT | | Yes | Yes | May | Transport protocol data transfer (DA = 225 is global, DA < 254 is point to point) |
| *26* | *Communication* | *60928* | *AC* | *2848* | *Yes* | *Yes* | *Yes - Not supported by single PGN sensors* | *NAME of controller application (Function [byte 6])* |
| Data Stream | | | | | | | | |
| 27 | Data Stream | 65260 | VI | 237 | May | May | May | Vehicle identification number (VIN)[1] |
| *28* | *Data Stream* | *65259* | *CI* | *588* | Yes | May | May | *Engine serial number (ESN)* |
| 29 | Data Stream | 54016 | DM19 | 1635<br>1634 | Yes | May | May - Not for compliance to regulation | Calibration identification and calibration verification number (CAL ID and CVN) |
| 30 | Data Stream | | DM56 | | Option | Option | No | Model year and certification engine family name |
| 31 | Data Stream | 49408 | DM21 | 3069 | Yes | May | No | Distance traveled while MIL is activated |

| Row # | Function | PGN | Acronym | SPN # | Primary OBD Device | Secondary OBD Device | SAE J1939 Contributor | Description |
|---|---|---|---|---|---|---|---|---|
| 32 | Data Stream | 49408 | DM21 | 3294 | Yes | May | No | Distance since diagnostic trouble codes cleared |
| 33 | Data Stream | 49408 | DM21 | 3295 | Yes | May | No | Minutes run by engine while MIL is activated |
| 34 | Data Stream | 49408 | DM21 | 3296 | Yes | May | No | Time since diagnostic trouble codes cleared |
| 35 | Data Stream | 64952 | DM26 | 3301 | Yes | May | No | Time since engine start |
| 36 | Data Stream | 64952 | DM26 | 3302 | Yes | May | No | Number of warm-ups since diagnostic trouble codes cleared |
| 37 | Data Stream | 65262 | ETI | 110 | Yes | May | No | Engine coolant temperature |
| 38 | Data Stream | 65265 | CCVS | 84 | Yes | May | May | Wheel-based vehicle speed |
| 39 | Data Stream | 65270 | IC1 | 102 | Yes | May | No | Engine intake manifold 1 pressure |
| 40 | Data Stream | 65270 | IC1 | 105 | Yes | May | No | Engine intake manifold 1 temperature |
| 41 | Data Stream | 61443 | EEC2 | 91 | Yes | May | May | Accelerator pedal position 1 |
| 42 | Data Stream | 61443 | EEC2 | 92 | Yes | May | No | Engine percent load at current speed |
| 43 | Data Stream | 61444 | EEC1 | 513 | Yes | May | No | Actual engine - percent torque |
| 44 | Data Stream | 61444 | EEC1 | 190 | Yes | May | No | Engine speed |
| 45 | Data Stream | 61444 | EEC1 | 899 | Yes | May | No | Engine torque mode |
| 46 | Data Stream | 65159 | IT6 | 1436 | Yes | May | No | Engine actual ignition timing |
| 47 | Data Stream | 65214 | EEC4 | 189 | Yes | No | No | Engine Rated Speed |
| 48 | Data Stream | 64711 | DM56 | 5845 | Yes | No | No | Certification Engine Family Name |
| 49 | Data Stream | 64841 | O2FT1 | 4236 | Yes | No | No | Short Term Fuel Trim - Bank 1 |
| 50 | Data Stream | 61693 | A2SCRDSR3 | 8442 | Yes | May | No | Aftertreatment 1 Diesel Exhaust Fluid Doser Valve 1 Command |
| 51 | Data Stream | 64488 | ASI2 | 12749 | Yes | May | No | Aftertreatment 1 DEF Dosing Mode |
| 52 | Data Stream | 65031 | ASI6 | 22719 | Yes | May | No | Aftertreatment 2 DEF Dosing Mode |
| 53 | Data Stream | 61475 | A1SCRDSI1 | 4331 | Yes | May | No | Aftertreatment 1 Diesel Exhaust Fluid Actual Dosing Quantity |
| 54 | Data Stream | 64488 | ASI2 | 12748 | Yes | May | No | DEF Dosing Mode |
| 55 | Data Stream | 64250 | ATNXMS1 | 12753 | Yes | May | No | Aftertreatment 1 SCR 1 Target NH3 Storage Level |
| 56 | Data Stream | 65250 | ATNXMS1 | 12752 | Yes | May | No | Aftertreatment 1 SCR 1 Modeled/Actual Storage Level |
| 57 | Data Stream | 64830 | A1SCREGT1 | 4360 | Yes | May | May | Aftertreatment 1 SCR Intake Temperature |
| 58 | Data Stream | 64830 | A1SCREGT1 | 4363 | Yes | May | May | Aftertreatment 1 SCR Outlet Temperature |
| 59 | Data Stream | 61454 | AT1IG1 | 3220 | Yes | May | May | Engine Exhaust 1 NOx 1 Reading Stable |
| 60 | Data Stream | 61457 | AT2OG1 | 3269 | Yes | May | May | Aftertreatment 2 Outlet NOx 1 Reading Stable |
| 61 | Data Stream | 37376 | HCDI1 | 5505 | Yes | May | No | Requested Fuel Mass Rate |
| 62 | Data Stream | 64488 | ASI2 | 12743 | Yes | May | No | Hydrocarbon Doser Duty Cycle |
| 63 | Data Stream | 64929 | AT1FC1 | 3480 | Yes | May | No | Aftertreatment 1 Fuel Pressure 1 |
| 64 | Data Stream | 64247 | AT1HI3 | 12765 | Yes | May | No | Aftertreatment 1 Historical Information 3 |
| 65 | Data Stream | 64247 | AT1HI3 | 12759 | Yes | May | No | Aftertreatment 1 DPF Regen Complete Starting Engine Vehicle Distance 1 |
| 66 | Data Stream | 64247 | AT1HI3 | 12760 | Yes | May | No | Aftertreatment 1 DPF Regen Complete Ending Engine Vehicle Distance 1 |

| Row # | Function | PGN | Acronym | SPN # | Primary OBD Device | Secondary OBD Device | SAE J1939 Contributor | Description |
|---|---|---|---|---|---|---|---|---|
| 67 | Data Stream | 64247 | AT1HI3 | 12761 | Yes | May | No | Aftertreatment 1 DPF Regen Complete Starting Engine Vehicle Distance 2 |
| 68 | Data Stream | 64247 | AT1HI3 | 12762 | Yes | May | No | Aftertreatment 1 DPF Regen Complete Ending Engine Vehicle Distance 2 |
| 69 | Data Stream | 64247 | AT1HI3 | 12763 | Yes | May | No | Aftertreatment 1 DPF Regen Complete Starting Engine Vehicle Distance 3 |
| 70 | Data Stream | 64247 | AT1HI3 | 12764 | Yes | May | No | Aftertreatment 1 DPF Regen Complete Ending Engine Vehicle Distance 3 |
| 71 | Data Stream | 64920 | AT1HI1 | 3525 | Yes | May | No | Aftertreatment 1 Total Number of Active Regenerations |
| 72 | Data Stream | 64920 | AT1HI1 | 3726 | Yes | May | No | Aftertreatment 1 Diesel Particulate Filter Total Number of Passive Regenerations |
| 73 | Data Stream | 64247 | AT1HI3 | 12766 | Yes | May | No | Aftertreatment 1 Total Time SCR Inlet Below 200 Celsius |

(1) Only one device shall support VIN. Most commonly, it will be the primary OBD device (e.g., engine), but it may be provided by another component.

# APPENDIX I - DIAGNOSTIC TROUBLE CODE LAMP STATUS REQUIREMENTS

## I.1 LOGIC TABLES FOR SETTING LAMP AND FLASH STATUS PARAMETER PAIRINGS

These tables list the proper setting of the lamp status and flash lamp status parameters to convey each of the different lamp operation conditions. These tables are intended to help the ECU designer select the proper lamp status and flash lamp status parameter settings to convey the proper meaning.

### I.1.1 MIL Lamp Control Logic Tables

These tables list the proper setting of the malfunction indicator lamp (SPN 1213 for DM1) and flash malfunction indicator lamp (SPN 3038 for DM1) to convey each of the different MIL conditions. Reserved or unused combinations in I.1.1.1 and I.1.1.2 shall result in the same behavior as Lamp Off.

#### I.1.1.1 ARB Regulations

| Condition to Convey | Malfunction Indicator Lamp (SPN 1213) | Flash Malfunction Indicator Lamp (SPN 3038) |
|---|---|---|
| MIL-Off (alternate) | $00_b$ (Lamp Off) | $00_b$ (Do Not Flash) |
| MIL-Off (preferred) | $00_b$ (Lamp Off) | $11_b$ (Do Not Flash) |
| MIL-On | $01_b$ (Lamp On) | $11_b$ (Do Not Flash) |
| MIL Slow Flash | $01_b$ (Lamp On) | $00_b$ (Slow Flash) |
| MIL Fast Flash | $01_b$ (Lamp On) | $01_b$ (Fast Flash) |
| Do Not Care | $11_b$ (Not Available) | $11_b$ (Do Not Flash) |

#### I.1.1.2 WWH Regulations and EURO VI

| Condition to Convey | Malfunction Indicator Lamp (SPN 1213) | Flash Malfunction Indicator Lamp (SPN 3038) | Annex 9B Activation Mode |
|---|---|---|---|
| MIL-Off (alternate) | $00_b$ (Lamp Off) | $00_b$ (Do Not Flash) | Mode 1 |
| MIL-Off (preferred) | $00_b$ (Lamp Off) | $11_b$ (Do Not Flash) | Mode 1 |
| MIL-On | $01_b$ (Lamp On) | $11_b$ (Do Not Flash) | Mode 4 |
| MIL Slow Flash | $01_b$ (Lamp On) | $00_b$ (Slow Flash) | Mode 4 (Slow Flash) [no regulated use] |
| MIL Fast Flash | $01_b$ (Lamp On) | $01_b$ (Fast Flash) | Mode 4 (Fast Flash) [no regulated use] |
| Class C DTC Active | $01_b$ (Lamp On) | $10_b$ (Class C DTC) | Mode 2 |
| Class C DTC Previously Active | $00_b$ (Lamp Off) | $10_b$ (Class C DTC) | Mode 1 [no regulated use] |
| Short MI Active | $10_b$ (Short MIL) | $01_b$ (Fast Flash) | Mode 3 |
| Short MI Previously Active | $10_b$ (Short MIL) | $00_b$ (Slow Flash) | Mode 3 for operating sequences 1-3, mode 1 for operating sequences >3 |
| Do Not Care | $11_b$ (Not Available) | $11_b$ (Do Not Flash) | No Function [no regulated use] |

I.1.2 RSL, AWL, Protect Lamp Control Logic Tables

These tables list the proper setting of the lamp status and flash lamp status parameters to convey each of the different lamp operation conditions.

| Condition to Convey | Lamp Status Parameters (SPNs 623, 624, 987) | Flash Lamp Status Parameters (SPNs 3039, 3040, 3041) |
|---|---|---|
| Lamp Off (alternate) | $00_b$ (Lamp Off) | $00_b$ (Do Not Flash) |
| Lamp Off (preferred) | $00_b$ (Lamp Off) | $11_b$ (Do Not Flash) |
| Lamp On | $01_b$ (Lamp On) | $11_b$ (Do Not Flash) |
| Slow Flash | $01_b$ (Lamp On) | $00_b$ (Slow Flash) |
| Fast Flash | $01_b$ (Lamp On) | $01_b$ (Fast Flash) |
| Do Not Care | $11_b$ (Not Available) | $11_b$ (Do Not Flash) |

I.2 LOGIC TABLES FOR INTERPRETING LAMP AND FLASH STATUS PARAMETER PAIRINGS

These tables list the requested behavior of the lamp based upon the lamp status and flash lamp status parameter values reported by an ECU. These tables are intended to help the lamp control component designer interpret the proper requested action based upon the reported lamp status and flash lamp status parameter values for a lamp.

I.2.1 MIL Lamp Control Logic Tables

These tables list the requested behavior (and meaning) of the lamp based upon the lamp status (SPN 1213 for DM1) and flash lamp status (SPN 3038 for DM1) parameter values reported by an ECU. DM1 is the driver for the dash lamps, but the other DMs that convey DTCs and lamps will also follow these conventions.

I.2.1.1 ARB Regulations

| | | Malfunction Indicator Lamp (SPN 1213) | | | |
|---|---|---|---|---|---|
| | | $00_b$ (Lamp Off) | $01_b$ (Lamp On) | $10_b$ (Short MIL) | $11_b$ (Not Available) |
| Flash MIL (SPN 3038) | $00_b$ (Slow Flash) | Lamp Off (alternate) | Lamp Slow Flash | SAE Reserved | SAE Reserved |
| | $01_b$ (Fast Flash) | Lamp Off | Lamp Fast Flash | SAE Reserved | SAE Reserved |
| | $10_b$ (Class C DTC) | SAE Reserved | SAE Reserved | SAE Reserved | SAE Reserved |
| | $11_b$ (Do Not Flash) | Lamp Off (preferred) | Lamp On | SAE Reserved | Do Not Care |

I.2.1.2 WWH Regulations and EURO VI

The table below shows the application of the WWH OBD activation modes adopted into Annex 9B of UN ECE R49 for all combinations of SPNs 1213 and 3038.

| | | Malfunction Indicator Lamp (SPN 1213) | | | |
|---|---|---|---|---|---|
| | | $00_b$ (Lamp Off) | $01_b$ (Lamp On) | $10_b$ (Short MIL) | $11_b$ (Not Available) |
| Flash MIL (SPN 3038) | $00_b$ (Slow Flash) | Lamp Off (Mode 1) | Lamp Slow Flash (Mode 4) | Short MI Previously Active (Mode 1) | SAE reserved (Mode 1) |
| | $01_b$ (Fast Flash) | Lamp Off (Mode 1) | Lamp Fast Flash (Mode 4) | Short MI Active (Mode 3) | SAE reserved (Mode 1) |
| | $10_b$ (Class C DTC) | Lamp Off, Class C DTC Previously Active (Mode 1) | Lamp Off, Class C DTC Active (Mode 2) | SAE Reserved (Mode 1) | SAE Reserved (Mode 1) |
| | $11_b$ (Do Not Flash) | Lamp Off (Mode 1) | Lamp On (Mode 1) | SAE Reserved (Mode 1) | Do Not Care (Mode 1) |

I.2.2 RSL, AWL, Protect Lamp Control Logic Tables

These tables list the proper setting of the lamp status and flash lamp status parameters to convey each of the different lamp operation conditions.

| | | Lamp Status Parameters (SPNs 623, 624, 987) | | | |
|---|---|---|---|---|---|
| | | $00_b$ (Lamp Off) | $01_b$ (Lamp On) | $10_b$ (Reserved) | $11_b$ (Not Available) |
| Flash Lamp (SPNs 3039, 3040, 3041) | $00_b$ (Slow Flash) | Lamp Off | Lamp Slow Flash | SAE Reserved | SAE Reserved |
| | $01_b$ (Fast Flash) | Lamp Off | Lamp Fast Flash | SAE Reserved | SAE Reserved |
| | $10_b$ (Reserved) | SAE Reserved | SAE Reserved | SAE Reserved | SAE Reserved |
| | $11_b$ (Do Not Flash) | Lamp Off | Lamp On | SAE Reserved | Do Not Care |

# APPENDIX J - OBD READINESS BIT CONSTRUCTION EXAMPLES

## J.1 LOGIC TABLES FOR CONSTRUCTION OF THE OBD READINESS BIT INFORMATION

The readiness code status bits (i.e., those in DM5) are used in part or in whole to ensure a vehicle (engine) is ready for an emission system inspection. The definition of the complete or incomplete status is provided in regulation(s), including the impact of specific service actions on readiness status. If a status bit for a particular component or system is set to complete, then the OBD system has had the opportunity to run all of the diagnostics relevant to that particular component or system. A component or system shall also indicate “complete” if, after the requisite number of decisions necessary for determining MIL status has been fully executed (i.e., producing a pass or fail decision), the monitor indicates a malfunction for the component or system. Normal vehicle shutdown (i.e., key off, engine off) shall not cause the status to indicate “not complete.”

For example, on engine start-up, it is likely that an EGR monitor might require operation at speed and torque points beyond the idle condition. Therefore, if the system had its diagnostic trouble codes erased and then the engine was restarted, the readiness code for EGR would not be set until the condition for the EGR monitor tests had been executed. It is also expected that other tests, not just the EGR monitors, would need to be performed before the readiness code could be set. Once set to complete, the readiness code status bits shall not change to incomplete each time the vehicle (engine) is powered down.

Definition:

A. The required malfunction for a required monitor (e.g., fuel pressure high, fuel pressure low, etc.).

B. For each required malfunction, there may be different technology across the industry. These different technologies are represented by related SPN.

R. Some monitors have a readiness status requirement, e.g., fuel system. This readiness status (i.e., readiness bit) is concluded based on the pass or fail conclusion of the specific monitors for a system. Those monitors, represented each by a DTC, must either all run and pass (e.g., a logical AND of all monitors is required to set the readiness bit for that system when all test results indicate that system is good). If, on the other hand, any are failing, then a logical OR is used to set the OBD readiness bit (e.g., if the EGR system has six sub-systems). If all of six sub-systems are monitored and a pass decision is made, then the readiness bit will be set to complete. If any one of the sub-system monitors declare a fail, then the EGR readiness bit will be set to complete. See Figures J1 and J2 for pictorial view.

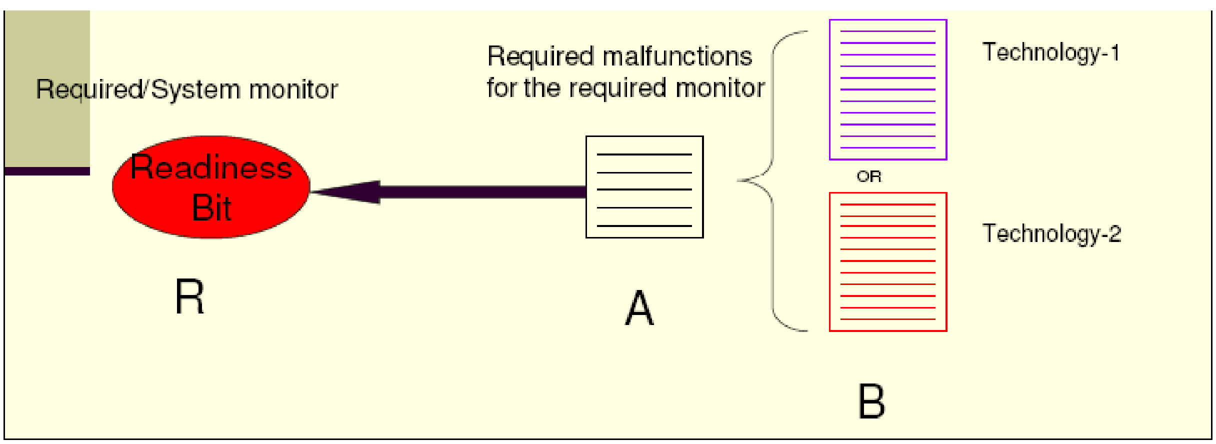


***Figure J1 - OBD readiness bit construction***

Example of EGR system readiness:

EGR system has one requirement to detect low flow malfunction. To detect such malfunction, there may be different technologies/approaches. Different technologies/approaches are illuminated in Figure J2 as B, for example. See Table J1 for EGR system readiness bit mapping. The intent of Table J1 is to make a point that how readiness bit will be set for required system monitors. As per ARB HD OBD regulations, some monitors do not map to readiness bit (e.g., EGR feedback control system).

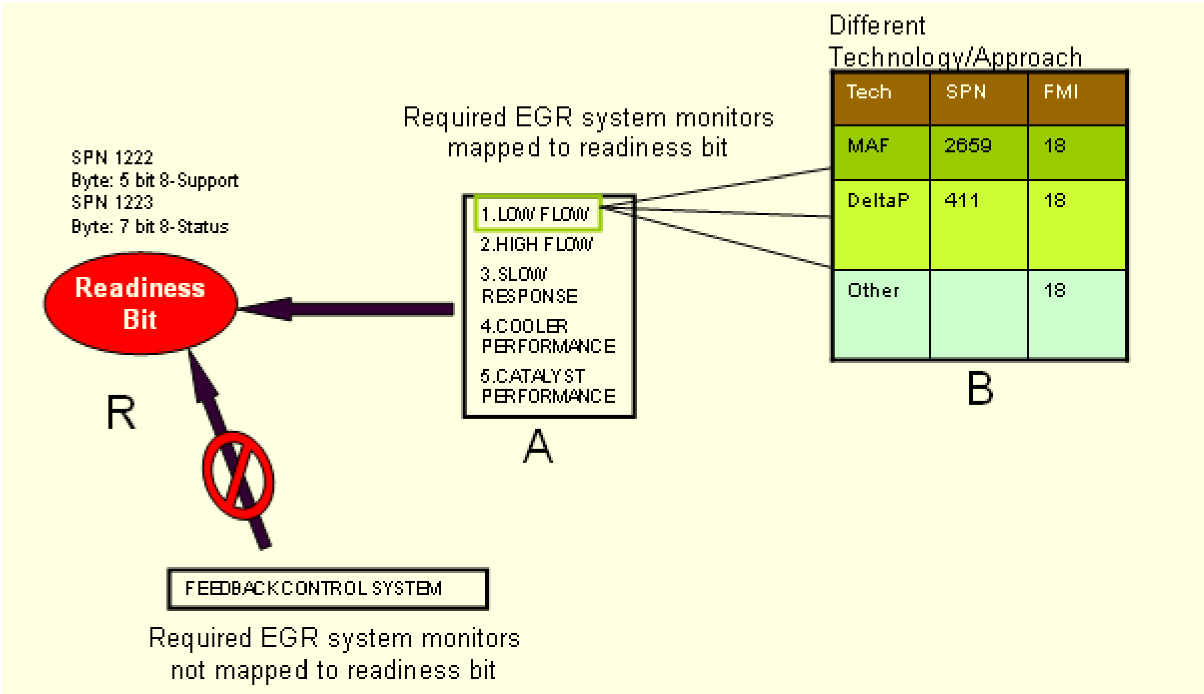


***Figure J2 - OBD readiness bit construction***

***Table J1 - Example EGR OBD readiness construction***

| Monitor Name | 1971.1 Reference | Monitoring | DTC SPN | DTC FMI | SPN-FMI Description |
|---|---|---|---|---|---|
| EGR | (e)3.2.1 | Low Flow | 3058 | 18 | EGR System Monitor -- Data Valid But Below Normal Operating Range - Moderately Severe Level |
| | (e)3.2.1 | Low Flow | 2659 | 18 | Engine Exhaust Gas Recirculation 1 (EGR1) Mass Flow Rate -- Data Valid But Below Normal Operating Range - Moderately Severe Level |
| | (e)3.2.1 | Low Flow | 411 | 18 | Engine Exhaust Gas Recirculation 1 Differential Pressure -- Data Valid But Below Normal Operating Range - Moderately Severe Level |
| | (e)3.2.2 | High Flow | 3058 | 16 | EGR System Monitor -- Data Valid But Above Normal Operating Range - Moderately Severe Level |
| | (e)3.2.2 | High Flow | 2659 | 16 | Engine Exhaust Gas Recirculation 1 (EGR1) Mass Flow Rate -- Data Valid But Above Normal Operating Range - Moderately Severe Level |
| | (e)3.2.2 | High Flow | 411 | 16 | Engine Exhaust Gas Recirculation 1 Differential Pressure -- Data Valid But Above Normal Operating Range - Moderately Severe Level |
| | (e)3.2.3 | Slow Response | 2791 | 10 | Engine Exhaust Gas Recirculation 1 (EGR1) Valve Control -- Abnormal Rate of Change |
| | (e)3.2.4 | Feedback Control - Time to Control | 3058 | 10 | EGR System Monitor -- Abnormal Rate of Change |
| | (e)3.2.4 | Feedback Control - Default or Open Loop | 3058 | 13 | EGR System Monitor -- Out of Calibration |
| | (e)3.2.4 | Feedback Control - Control Limits | 3058 | 7 | EGR System Monitor -- Mechanical system not responding or out of adjustment |
| | (e)3.2.5 | Cooler performance | 4752 | 18 | Engine Exhaust Gas Recirculation 1 (EGR1) Cooler Efficiency -- Data Valid But Below Normal Operating Range - Moderately Severe Level |
| | (e)3.2.6 | Catalyst Performance | 5929 | 18 | Engine Exhaust Gas Recirculation 1 Catalyst 1 Performance Monitor -- Data Valid But Below Normal Operating Range - Moderately Severe Level |

# APPENDIX K - EURO VI DEFINITIONS

The Euro VI definitions provided in Annex 9B of UN/ECE R49 leverage the WWH OBD discriminatory display system concept. Table K1 shows the SPN value settings for selected DM36 HRWV, DM37 HRWS, and DM1 SPNs by UN/ECE R49-05 Annex 9B MI activation modes. All data values in Table K1 are given in base 2 representation. Continuous malfunction indicator SPNs will only indicate a continuous MI when MI Activation Mode 4 is active. The malfunction indicator display strategy SPNs are shown with the discriminatory MI strategy setting, $01_b$. Local regulations may require the non-discriminatory strategy, which would be indicated by value $00_b$. Appendix I shows the inferred MI activation modes for value combinations of SPN 1213 and 3038 not shown in Table K1.

***Table K1 - DM36 HRWV, DM37 HRWS, and DM1 active DTCs content by MI activation mode for Euro VI***

| | DM36 HRWV PGN 64868 Content | | | DM37 HRWS PGN 64867 Content | | | DM1 Active DTCs PGN 65226 Content | |
|---|---|---|---|---|---|---|---|---|
| UN/ECE R49-05 Annex 9B Paragraph 4.6<br><br>MI Activation Mode | Vehicle Continuous Malfunction Indicator SPN 4134 | Vehicle Malfunction Indicator Display Strategy SPN 4135 | Vehicle Malfunction Indicator Activation Mode SPN 4136 | System Continuous Malfunction Indicator SPN 4140 | System Malfunction Indicator Display Strategy SPN 4141 | System Malfunction Indicator Activation Mode SPN 4142 | Malfunction Indicator Lamp SPN 1213 | Flash Malfunction Indicator Lamp SPN 3038 |
| Mode 1 Off | $00_b$ | $01_b$ | $000_b$ | $00_b$ | $01_b$ | $000_b$ | $00_b$ | $11_b$ |
| Mode 2 On Demand MI | $00_b$ | $01_b$ | $001_b$ | $00_b$ | $01_b$ | $001_b$ | $01_b$ | $10_b$ |
| Mode 3 Short MI | $00_b$ | $01_b$ | $010_b$ | $00_b$ | $01_b$ | $010_b$ | $10_b$ | $01_b$ |
| Mode 4 Continuous MI | $01_b$ | $01_b$ | $011_b$ | $01_b$ | $01_b$ | $011_b$ | $01_b$ | $11_b$ |

Vehicles must support both DM36 HRWV PGN 64868 and DM37 HRWS PGN 64867. The content in DM36 HRWV shall show the aggregated status of all individual DM37 HRWS messages from compliant emissions control (sub-) systems. Thus, if one system reports Mode 4 in SPN 4142, the vehicle must report Mode 4 in SPN 4144. Priority rules for aggregating MI activation modes are given in paragraph 4.6.4.2 of Annex 9B. Future legislation may extend the aggregation to other safety related systems. Refer to SAE J1939-03 for additional discussion regarding the DM36 and DM37 messages.

For WWH OBD and EURO VI fault reporting, the settings for malfunction indicator lamp (SPN 1213) and flash malfunction indicator lamp (SPN 3038) apply for all fault code displays, including all those defined in Table 27. There is one setting for malfunction indicator lamp SPN 1213 and flash malfunction indicator SPN 3038 for Activation Mode 3, Short MI given in the definitions for SPN 1213 and SPN 3038. The value pair ($10_b$, $01_b$) is used when there is an active B1 fault. The value pair ($00_b$, $10_b$) indicates there is an active C1 fault that requires an Activation Mode 2 display.

UN/ECE R49-05 Annex 9B paragraph 4.6 defines MI activation modes that govern the MI illumination behavior patterns given in Figures B1 and B2 of paragraph 4.6.2. The patterns also vary by engine operating mode and vehicle readiness (defined by SPN 4143, Vehicle Incomplete Monitor Count). Figures C1 and C2 in Annex 9B show MI counter and B1 counter behavior for DM39 and DM40 message content. Figures 1 through 5 of Annex 9B, Appendix 2, show the relative signal timing for pending, active and confirmed, and previously active DTC status classes and additional counter behavior.